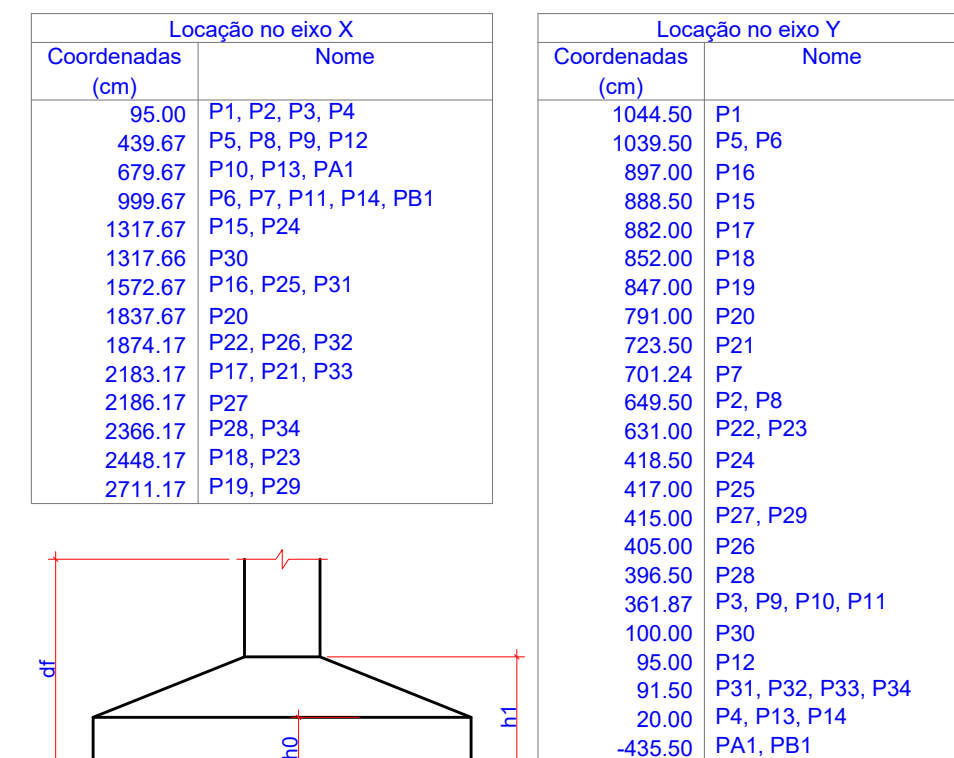
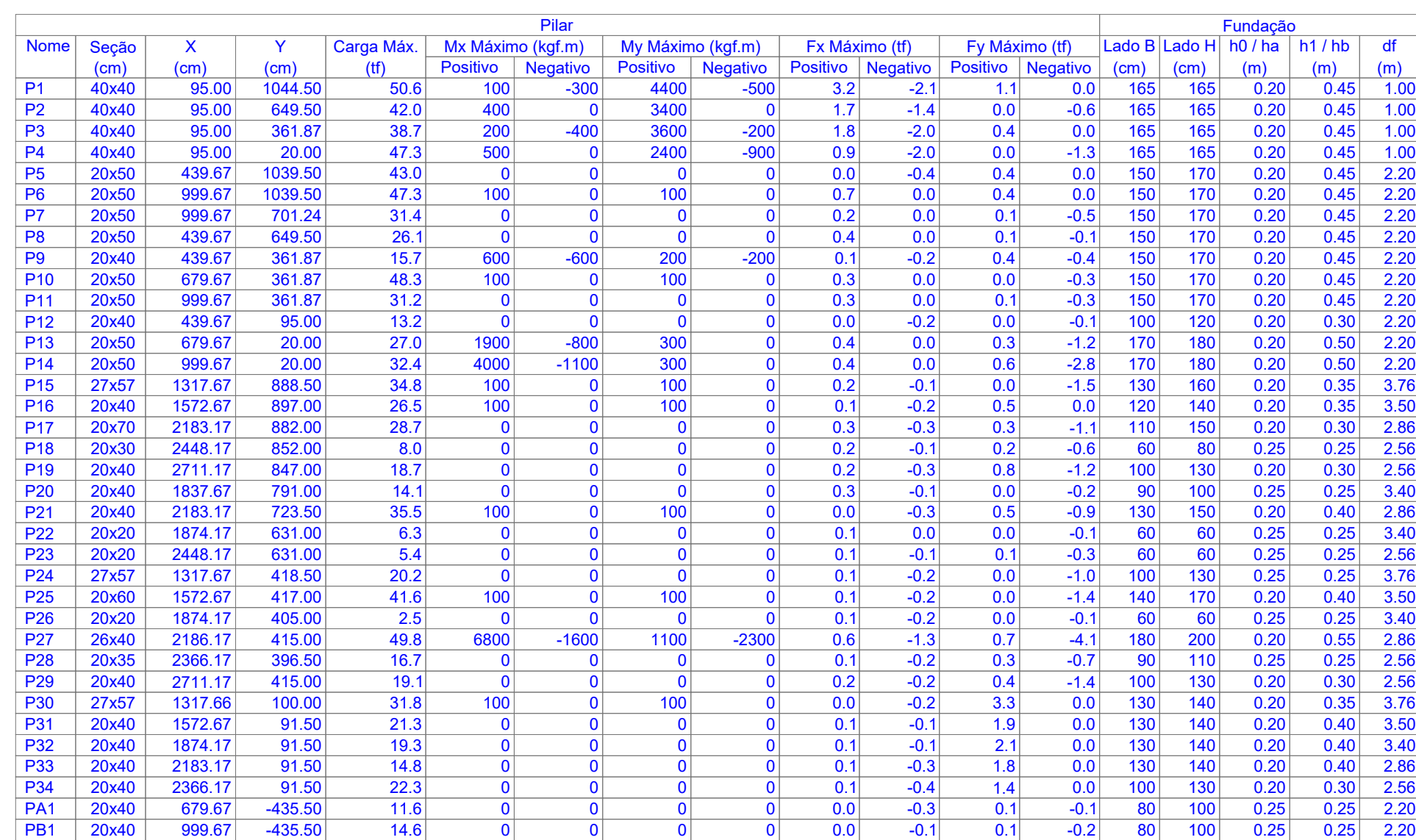
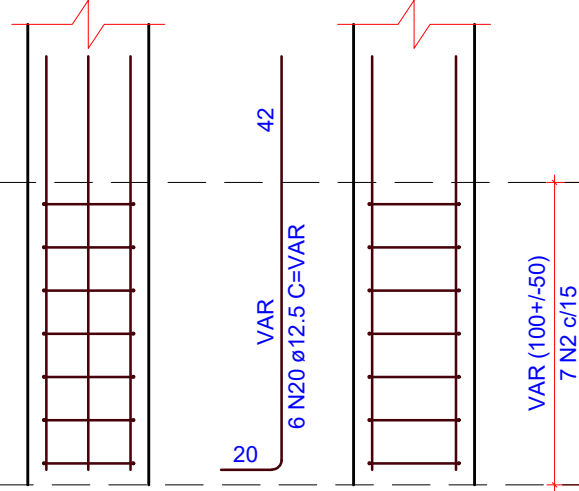
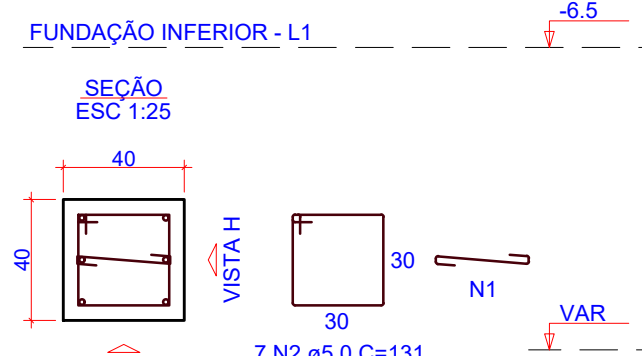
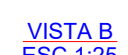
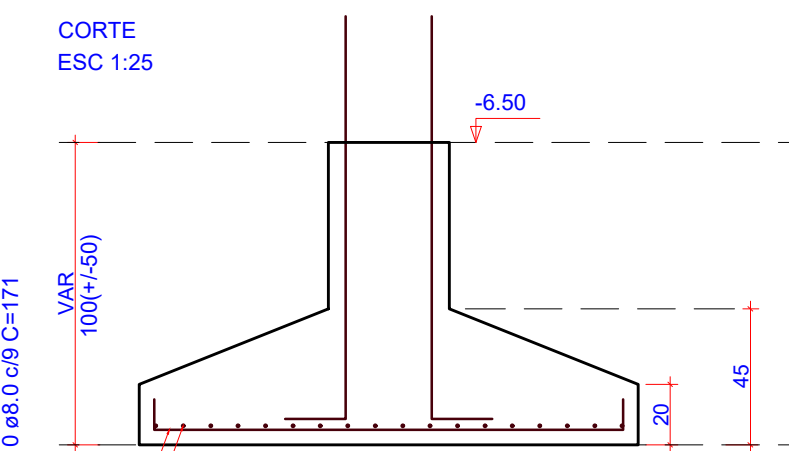




Nº	REVISÃO	DATA	CLIENTE JULIANO CORBELLINI			
A	EMISSÃO INICIAL	15/12/2025	LOCALIDADE – MUNICÍPIO PORTO ALEGRE - RS			
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO RUA ALCEBÍADES ANTÔNIO DOS SANTOS- 356 - LOTE 207			
			PROJETO ESTRUTURAL PROJETO ESTRUTURAL FUNDAÇÕES PLANTA DE LOCAÇÃO			
			PROJETO Engº Rafael Buss	DESENHO MMD	DATA 15/12/2025	FOLHA Nº EST-0010-A
			SERIAL 133.073-9	ESCALA INDICADA	DATA TOP:	

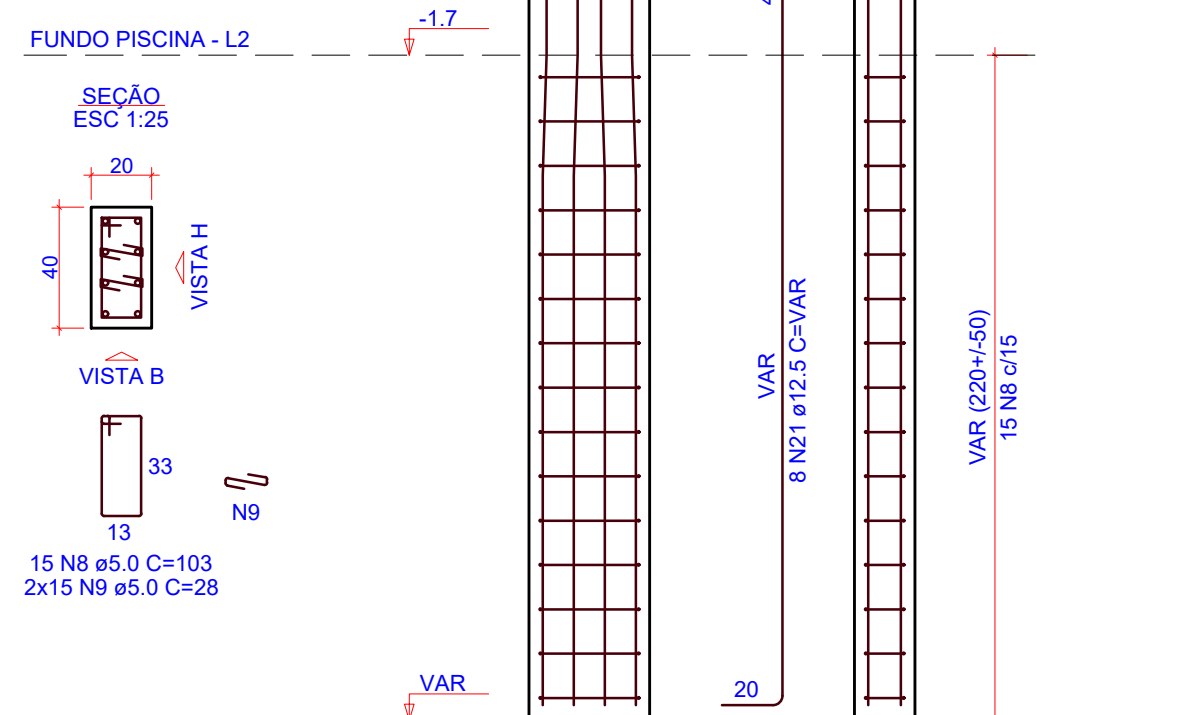
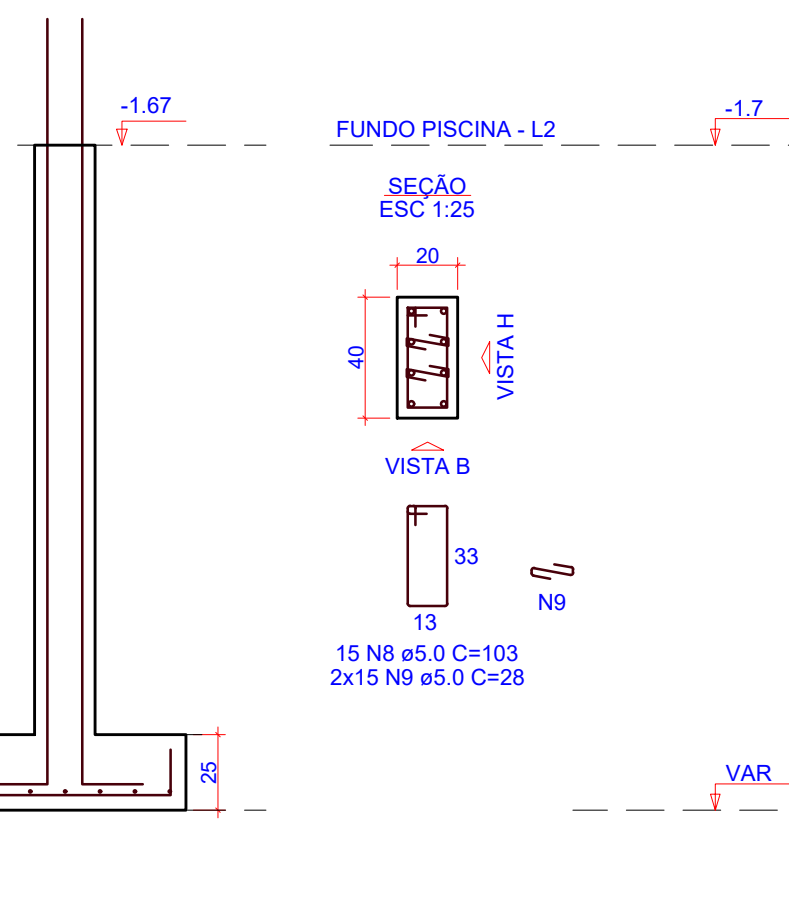
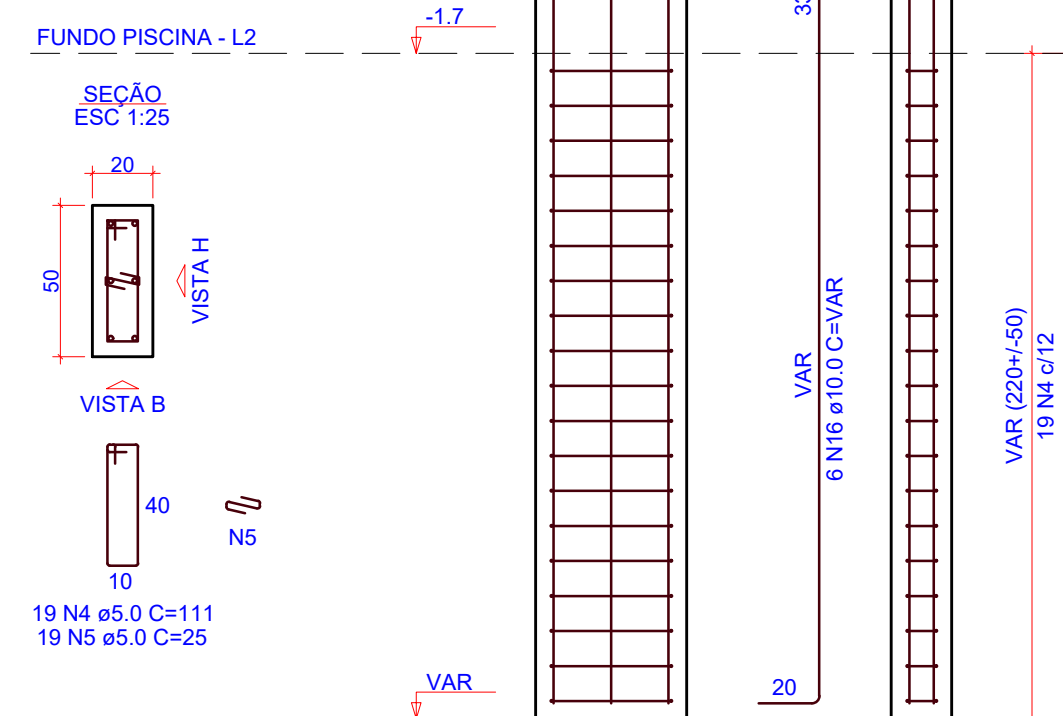
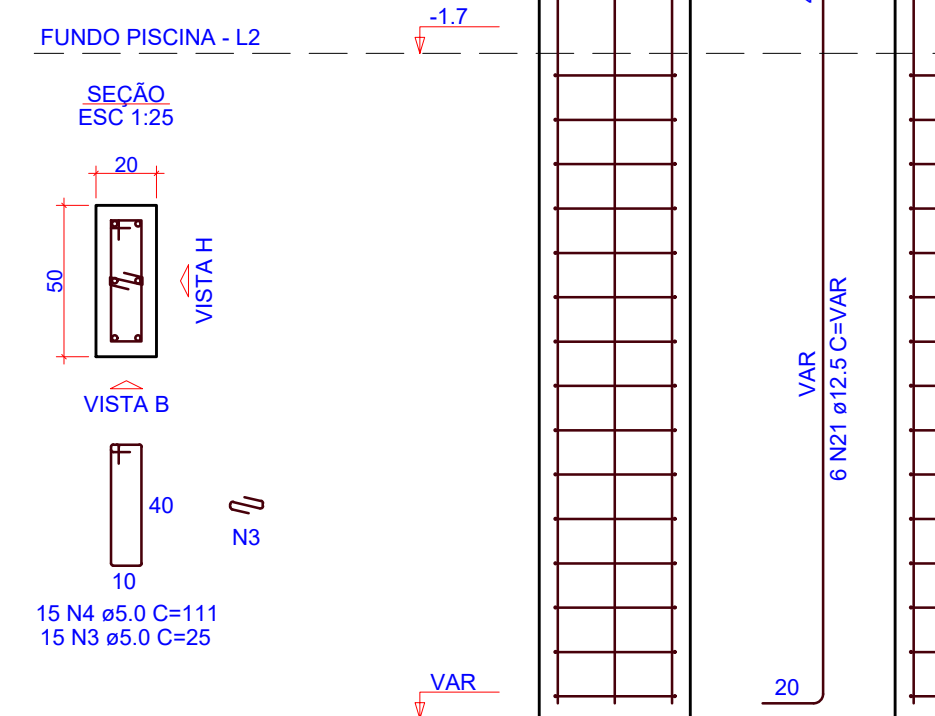
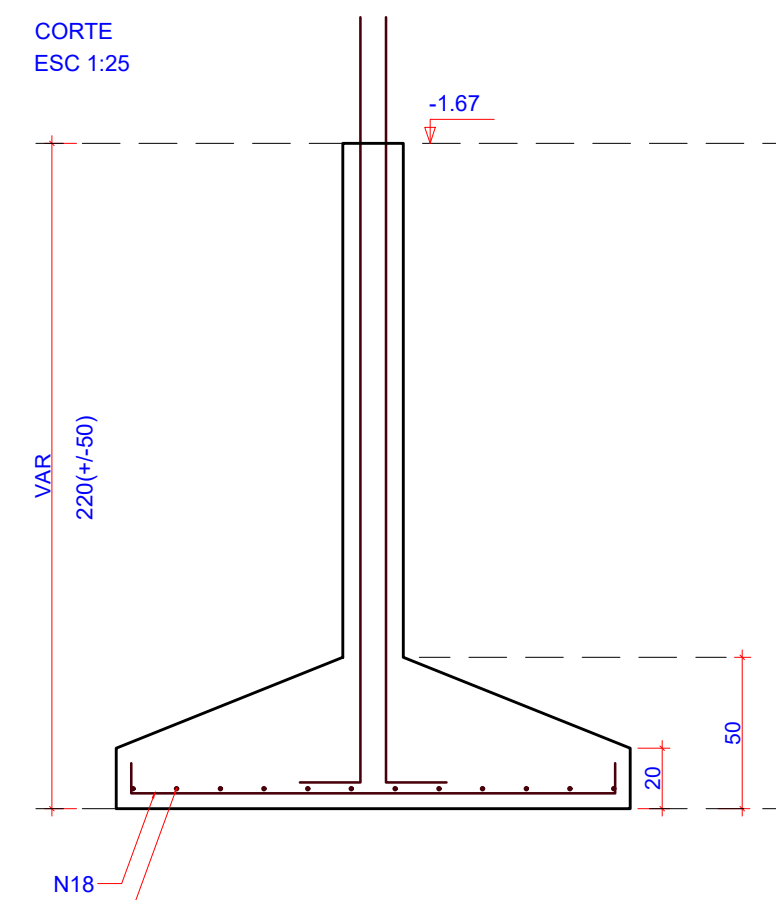
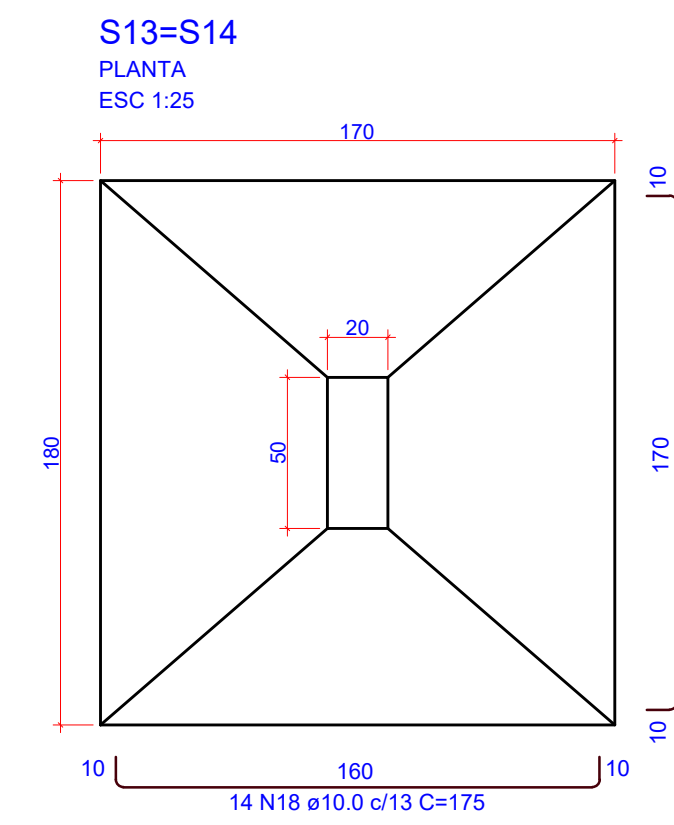
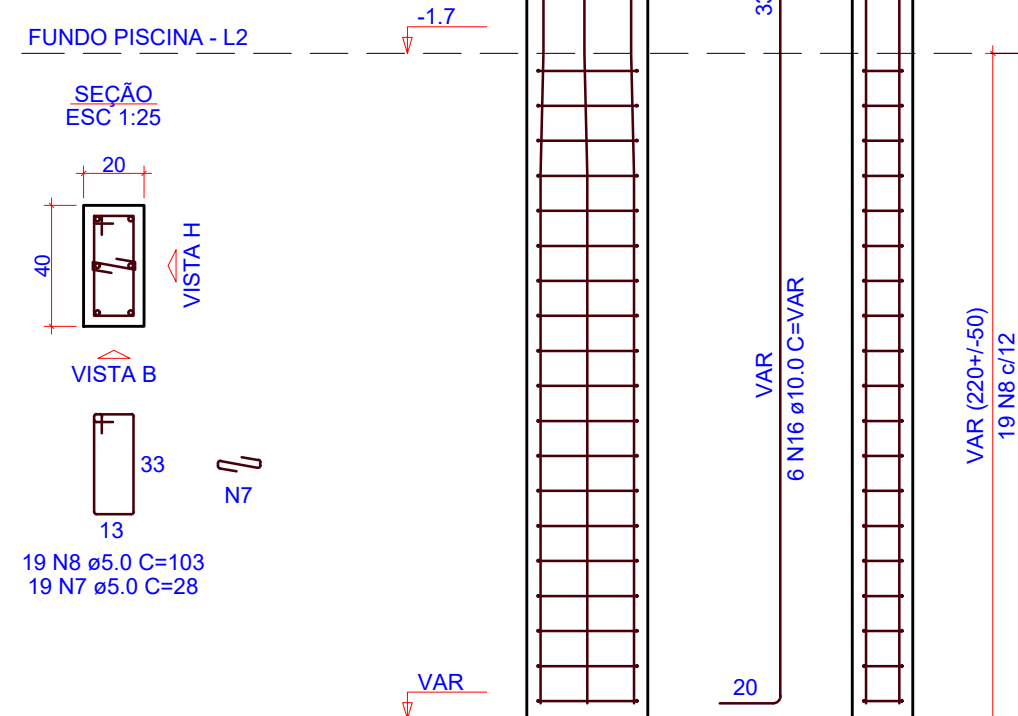
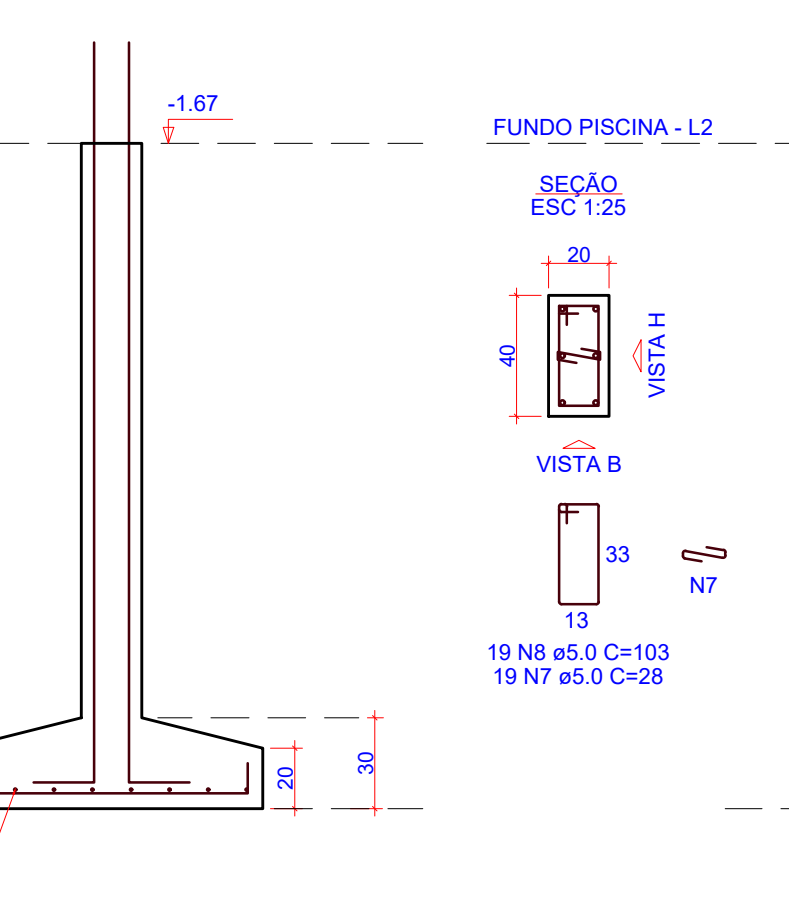
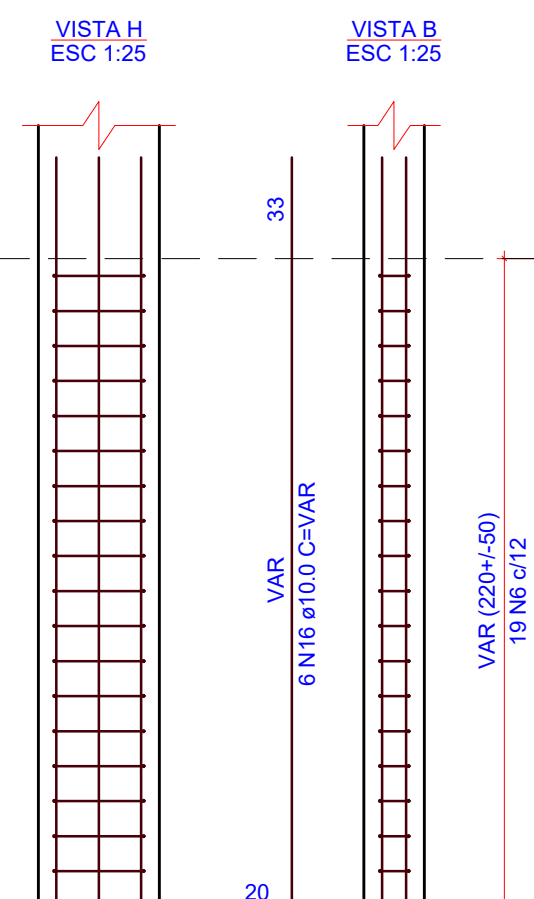
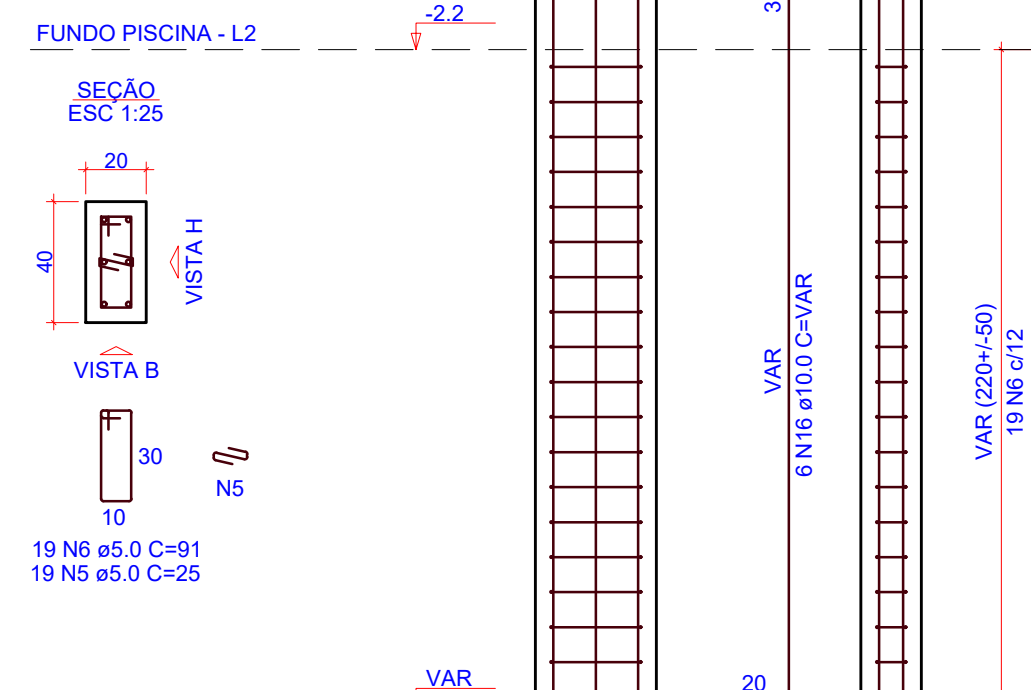
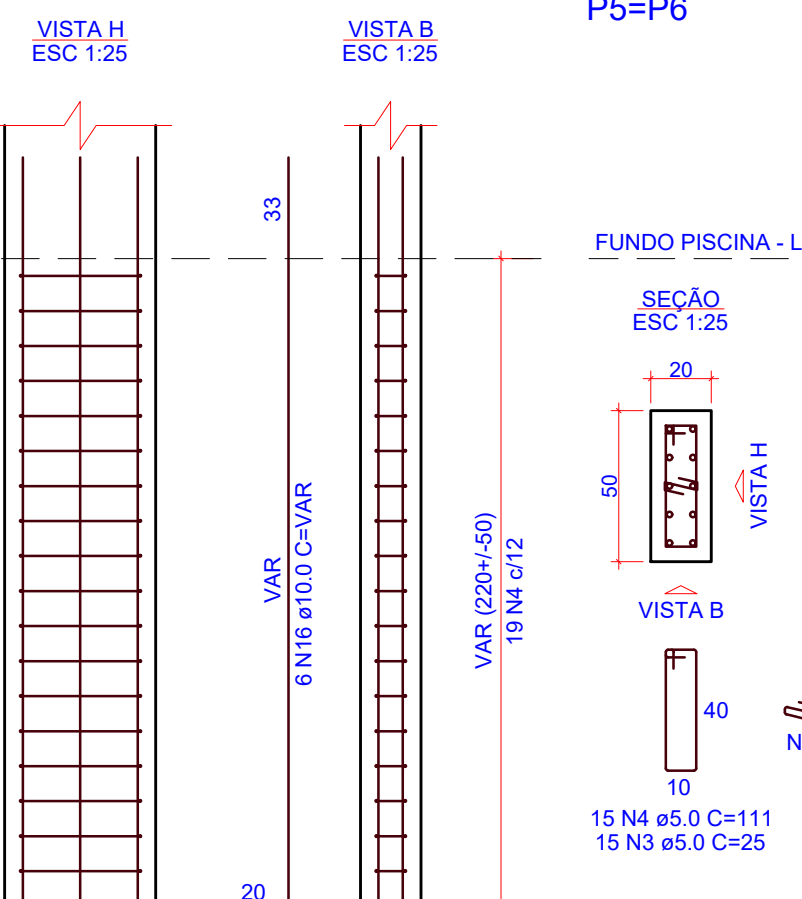
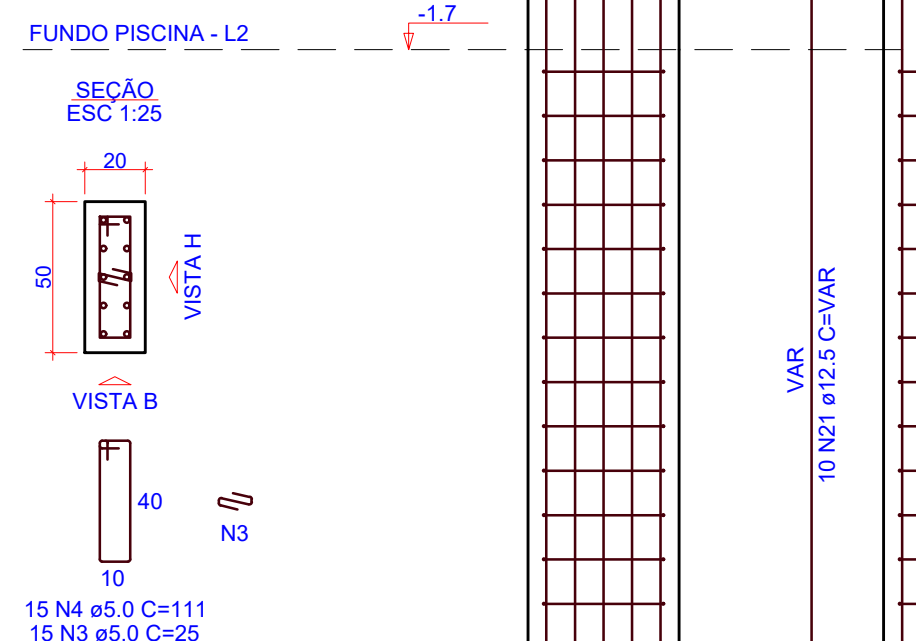
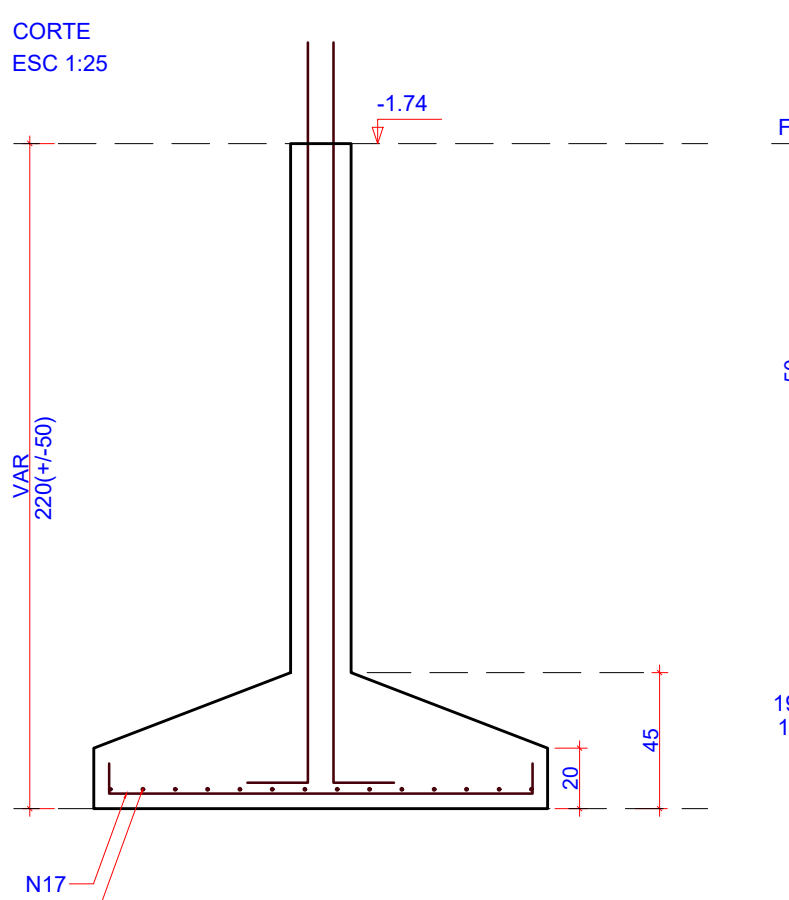


RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (t)	QUANT + 10% (Barras)	PESO + 10% (kg)
CA50	8.0	472.9	44	205.3
	10.0	335.3	31	227.4
	12.5	152	14	161.1
CA60	5.0	355.4	-	60.3
PESO TOTAL (kg)				
CA50	593.8			
CA60	60.3			

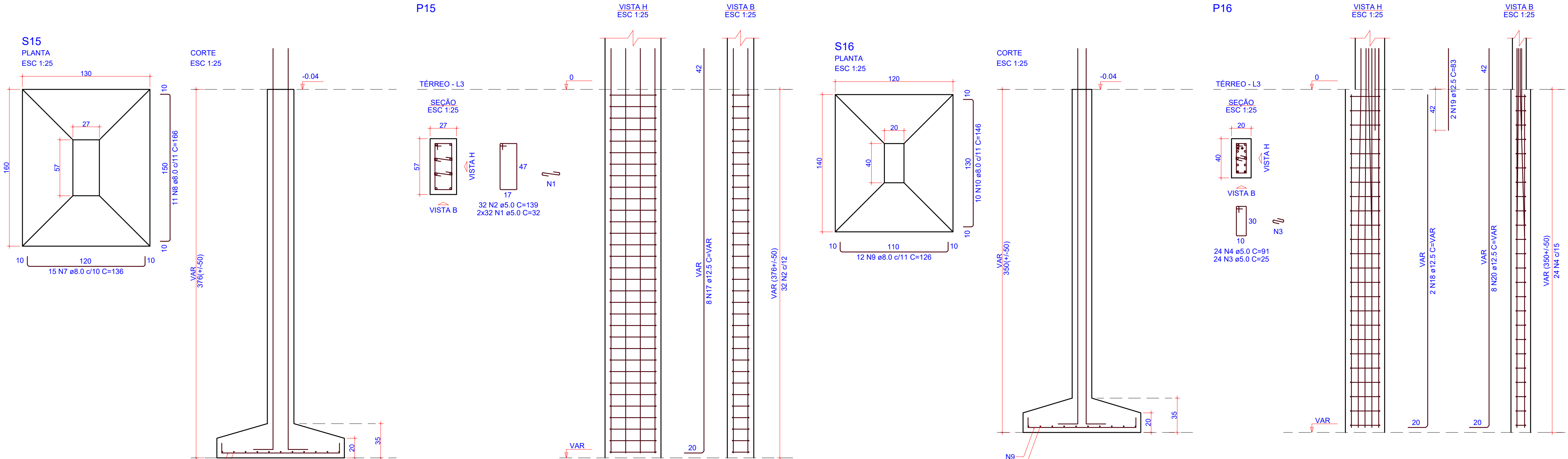
Volume de concreto (C-30) = 15.74 m³

Área de forma = 51.33 m²



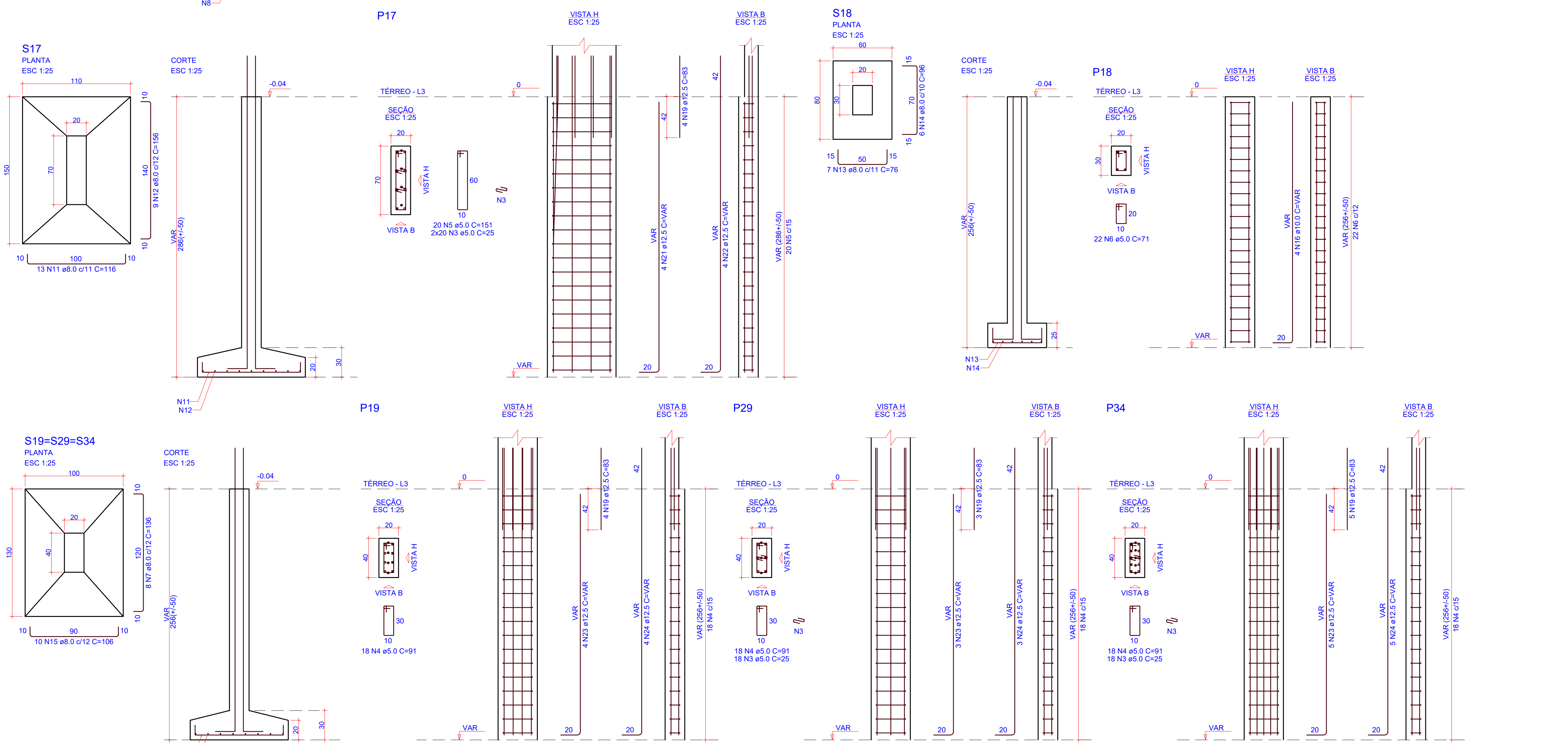
ESCALA 1:25

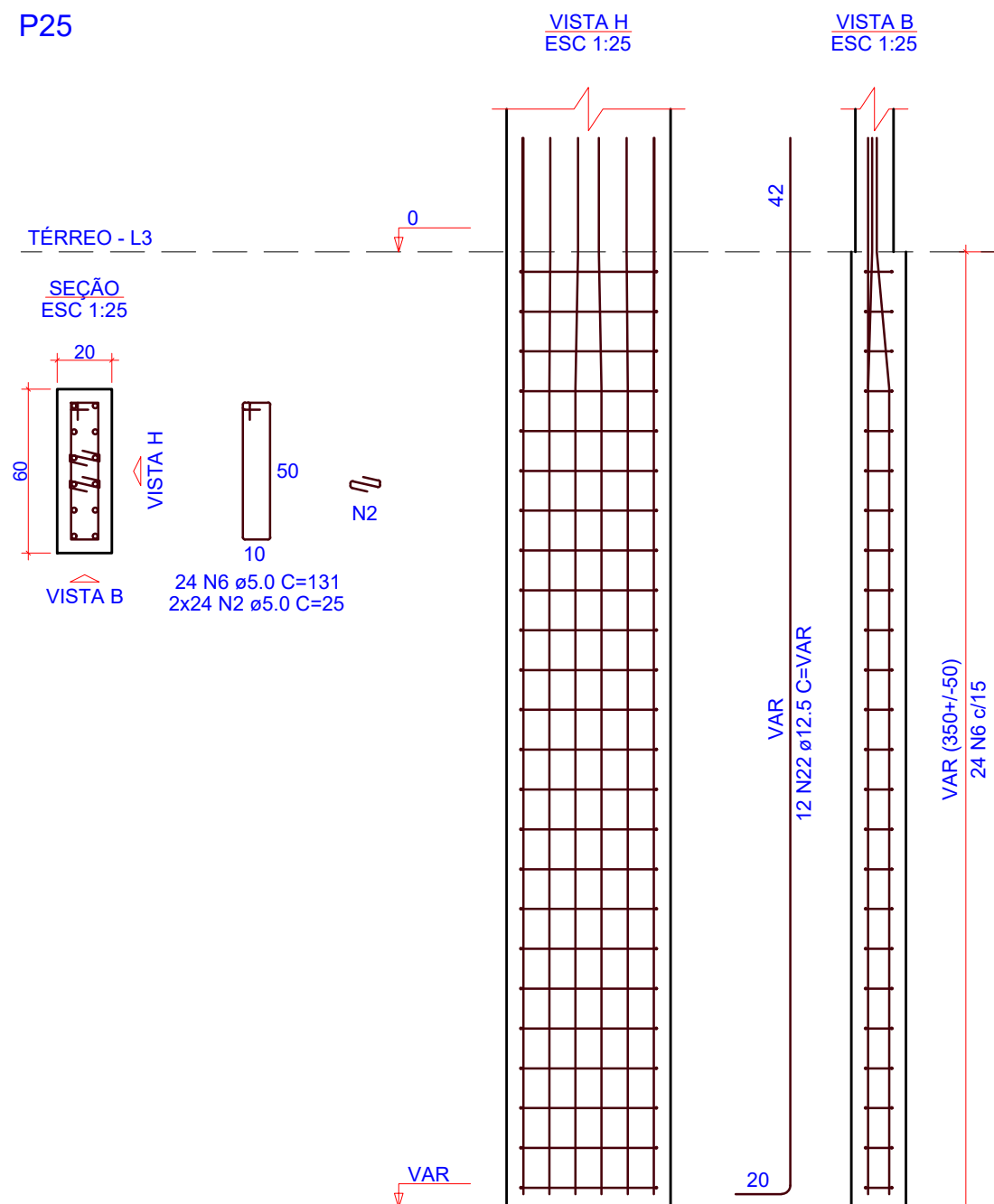
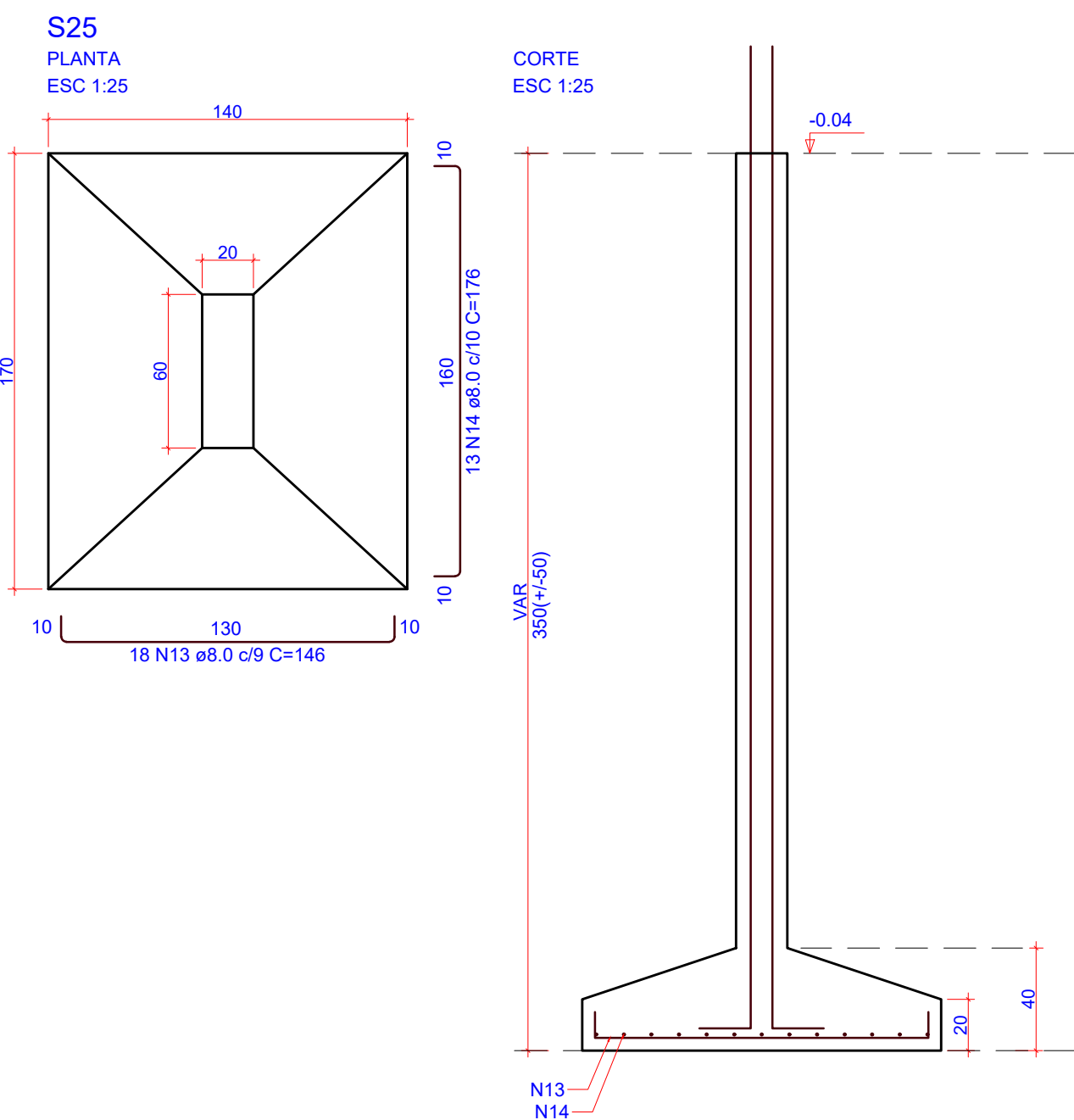
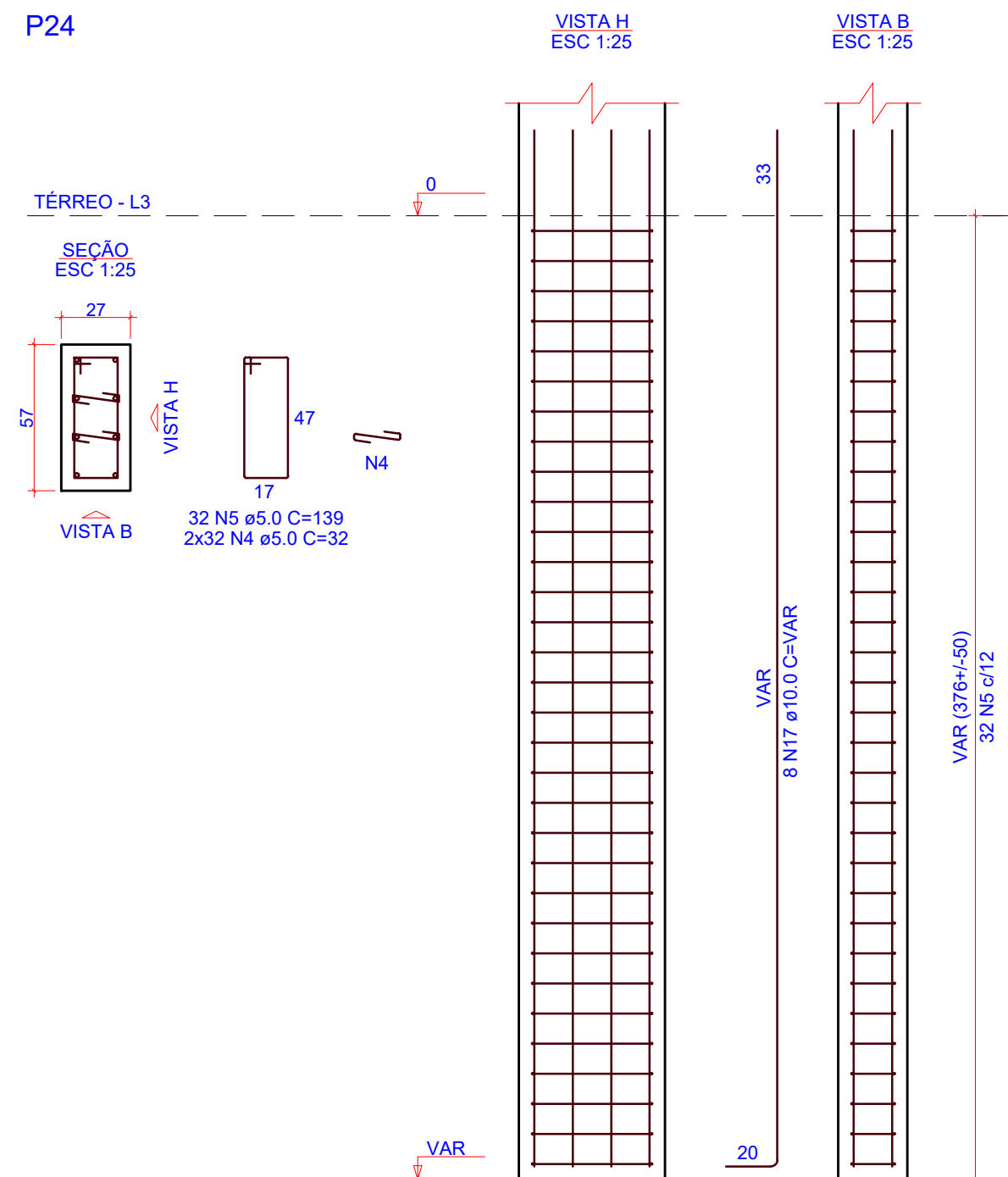
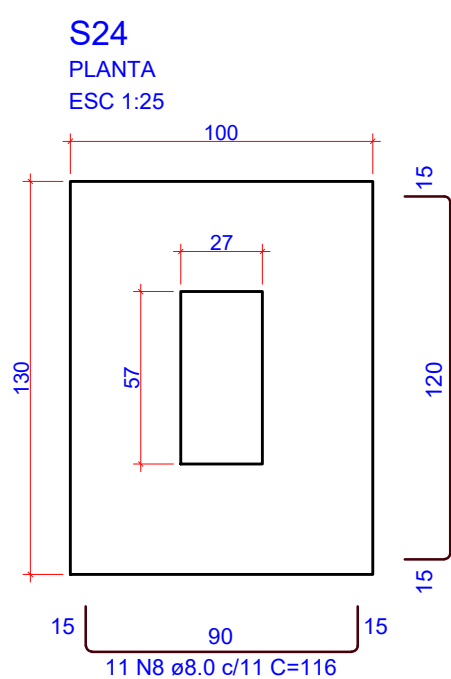
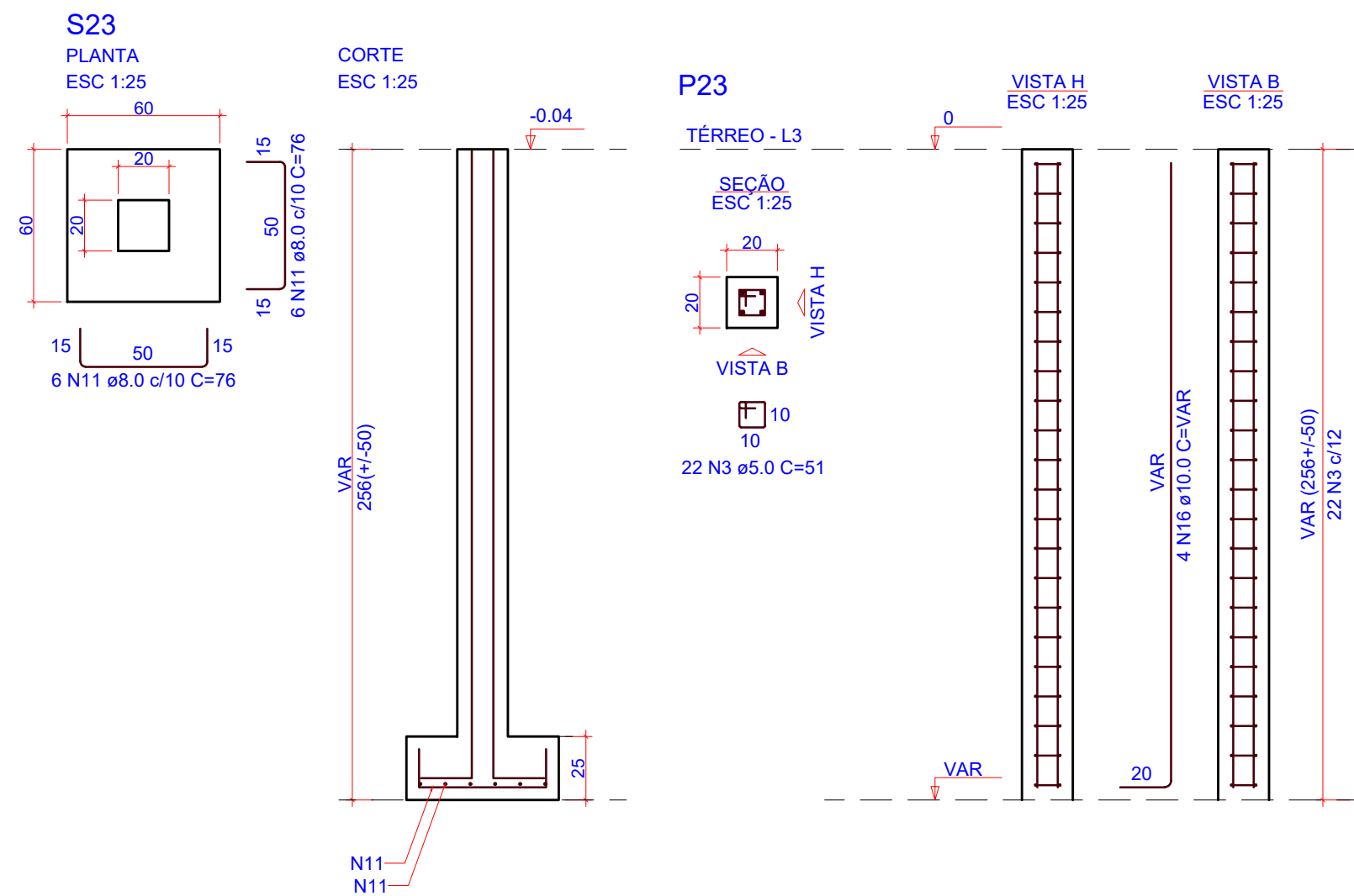
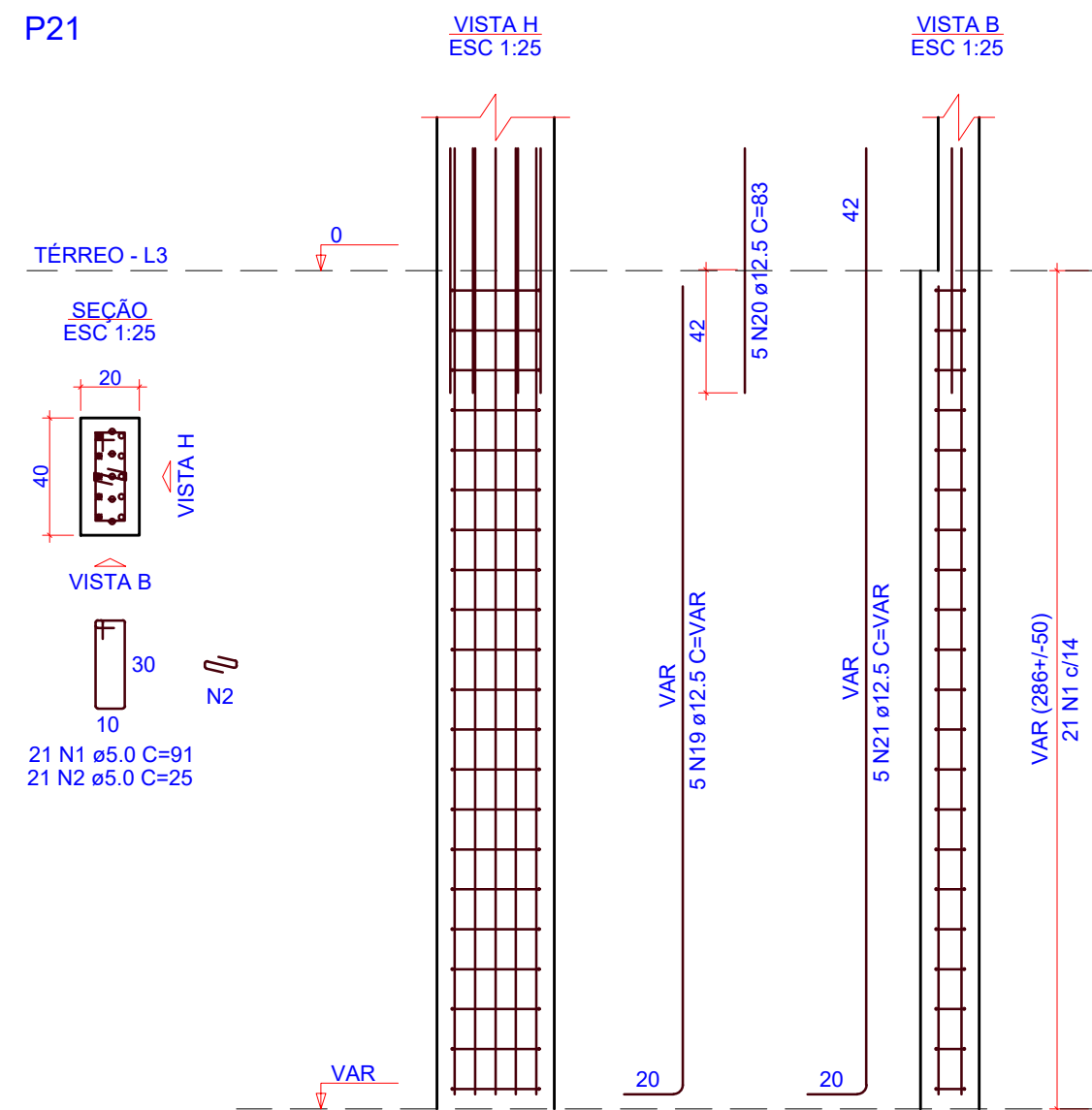
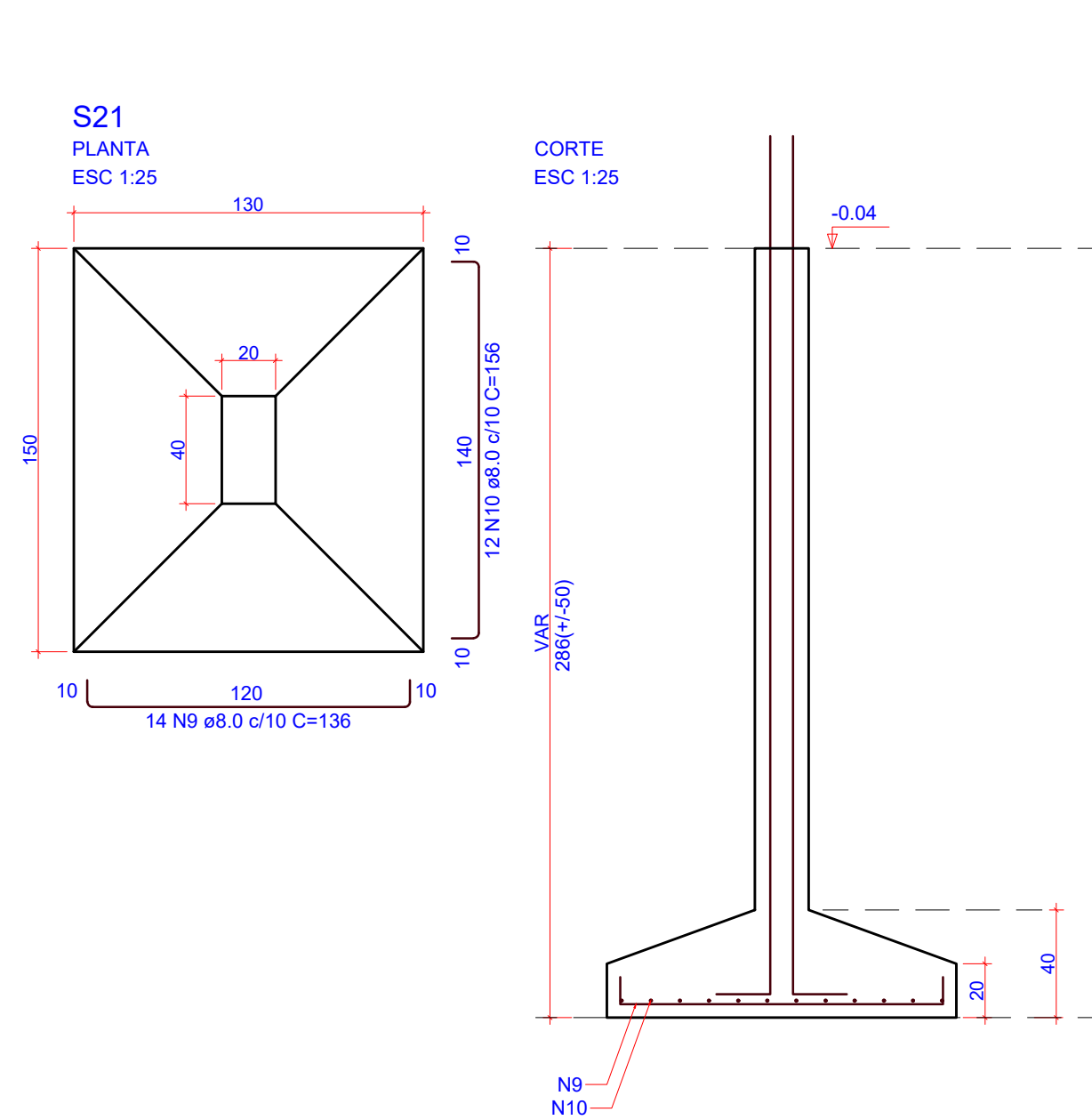
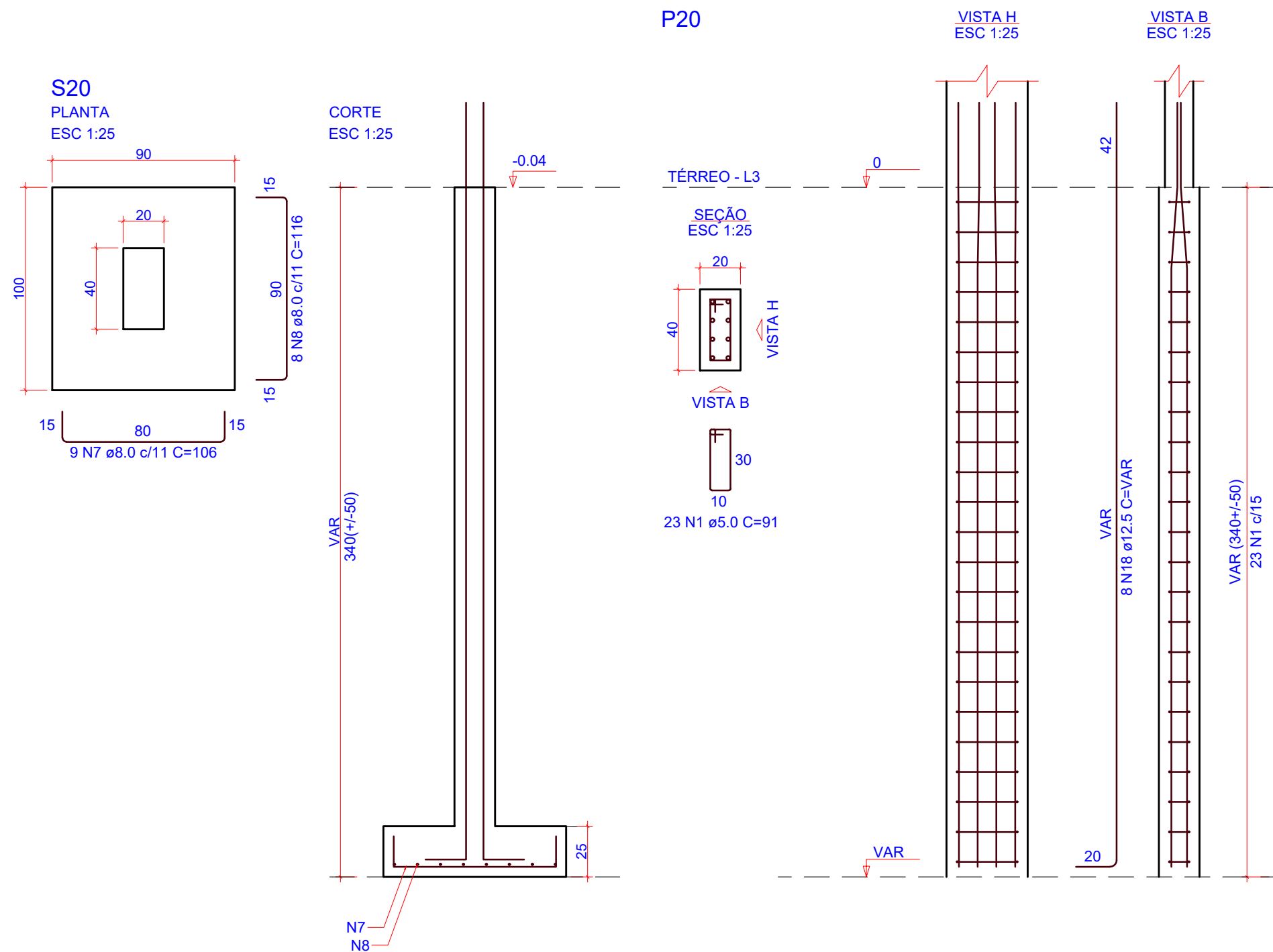
REVISÃO		DATA	CLIENTE JULIANO CORBELLINI			
A	EMIÇÃO INICIAL	15/12/2025	LOCALIDADE – MUNICÍPIO PORTO ALEGRE - RS			
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO RUA ALCEBIADES ANTÔNIO DOS SANTOS- 356 - LOTE 207			
			PROJETO ESTRUTURAL PROJETO ESTRUTURAL FUNDAÇÕES DETALHAMENTO FUNDAÇÕES E PILARES DE ARRANQUE (1/4)			
			PROJETO Engº Rafael Buss CREA/SC 131.013-9	DESENHO MMD ESCALA INDICADA	DATA 15/12/2025 DATA TOP:	FOLHA Nº EST-0020-B



RELAÇÃO DO AÇO					
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	64	32	2048
	2	5.0	32	139	4448
	3	5.0	100	25	2500
	4	5.0	78	91	7098
	5	5.0	20	151	3020
CA50	6	5.0	22	71	1562
	7	8.0	39	136	5304
	8	8.0	11	186	1626
	9	8.0	12	126	1512
	10	8.0	10	146	1460
	11	8.0	13	116	1508
	12	8.0	9	156	1404
	13	8.0	7	76	532
	14	8.0	6	96	576
	15	8.0	30	106	3180
CA50	16	10.0	4	VAR	VAR
	17	12.5	8	VAR	VAR
	18	12.5	2	VAR	VAR
	19	12.5	18	83	1494
	20	12.5	8	VAR	VAR
CA60	21	12.5	4	VAR	VAR
	22	12.5	4	VAR	VAR
	23	12.5	12	VAR	VAR
	24	12.5	12	VAR	VAR

RESUMO DO AÇO				
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10% (Barras)	PESO + 10% (kg)
CA50	8.0	173	16	75.1
CA50	10.0	10.6	1	7.2
CA60	12.5	182.9	17	193.8
CA60	5.0	206.8	-	35.1
PESO TOTAL (kg)				
CA50	276.1			
CA60	35.1			
Volume de concreto (C-30) = 4.29 m³				
Área de forma = 31.26 m²				





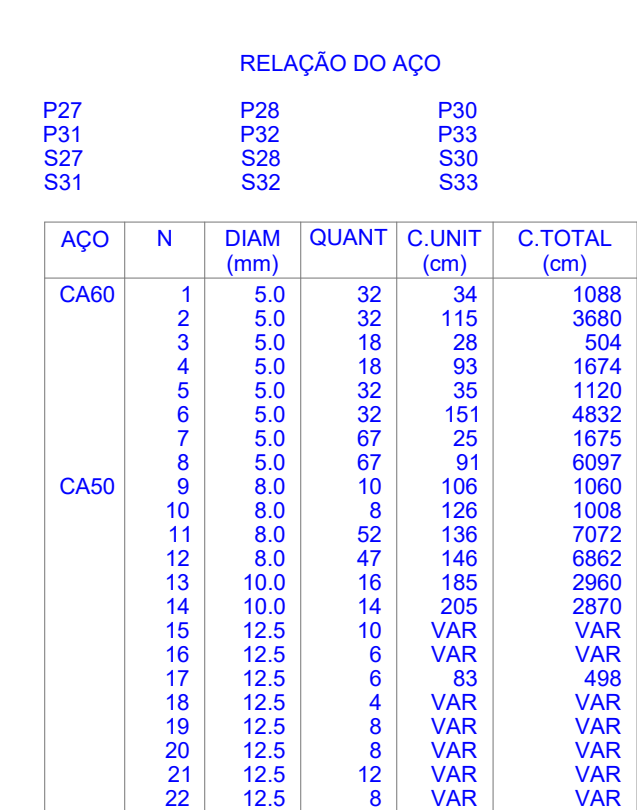
1 DETALHAMENTO FUNDAÇÕES E PILARES DE ARRANQUE (3/4)

RELAÇÃO DO AÇO					
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
P20	1	5.0	44	91	4004
P23	2	5.0	69	25	1725
S20	3	5.0	80	51	4080
S23	4	5.0	64	32	2048
	5	5.0	32	139	4448
	6	5.0	24	131	3144
	7	8.0	9	106	954
	8	8.0	19	116	2204
	9	8.0	14	136	1904
	10	8.0	12	156	1872
	11	8.0	36	76	2736
	12	8.0	9	146	1314
	13	8.0	18	146	2628
	14	8.0	13	176	2288
	15	10.0	8	VAR	VAR
	16	10.0	4	VAR	VAR
	17	10.0	7	VAR	VAR
	18	12.5	8	VAR	VAR
	19	12.5	5	VAR	VAR
	20	12.5	5	83	415
	21	12.5	5	VAR	VAR
	22	12.5	12	VAR	VAR

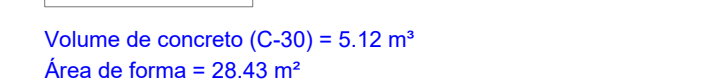
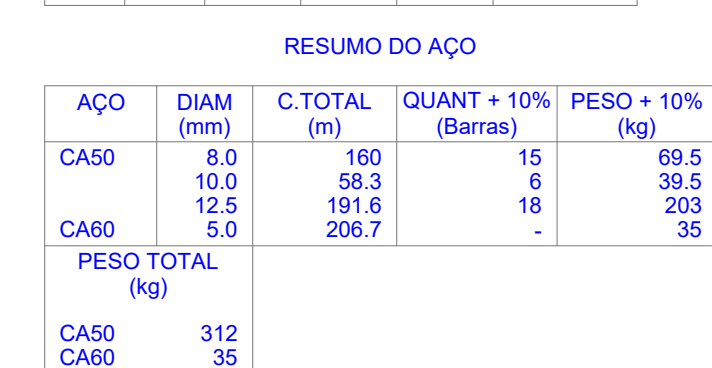
RESUMO DO AÇO				
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10% (Barras)	PESO + 10% (kg)
CA50	8.0	159	15	69
	10.0	72.2	7	48.9
CA60	5.0	115.8	11	122.7
		194.5	-	33
PESO TOTAL (kg)				
CA50	240.7			
CA60	33			

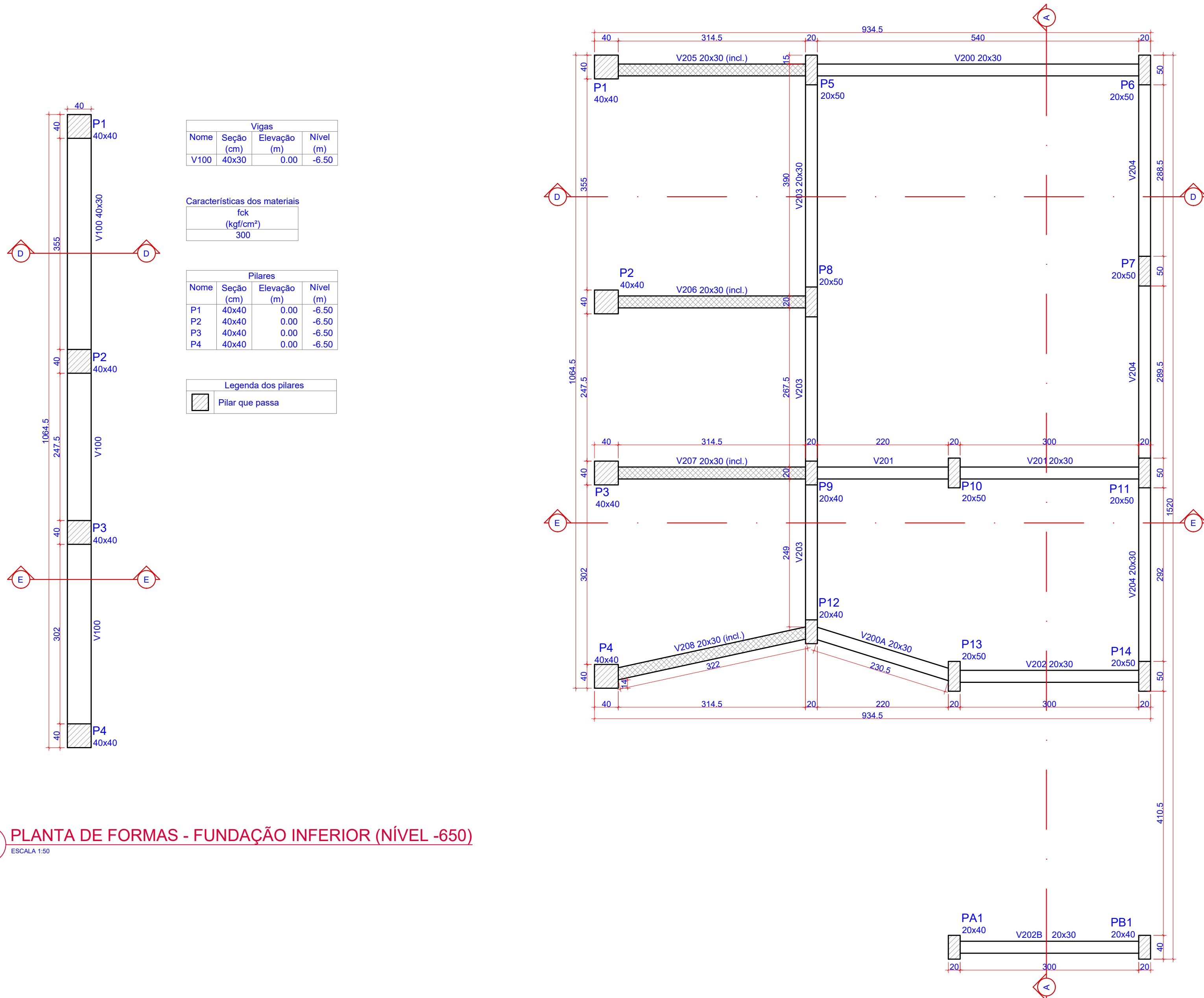
Volume de concreto (C-30) = 3.76 m³
Área de forma = 30.74 m²

Nº	REVISÃO	DATA	CLIENTE
A	EMIÇÃO INICIAL	15/12/2025	JULIANO CORBELLINI
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	PORTO ALEGRE - RS
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PROJETO ESTRUTURAL
			FUNDAÇÕES
			DETALHAMENTO FUNDAÇÕES E PILARES DE ARRANQUE (3/4)
			PROJETO
			Engº Rafael Buss
			CREA/SC 133.073-9
			DESIGNO
			MIO
			15/12/2025
			ESCALA
			INDICADA
			DATA TOP.
			FOLHA Nº
			EST-0040-B



Volume de concreto (C-30) = 5.12 m³
Área de forma = 28.43 m²

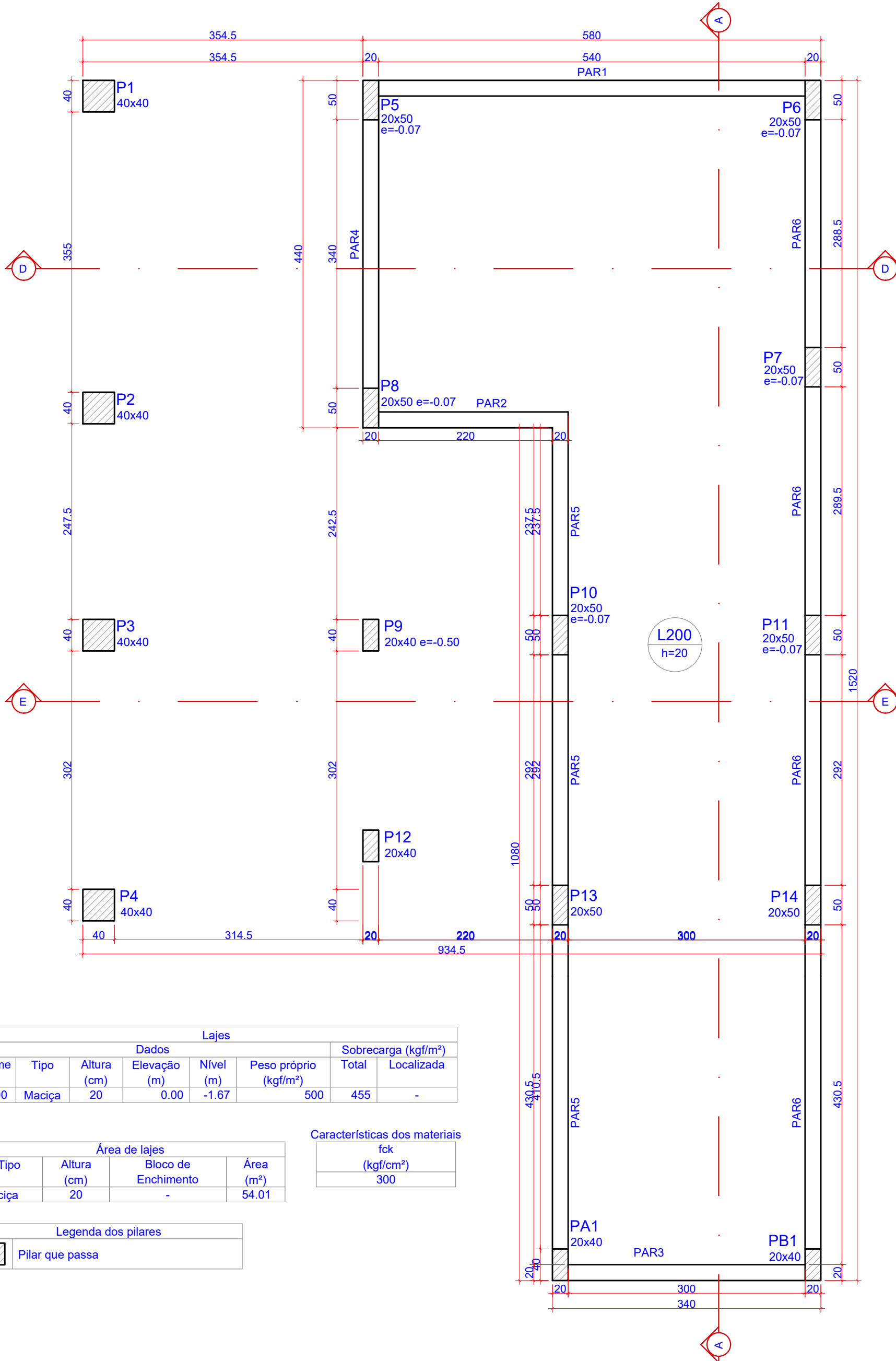




Pilares			
Nome	Seção (cm)	Elevação (m)	Nível (m)
P1	40x40	0.00	-3.50
P2	40x40	0.00	-3.50
P3	40x40	0.00	-3.50
P4	40x40	0.00	-3.50
P5	20x50	0.00	-3.50
P6	20x50	0.00	-3.50
P7	20x50	0.00	-3.50
P8	20x50	0.00	-3.50
P9	20x40	0.00	-3.50
P10	20x50	0.00	-3.50
P11	20x50	0.00	-3.50
P12	20x40	0.00	-3.50
P13	20x50	0.00	-3.50
P14	20x50	0.00	-3.50
PA1	20x40	0.00	-3.50
PB1	20x40	0.00	-3.50

Vigas			
Nome	Seção (cm)	Elevação (m)	Nível (m)
V200	20x30	0.00	-3.50
V200A	20x30	0.00	-3.50
V201	20x30	0.00	-3.50
V202	20x30	0.00	-3.50
V202B	20x30	0.00	-3.50
V203	20x30	0.00	-3.50
V204	20x30	0.00	-3.50
V205	20x30	-0.80 / -2.00	+4.30 / -5.50
V206	20x30	-0.80 / -2.00	+4.30 / -5.50
V207	20x30	-0.80 / -2.00	+4.30 / -5.50
V208	20x30	-0.80 / -2.00	+4.30 / -5.50

Características dos materiais	
fck	(kgf/cm²)
300	
Legenda dos pilares	
	Pilar que passa
Legenda das vigas e paredes	
	Viga
	Viga inclinada

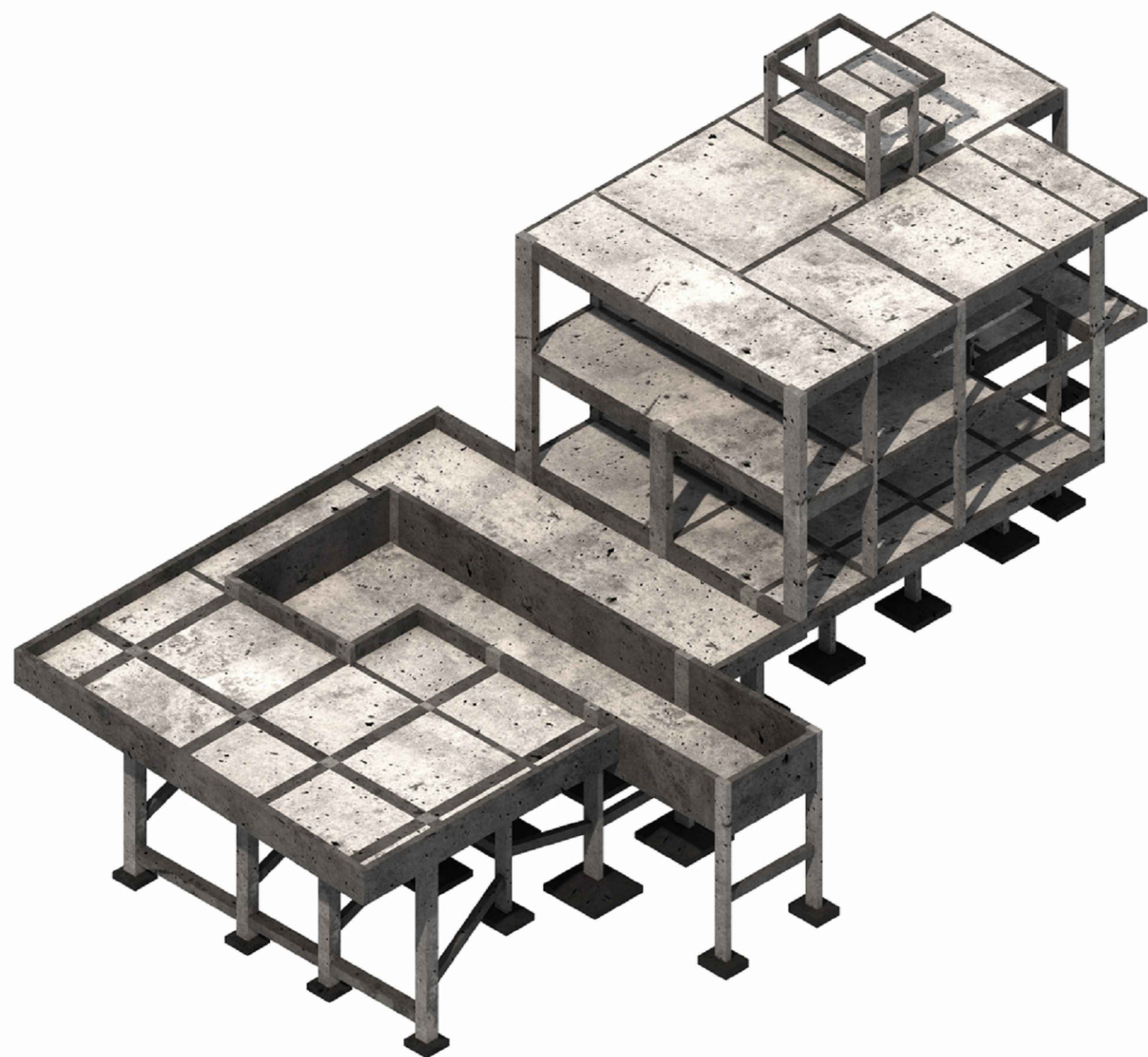


Pilares			
Nome	Seção (cm)	Elevação (m)	Nível (m)
P1	40x40	0.00	-1.67
P2	40x40	0.00	-1.67
P3	40x40	0.00	-1.67
P4	40x40	0.00	-1.67
P5	20x50	-0.07	-1.74
P6	20x50	-0.07	-1.74
P7	20x50	-0.07	-1.74
P8	20x50	-0.07	-1.74
P9	20x40	-0.50	-2.17
P10	20x50	-0.07	-1.74
P11	20x50	-0.07	-1.74
P12	20x40	0.00	-1.67
P13	20x50	0.00	-1.67
P14	20x50	0.00	-1.67
PA1	20x40	0.00	-1.67
PB1	20x40	0.00	-1.67

Lajes					
Nome	Tipo	Altura (cm)	Elevação (m)	Nível (m)	Sobrecarga (kgf/m²)
L200	Maciça	20	0.00	-1.67	500

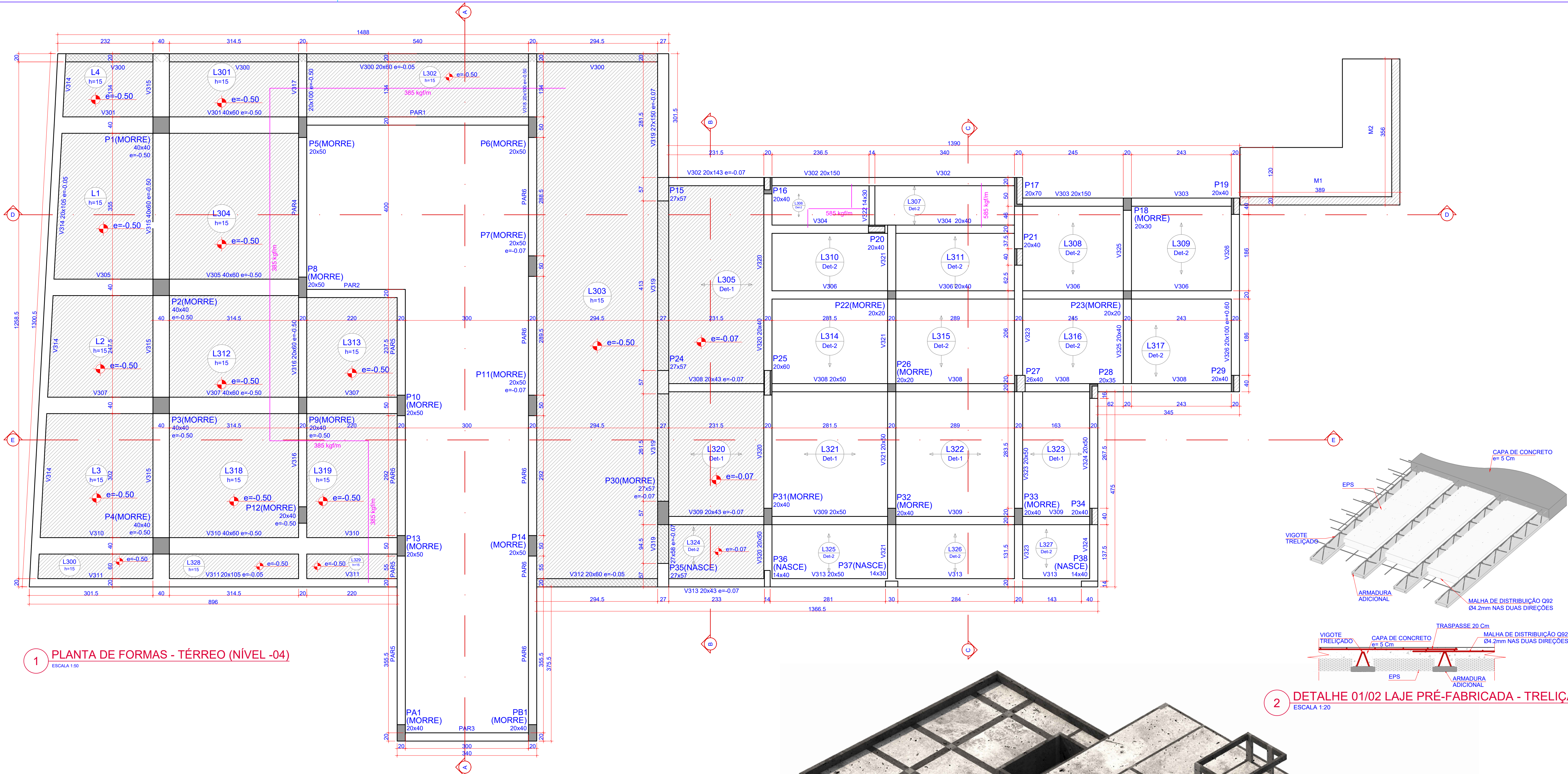
Características dos materiais			
fck	(kgf/cm²)	300	
Legenda dos pilares			
	Pilar que passa		

3 PLANTA DE FORMAS - FUNDO PISCINA (NÍVEL -167)



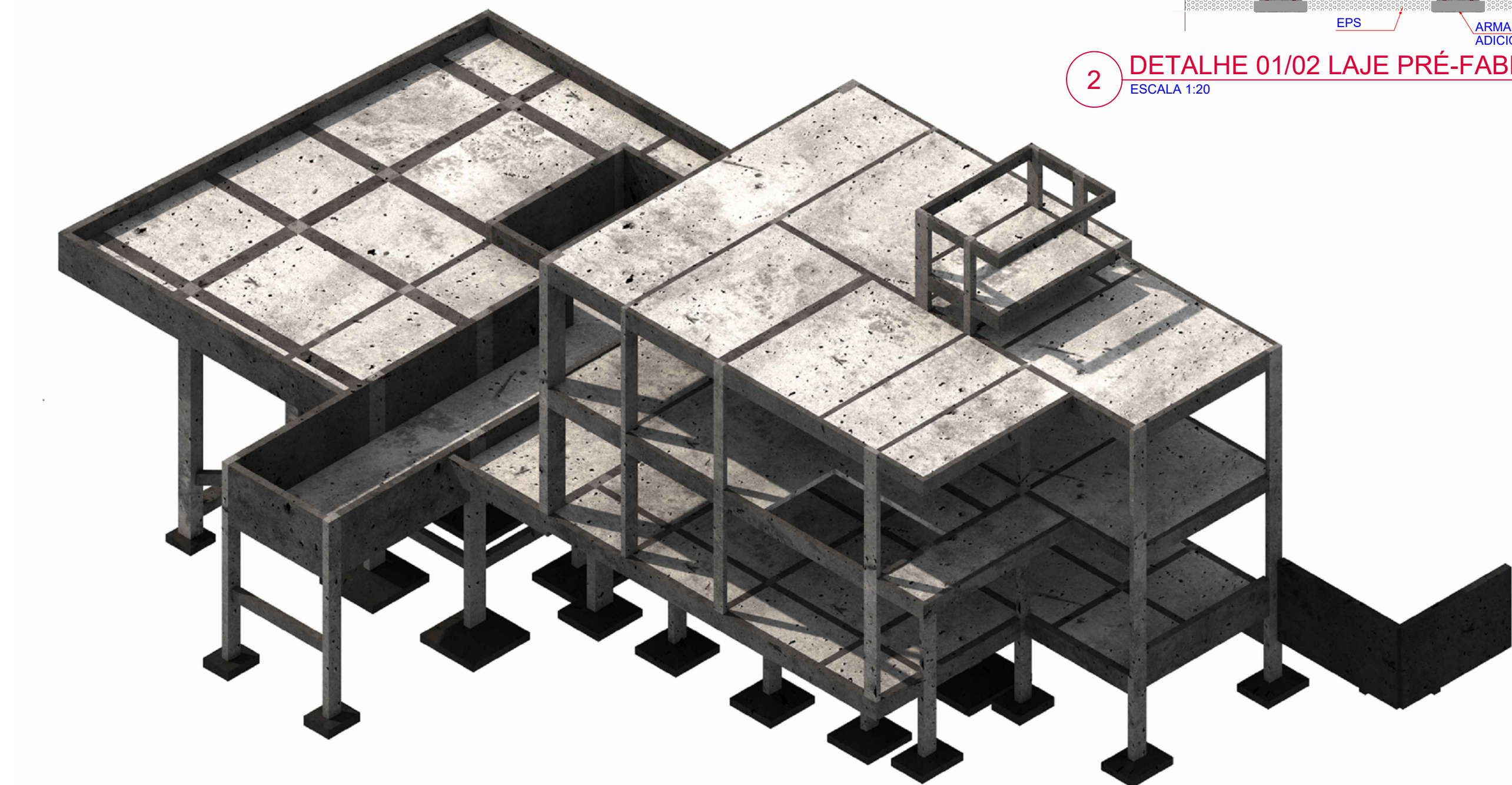
4 PERSPECTIVA

Nº	REVISÃO	DATA	CLIENTE
A	EMIÇÃO INICIAL	15/12/2025	JULIANO CORBELLINI
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	PORTO ALEGRE - RS
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE FORMAS
			PLANTA BAIXAS E PERSPECTIVAS
			PROJETO
			Engº Rafael Buss
			CREA/SC
			133.073-9
			DESIGNO
			MUD.
			ESCALA
			INDICADA
			DATA
			15/12/2025
			DATA TOP.
			FOLHA Nº
			EST-0060-B



1 PLANTA DE FORMAS - TÉRREO (NÍVEL -04)
ESCALA 1:50

2 DETALHE 01/02 LAJE PRÉ-FABRICADA - TRELIÇADA
ESCALA 1:20



3 PERSPECTIVA
ESCALA REPRESENTATIVA

Pilares			
Nome	Seção (cm)	Elevação (m)	Nível (m)
P1	40x40	-0.50	-0.54
P2	40x40	-0.50	-0.54
P3	40x40	-0.50	-0.54
P4	40x40	-0.50	-0.54
P5	20x50	0.00	-0.04
P6	20x50	0.00	-0.04
P7	20x50	-0.07	-0.11
P8	20x50	0.00	-0.04
P9	20x40	-0.50	-0.54
P10	20x50	0.00	-0.04
P11	20x50	-0.07	-0.11
P12	20x40	-0.50	-0.54
P13	20x50	0.00	-0.04
P14	20x50	0.00	-0.04
P15	27x57	0.00	-0.04
P16	20x40	0.00	-0.04
P17	20x70	0.00	-0.04
P18	20x30	0.00	-0.04
P19	20x40	0.00	-0.04
P20	20x40	0.00	-0.04
P21	20x40	0.00	-0.04
P22	20x20	0.00	-0.04
P23	20x20	0.00	-0.04
P24	27x57	0.00	-0.04
P25	20x60	0.00	-0.04
P26	20x20	0.00	-0.04
P27	26x40	0.00	-0.04
P28	20x35	0.00	-0.04
P29	20x40	0.00	-0.04
P30	27x57	0.00	-0.11
P31	20x40	0.00	-0.04
P32	20x40	0.00	-0.04
P33	20x40	0.00	-0.04
P34	20x40	0.00	-0.04
P35	27x57	-0.07	-0.11
P36	14x40	0.00	-0.04
P37	14x30	0.00	-0.04
P38	14x40	0.00	-0.04
PA1	20x40	0.00	-0.04
PB1	20x40	0.00	-0.04

Vigas			
Nome	Seção (cm)	Elevação (m)	Nível (m)
V300	20x60	-0.05	-0.09
V301	40x60	-0.50	-0.54
V302	20x143	-0.07	-0.11
V303	20x150	0.00	-0.04
V304	20x150	0.00	-0.04
V305	40x60	-0.50	-0.54
V306	20x40	0.00	-0.04
V307	40x60	-0.50	-0.54
V308	20x43	-0.07	-0.11
V309	20x43	-0.07	-0.11
V310	40x60	-0.50	-0.54
V311	20x105	-0.05	-0.09
V312	20x60	-0.05	-0.09
V313	20x70	-0.07	-0.11
V314	20x105	-0.05	-0.09
V315	40x60	-0.50	-0.54
V316	20x60	-0.50	-0.54
V317	20x100	-0.50	-0.54
V318	20x100	-0.50	-0.54
V319	27x57	-0.07	-0.11
V320	20x50	0.00	-0.04
V321	20x40	0.00	-0.04
V322	20x50	0.00	-0.04
V323	20x50	0.00	-0.04
V324	20x50	0.00	-0.04
V325	20x40	0.00	-0.04
V326	20x100	0.60	0.56

Lajes					
Nome	Tipo	Altura (cm)	Elevação (m)	Nível (m)	Peso próprio (kgf/m²)
L1	Maciça	15	-0.50	-0.54	375
L2	Maciça	15	-0.50	-0.54	375
L3	Maciça	15	-0.50	-0.54	375
L4	Maciça	15	-0.50	-0.54	375
L300	Maciça	15	-0.50	-0.54	375
L301	Maciça	15	-0.50	-0.54	375
L302	Maciça	15	-0.50	-0.54	375
L303	Maciça	15	-0.50	-0.54	375
L304	Maciça	15	-0.50	-0.54	375
L305	Treliçada 1D	13	-0.07	-0.11	172
L306	Treliçada 1D	13	0.00	-0.04	172
L307	Treliçada 1D	13	0.00	-0.04	172
L308	Treliçada 1D	13	0.00	-0.04	172
L309	Treliçada 1D	13	0.00	-0.04	172
L310	Treliçada 1D	13	0.00	-0.04	172
L311	Treliçada 1D	13	0.00	-0.04	172
L312	Maciça	15	-0.50	-0.54	375
L313	Maciça	15	-0.50	-0.54	375
L314	Treliçada 1D	13	0.00	-0.04	172
L315	Treliçada 1D	13	0.00	-0.04	172
L316	Treliçada 1D	13	0.00	-0.04	172
L317	Treliçada 1D	13	0.00	-0.04	172
L318	Treliçada 1D	13	0.00	-0.04	172
L319	Maciça	15	-0.50	-0.54	375
L320	Treliçada 1D	13	-0.07	-0.11	172
L321	Treliçada 1D	13	0.00	-0.04	172
L322	Treliçada 1D	13	0.00	-0.04	172
L323	Treliçada 1D	13	0.00	-0.04	172
L324	Treliçada 1D	13	-0.07	-0.11	172
L325	Treliçada 1D	13	0.00	-0.04	172
L326	Treliçada 1D	13	0.00	-0.04	172
L327	Treliçada 1D	13	0.00	-0.04	172
L328	Maciça	15	-0.50	-0.54	375
L329	Maciça	15	-0.50	-0.54	375

Legenda dos pilares	
	Pilar que morre
	Pilar que passa
	Pilar que nasce
	Pilar com mudança de seção
Legenda das vigas e paredes	
	Viga
	Viga chata ou invertida
	Parede de concreto
Legenda das lajes	
	Laje
	Laje com elevação negativa

Área de lajes			
Tipo	Altura (cm)	Bloco de Enchimento	Área (m²)
Maciça	15	-	119.19
Treliçada 1D	13	88/30/100	98.53

Características dos materiais	
fck	(kgf/cm²)
300	

Blocos de enchimento				
Detalhe	Tipo	Nome	Dimensões (cm)	Quantidade
1/2	EPS Unidirecional	88/30/100	8 30 100	356

Nº	REVISÃO	DATA	CLIENTE
A	EMIÇÃO INICIAL	15/12/2025	JULIANO CORBELLINI
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	PORTO ALEGRE - RS
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE FORMAS
			PLANTA BAIXAS E PERSPECTIVAS
			PROJETO
			Engº Rafael Buss
			CREA/SC 133.073-9
			DESENHO
			15/12/2025
			FOLHA Nº
			EST-0070-B



VIGOTE TRELIÇADO

CAPA DE CONCRETO $e=5\text{ cm}$

TRASPASSE 20 cm

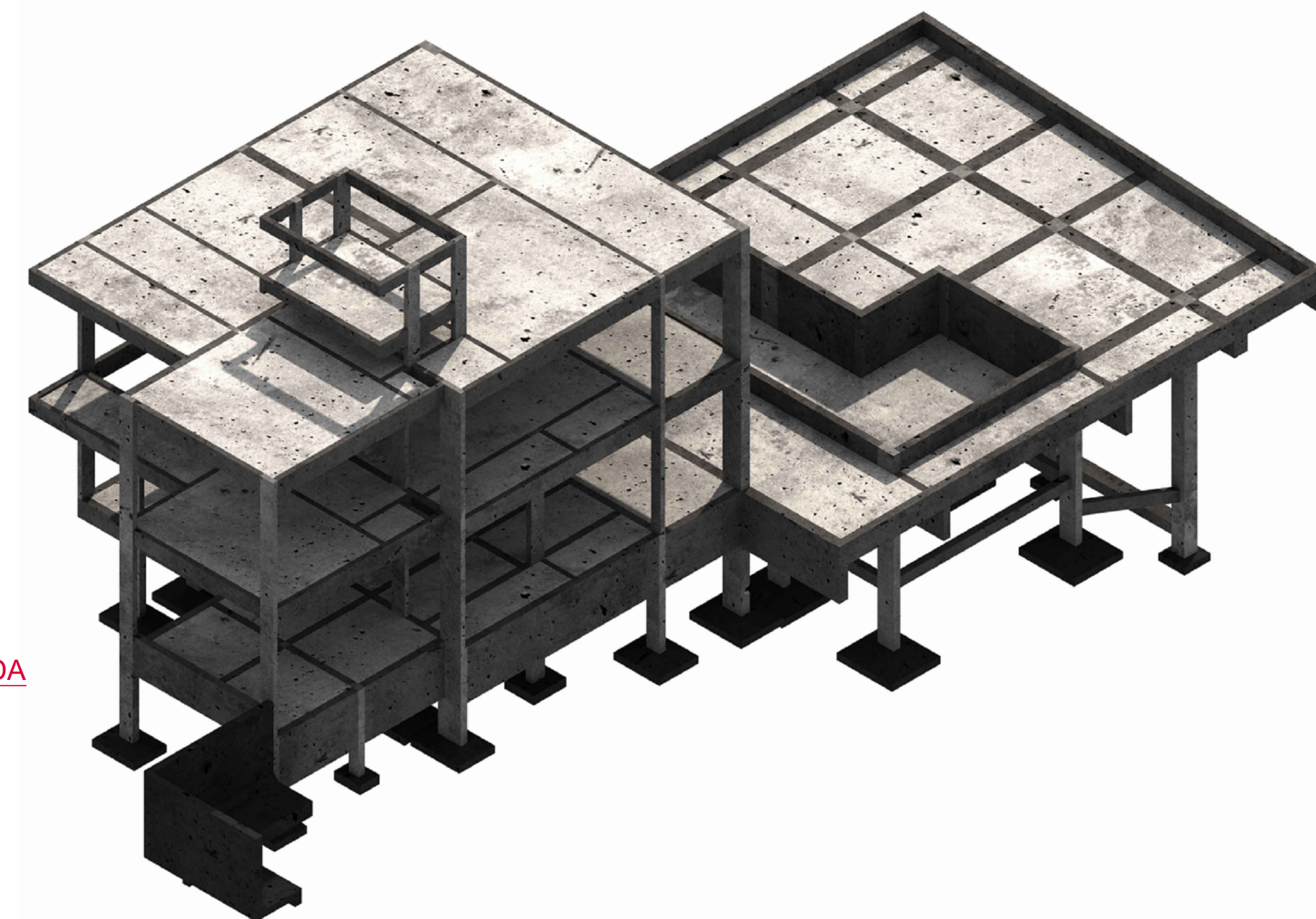
MALHA DE DISTRIBUIÇÃO Ø8 e Ø4.2mm NAS DUAS DIREÇÕES

EPS

ARMADURA

3 DETALHE 01/02/03/04 LAJE PRÉ-FABRICADA - TRELIÇADA
ESCALA 1:20

Fck (MPa)
30

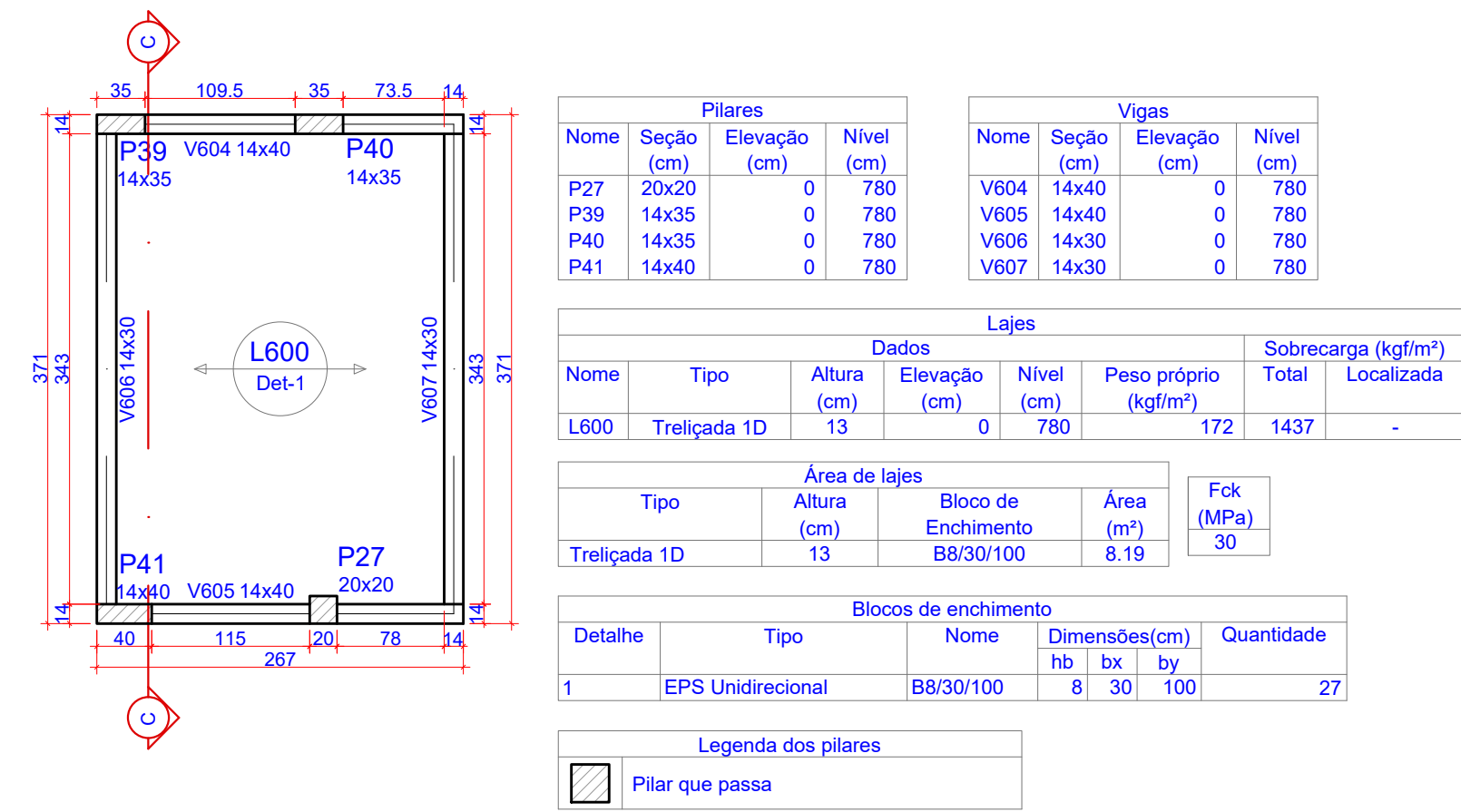


4 PERSPECTIVA
ESCALA REPRESENTATIVA

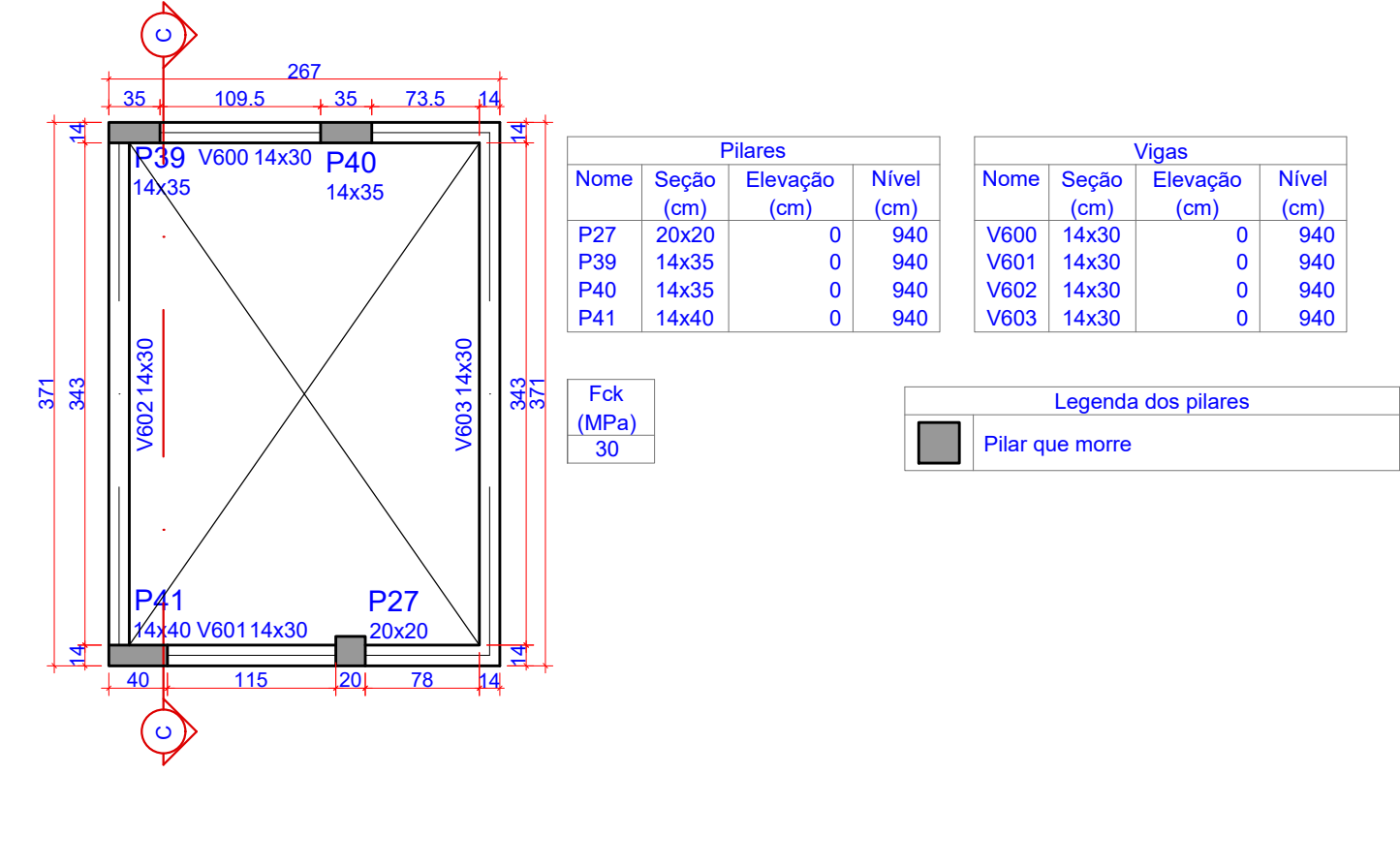
Legenda dos pilares	
	Pilar que morre
	Pilar que nasce
	Pilar com mudança de seção

Fck (MPa)	30
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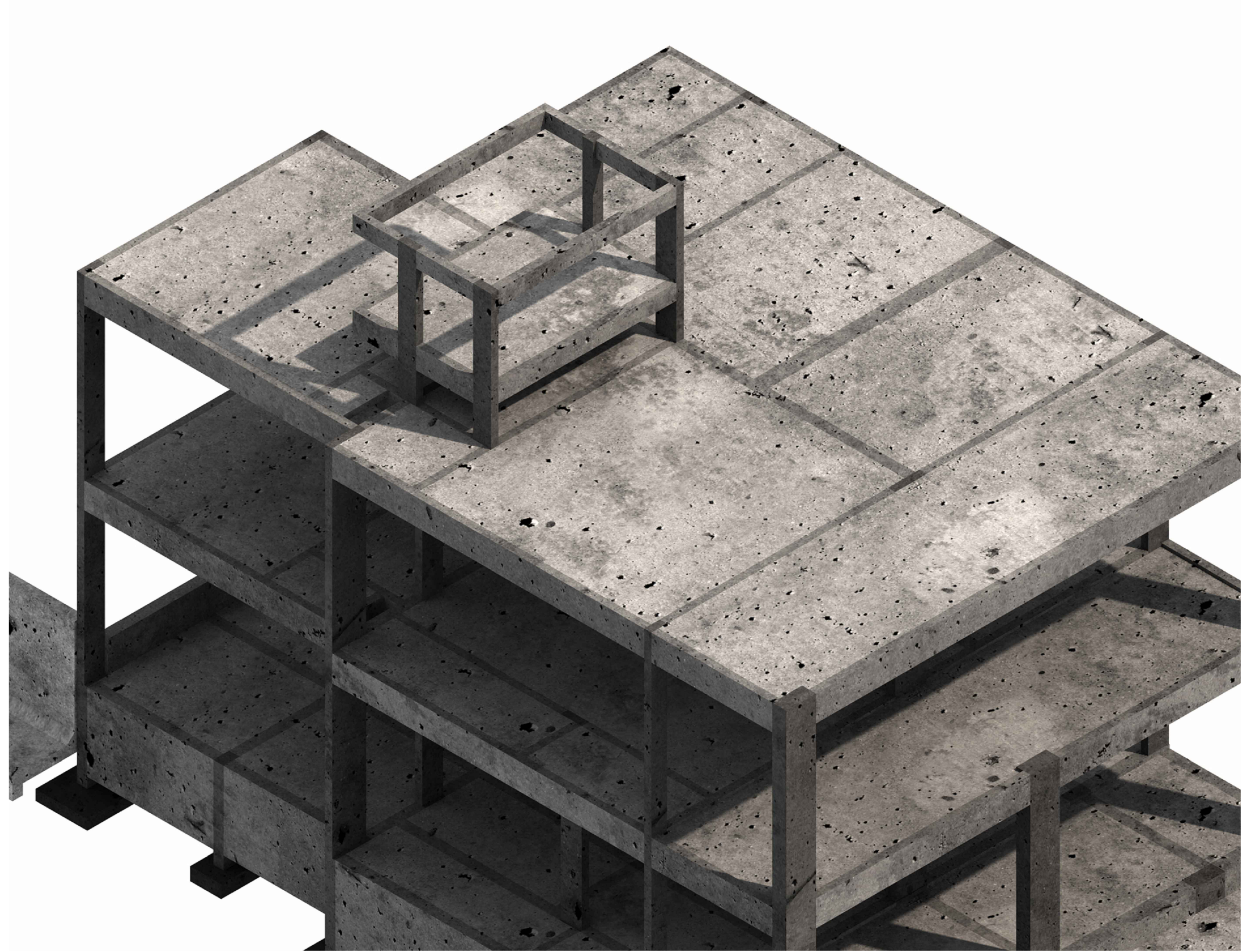
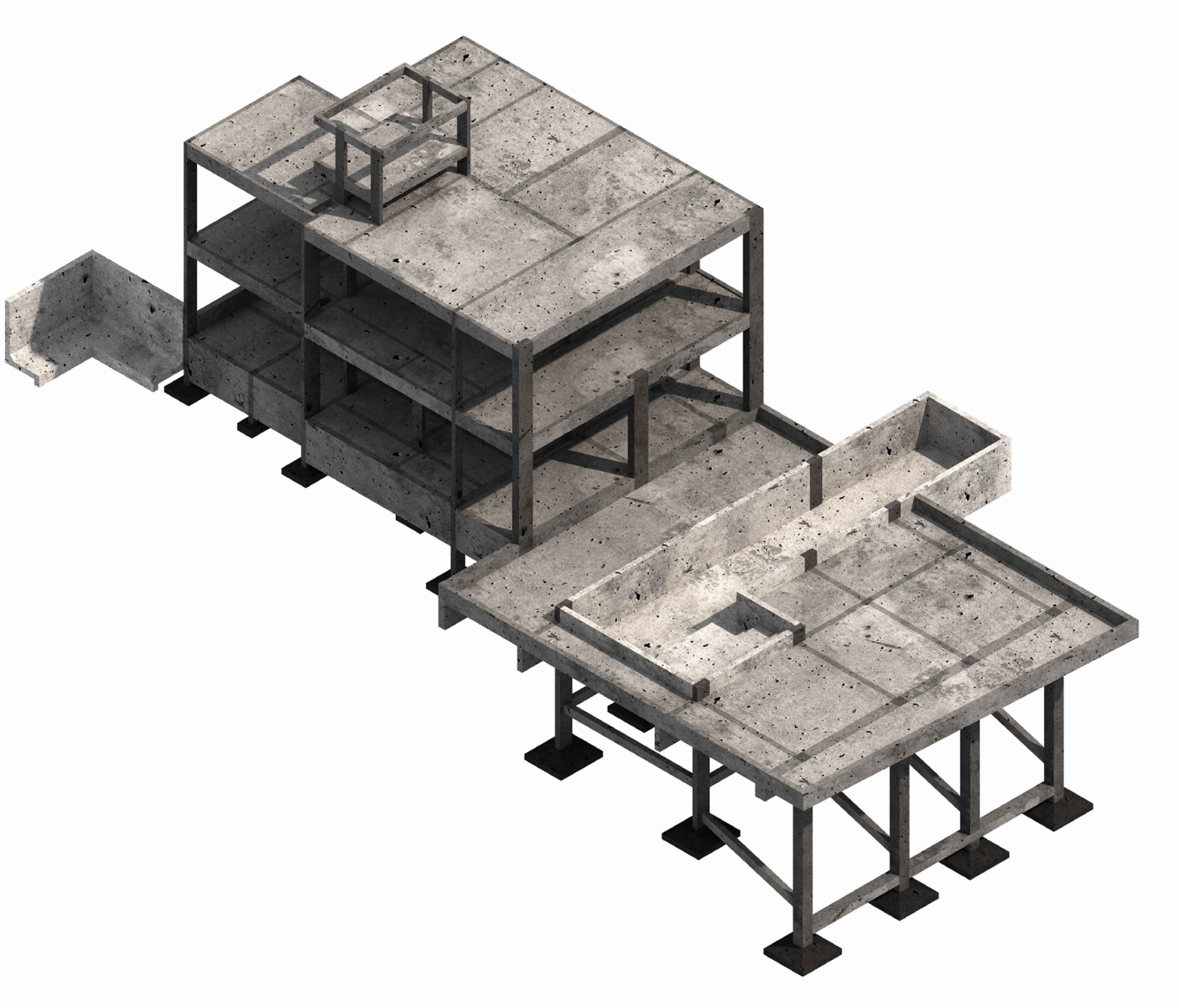
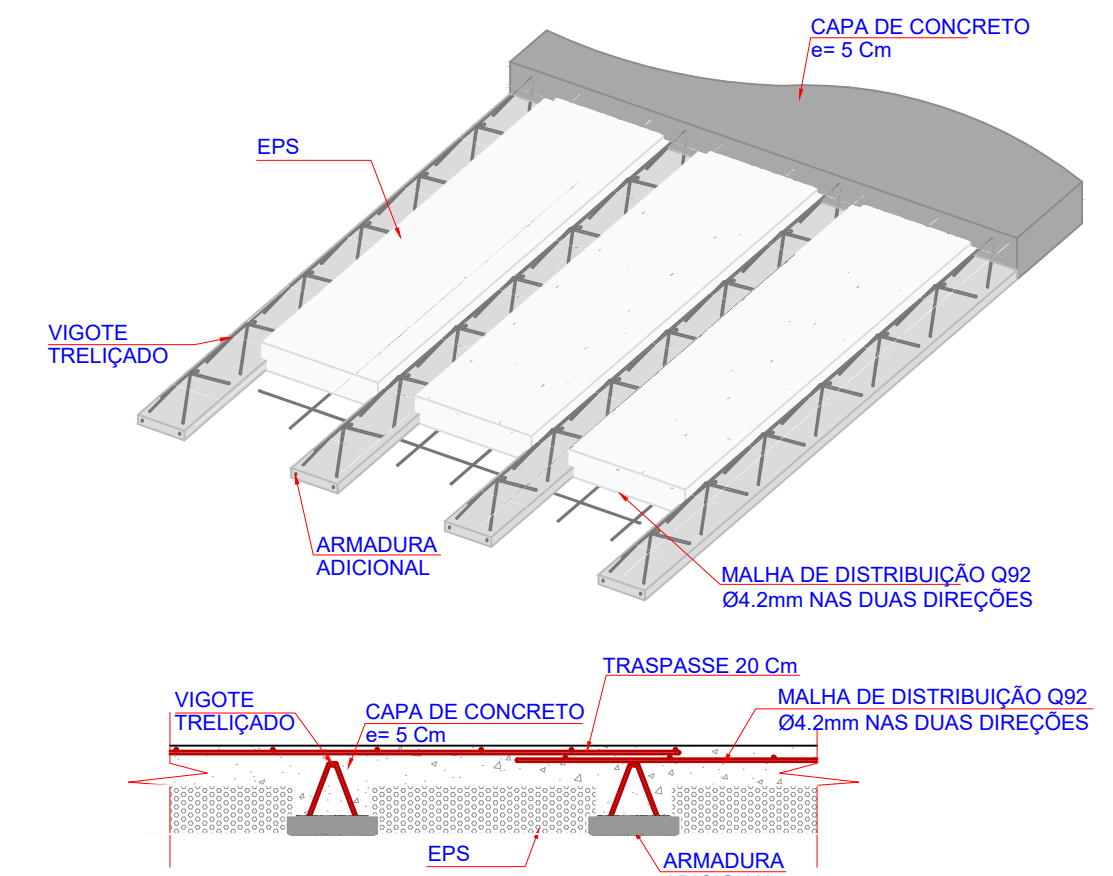
Nº		REVISÃO	DATA	CLIENTE							
A		EMIÇÃO INICIAL	15/12/2025	LOCALIDADE - MUNICÍPIO							
B		REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO							
				RUA ALCEBÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207							
				PROJETO ESTRUTURAL							
				PROJETO ESTRUTURAL							
				PLANTA DE FORMAS							
				PLANTA BAIXAS E PERSPECTIVAS							
				PROJETO		DESENHO		DATA		FOLHA Nº	
				Engº Rafael Buss		MMB		15/12/2025		EST-0080-B	
				ORLA/SO		ESCALA		DATA TOP.			
				113.002-9		INDICADA					



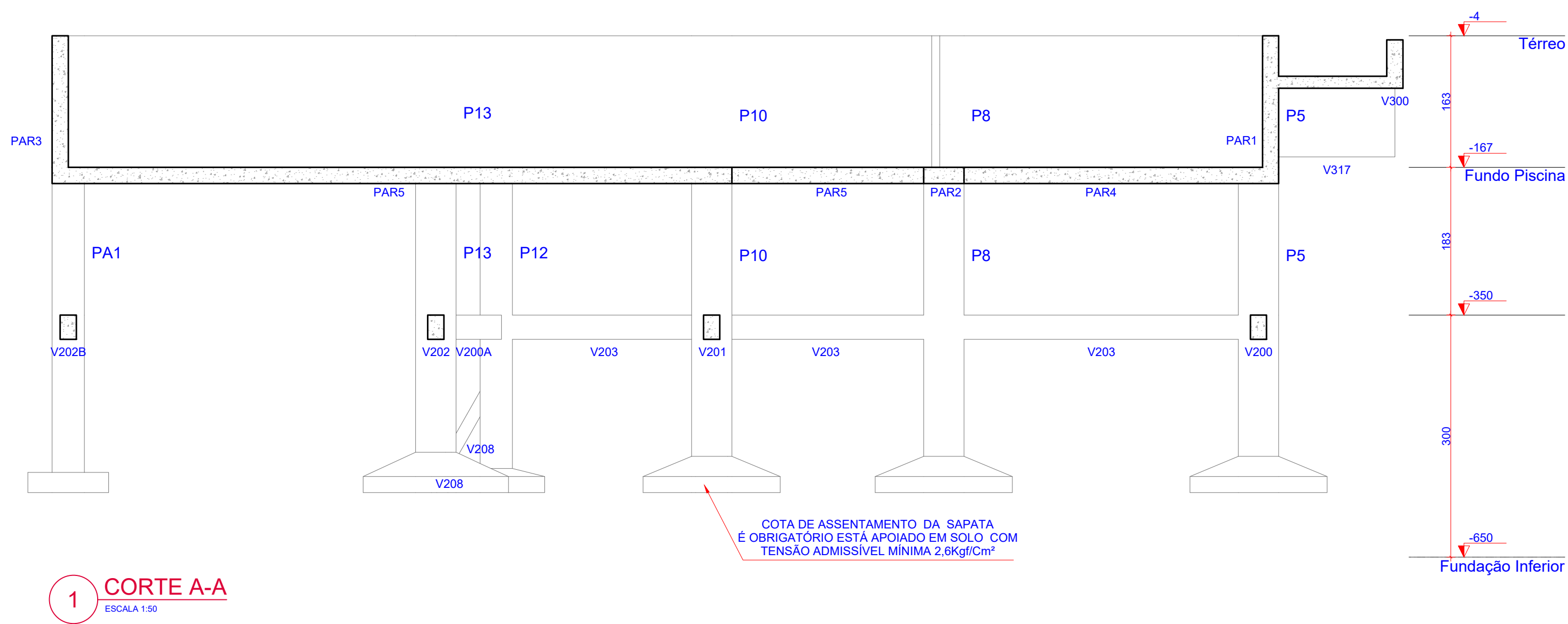
1 PLANTA DE FORMAS - INTERMEDIÁRIA RESERVATÓRIO (NÍVEL 780)
ESCALA 1:50



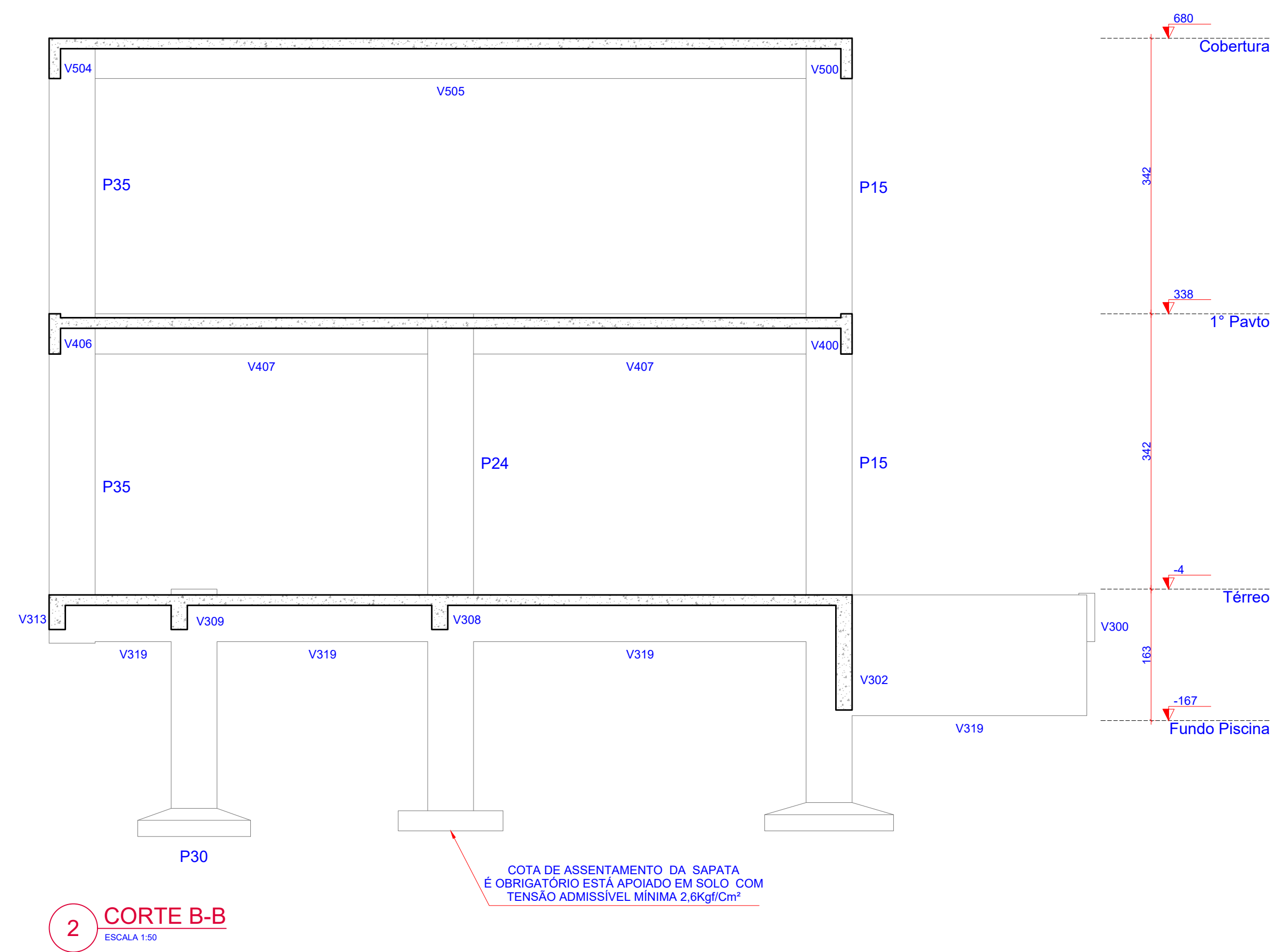
2 PLANTA DE FORMAS - TOPO RESERVATÓRIO (NÍVEL 940)
ESCALA 1:50



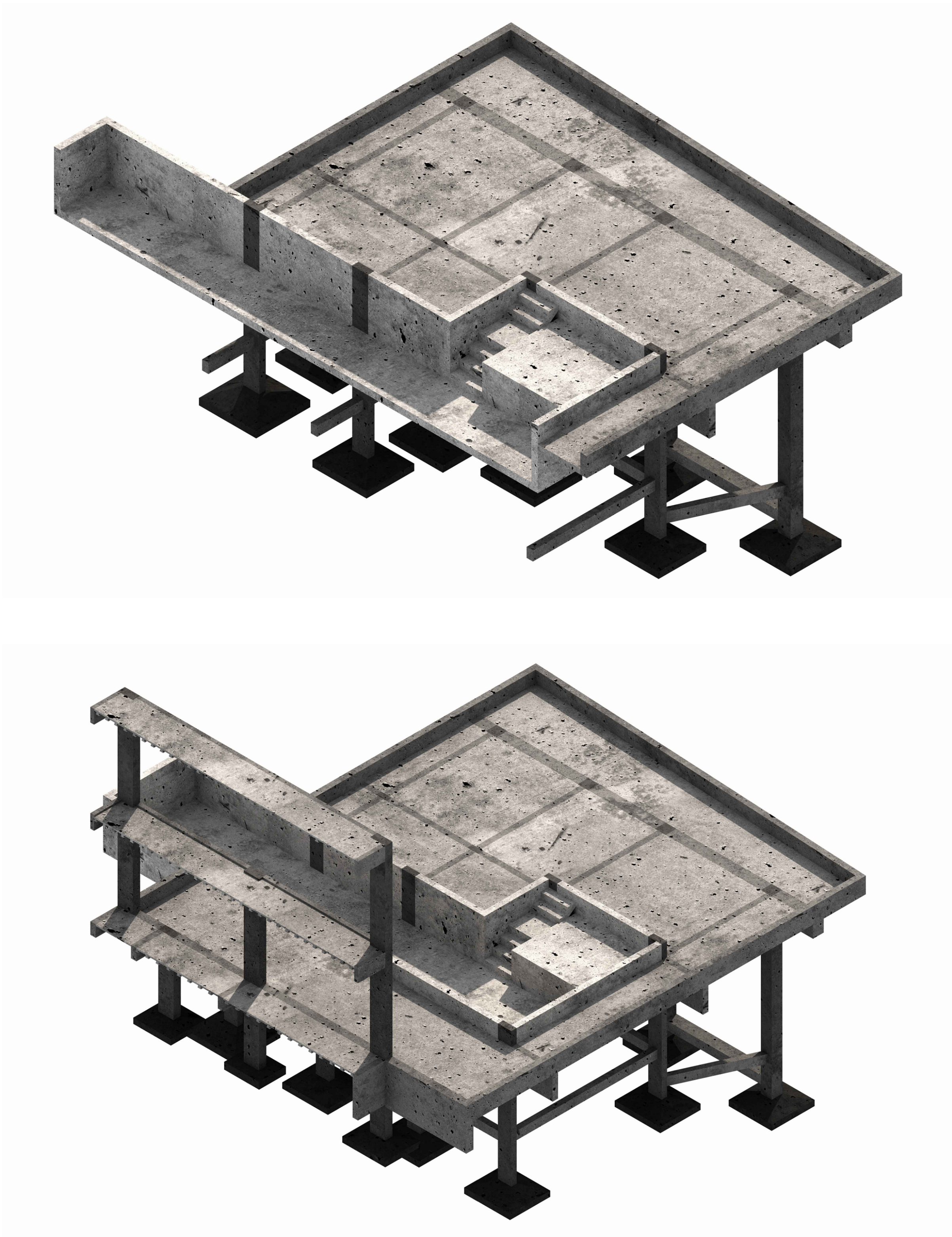
Nº	REVISÃO	DATA	CLIENTE
A	EMIÇÃO INICIAL	15/12/2025	JULIANO CORBELLINI
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	PORTO ALEGRE - RS
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE FORMAS
			PLANTA BAIXAS E PERSPECTIVAS
			PROJETO
			Engº Rafael Buss
			CREA/SC
			133.073-9
			DESIGNO
			MDO
			15/12/2025
			ESCALA
			INDICADA
			FOLHA Nº
			EST-0090-B



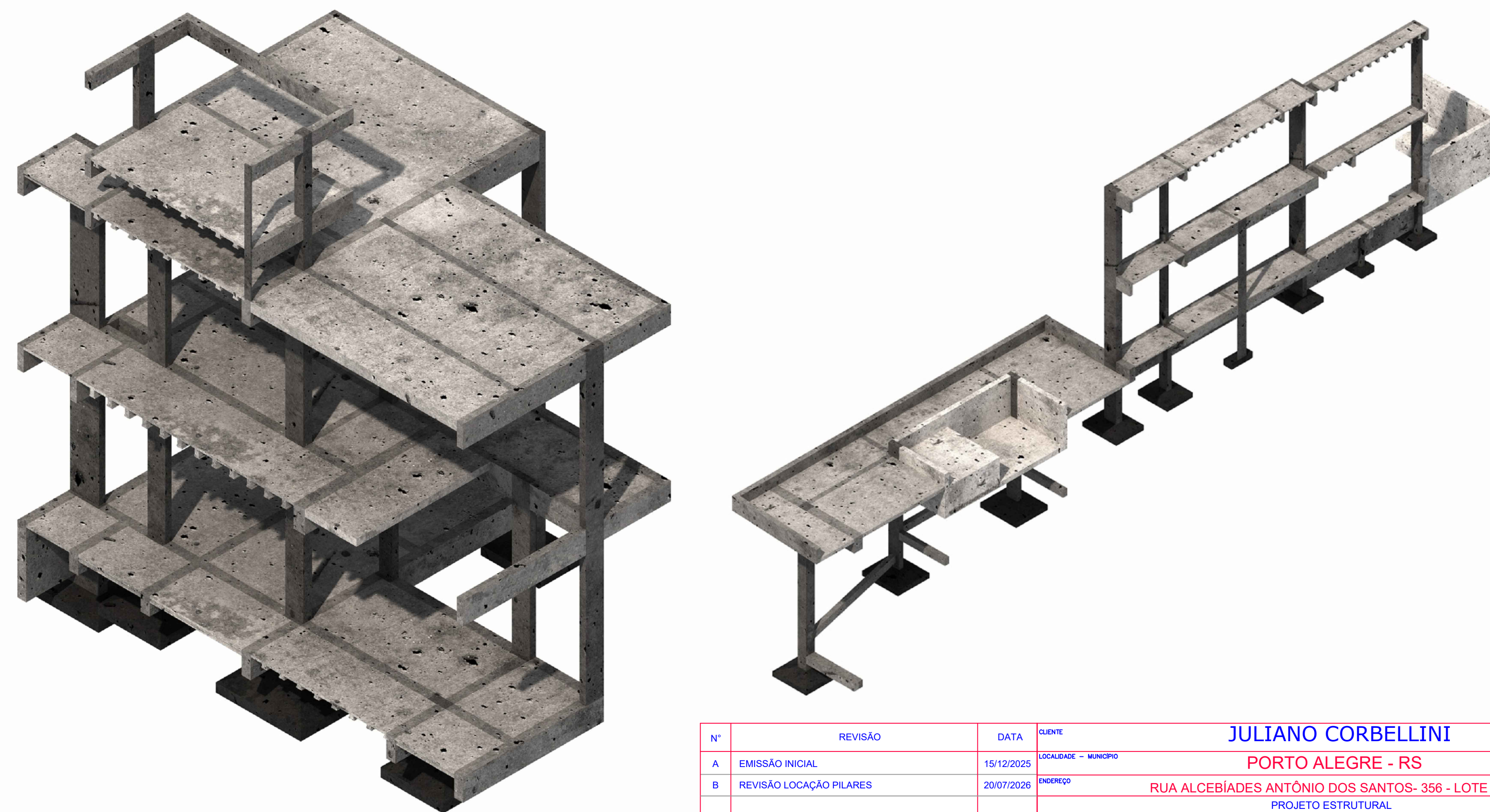
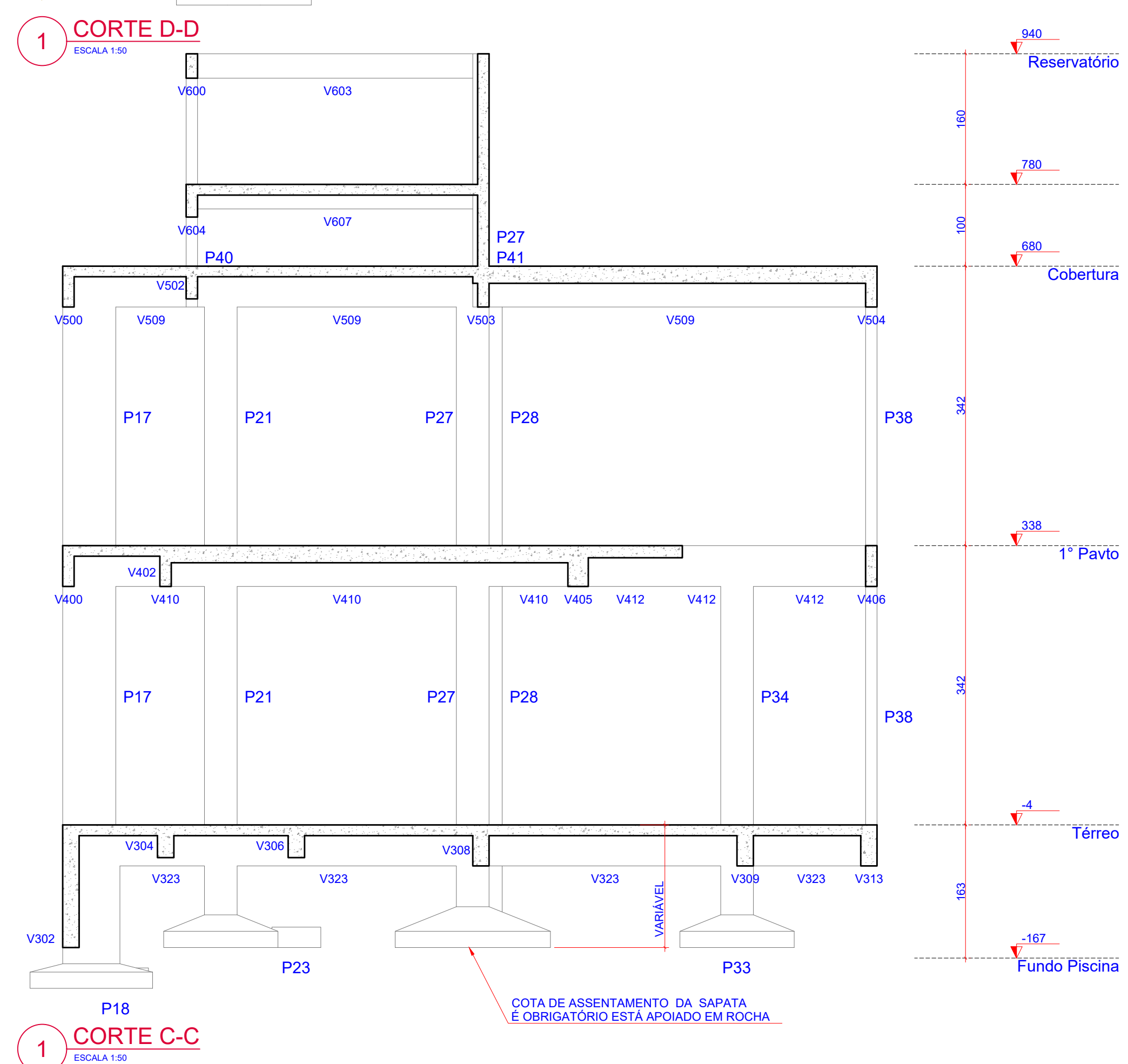
1 CORTE A-A
ESCALA 1:50



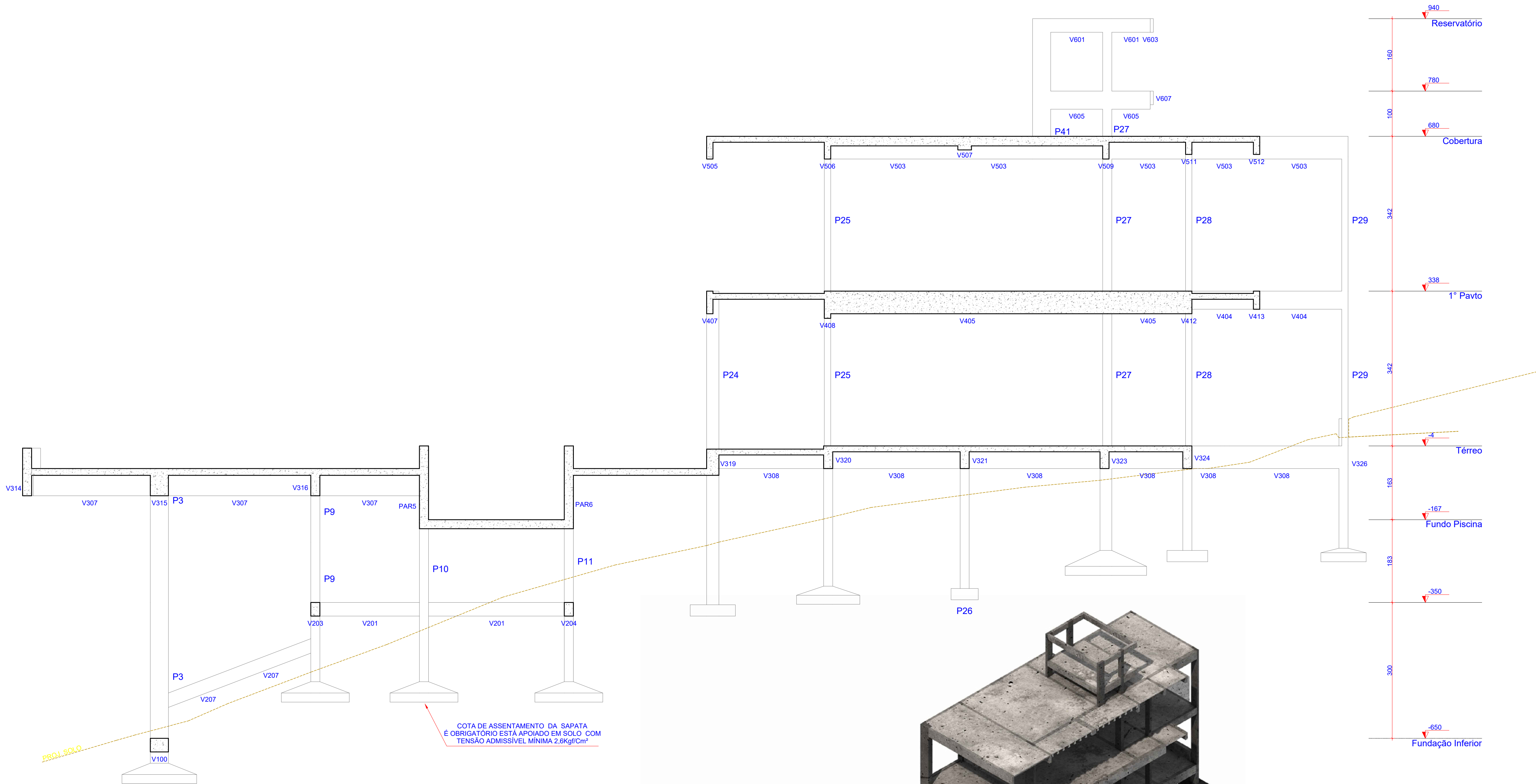
2 CORTE B-B
ESCALA 1:50



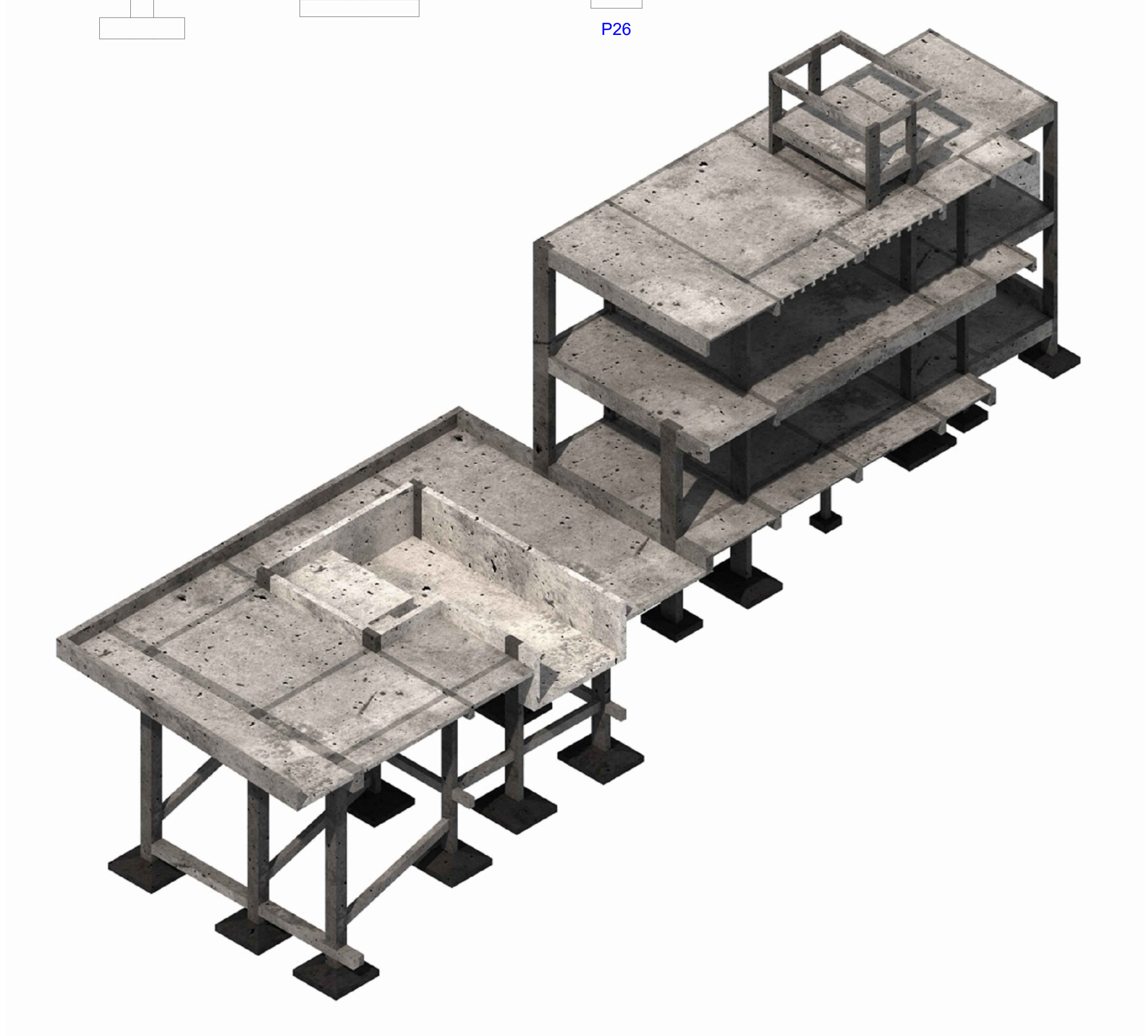
Nº	REVISÃO	DATA	CLIENTE JULIANO CORBELLINI			
A	EMIÇÃO INICIAL	15/12/2025	LOCALIDADE - MUNICÍPIO PORTO ALEGRE - RS			
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO RUA ALCEBIADES ANTÔNIO DOS SANTOS- 356 - LOTE 207			
			PROJETO ESTRUTURAL			
			PROJETO ESTRUTURAL			
			PLANTA DE FORMAS			
			CORTES E PERSPECTIVAS			
			PROJETO MUD. CREA/SC 133.073-9	DESENHO MUD. ESCALA INDICADA	DATA 15/12/2025 DATA TOP.	FOLHA Nº EST-0100-B



Nº	REVISÃO	DATA	CLIENTE JULIANO CORBELLINI					
A	EMISSÃO INICIAL	15/12/2025	LOCALIDADE -- MUNICÍPIO PORTO ALEGRE - RS					
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO RUA ALCEBIADES ANTÔNIO DOS SANTOS- 356 - LOTE 207					
			PROJETO ESTRUTURAL PROJETO ESTRUCTURAL PLANTA DE FORMAS CORTES E PERSPECTIVAS					
			PROJETO Engº Rafael Buss	DESENHO MMD	DATA 15/12/2025	FOLHA Nº	EST-0110-B	
			ORÇ/SC 153.003-9	ESCALA	DATA TOP.			

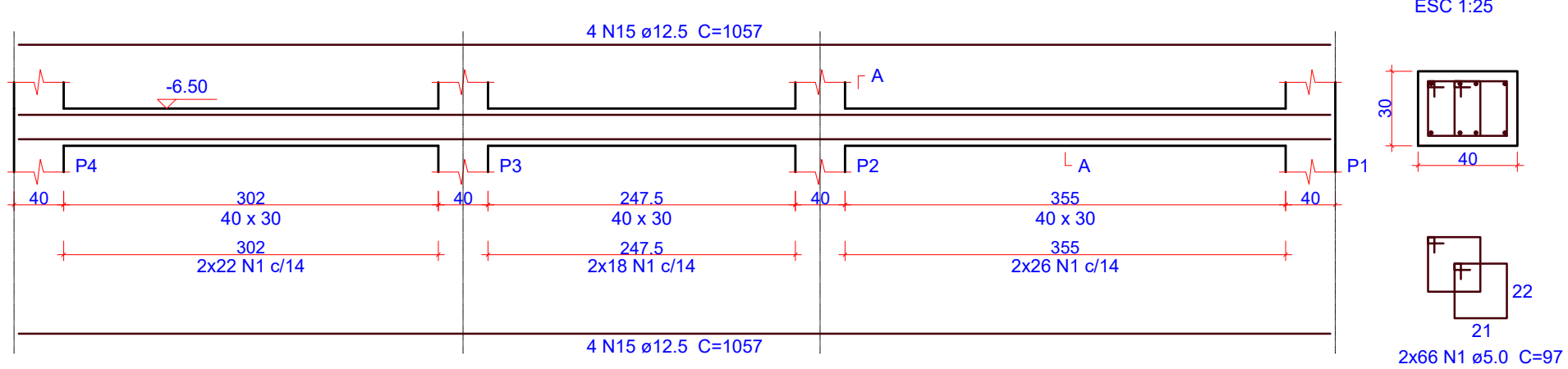


1 CORTE E-E
ESCALA 1:50

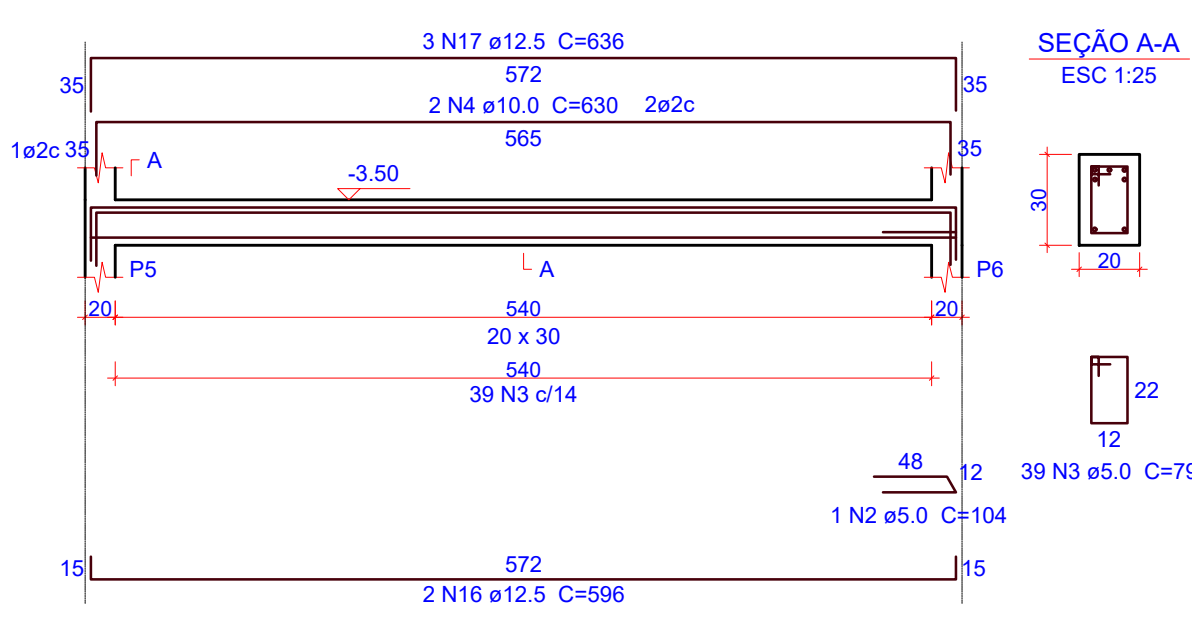


Nº	REVISÃO	DATA	CLIENTE JULIANO CORBELLINI			
A	EMIÇÃO INICIAL	15/12/2025	LOCALIDADE - MUNICÍPIO PORTO ALEGRE - RS			
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO RUA ALCEBIADES ANTÔNIO DOS SANTOS- 356 - LOTE 207			
			PROJETO ESTRUTURAL			
			PLANTA DE FORMAS			
			CORTES E PERSPECTIVAS			
			PROJETO Engº Rafael Buss	DESENHO MDO	DATA 15/12/2025	FOLHA Nº
			CREA/SC 133.073-9	ESCALA INDICADA	DATA TOP.	EST-0120-B

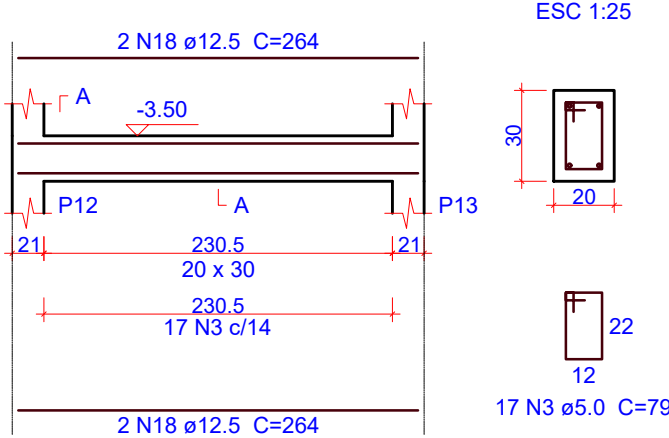
V100



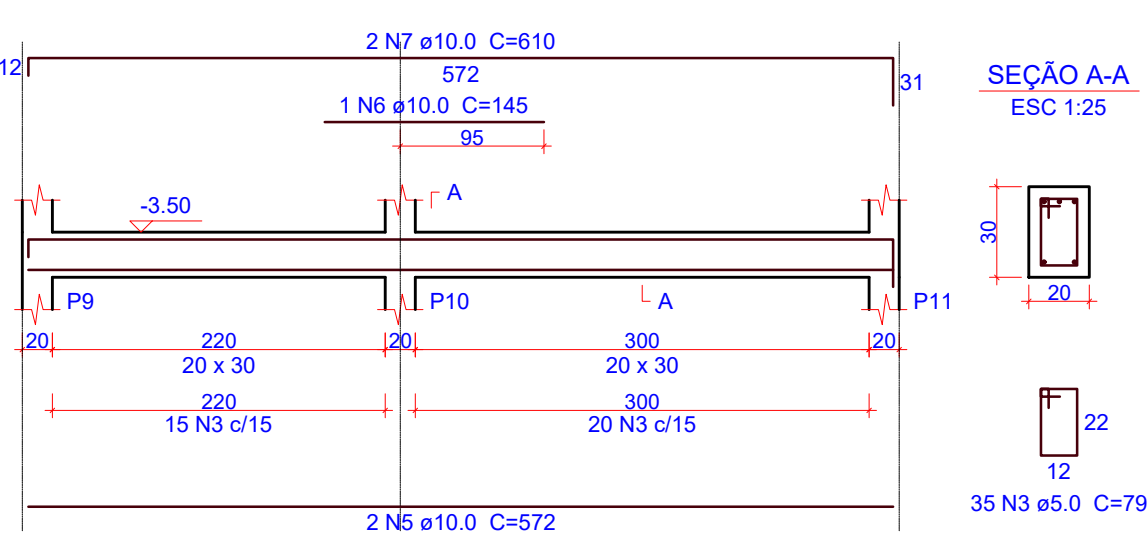
V200



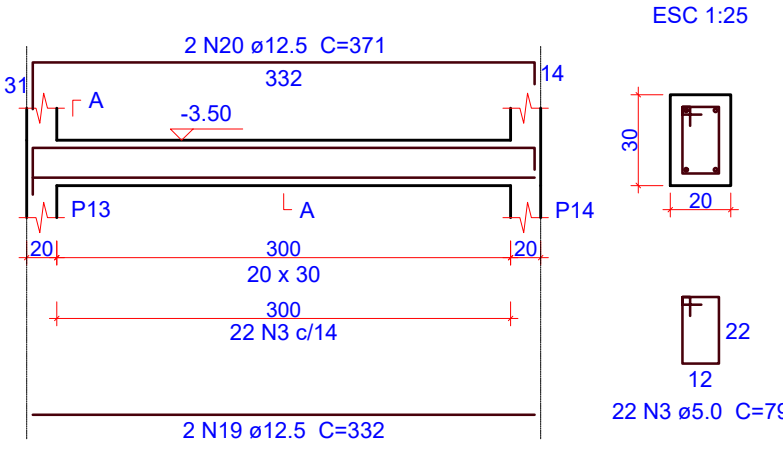
V200A



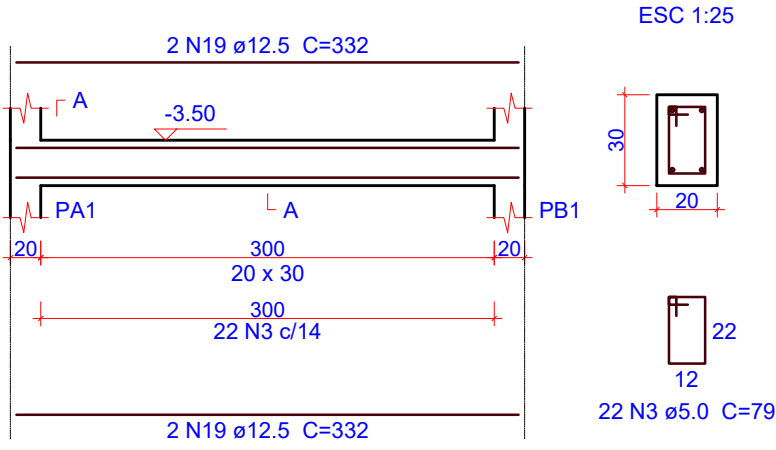
V201



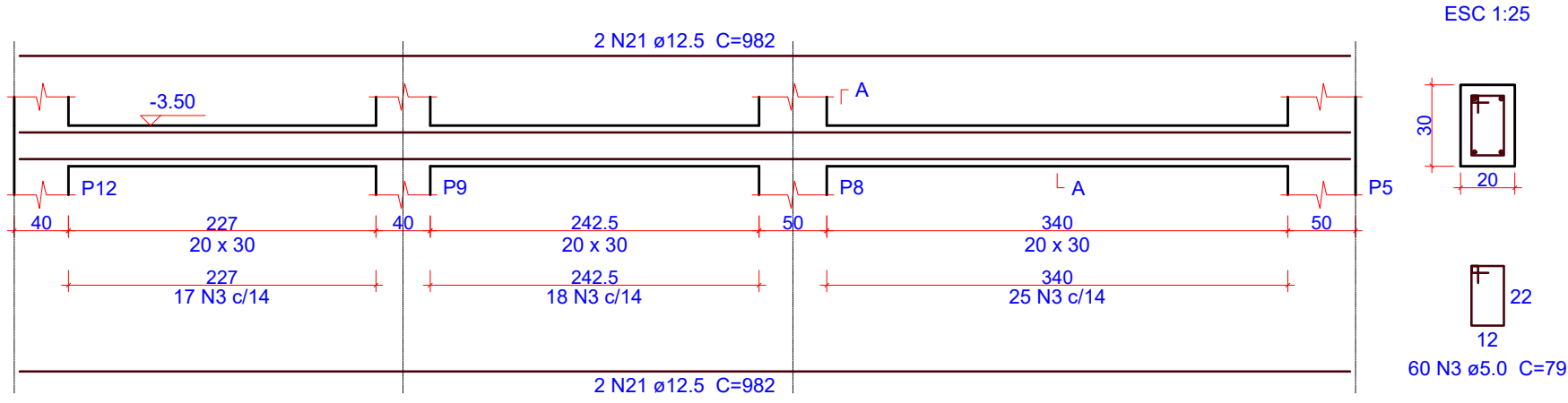
V202



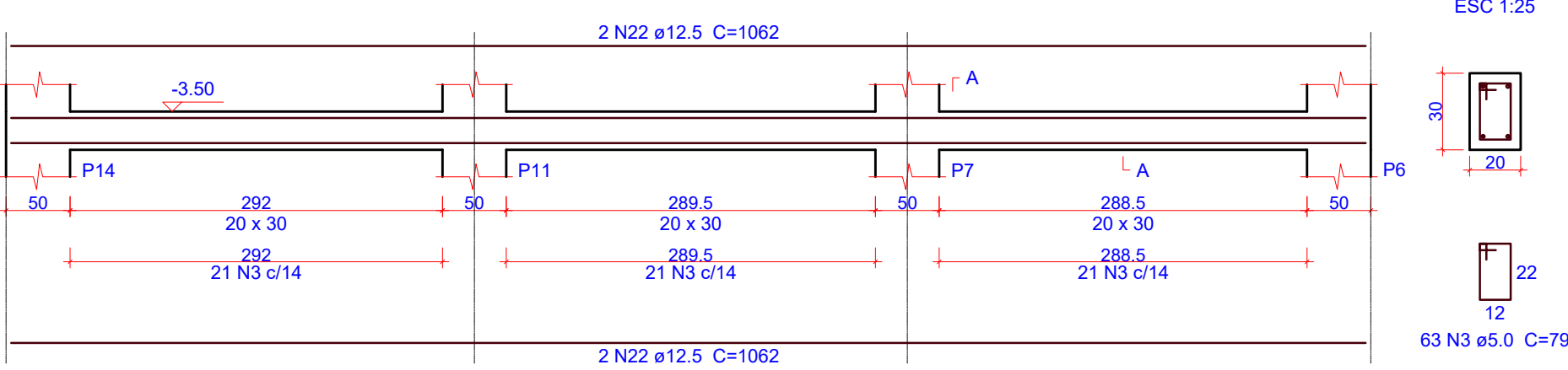
V202B



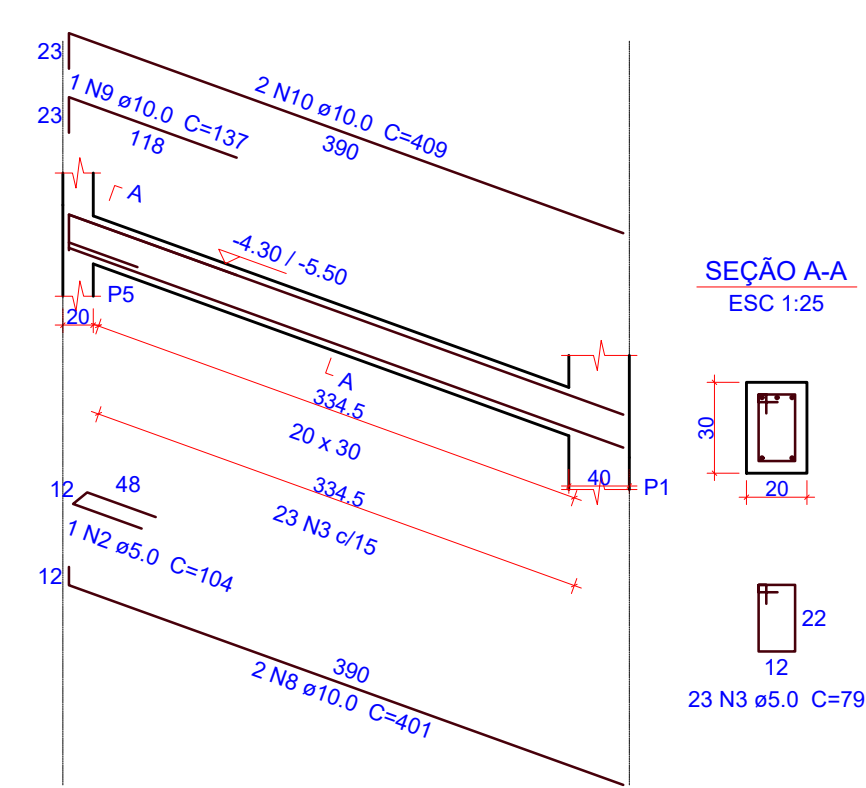
V203



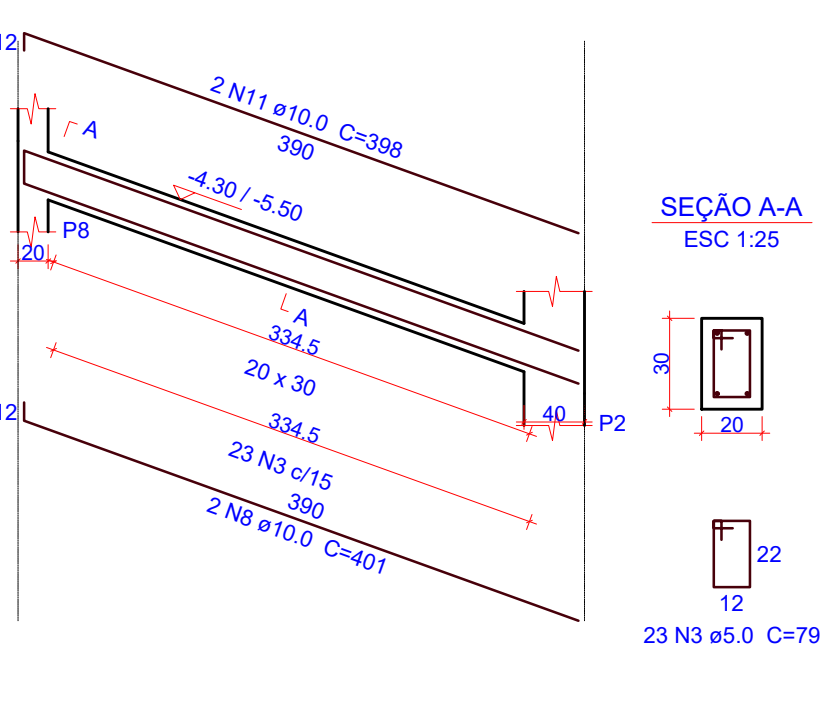
V204



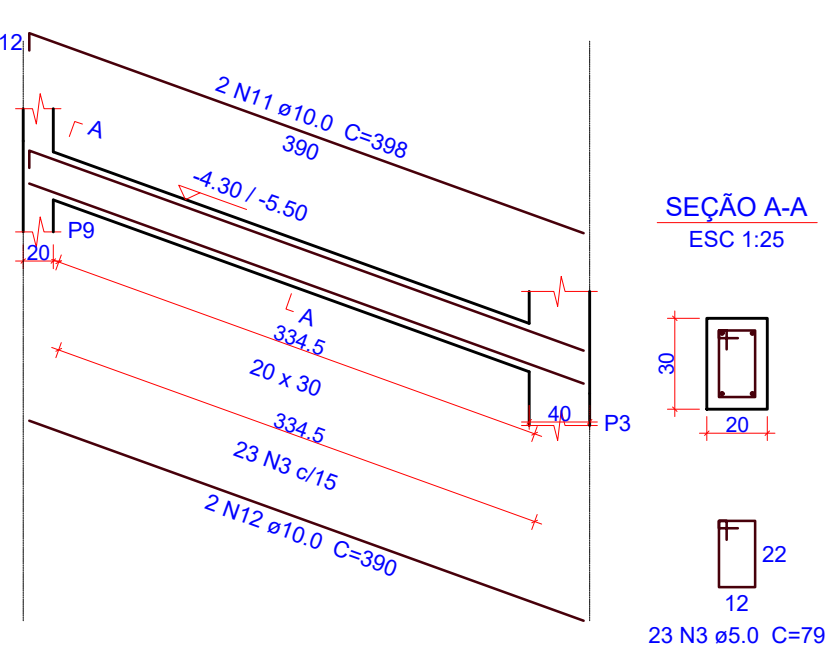
V205



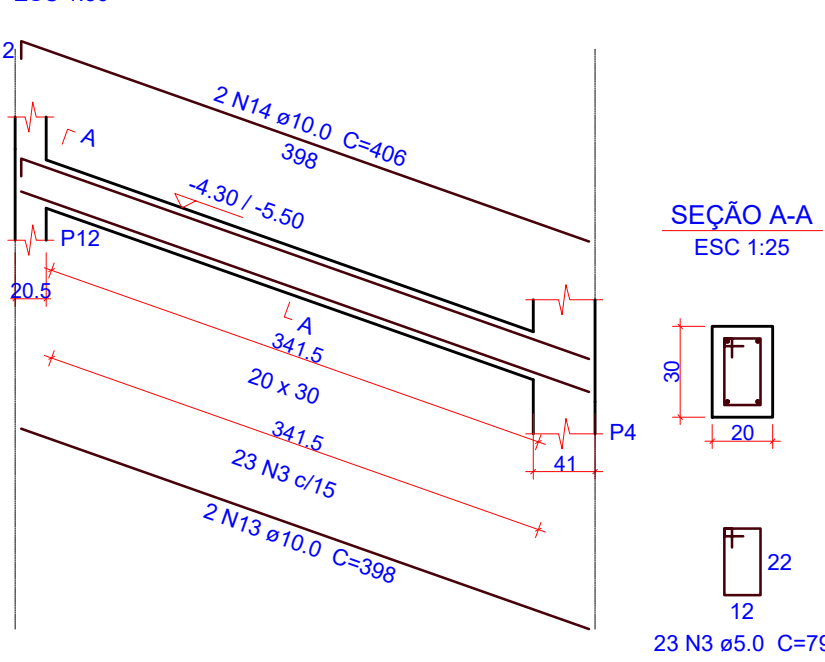
V206



V207



V208



Relação do aço

Fundação Inferior:	V100	
Fundo Piscina:	V200	V200A
	V201	V202
	V202B	V203
	V204	V205
	V206	V207
	V208	

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	132	97	12804
	2	5.0	2	104	208
	3	5.0	350	79	27650
CA50	4	10.0	2	630	1260
	5	10.0	2	672	1344
	6	10.0	1	145	145
	7	10.0	2	610	1220
	8	10.0	4	401	1604
	9	10.0	1	137	137
	10	10.0	2	409	818
	11	10.0	4	398	1592
	12	10.0	2	390	780
	13	10.0	2	398	796
	14	10.0	2	406	812
	15	12.5	8	1057	8456
	16	12.5	2	596	1192
	17	12.5	3	636	1908
	18	12.5	4	264	1056
	19	12.5	6	332	1992
	20	12.5	2	371	742
	21	12.5	4	982	3928
	22	12.5	4	1062	4248

Resumo do aço

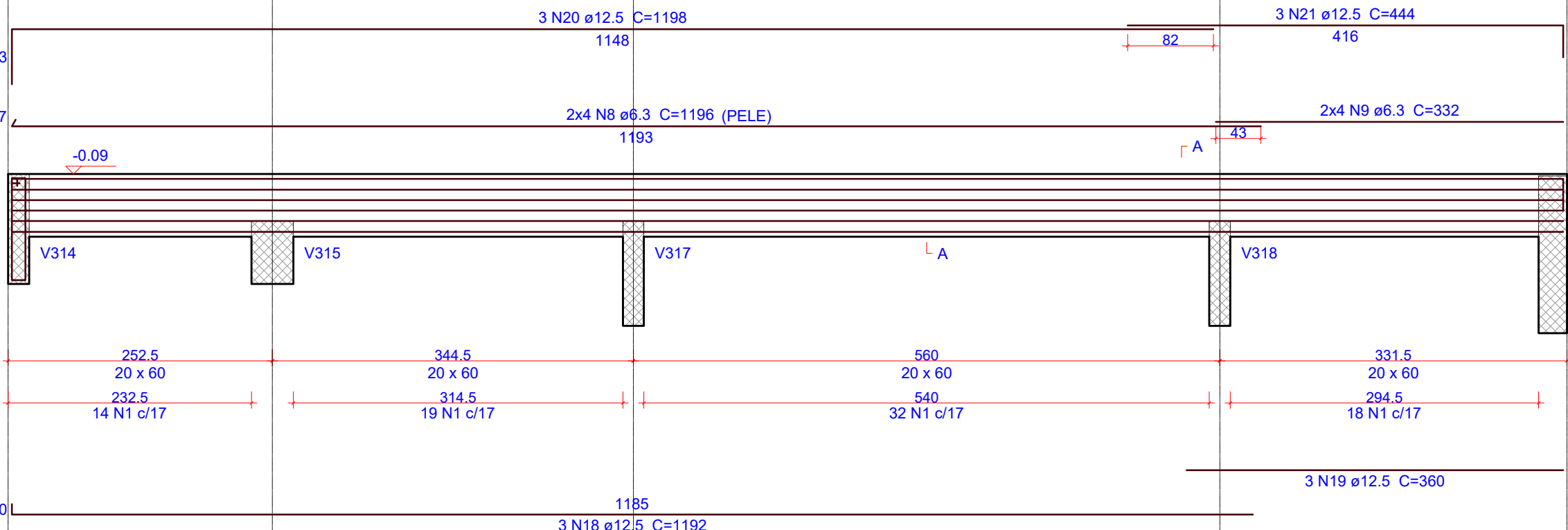
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	10.0	103.1	10	69.9
	12.5	235.3	22	249.3
CA60	5.0	406.7	38	66.9
PESO TOTAL (kg)				
CA50		319.2		
CA60		68.9		

Volume de concreto (C-30) = 4.04 m³
Área de forma = 48.48 m²

Nº	REVISÃO	DATA	CLIENTE	JULIANO CORBELLINI
A	EMISSÃO INICIAL	15/12/2025	LOCALIDADE - MUNICÍPIO	PORTO ALEGRE - RS
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO	RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL	PROJETO ESTRUTURAL
				PLANTA DE ARMADURAS
				DETALHAMENTO VIGAS - FUNDAÇÃO INFERIOR E FUNDO PISCINA
			PROJETO	
			Engº Rafael Buss	
			CREA/SC	
			133.073-9	
			DESENHO	
			MÃO	
			ESCALA	
			INDICADA	
			DATA	
			15/12/2025	
			DATA TOP.	
			FOLHA Nº	EST-0130-B

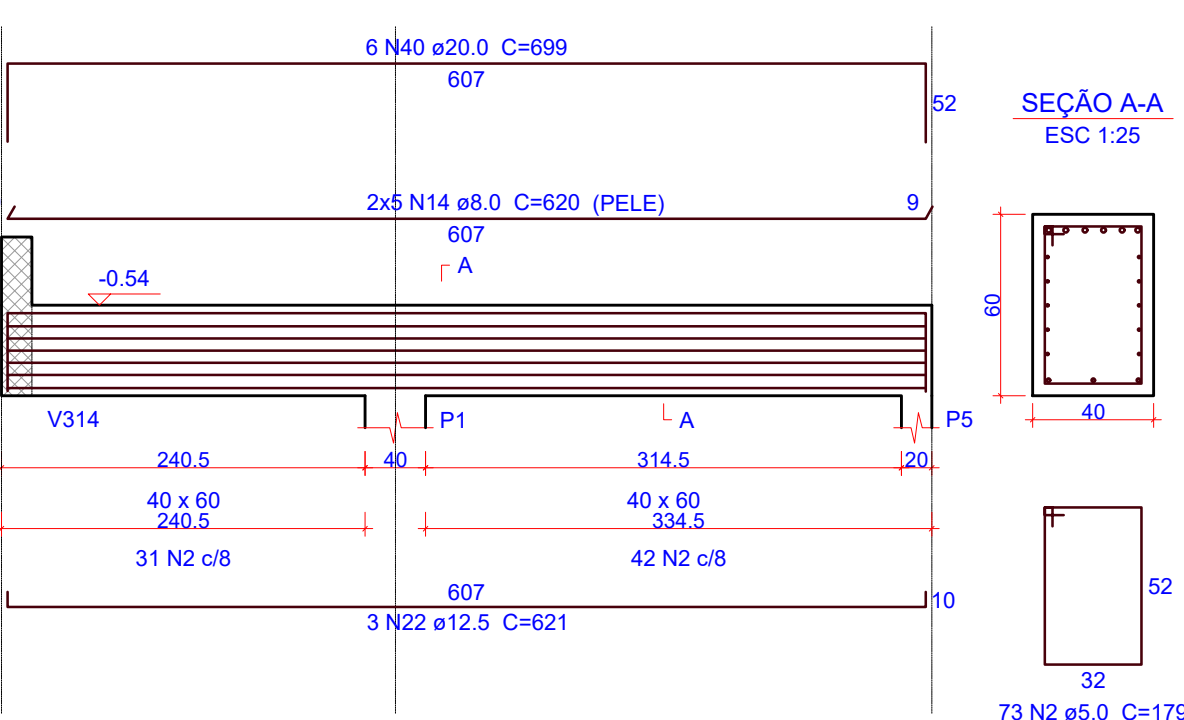
V300

ESC 1:50



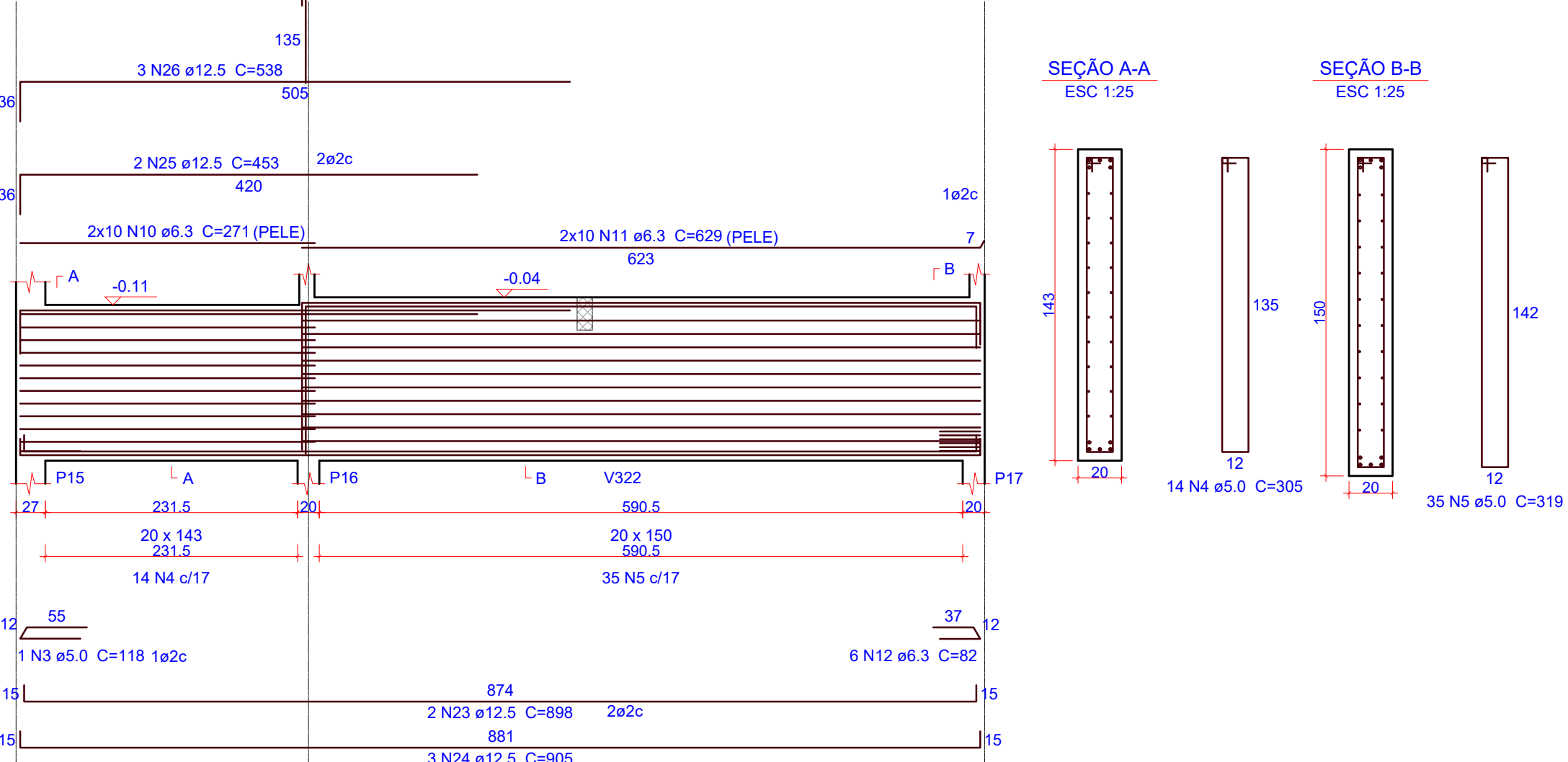
V301

ESC 1:50



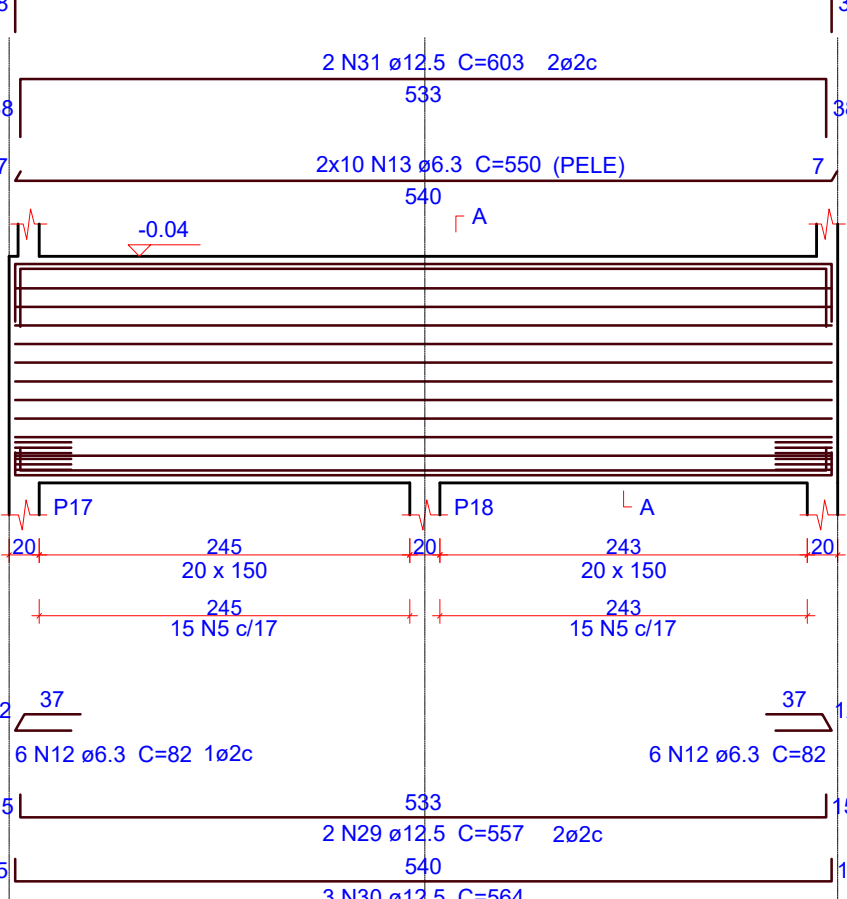
V302

ESC 1:50



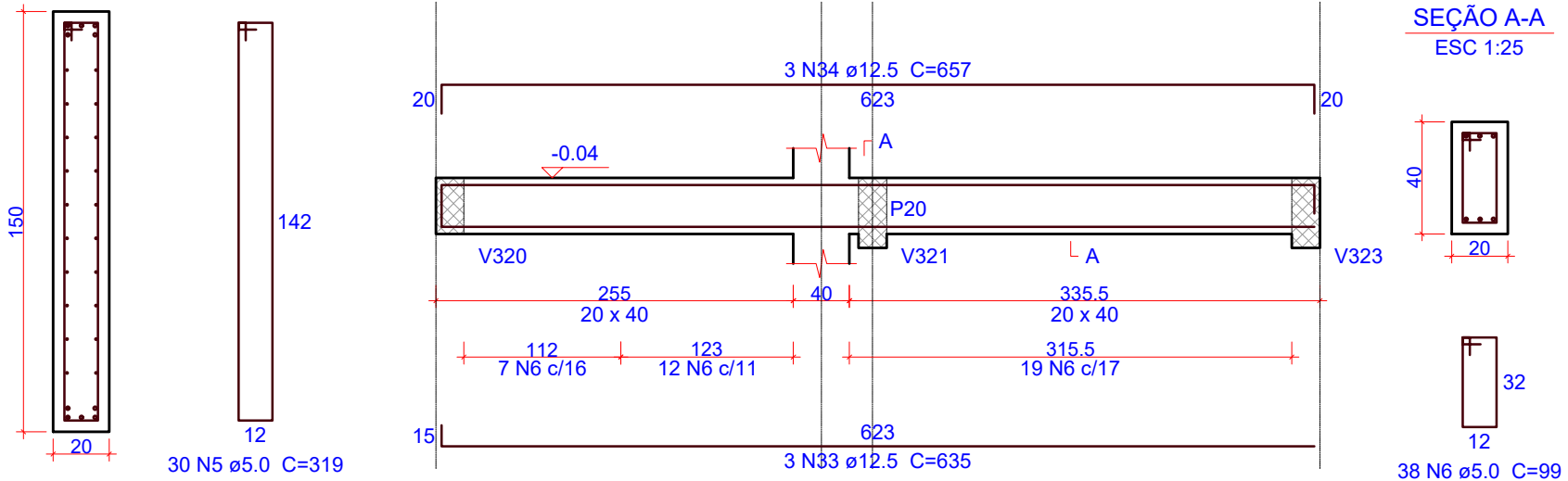
V303

ESC 1:50



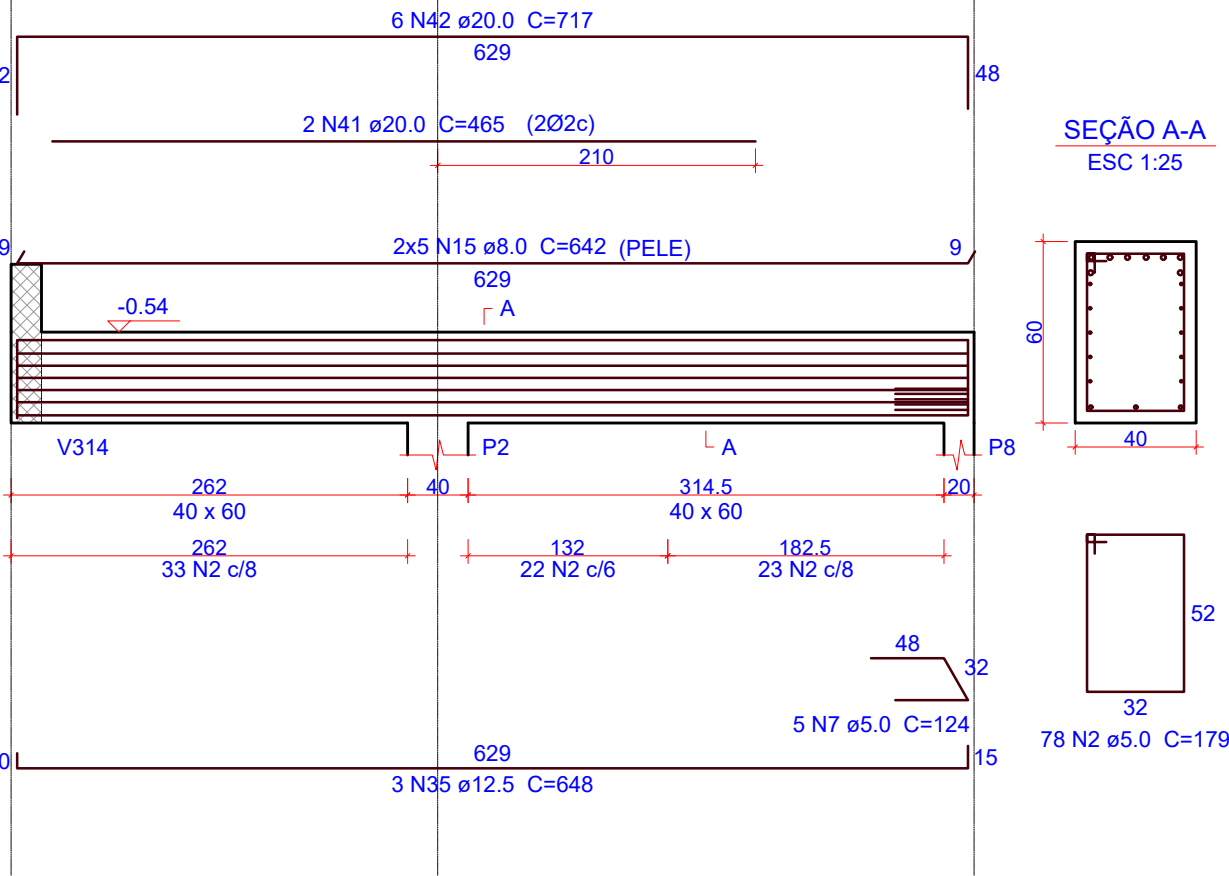
V304

ESC 1:50



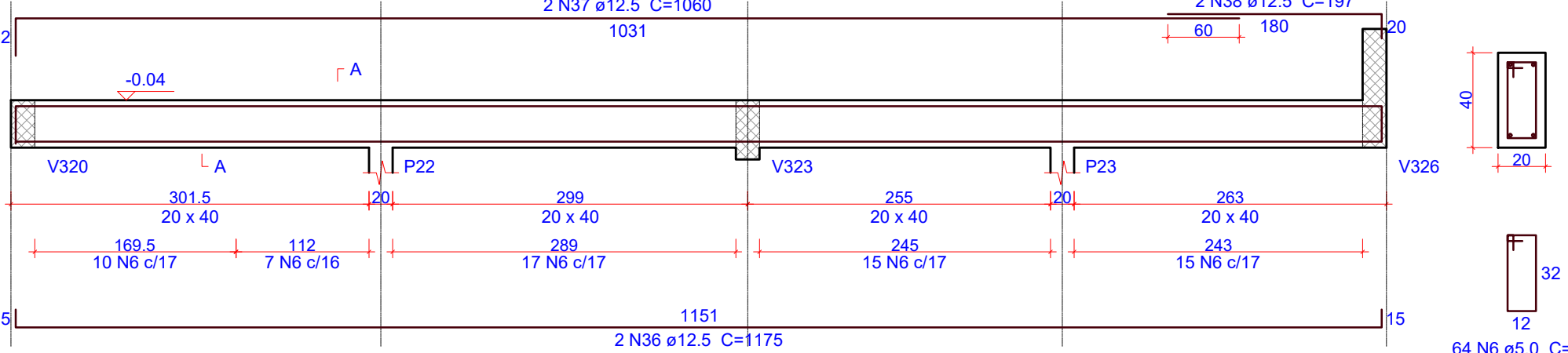
V305

ESC 1:50



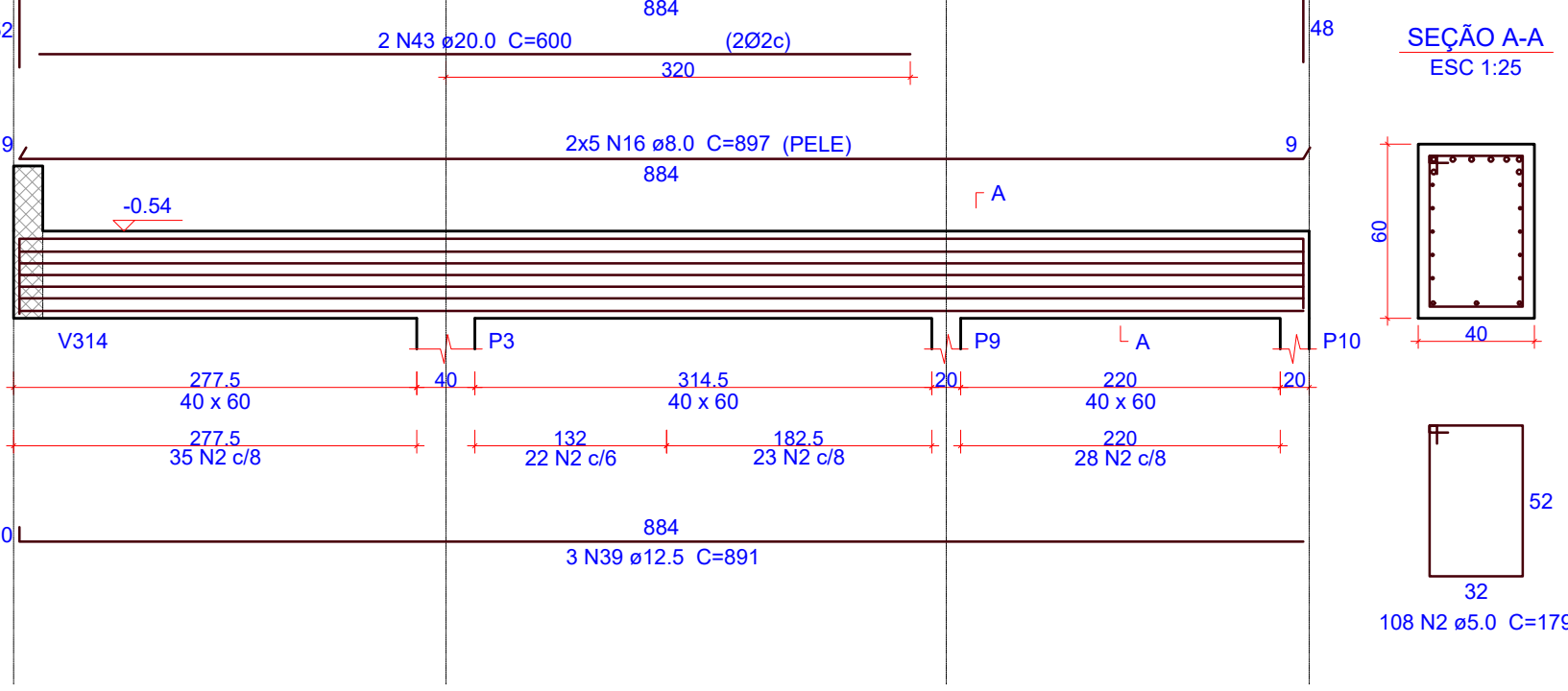
V306

ESC 1:50



V307

ESC 1:50



Relação do aço

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	83	143	11869
	2	5.0	259	179	46361
	3	5.0	1	118	118
	4	5.0	14	305	4270
	5	5.0	65	319	20735
	6	5.0	102	99	10098
	7	5.0	5	124	620
	8	6.3	8	1196	9568
	9	6.3	8	332	2656
	10	6.3	20	271	5420
	11	6.3	20	629	12580
	12	6.3	18	82	1476
	13	6.3	20	550	11000
	14	8.0	10	620	6200
	15	8.0	10	642	6420
	16	8.0	10	897	8970
	17	10.0	1	235	235
	18	12.5	3	1192	3576
	19	12.5	3	360	1080
	20	12.5	3	1198	3594
	21	12.5	3	444	1332
	22	12.5	3	621	1863
	23	12.5	2	898	1796
	24	12.5	3	905	2715
	25	12.5	2	453	906
	26	12.5	3	538	1614
	27	12.5	1	782	782
	28	12.5	3	790	2370
	29	12.5	2	557	1114
	30	12.5	3	564	1692
	31	12.5	2	603	1206
	32	12.5	3	610	1830
	33	12.5	3	635	1905
	34	12.5	3	657	1971
	35	12.5	3	648	1944
	36	12.5	2	1175	2350
	37	12.5	2	1060	2120
	38	12.5	2	197	394
	39	12.5	3	891	2673
	40	20.0	6	699	4194
	41	20.0	2	465	930
	42	20.0	6	717	4302
	43	20.0	2	600	1200
	44	20.0	6	972	5832

Resumo do aço

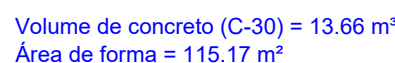
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	427	40	114.9
	8.0	215.9	20	93.7
	10.0	2.4	1	1.6
	12.5	408.3	38	432.6
	20.0	164.6	16	446.5
CA60	5.0	940.8	87	159.5
PESO TOTAL (kg)				
CA50	1089.3			
CA60	159.5			

Volume de concreto (C-30) = 11.59 m³
Área de forma = 95.91 m²

1 DETALHAMENTO VIGAS - TÉRREO (1/3)

Nº	REVISÃO	DATA	CLIENTE
A	EMISSÃO INICIAL	15/12/2025	JULIANO CORBELLINI
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	PORTO ALEGRE - RS
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			DETALHAMENTO VIGAS - TÉRREO (1/3)
			PROJETO
			Engº Rafael Buss
			CREA/SC 133.073-9
			DESIGNO
			ESCALA INDICADA
			DATA
			DATA TOP.
			FOLHA Nº
			EST-0140-B

ESC 1:50

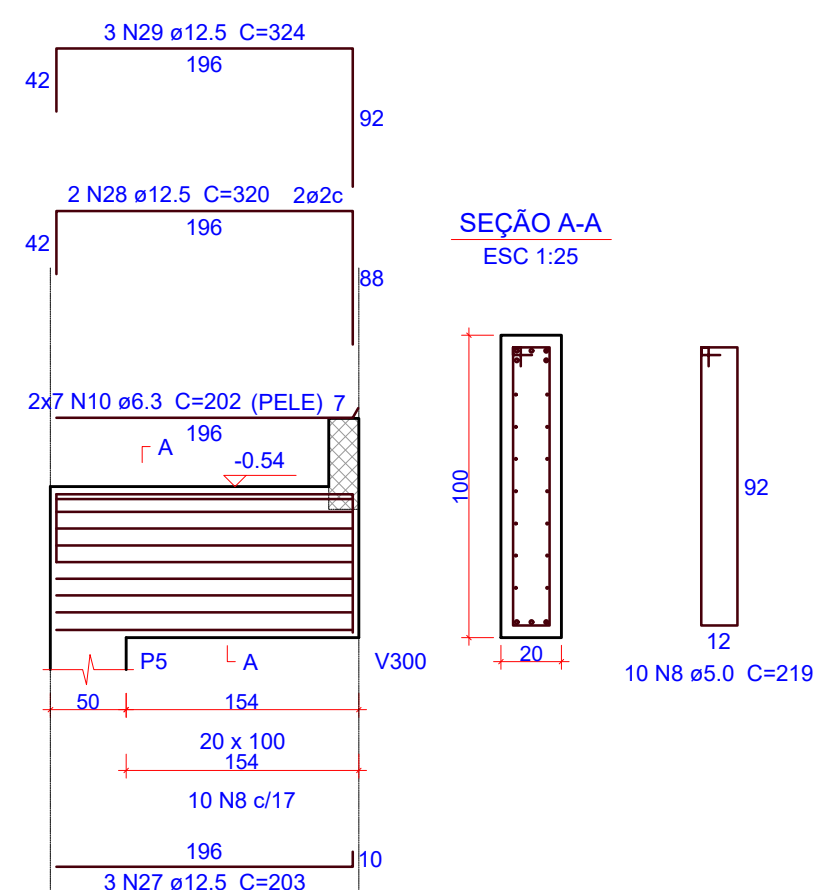


ESC 1:50

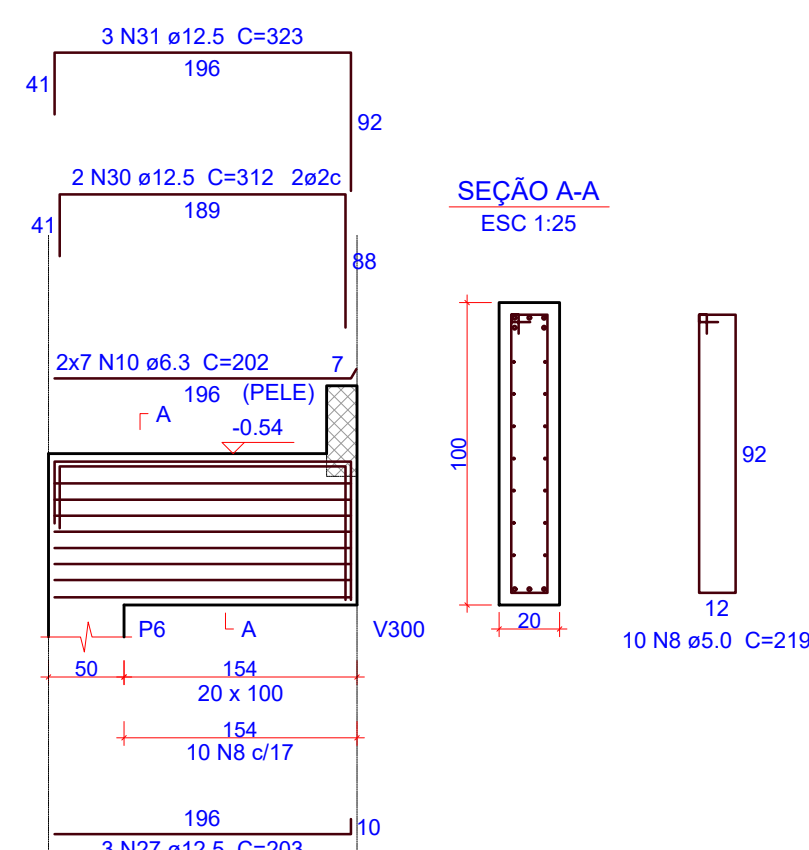


Nº		REVISÃO	DATA	CLIENTE			
A		EMISSÃO INICIAL	15/12/2025	LOCALIDADE = MUNICÍPIO			
B		REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO			
				RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207			
				PROJETO ESTRUTURAL			
				PROJETO ESTRUTURAL			
				PLANTA DE ARMADURAS			
				DETALHAMENTO VIGAS - TÉRREO (2/3)			
				PROJETO	DESENHO	DATA	FOLHA Nº
				Engº Rafael Buss	MMD	15/12/2025	
				133.073-9	ESCALA	DATA TOP:	EST-0150-B
					INDICADA		

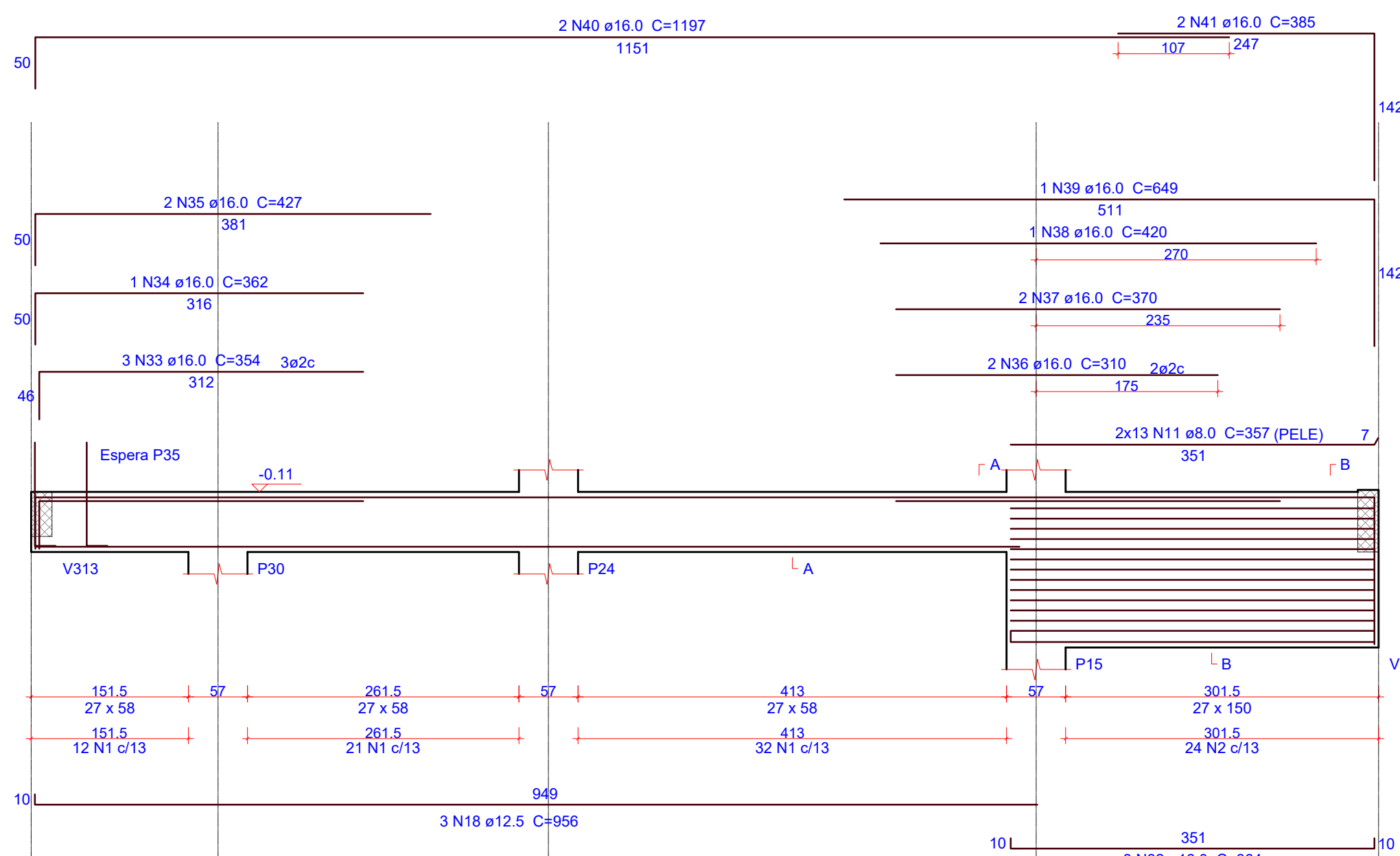
ESC 1:50



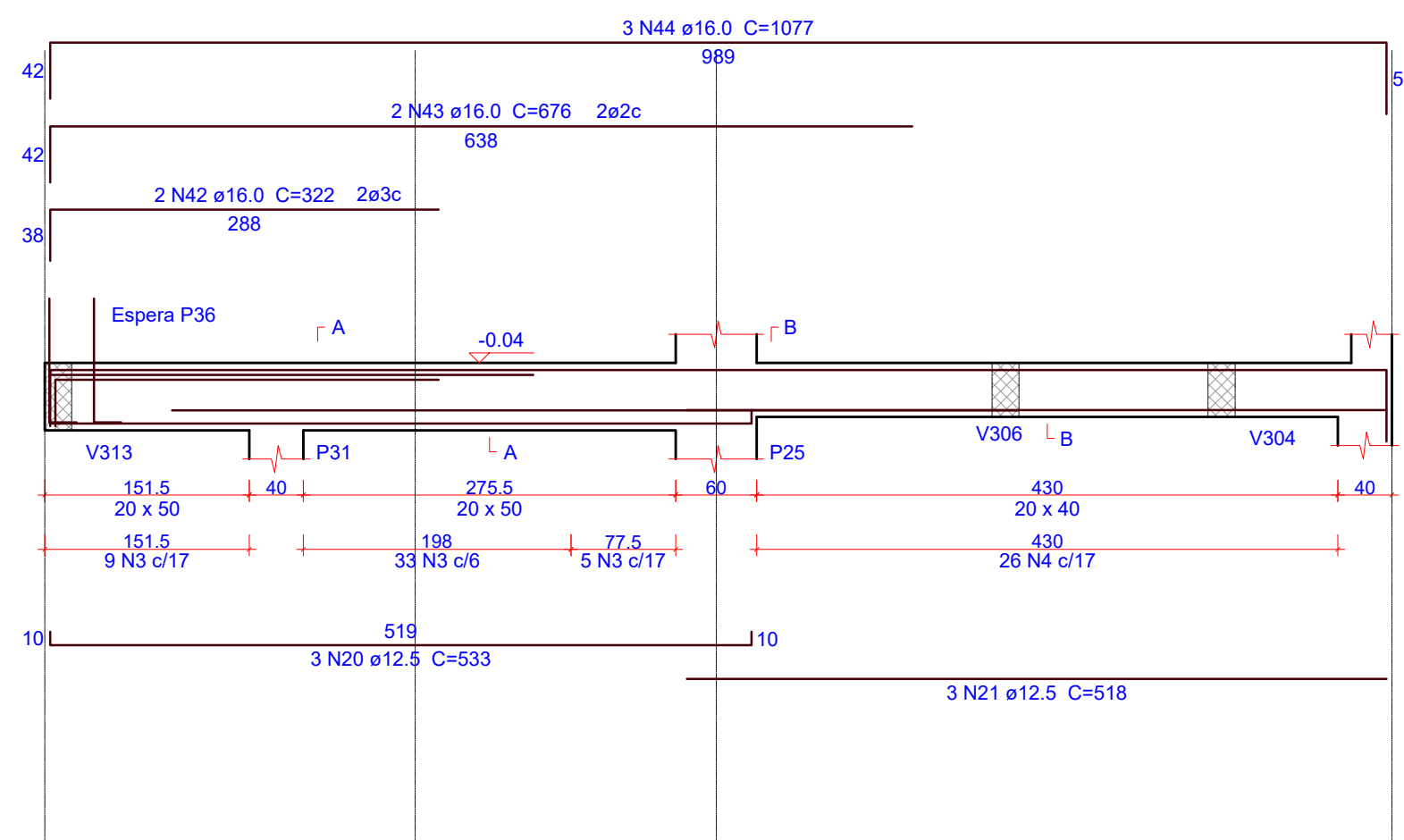
ESC 1:50



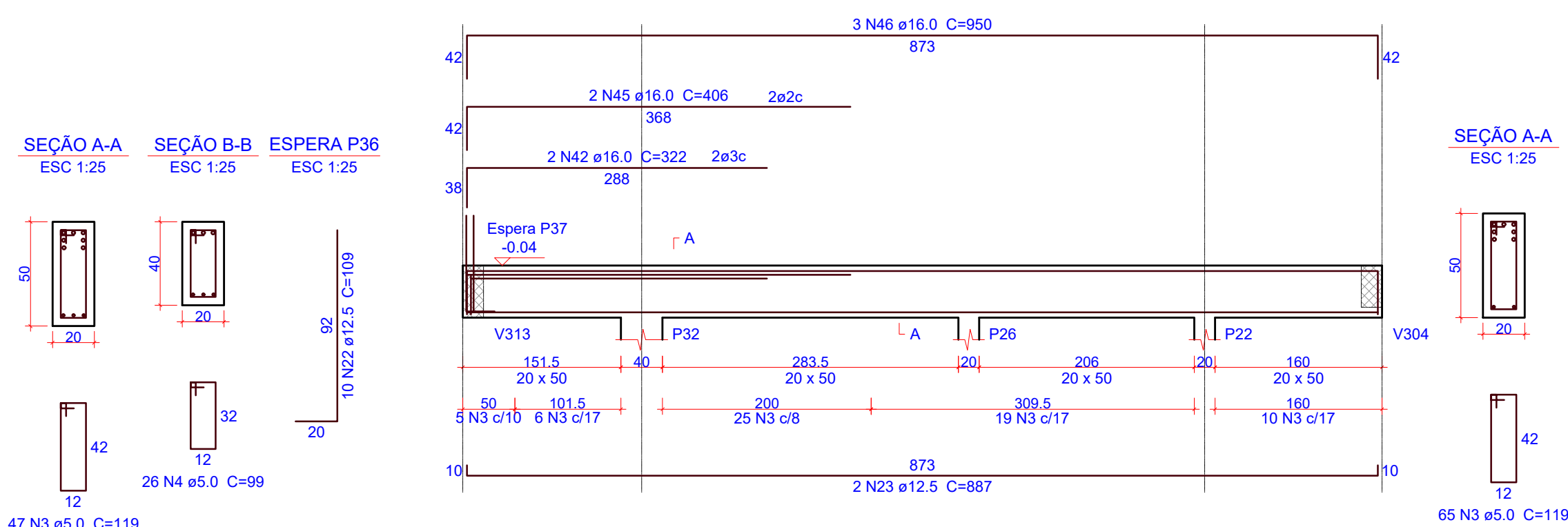
ESC 1:50



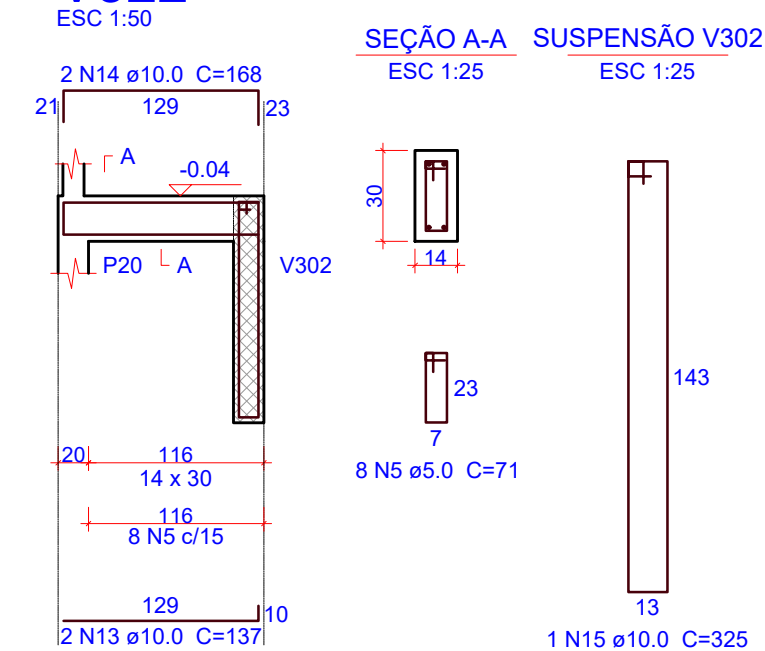
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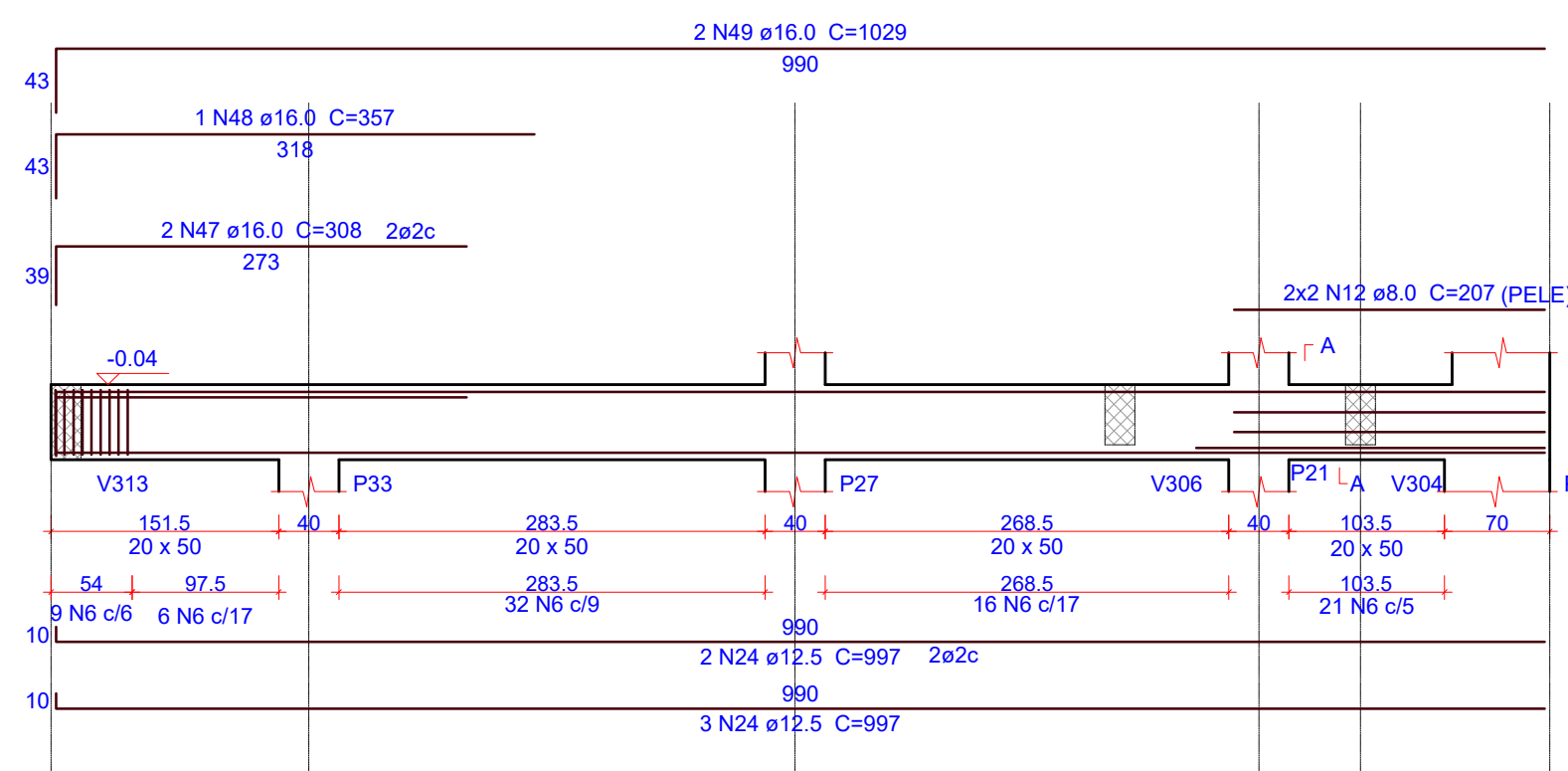
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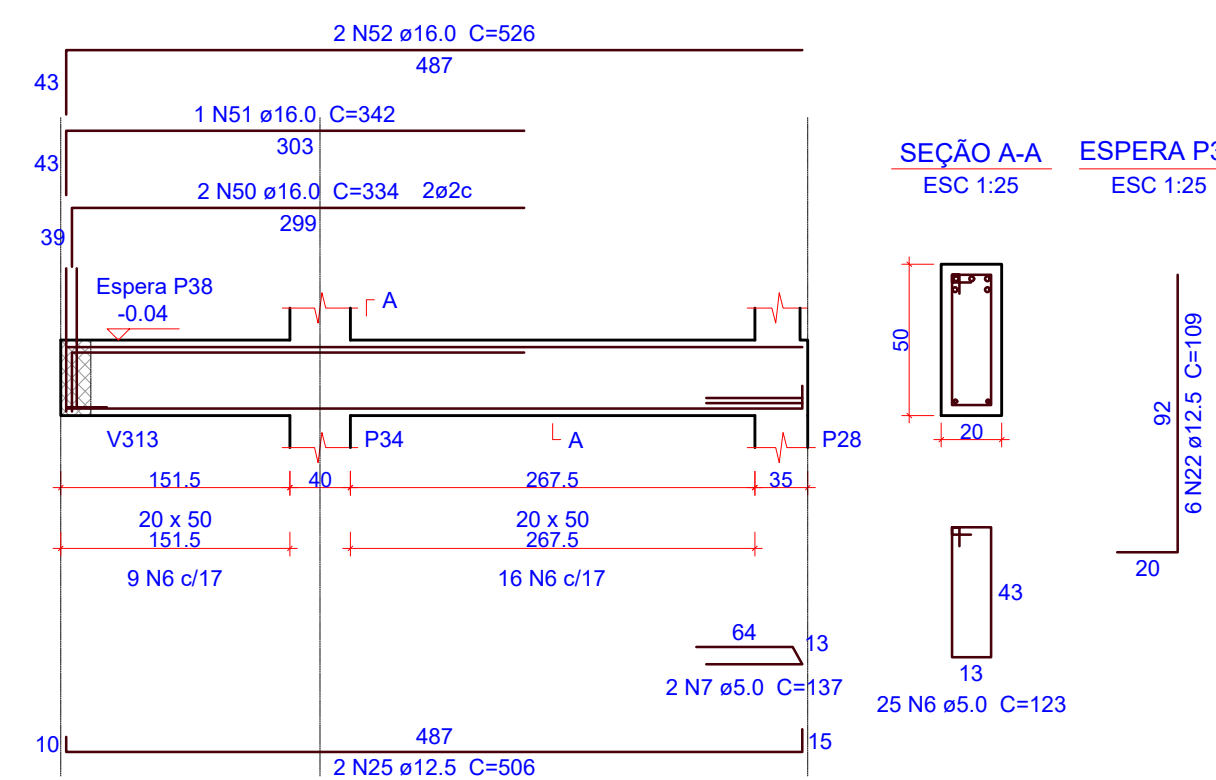
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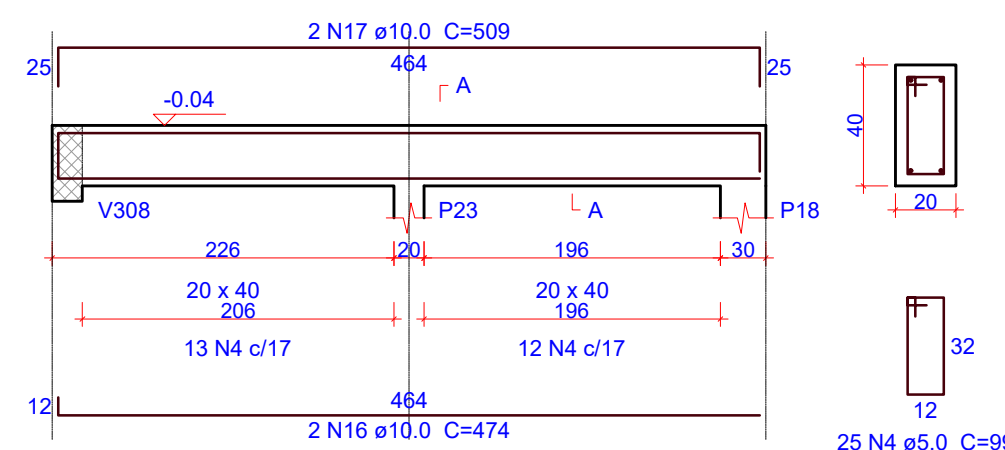
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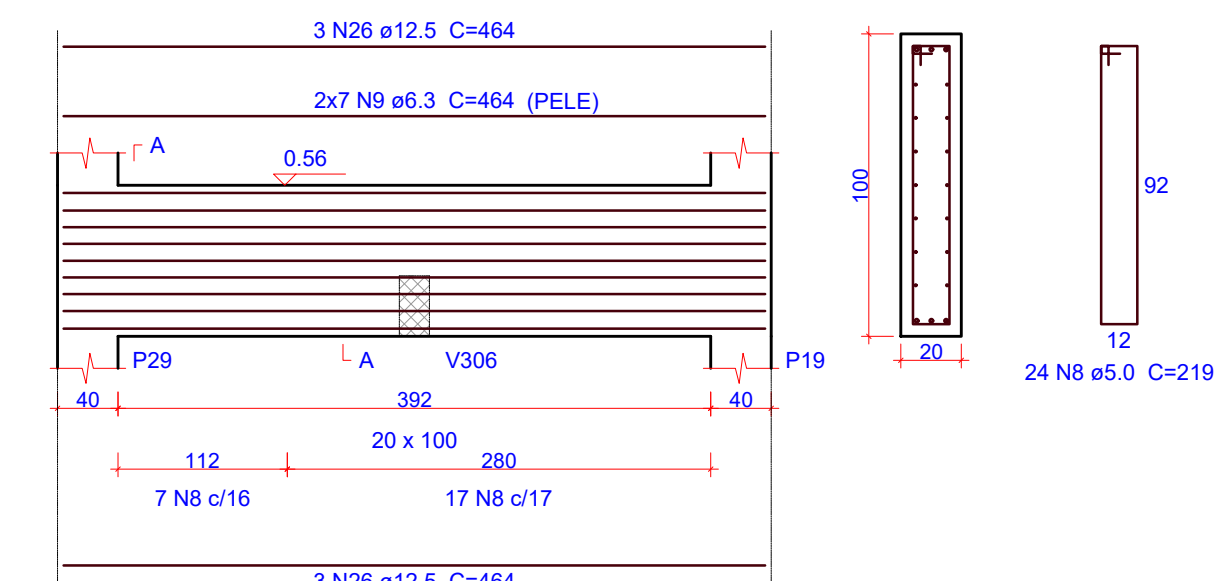
ESC 1:50



ESC 1:50



ESC 1:50



14010

V317	V318	V319			
V320	V321	V322			
V323	V324	V325			
V326					
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5,0	65	149	9685
	2	5,0	24	333	7992
	3	5,0	112	119	13320
	4	5,0	5	99	5045
	5	5,0	8	71	5685
CA50	6	5,0	109	123	13407
	7	5,0	2	137	274
	8	5,0	44	219	9636
	9	6,3	14	454	6490
	10	6,3	28	202	5655
	11	8,0	26	357	9282
	12	8,0	4	207	828
	13	10,0	2	137	274
	14	10,0	2	168	336
	15	10,0	1	325	325
	16	10,0	2	474	948
	17	10,0	2	509	1018
	18	12,5	6	956	5736
	19	12,5	8	936	117
	20	12,5	3	533	1599
	21	12,5	3	518	1554
	22	12,5	24	109	2616
	23	12,5	2	887	1774
	24	12,5	5	997	4985
	25	12,5	2	506	1012
	26	12,5	6	464	2784
	27	12,5	6	203	1218
	28	12,5	2	320	640
	29	12,5	3	324	972
	30	12,5	2	625	1250
	31	12,5	3	323	969
	32	16,0	3	364	1092
33	16,0	3	354	1062	
34	16,0	1	362	362	
35	16,0	1	427	854	
36	16,0	2	310	620	
37	16,0	2	370	740	
38	16,0	1	420	420	
39	16,0	1	649	649	
40	16,0	2	1197	2394	
41	16,0	2	385	770	
42	16,0	4	322	1288	
43	16,0	2	676	1352	
44	16,0	3	1077	3231	
45	16,0	2	406	812	
46	16,0	2	550	2850	
47	16,0	2	308	616	
48	16,0	1	357	357	
50	16,0	2	1029	2058	
51	16,0	2	334	668	
52	16,0	1	342	342	
53	16,0	2	526	1052	

TOTAL	QUANT : 10
-------	------------

AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	121.6	12	32.7
	8.0	101.1	10	43.9
	10.0	29.1	3	19.7
	12.5	274.2	26	290.6
	16.0	235.9	22	409.5
CA60	5.0	599.4	55	101.6

Volume de concreto (C-30) = 7.09 m³
Área de forma = 63.22 m²

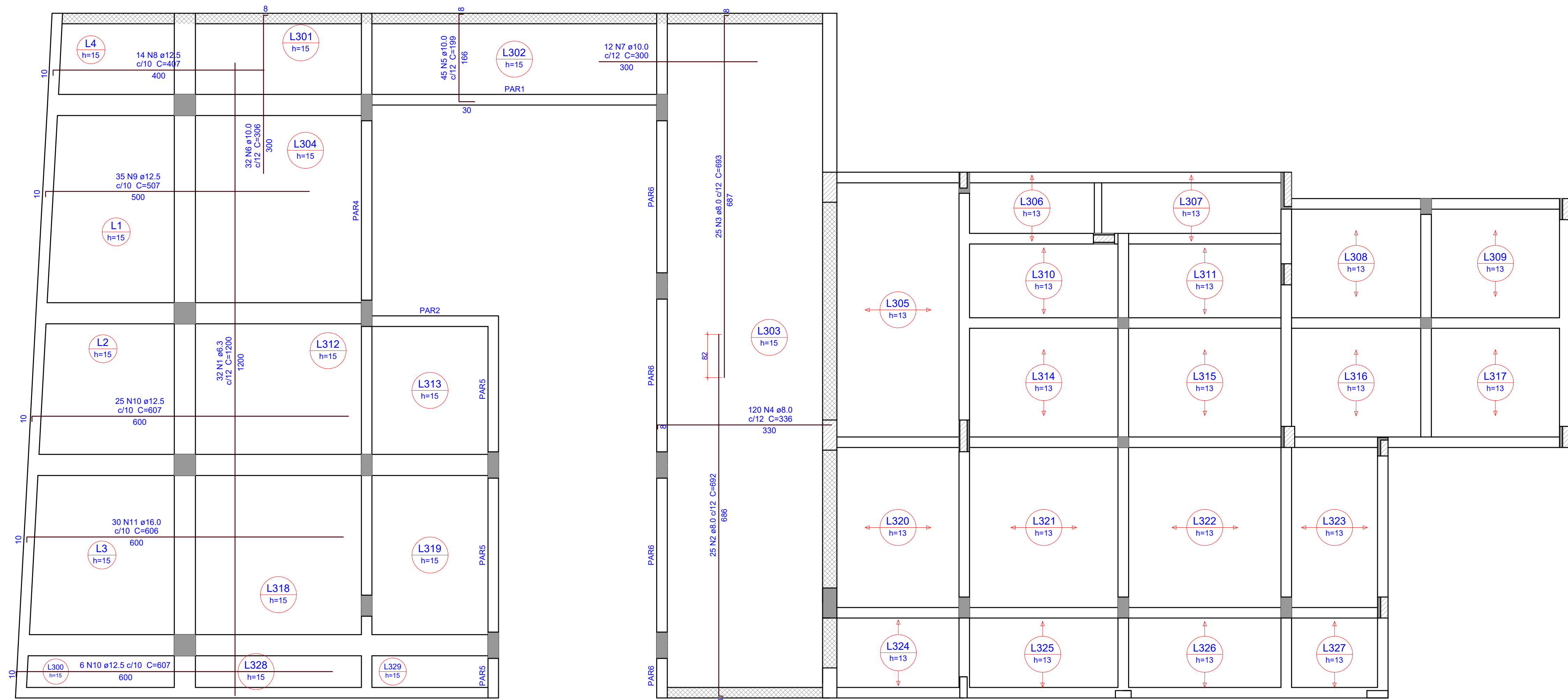
Relação do aço

Negativos

ÇÖ	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA50	1	6.3	32	1200	38400
	2	8.0	25	692	17300
	3	8.0	25	693	17325
	4	8.0	120	336	40320
	5	10.0	45	199	8955
	6	10.0	32	306	9792
	7	10.0	12	300	3600
	8	12.5	14	407	5698
	9	12.5	35	507	17745
	10	12.5	31	607	18817
	11	16.0	30	606	18180

Resumo do aço

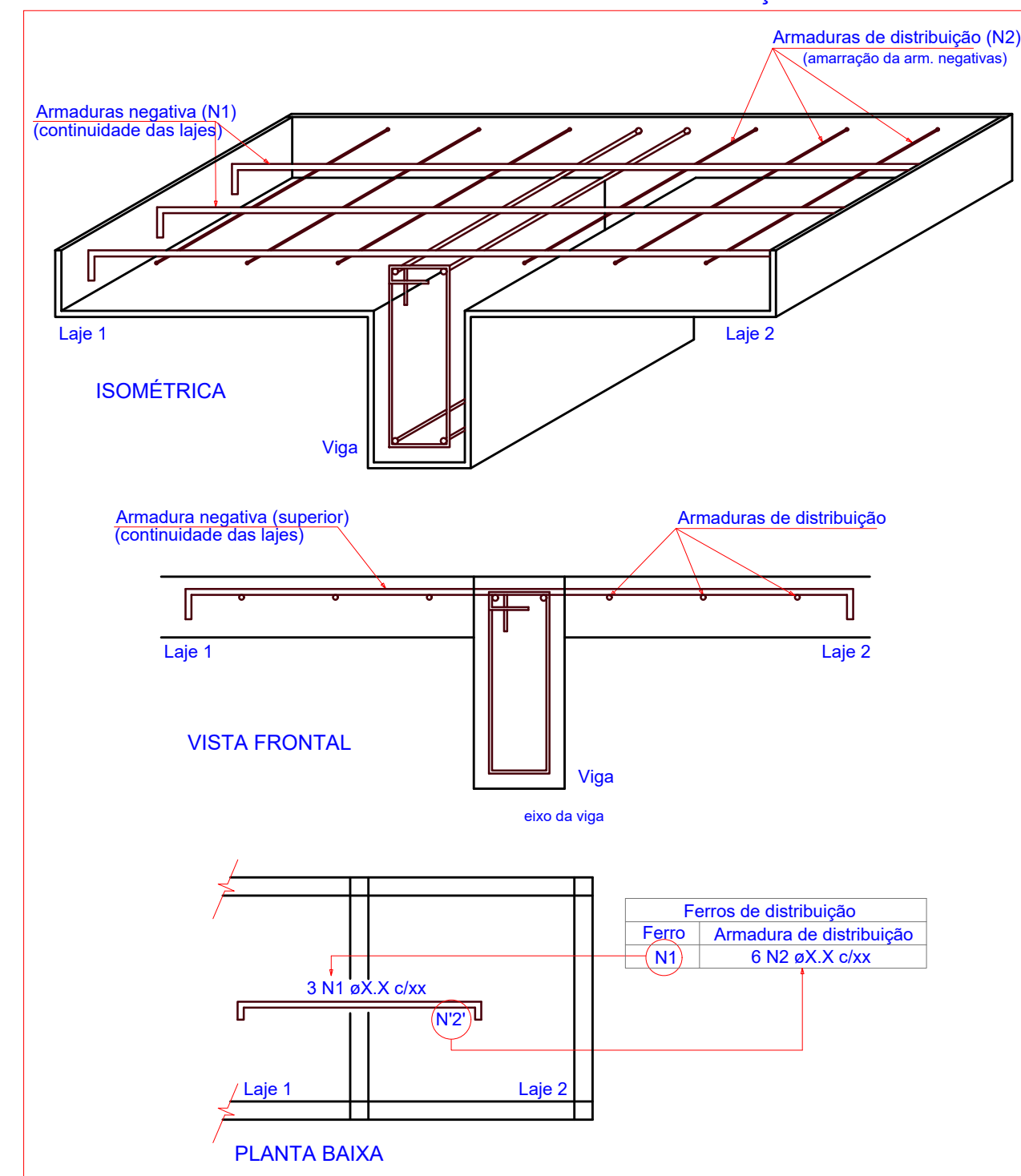
CAÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	384	36	103.4
	8.0	749.5	69	325.3
	10.0	223.5	21	151.6
	12.5	422.6	39	447.8
	16.0	161.8	17	315.6
PESO TOTAL (kg)				
CA50	1343.7			



1 ARMAÇÃO NEGATIVA DAS LAJES - TÉRREO

ESCALA 1:50

DETALHE DA ARMADURA DE SUPERIOR DE CONTINUIDADE DA LAJE
E MONTAGEM DA ARMADURA DE DISTRIBUIÇÃO



Nº	REVISÃO	DATA	CLIENTE JULIANO CORBELLINI			
A	EMISSÃO INICIAL	15/12/2025	LOCALIDADE = MUNICÍPIO PORTO ALEGRE - RS			
B	REVISÃO LOCAÇÃO PILARES	20/07/2026	ENDEREÇO RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207			
			PROJETO ESTRUTURAL PROJETO ESTRUTURAL PLANTA DE ARMADURAS ARMAÇÃO NEGATIVA DAS LAJES - TÉRREO			
			PROJETO Engº Rafael Buss	DESENHO MUD	DATA 15/12/2025	FOLHA Nº EST-0180-B
			ORÇ.VIS. 133.073-9	ESCALA INDICADA	DATA TOP.	



1 PLANTA DE VIGOTAS PRÉ-MOLDADAS - TÉRREO

ESCALA 1:50

Nº		REVISÃO	DATA	CLIENTE	JULIANO CORBELLINI		
A	EMISSÃO INICIAL		15/12/2025	LOCALIDADE - MUNICÍPIO	PORTO ALEGRE - RS		
B	REVISÃO LOCAÇÃO PILARES		20/07/2026	ENDEREÇO	RUA ALCEBÍADES ANTÔNIO DOS SANTOS- 356 - LOTE 207		
				PROJETO ESTRUTURAL			
				PROJETO ESTRUTURAL			
				PLANTA DE ARMADURAS			
				PLANTA DE VIGOTAS PRÉ-MOLDADAS - TÉRREO			
				PROJETO Engº Rafael Buss	DESENHO MÃO	DATA 15/12/2025	FOLHA Nº
				CREA/SC 133.073-9	ESCALA INDICADA	DATA E TOP.	EST-0190-B

ESC 1:50



ESC 1:50



ESC 1:50



Relação do aço

ESC 1:50



ESC 1:50



ESC 1:50



ESC 1:50



ESC 1:50



JULIANO CORBELLINI

PORTO ALEGRE - RS

RUA ALCEBÍADES ANTÔNIO DOS SANTOS- 356 - LOTE 207

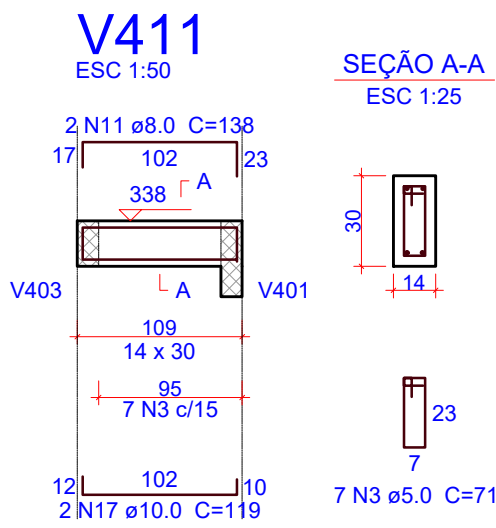
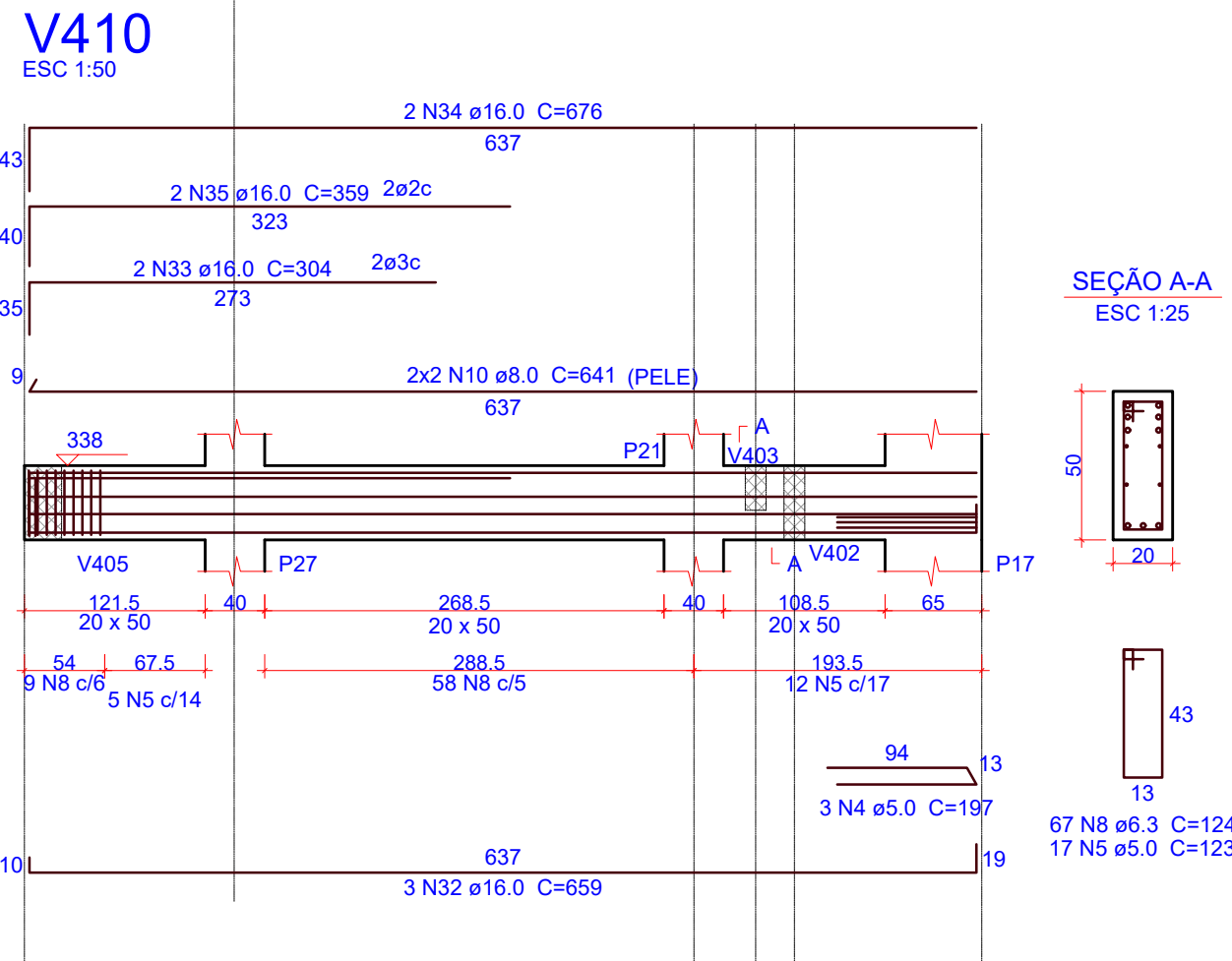
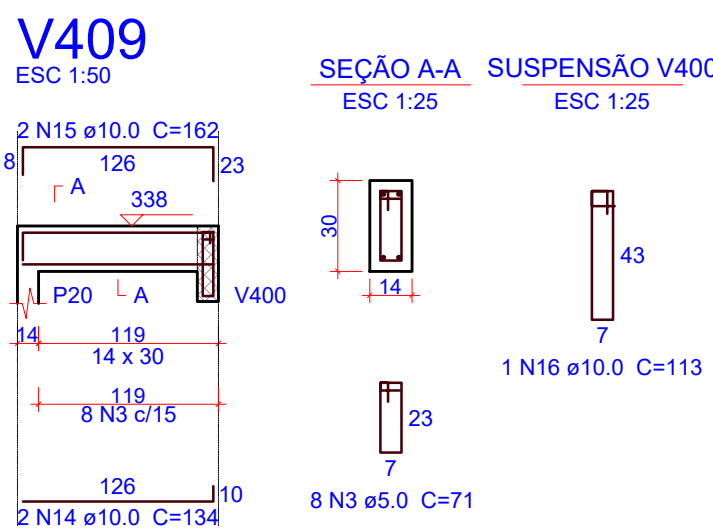
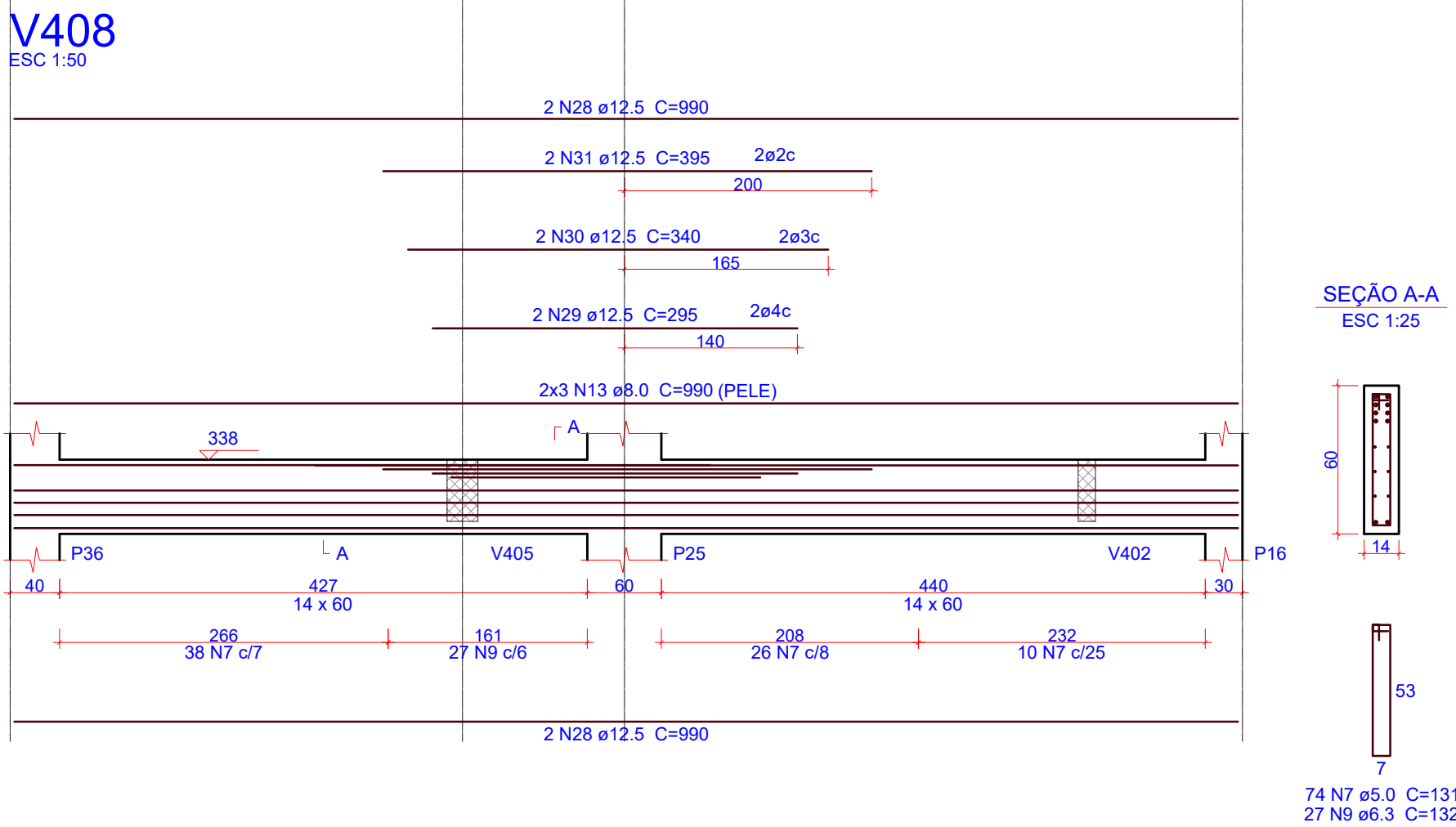
PROJETO ESTRUTURAL
PROJETO ESTRUTURAL
PLANTA DE ARMADURAS
DETALHAMENTO VIGAS - 1º PAVIMENTO (1/2)

PROJETO
Engº Rafeol Buss
CREA/SC
133.073-9

DESENHO MMD	
ESCALA INDICADA	

DATA	15/12/2
DATA TOP	

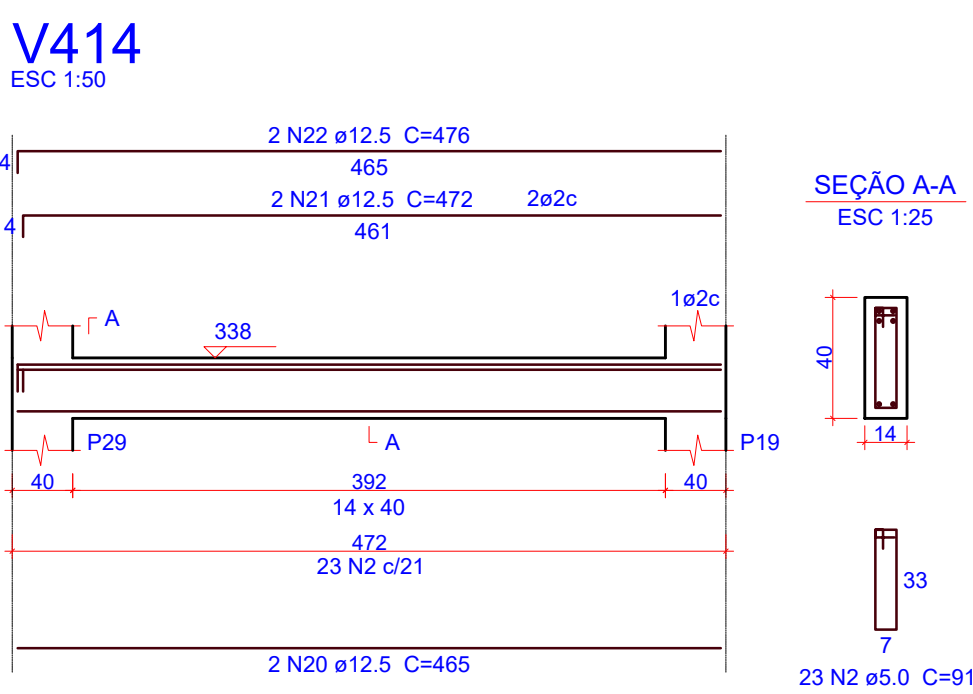
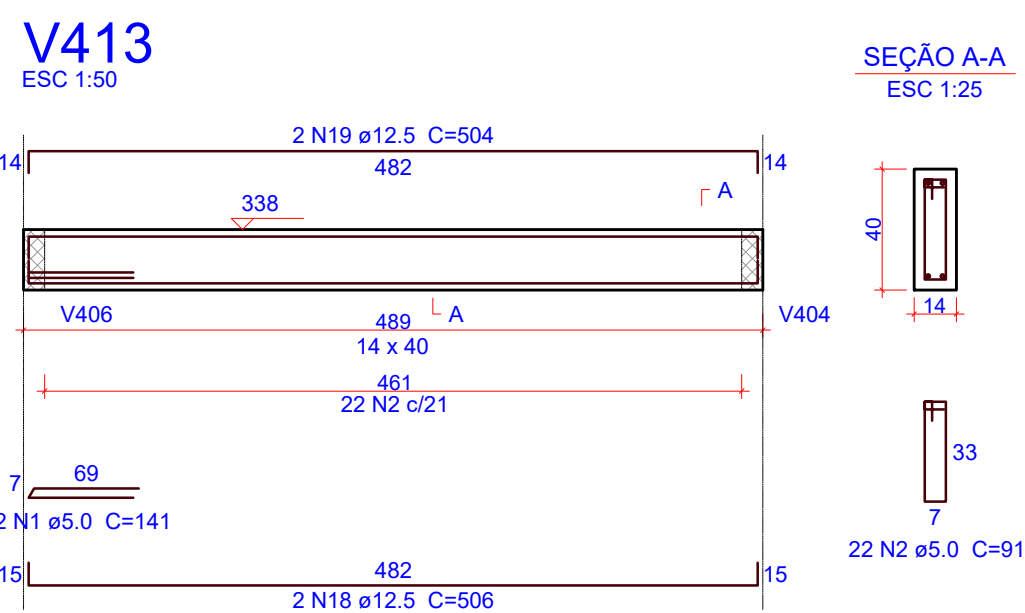
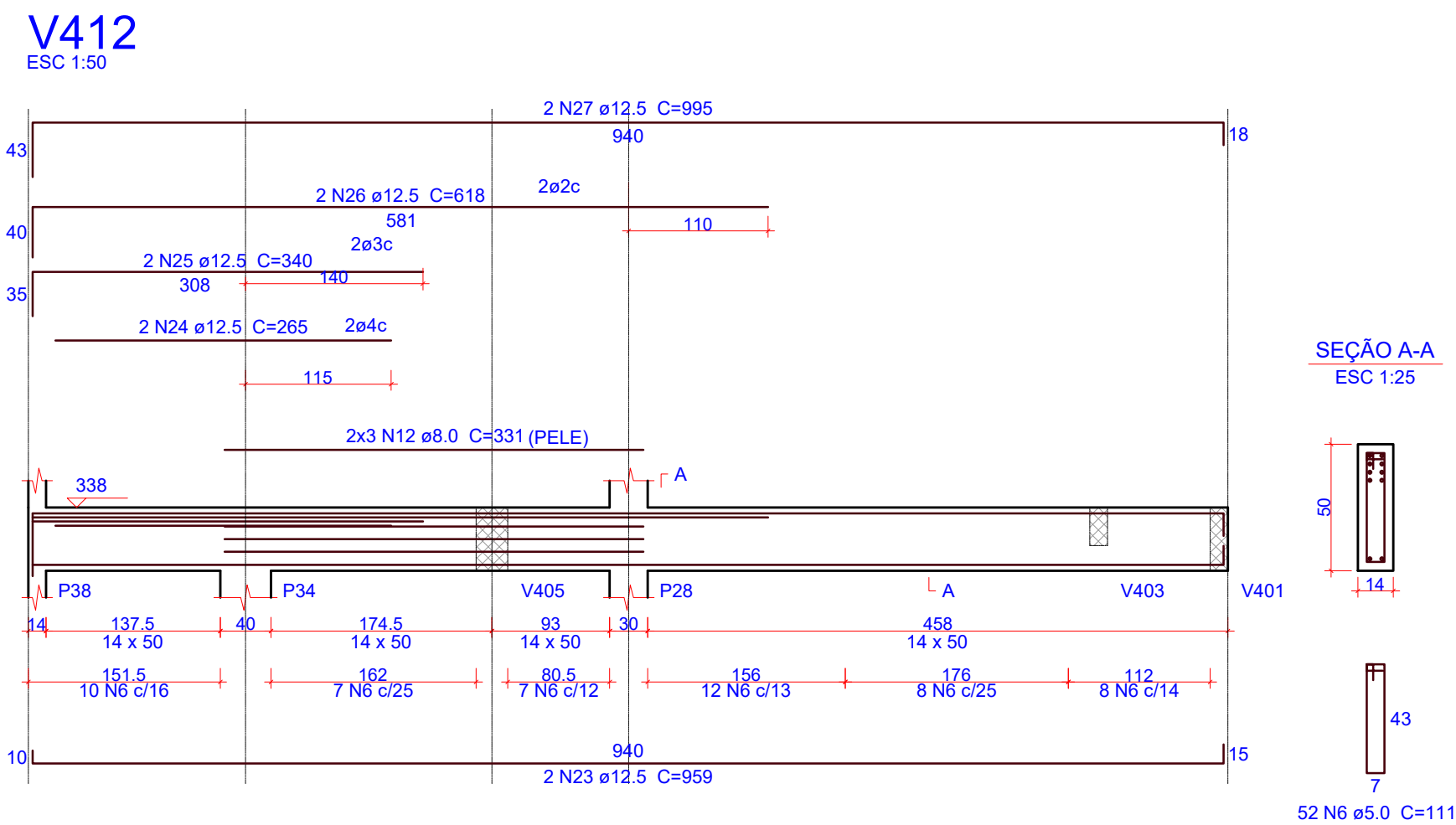
EST-0200-A



Relação do aço					
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	2	141	282
	2	5.0	45	91	4095
	3	5.0	15	71	1065
	4	5.0	3	197	591
	5	5.0	17	123	2091
CA50	6	5.0	52	111	5772
	7	5.0	74	131	9694
	8	6.3	67	124	8308
	9	6.3	27	132	3564
	10	8.0	4	641	2564
	11	8.0	2	138	276
	12	8.0	6	331	1986
	13	8.0	6	990	5940
	14	10.0	2	134	268
	15	10.0	2	162	324
	16	10.0	1	113	113
	17	10.0	2	119	238
	18	12.5	2	506	1012
	19	12.5	2	504	1008
	20	12.5	2	465	930
	21	12.5	2	472	944
	22	12.5	2	476	952
	23	12.5	2	959	1918
	24	12.5	2	265	530
	25	12.5	2	340	680
CA60	26	12.5	2	618	1236
	27	12.5	2	995	1990
	28	12.5	4	990	3960
	29	12.5	2	295	590
	30	12.5	2	340	680
	31	12.5	2	395	790
	32	16.0	3	659	1977
	33	16.0	2	304	608
	34	16.0	2	676	1352
	35	16.0	2	359	718

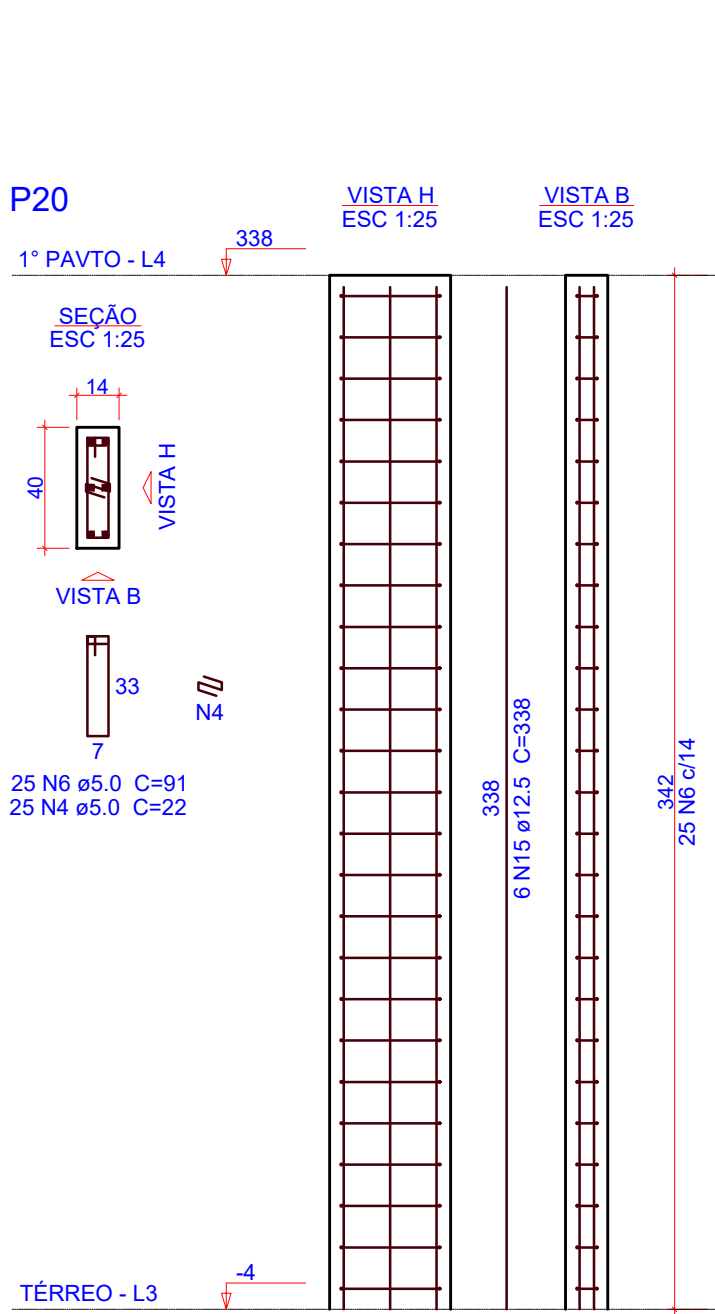
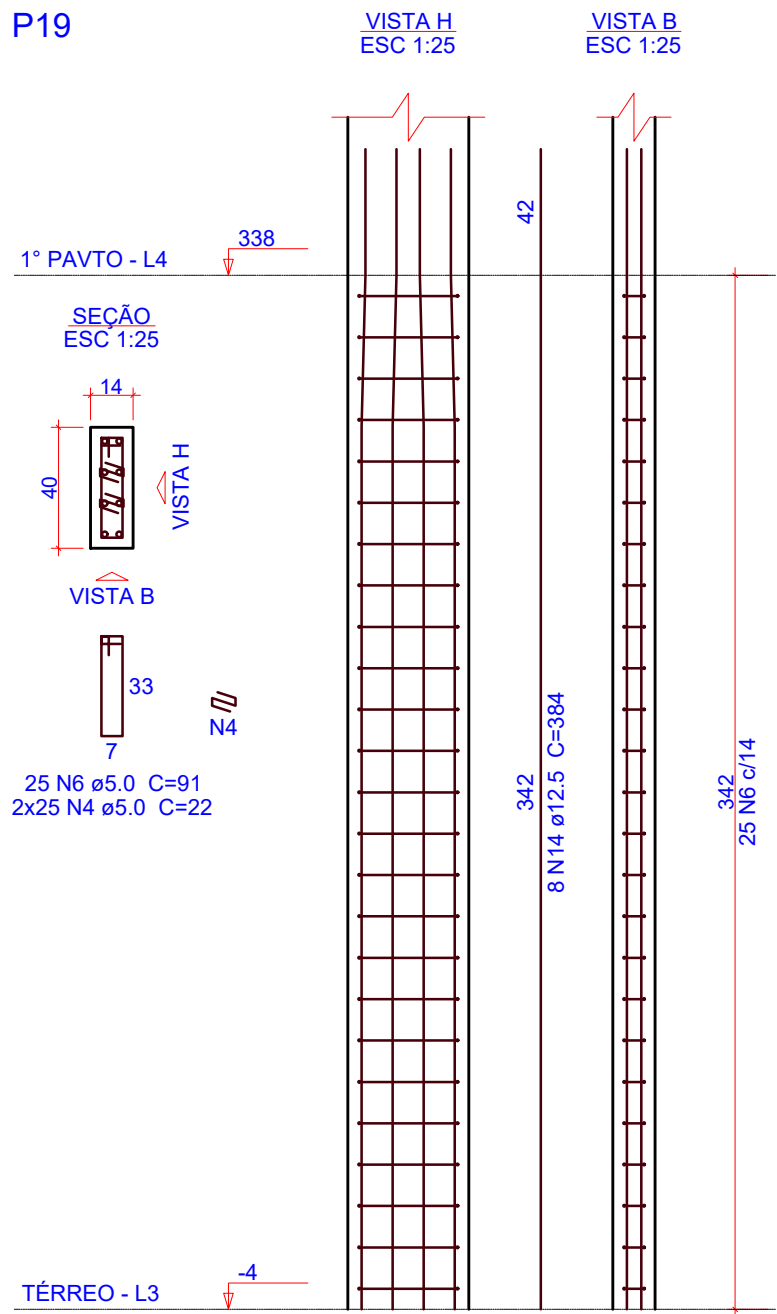
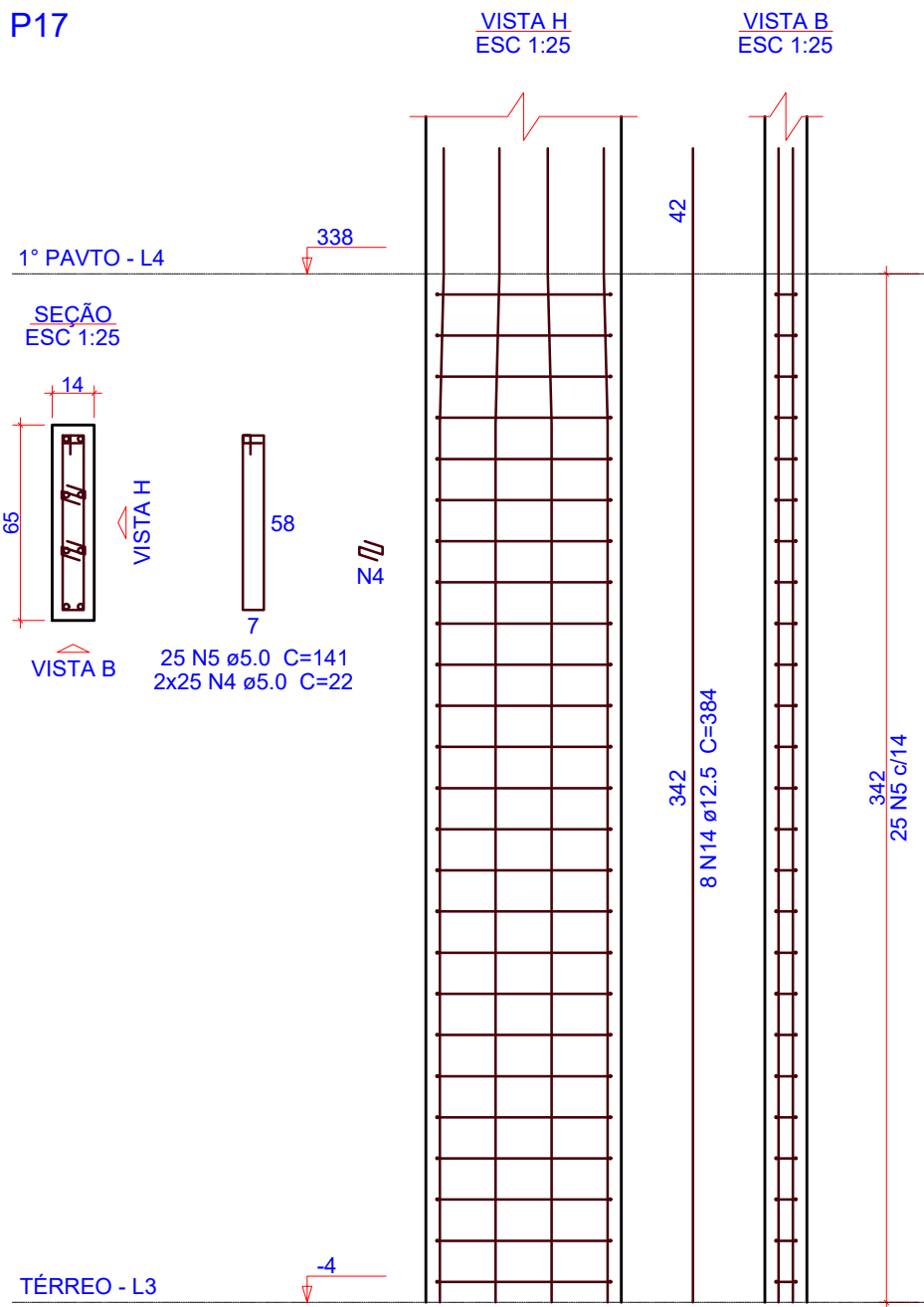
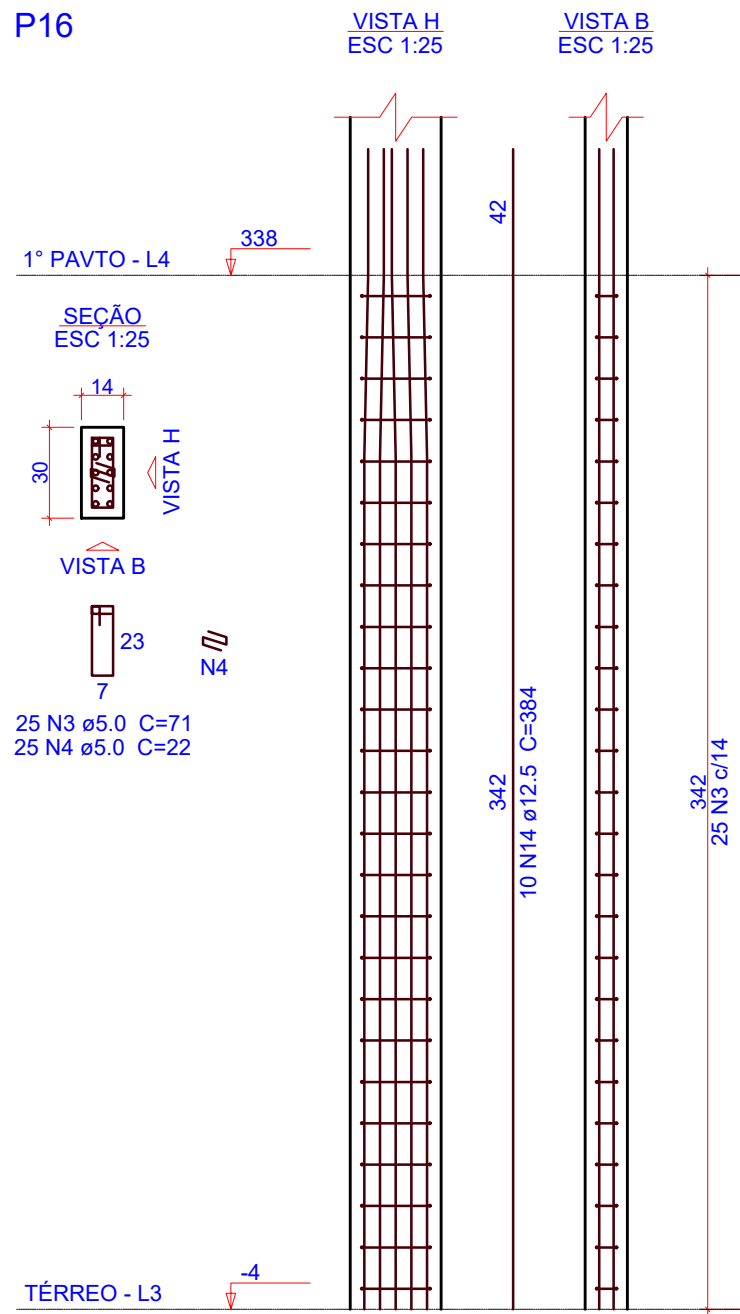
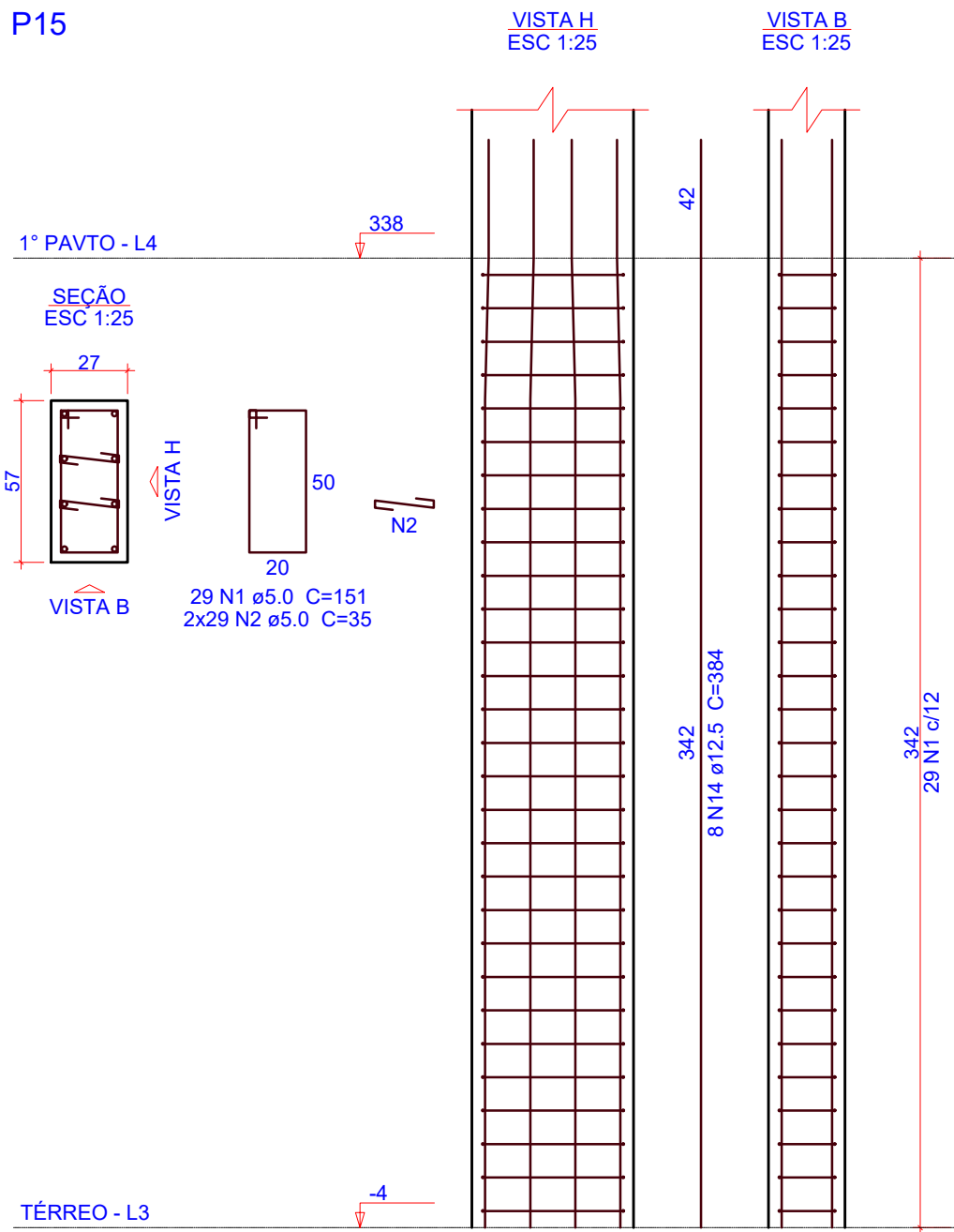
Resumo do aço				
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	118.8	11	32
	8.0	107.7	10	46.7
	10.0	9.5	1	6.4
	12.5	172.2	16	182.5
CA60	16.0	46.6	5	80.8
CA60	5.0	235.9	22	40
PESO TOTAL (kg)				
CA50	348.4			
CA60	40			

Volume de concreto (C-30) = 2.42 m³
Área de forma = 29.37 m²



1 DETALHAMENTO VIGAS - 1º PAVIMENTO (2/2)
ESCALA INDICADA

Nº	REVISÃO	DATA	CLIENTE
A	EMISSION INICIAL	15/12/2025	JULIANO CORBELLINI
			LOCALIDADE - MUNICÍPIO
			ENDEREÇO
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			DETALHAMENTO VIGAS - 1º PAVIMENTO (2/2)
			PROJETO
			DESIGNO
			DATA
			FOLHA Nº
			EST-0210-A



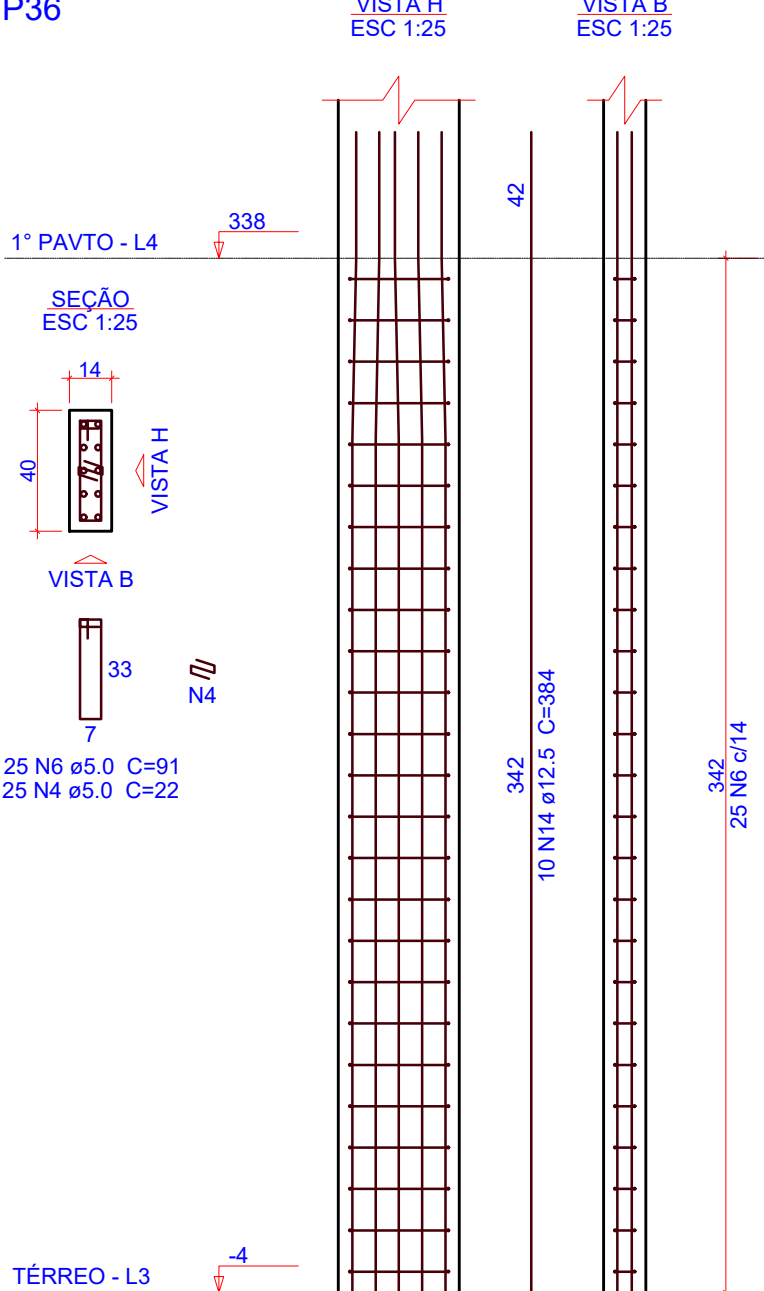
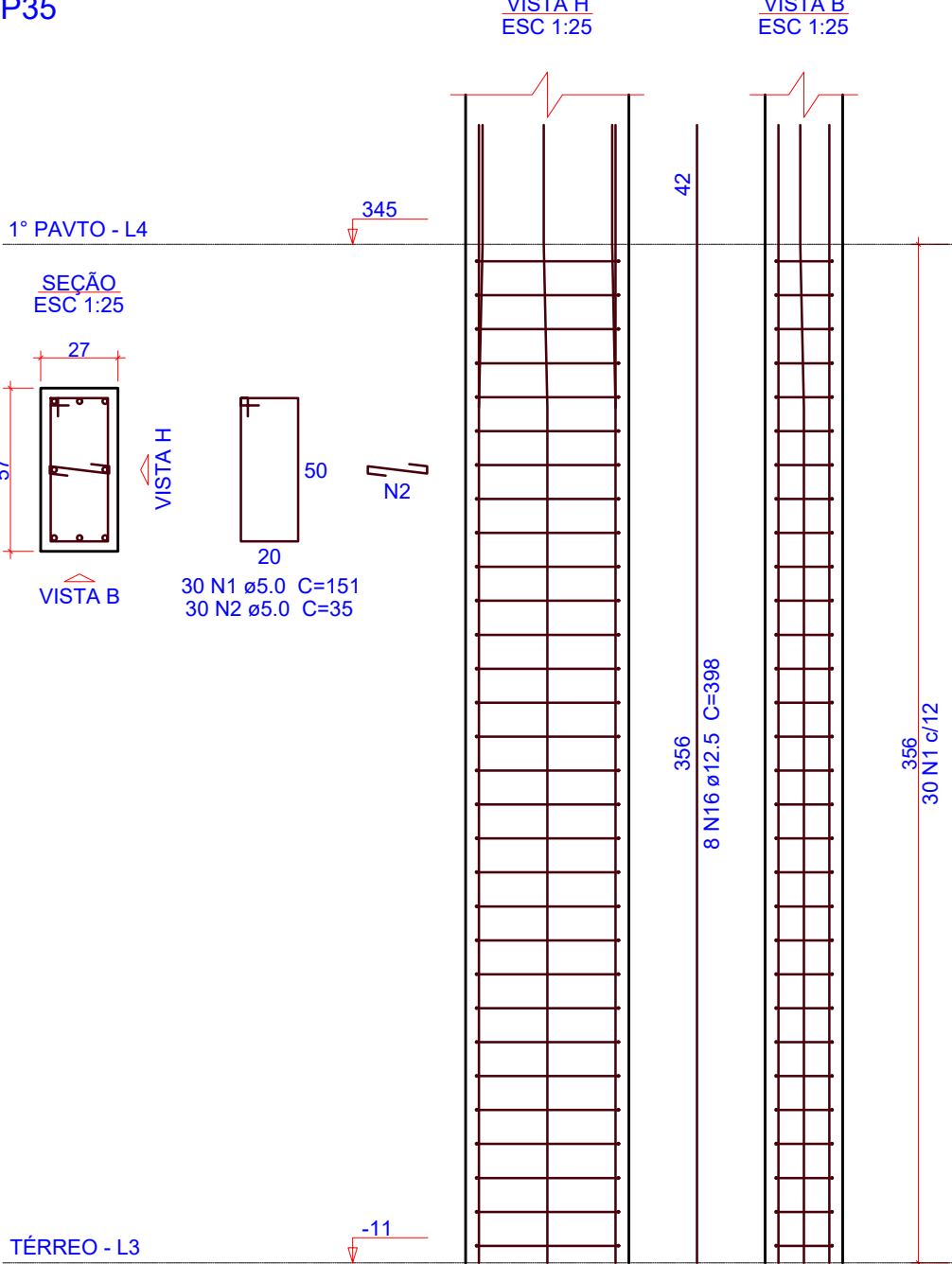
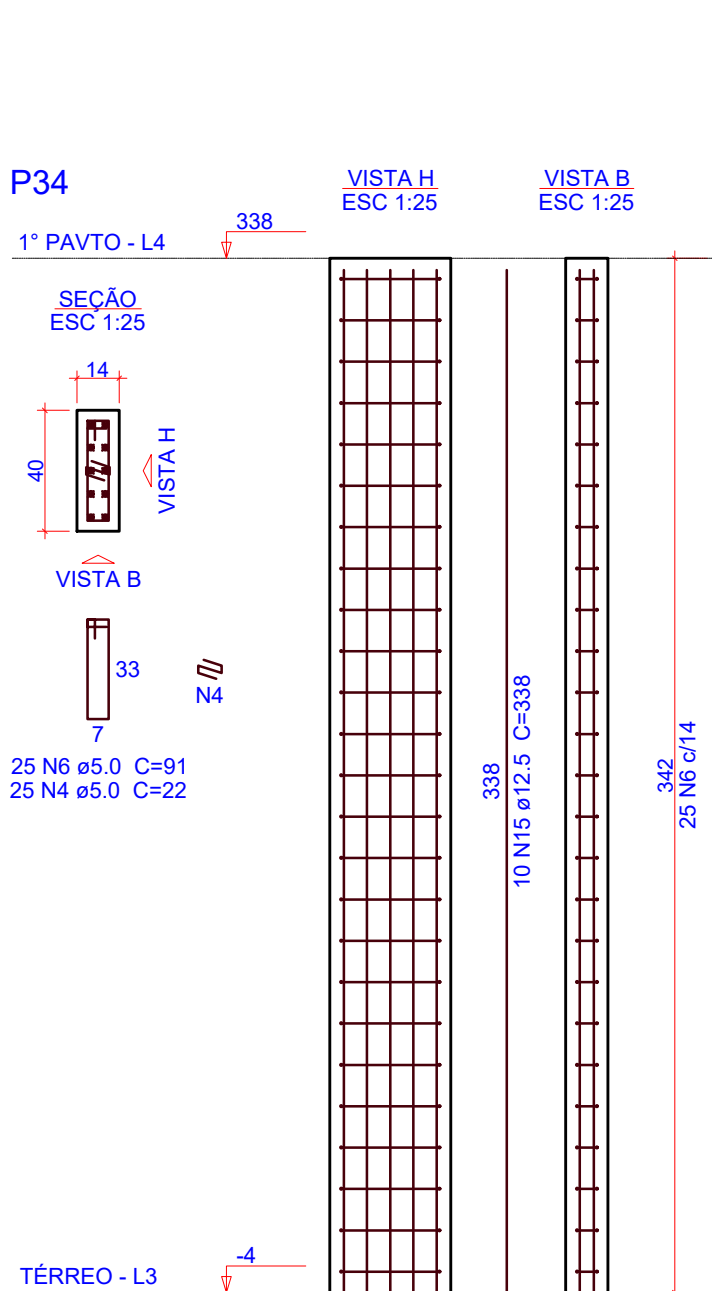
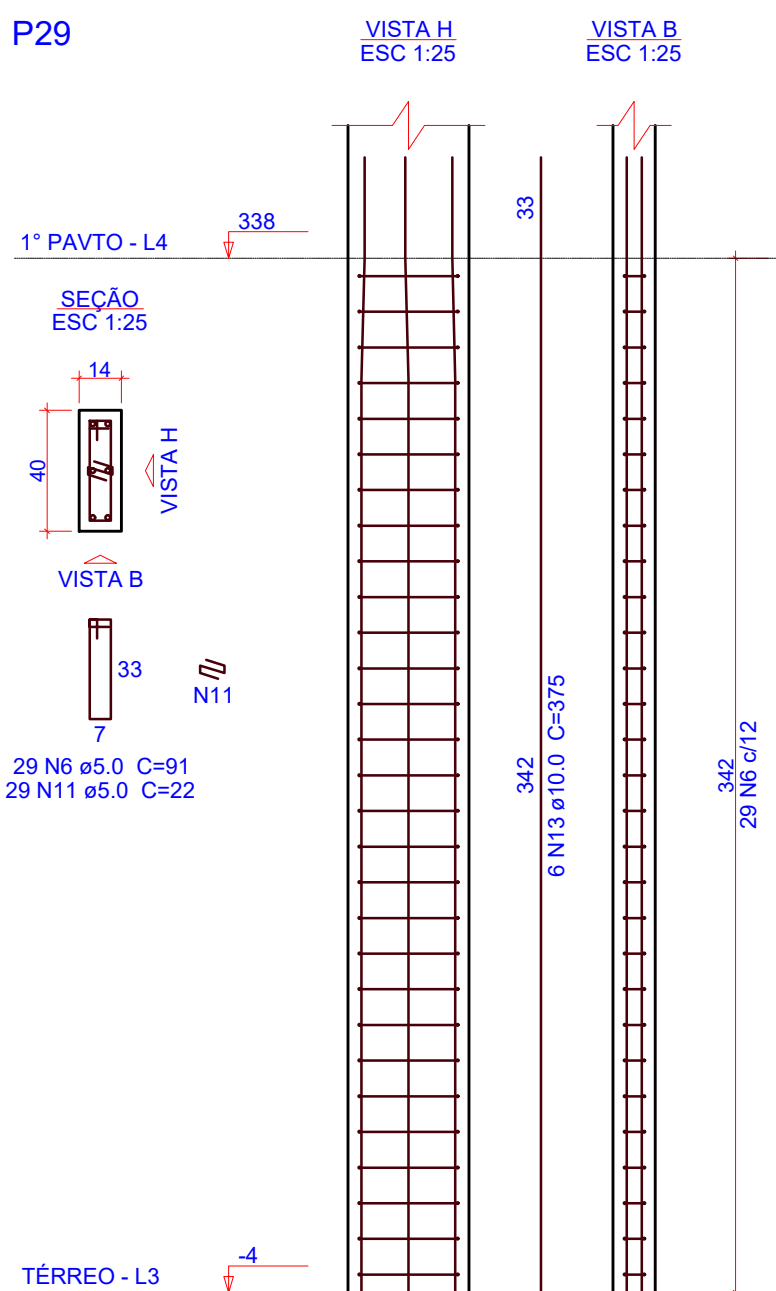
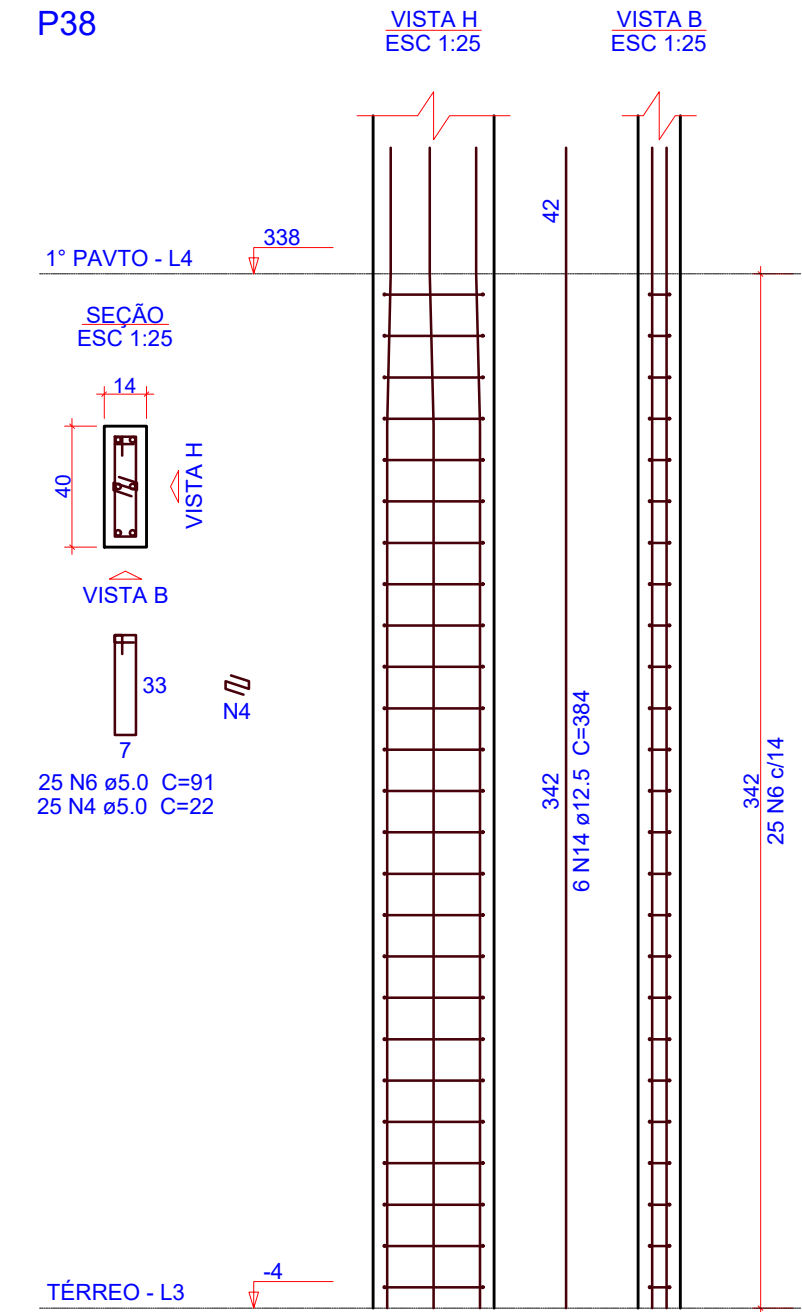
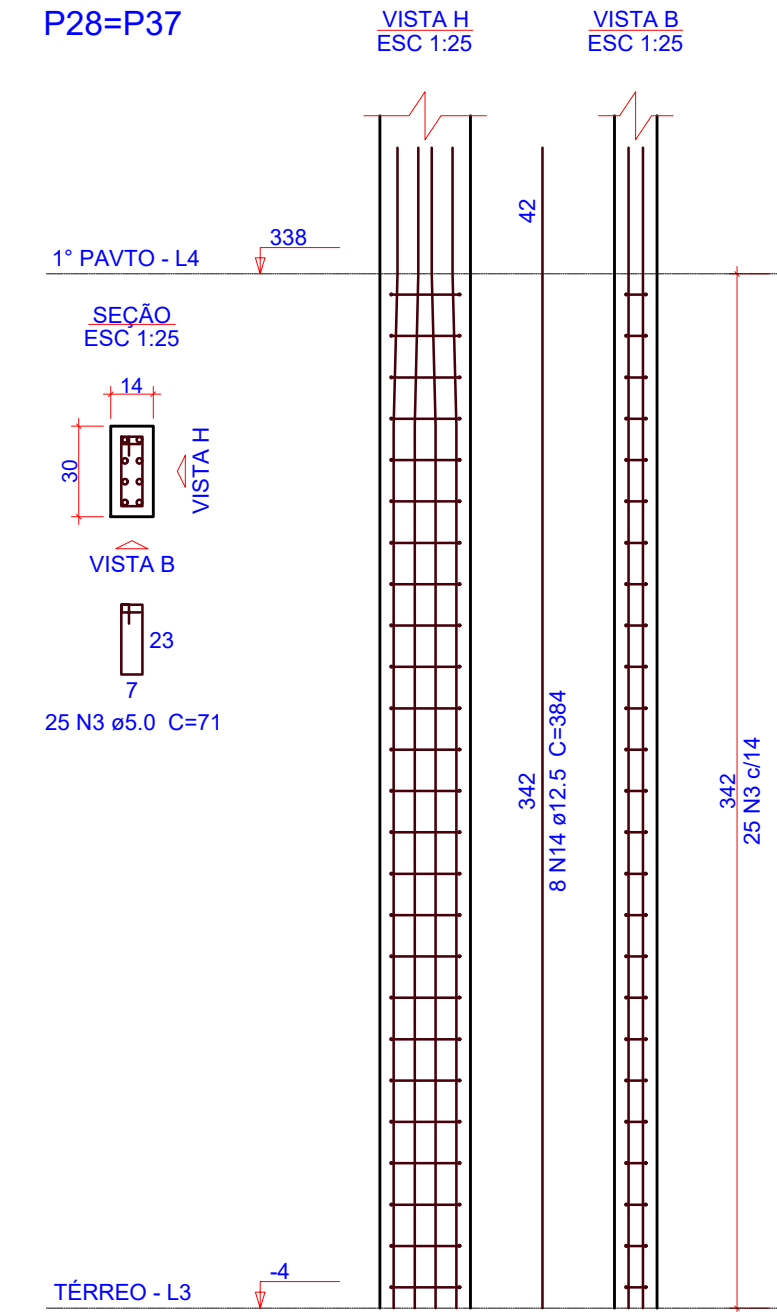
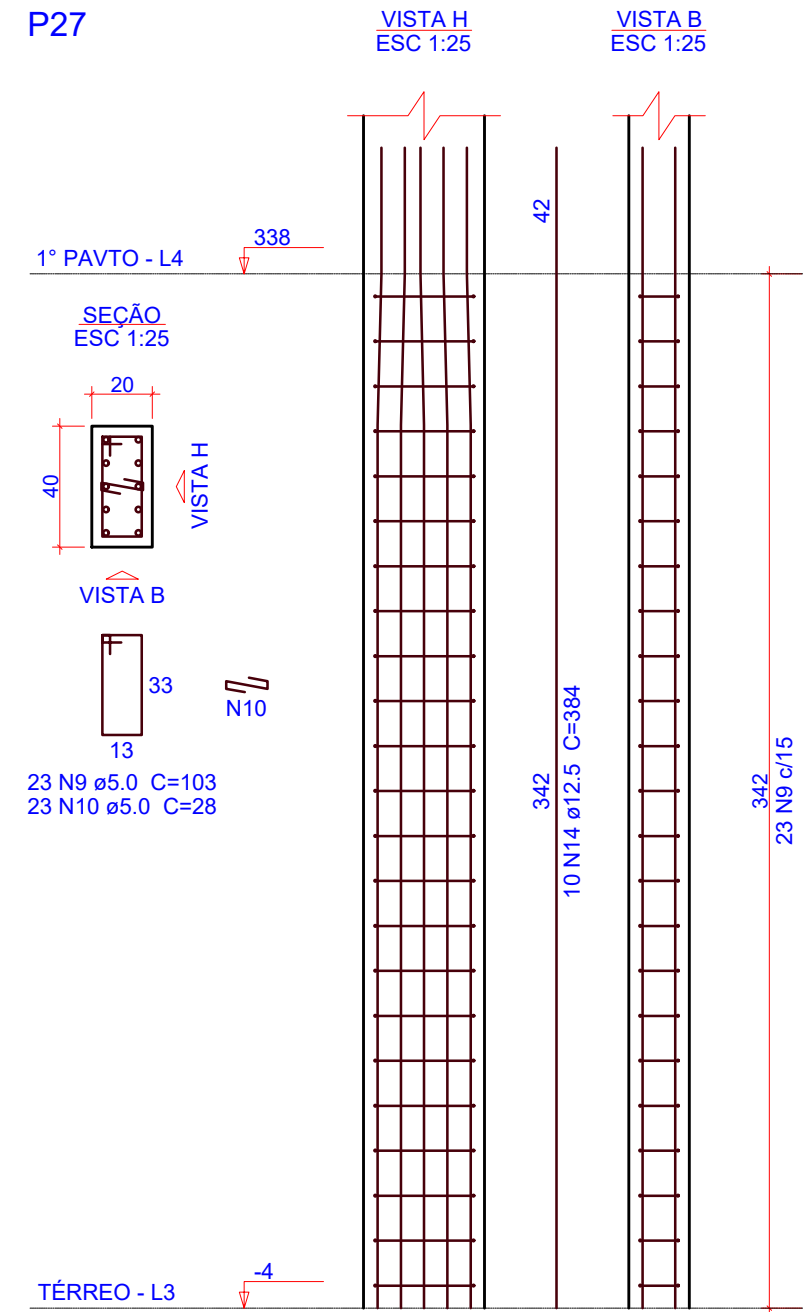
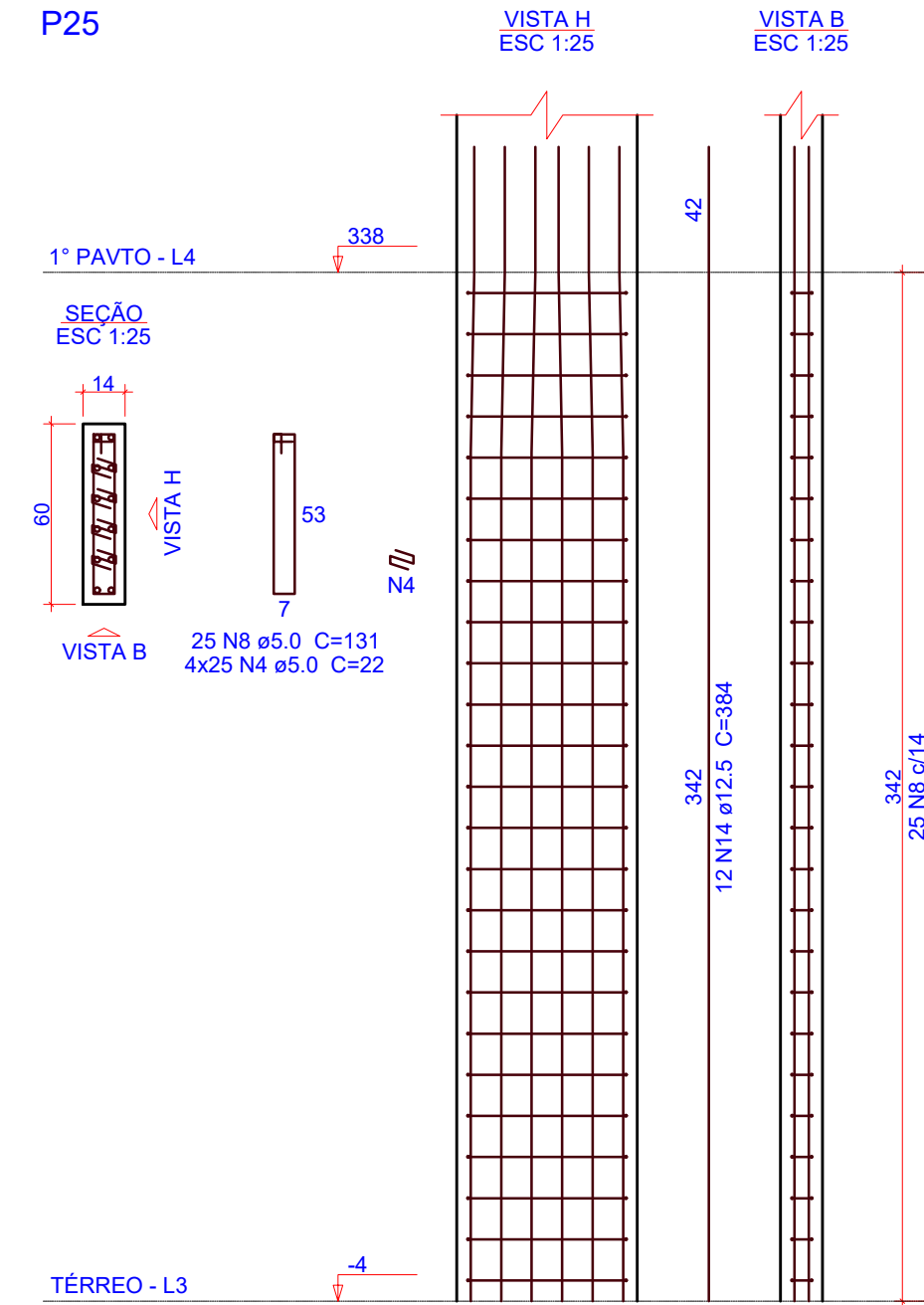
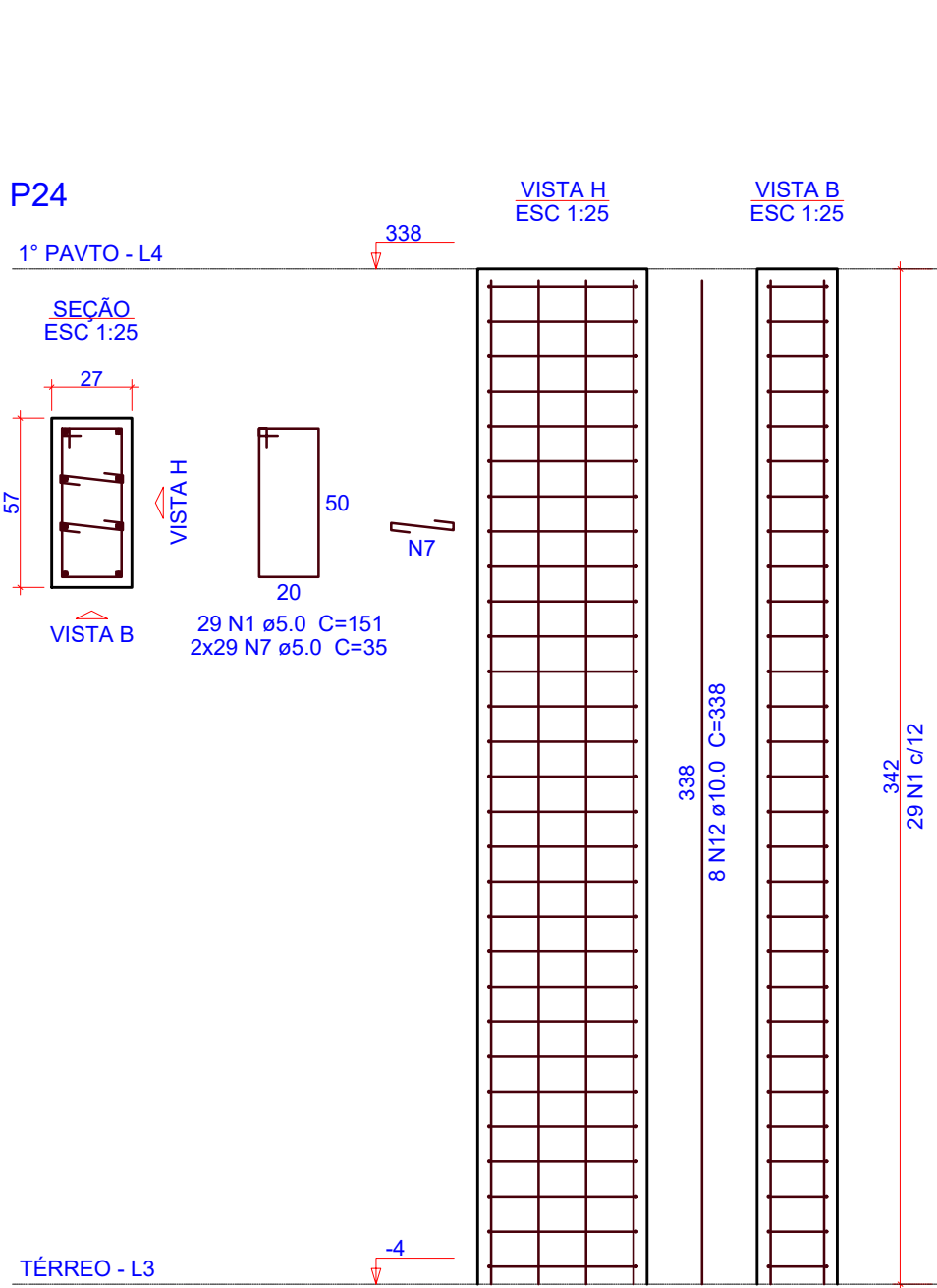
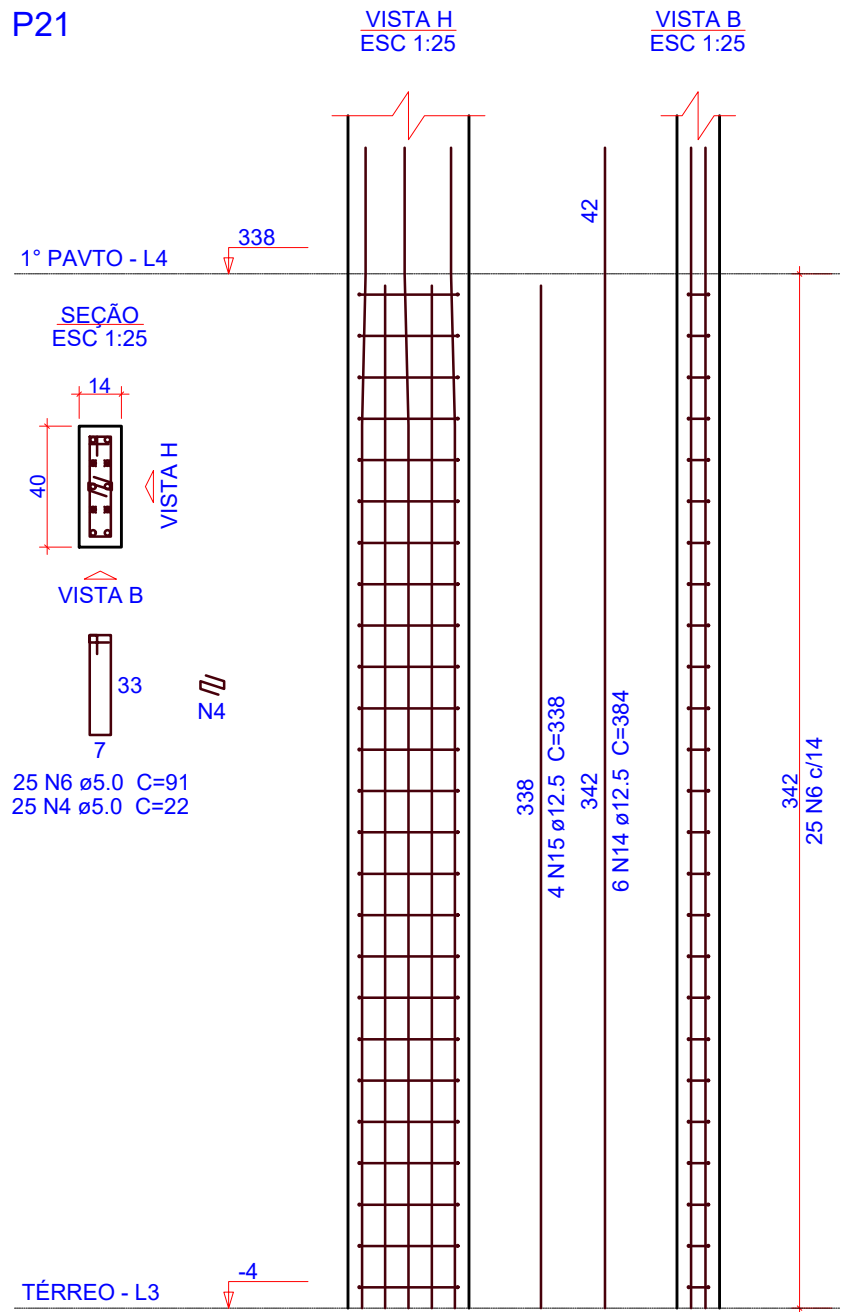
Relação do aço

ACO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	88	151	13288
	2	5.0	88	35	3080
	3	5.0	75	71	5325
	4	5.0	350	22	7700
	5	5.0	25	141	3525
	6	5.0	179	91	16289
	7	5.0	58	35	2030
	8	5.0	25	131	3275
	9	5.0	23	103	2369
	10	5.0	23	28	644
CA50	11	5.0	29	22	638
	12	10.0	8	338	2704
	13	10.0	6	375	2250
	14	12.5	94	384	36096
	15	12.5	20	338	6760
	16	12.5	8	398	3184

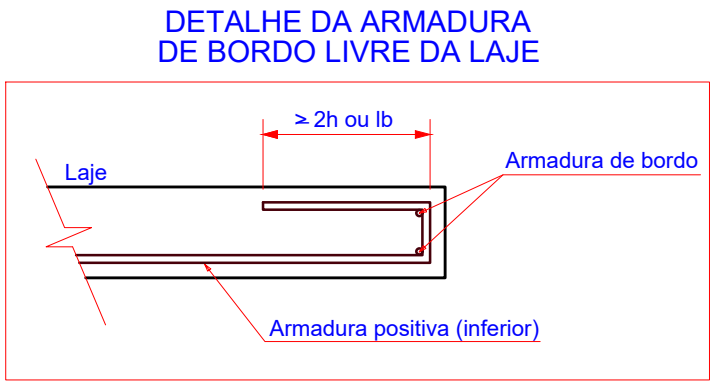
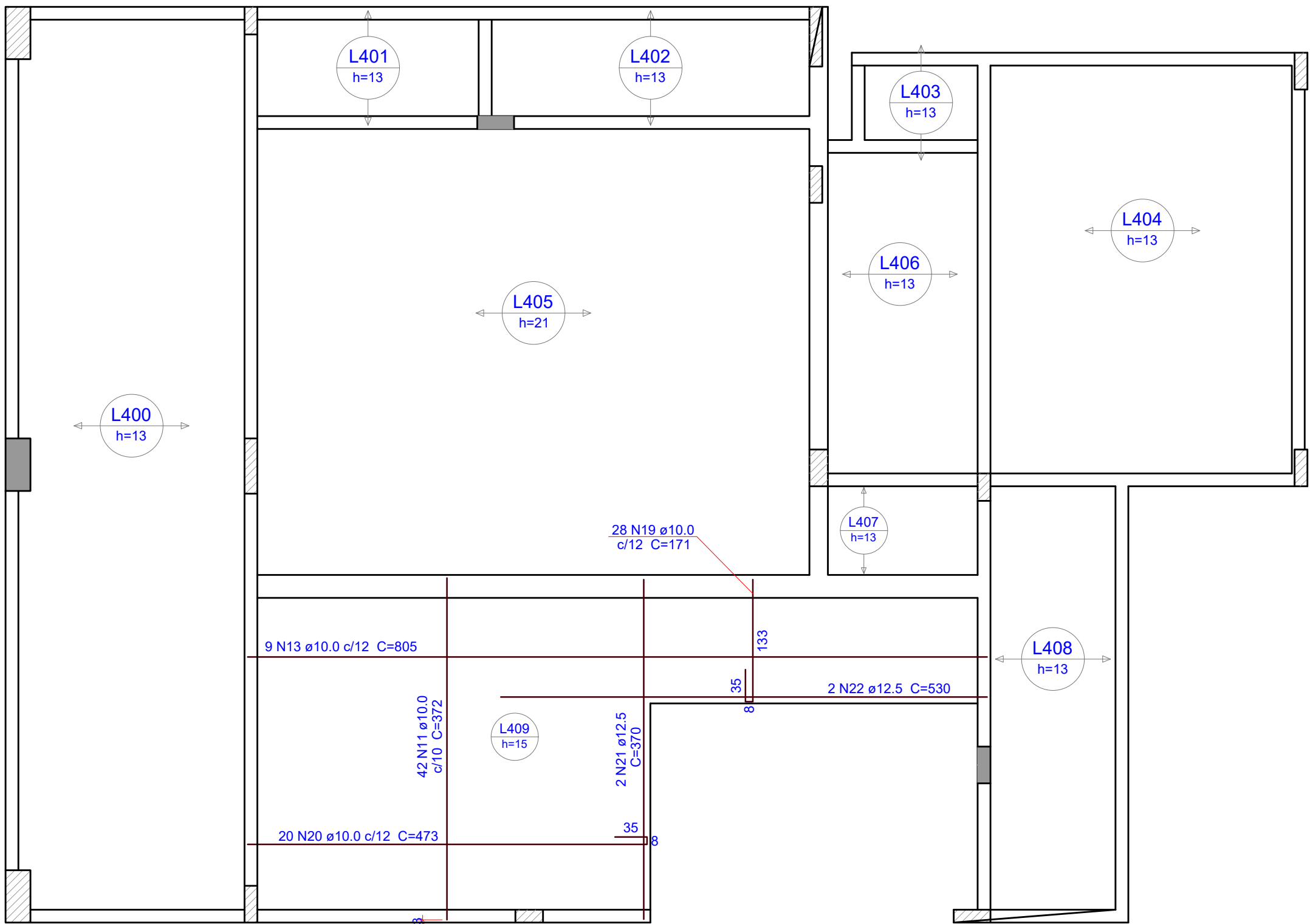
Resumo do aço

ACO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	10.0	49.6	5	33.6
CA60	12.5	460.4	43	487.9
CA60	5.0	561.7	54	96.6
PESO TOTAL (kg)				
CA50	521.5			
CA60	98.6			

Volume de concreto (C-30) = 4.24 m³
Área de forma = 66.93 m²



Nº	REVISÃO	DATA	CLIENTE
A	EMISSÃO INICIAL	15/12/2025	JULIANO CORBELLINI
			LOCALIDADE - MUNICÍPIO
			ENDEREÇO
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			DETALHAMENTO PILARES - 1º PAVIMENTO
			PROJETO
			MUD.
			CREA/SC
			133.073-9
			DESIGNO
			MUD.
			ESCALA
			INDICADA
			DATA
			TOP.
			FOLHA Nº
			EST-0220-A



Relação do aço

Negativos		Positivos			
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA50	16	8.0	40	188	7520
	17	10.0	30	338	10140
	18	10.0	36	400	14400
	19	10.0	28	171	4788
	11	10.0	42	372	15624
	20	10.0	20	473	9460
	13	10.0	9	805	7245
	21	12.5	2	370	740
	22	12.5	2	530	1060

Resumo do aço

AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	8.0	75.2	7	32.6
	10.0	616.6	57	418.2
	12.5	18	2	19.1

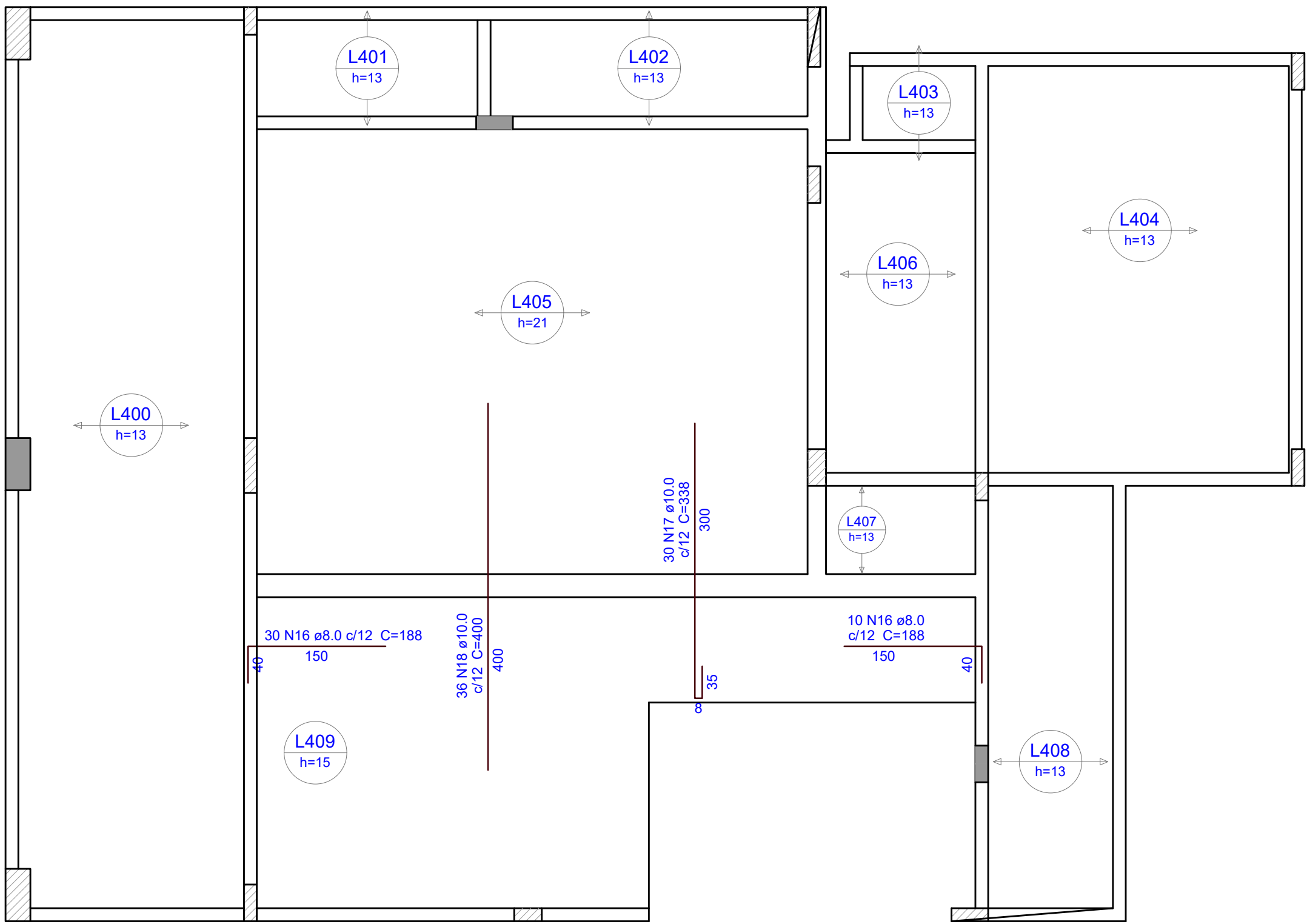
PESO TOTAL (kg)

CA50 469.9

Volume de concreto (C-30) = 3.23 m³
Área de forma = 19.5 m²

1 ARMAÇÃO POSITIVA DAS LAJES - 1º PAVIMENTO

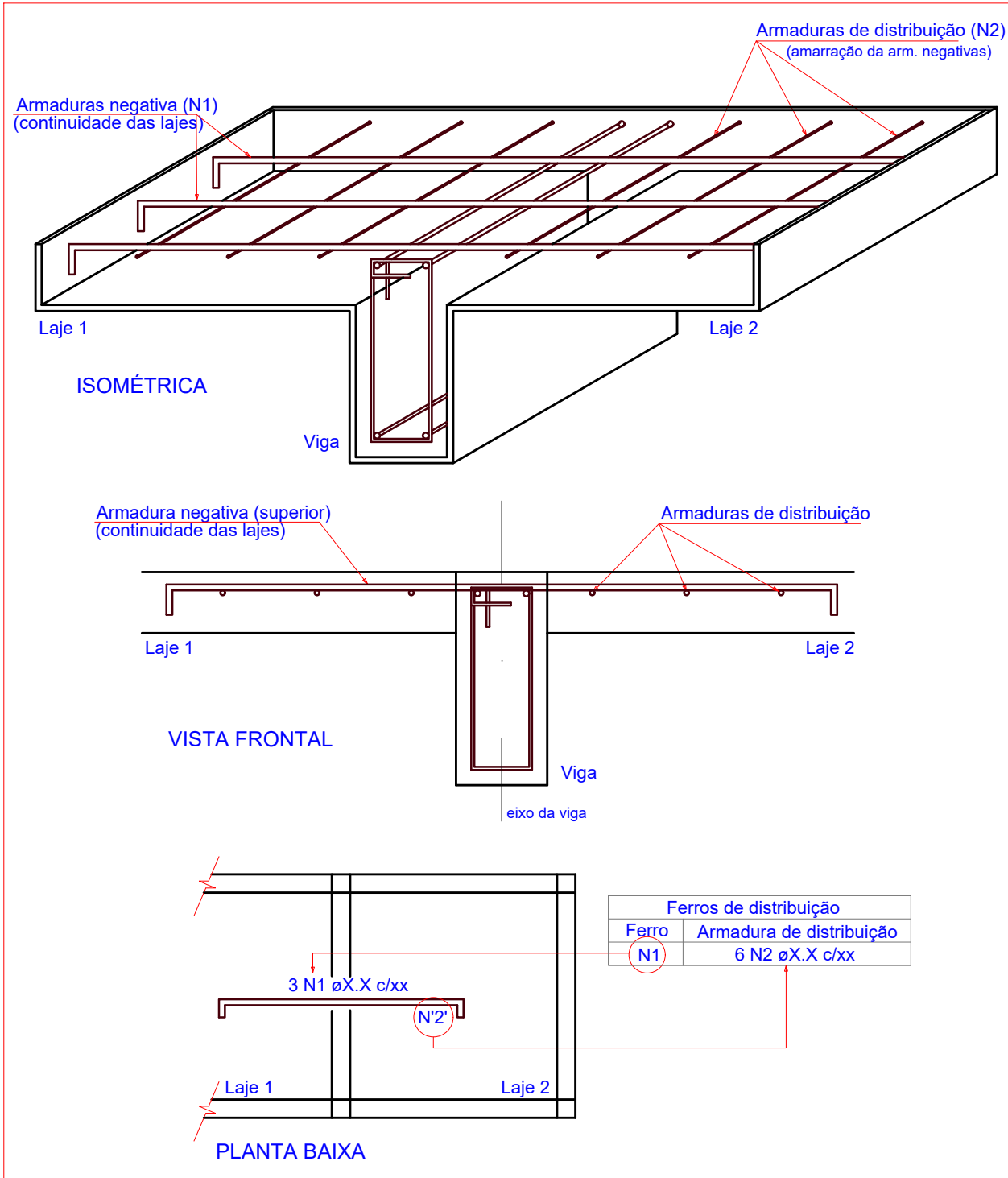
ESCALA 1:50



1 ARMAÇÃO NEGATIVA DAS LAJES - 1º PAVIMENTO

ESCALA 1:50

DETALHE DA ARMADURA DE SUPERIOR DE CONTINUIDADE DA LAJE E MONTAGEM DA ARMADURA DE DISTRIBUIÇÃO



Ferros de distribuição
Ferro N1
Armadura de distribuição N2
6 N2 øX.X c/xx

3 N1 øX.X c/xx

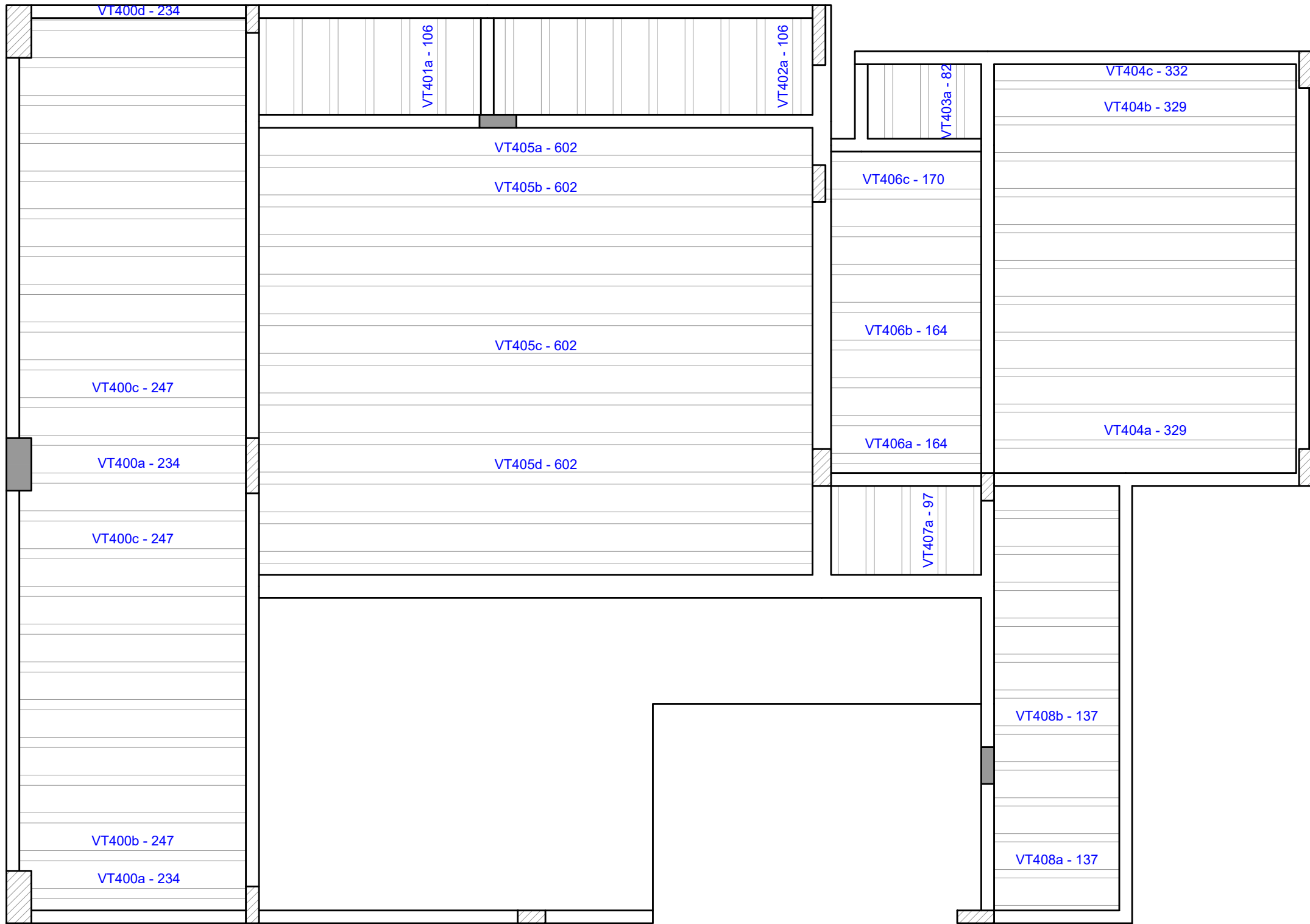
N2'

Laje 1

Laje 2

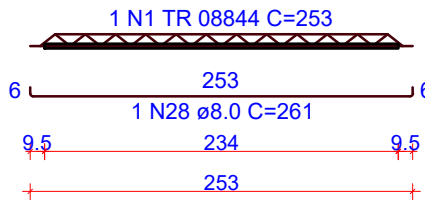
PLANTA BAIXA

Nº	REVISÃO	DATA	CLIENTE
A	EMIÇÃO INICIAL	15/12/2025	JULIANO CORBELLINI
			LOCALIDADE - MUNICÍPIO
			PORTO ALEGRE - RS
			ENDEREÇO
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			ARMAÇÃO POSITIVA E NEGATIVA DAS LAJES - 1º PAVIMENTO
			PROJETO
			Engº Rafael Buss
			CREA/SC 133.073-9
			DESIGNO
			MID.
			ESCALA INDICADA
			DATA
			15/12/2025
			DATA TOP.
			FOLHA Nº
			EST-0230-A

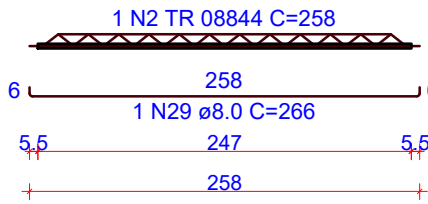


1 PLANTA DE VIGOTAS PRÉ-MOLDADAS - 1º PAVIMENTO
ESCALA 1:50

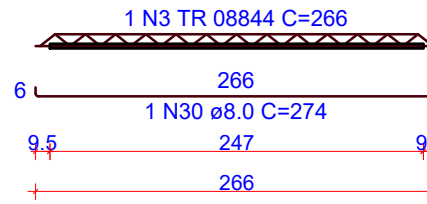
VT400a (3 unidades)
(L400)
ESC 1:50



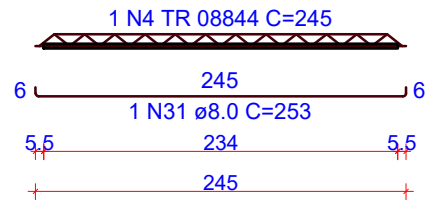
VT400b (8 unidades)
(L400)
ESC 1:50



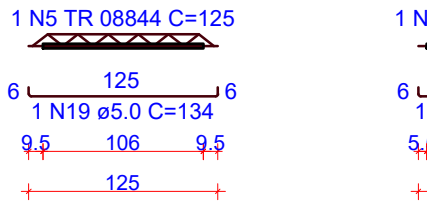
VT400c (12 unidades)
(L400)
ESC 1:50



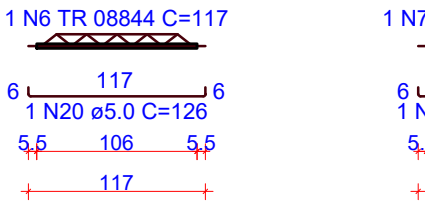
VT400d (1 unidades)
(L400)
ESC 1:50



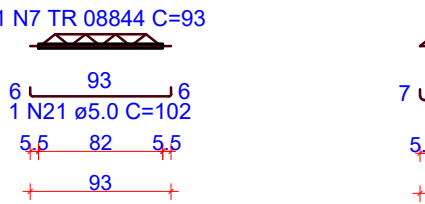
VT401a (5 unidades)
(L401)
ESC 1:50



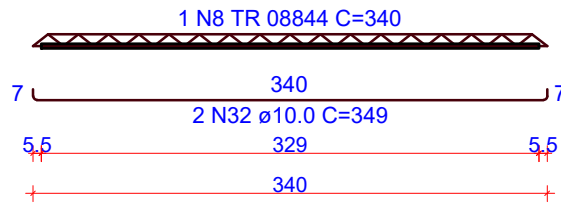
VT402a (9 unidades)
(L402)
ESC 1:50



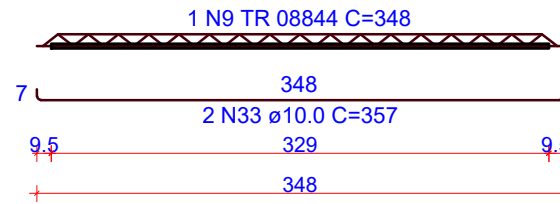
VT403a (3 unidades)
(L403)
ESC 1:50



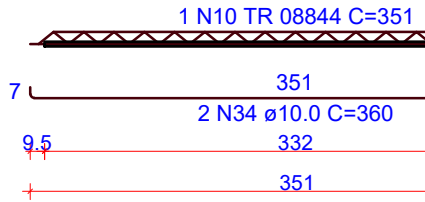
VT404a (9 unidades)
(L404)
ESC 1:50



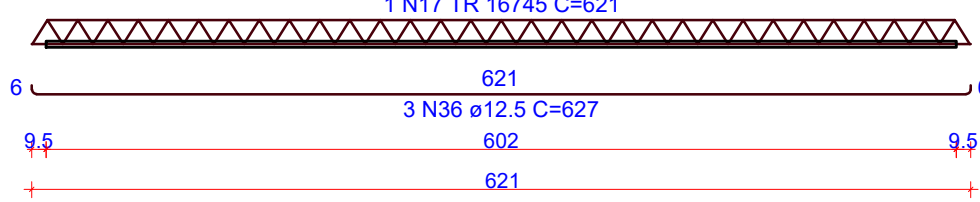
VT404b (1 unidades)
(L404)
ESC 1:50



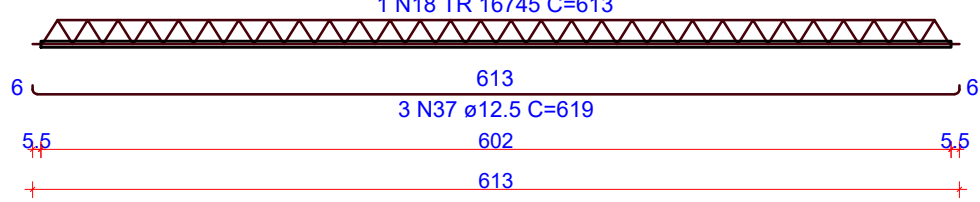
VT404c (1 unidades)
(L404)
ESC 1:50



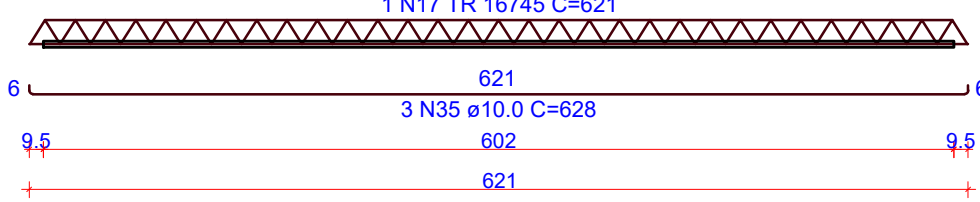
VT405a (1 unidades)
(L405)
ESC 1:50



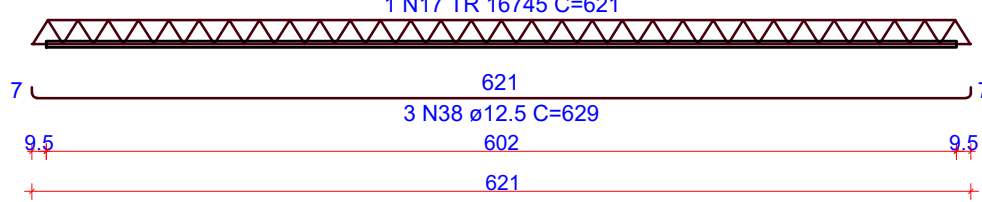
VT405b (4 unidades)
(L405)
ESC 1:50



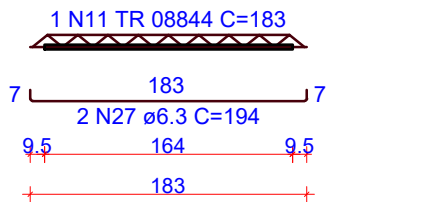
VT405c (3 unidades)
(L405)
ESC 1:50



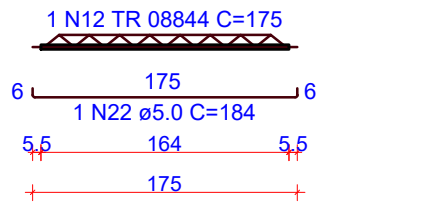
VT405d (3 unidades)
(L405)
ESC 1:50



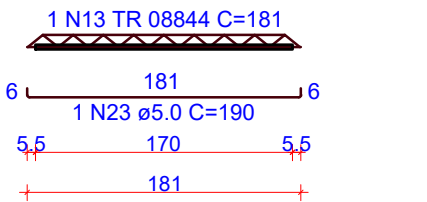
VT406a (3 unidades)
(L406)
ESC 1:50



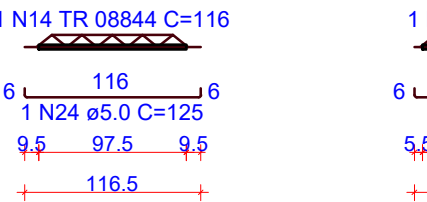
VT406b (4 unidades)
(L406)
ESC 1:50



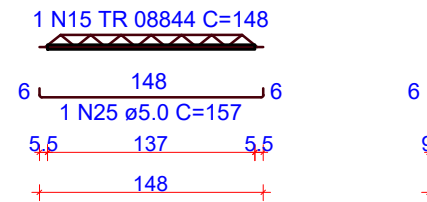
VT406c (1 unidades)
(L406)
ESC 1:50



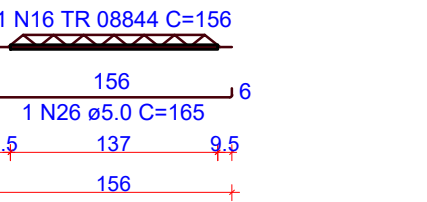
VT407a (3 unidades)
(L407)
ESC 1:50



VT408a (4 unidades)
(L408)
ESC 1:50



VT408b (7 unidades)
(L408)
ESC 1:50



2 DETALHAMENTO DE VIGOTAS PRÉ-MOLDADAS - 1º PAVIMENTO
ESCALA 1:50

RELAÇÃO DO AÇO

3xVT400a	8xVT400b	12xVT400c
VT400d	5xVT401a	9xVT402a
3xVT403a	9xVT404a	VT404b
VT404c	VT405a	4xVT405b
3xVT405c	3xVT405d	3xVT406a
4xVT406b	VT406c	3xVT407a
4xVT408a	7xVT408b	

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	TR 08844	3	253	759
	2	TR 08844	8	258	2064
	3	TR 08844	12	266	3192
	4	TR 08844	1	245	245
	5	TR 08844	5	125	625
	6	TR 08844	9	117	1053
	7	TR 08844	3	93	279
	8	TR 08844	9	340	3060
	9	TR 08844	1	348	348
	10	TR 08844	1	351	351
	11	TR 08844	3	183	549
	12	TR 08844	4	175	700
	13	TR 08844	1	181	181
	14	TR 08844	3	116	348
	15	TR 08844	4	148	592
	16	TR 08844	7	156	1092
	17	TR 16745	7	621	4347
	18	TR 16745	4	613	2452
	19	5.0	5	134	670
	20	5.0	9	126	1134
	21	5.0	3	102	306
	22	5.0	4	184	736
	23	5.0	1	190	190
	24	5.0	3	125	375
	25	5.0	4	157	628
	26	5.0	7	165	1155
	27	6.3	6	194	1164
	28	8.0	3	261	783
	29	8.0	8	266	2128
	30	8.0	12	274	3288
	31	8.0	1	253	253
	32	10.0	18	349	6282
	33	10.0	2	357	714
	34	10.0	2	360	720
	35	10.0	9	628	5652
	36	12.5	3	627	1881
	37	12.5	12	619	7428
	38	12.5	9	629	5661
CA50	27	6.3	6	194	1164
	28	8.0	3	261	783
	29	8.0	8	266	2128
	30	8.0	12	274	3288
	31	8.0	1	253	253
	32	10.0	18	349	6282
PESO TOTAL (kg)	CA50	280.4			
	CA60	240			

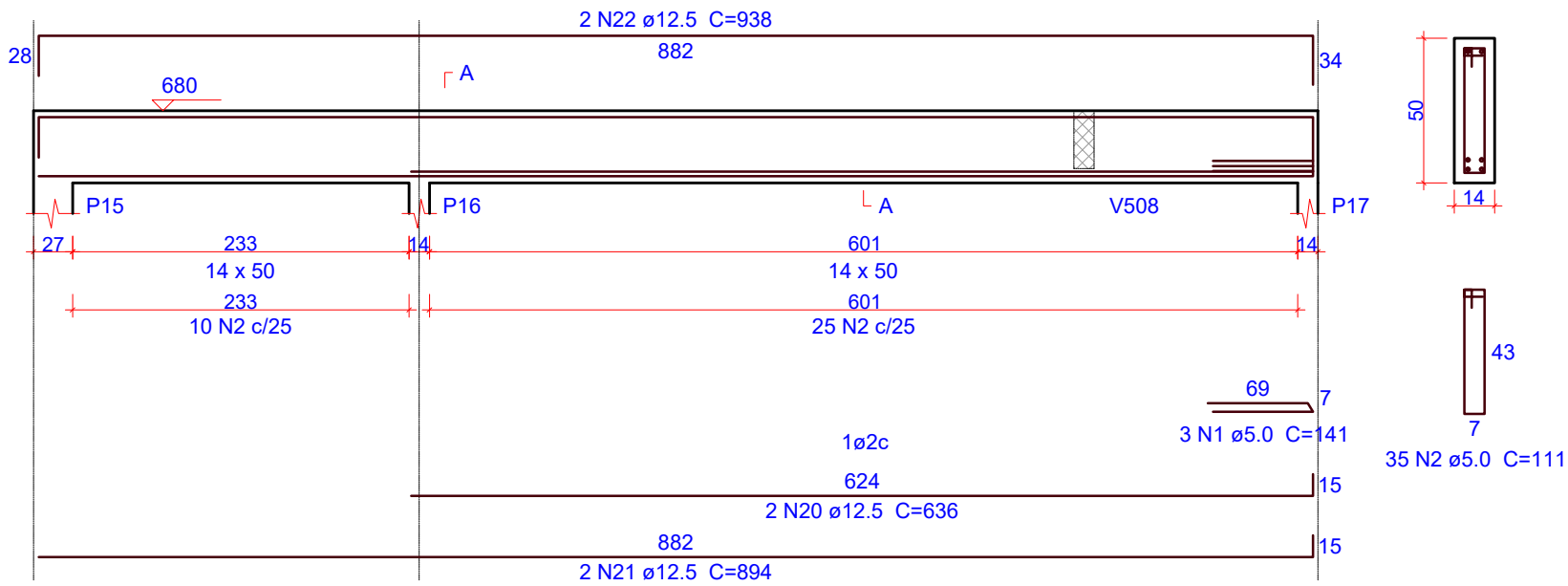
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10% (Barras)	PESO + 10% (kg)
CA50	6.3	11.6	2	3.1
	8.0	64.5	6	28
	10.0	133.7	13	90.7
	12.5	149.7	14	158.6
CA60	5.0	51.9	-	8.8
	TR 08844	154.4	-	154
PESO TOTAL (kg)	TR 16745	68	-	77.2

Volume de concreto (C-30) = 0.00 m³
Área de forma = 0.00 m²

Nº	REVISÃO	DATA	CLIENTE
A	EMISSION INICIAL	15/12/2025	JULIANO CORBELLINI
			LOCALIDADE - MUNICÍPIO
			ENDEREÇO
			PROJETO ESTRUTURAL
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			PLANTA E DETALHAMENTO VIGOTAS PRÉ-MOLDADAS - 1º PAVIMENTO
			PROJETO
			MIQ.
			CREA/SC
			133.073-9
			DESENHO
			INDICADA
			DATA
			15/12/2025
			DATA TOP.
			FOLHA Nº
			EST-0240-A

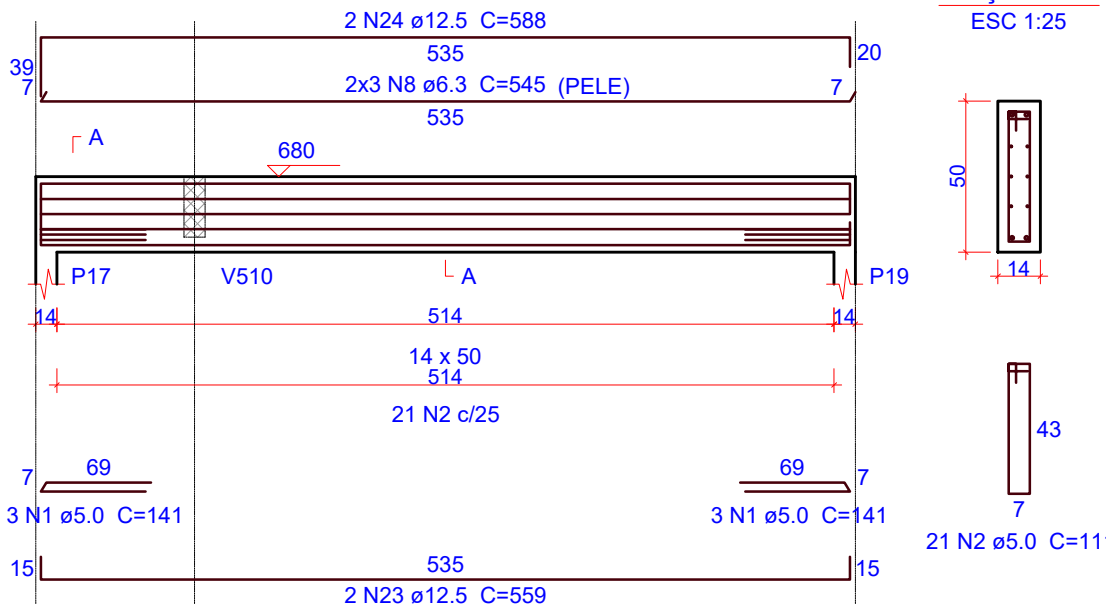
V500

ESC 1:50



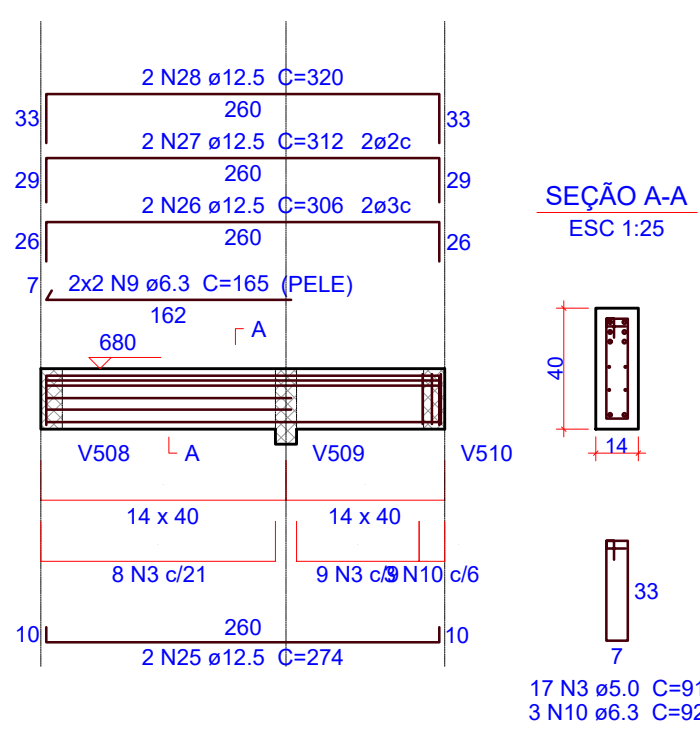
V501

ESC 1:50



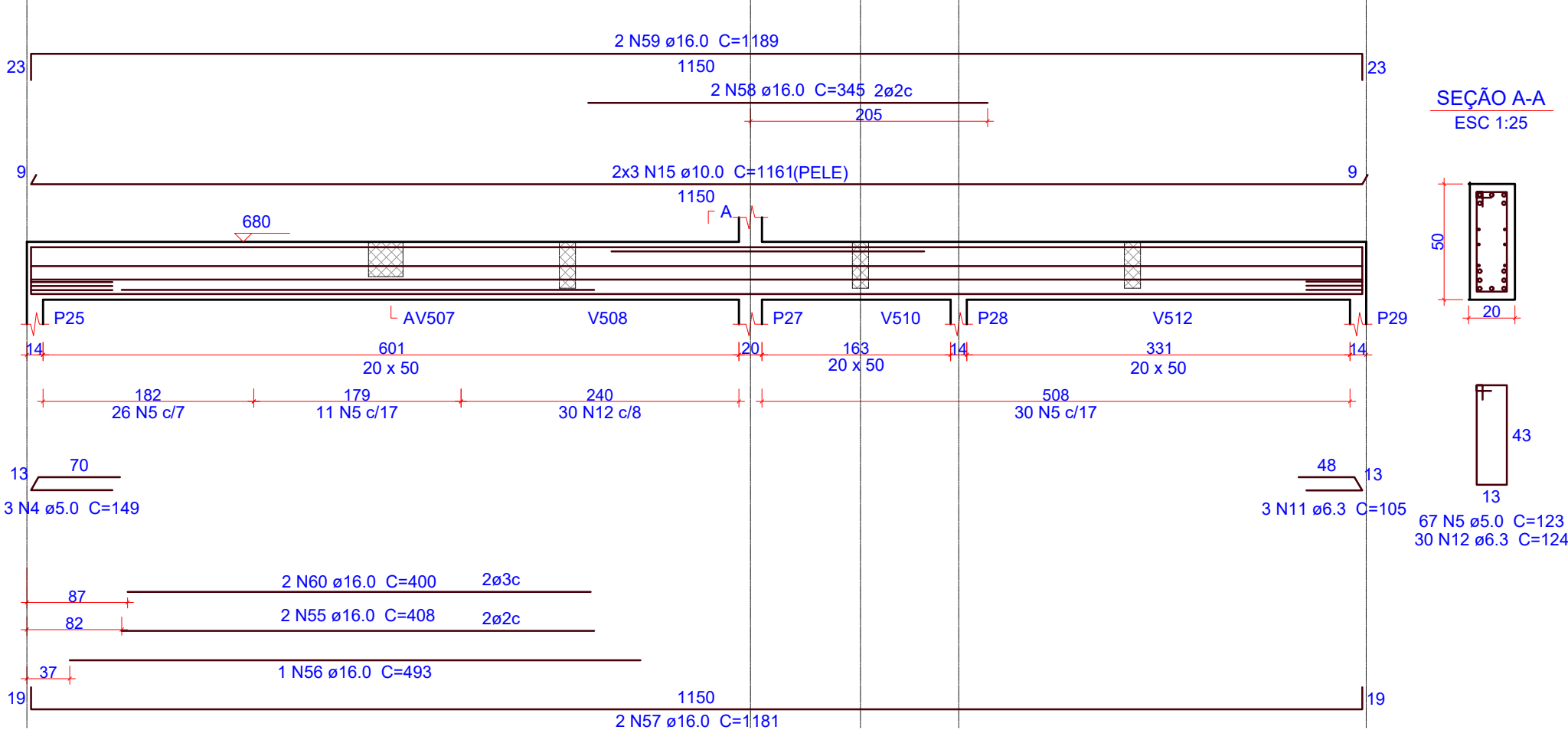
V502

ESC 1:50



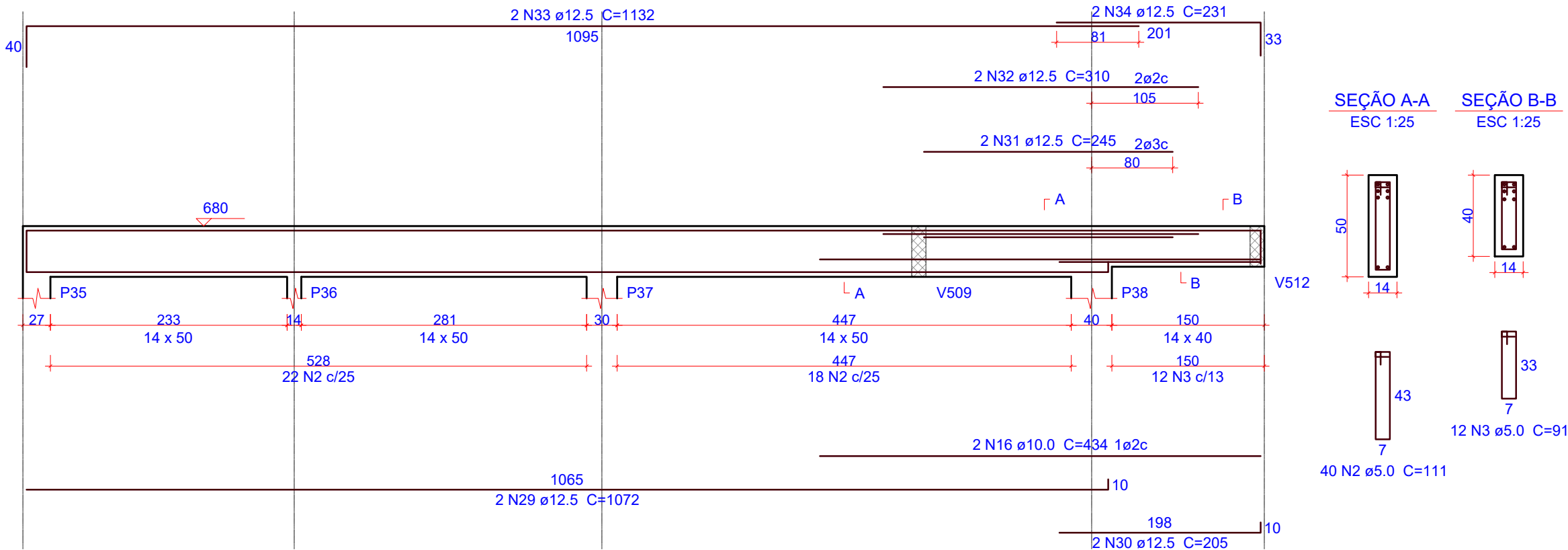
V503

ESC 1:50



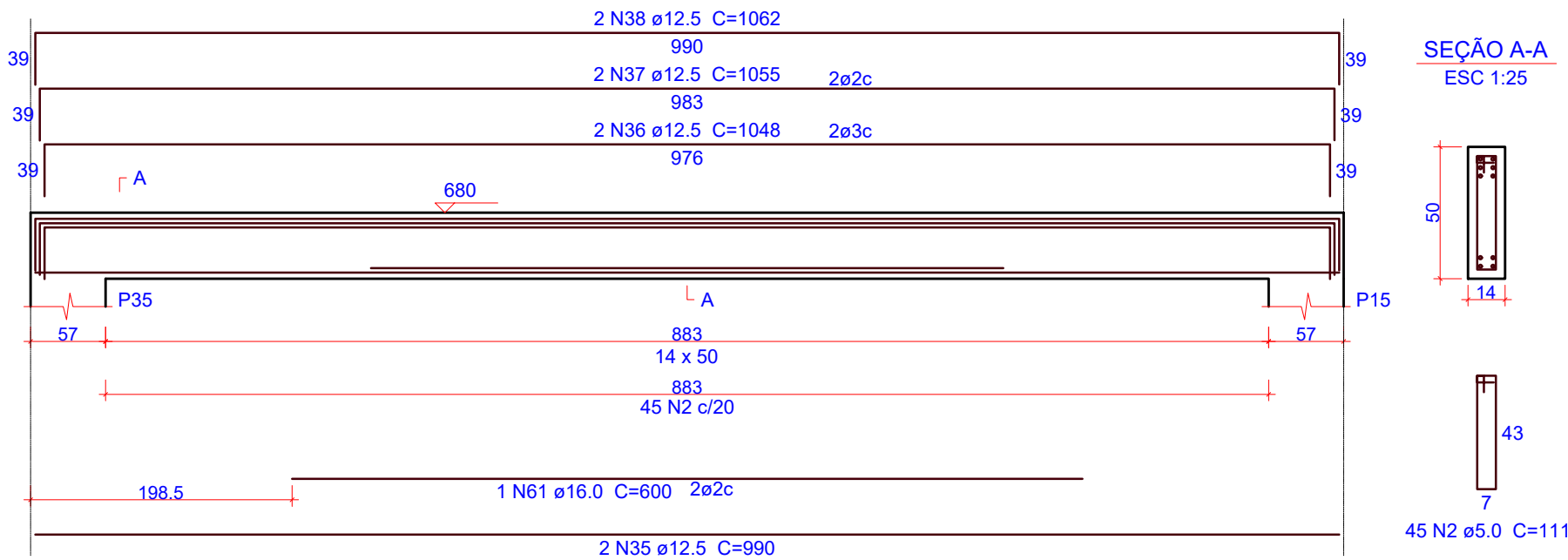
V504

ESC 1:50



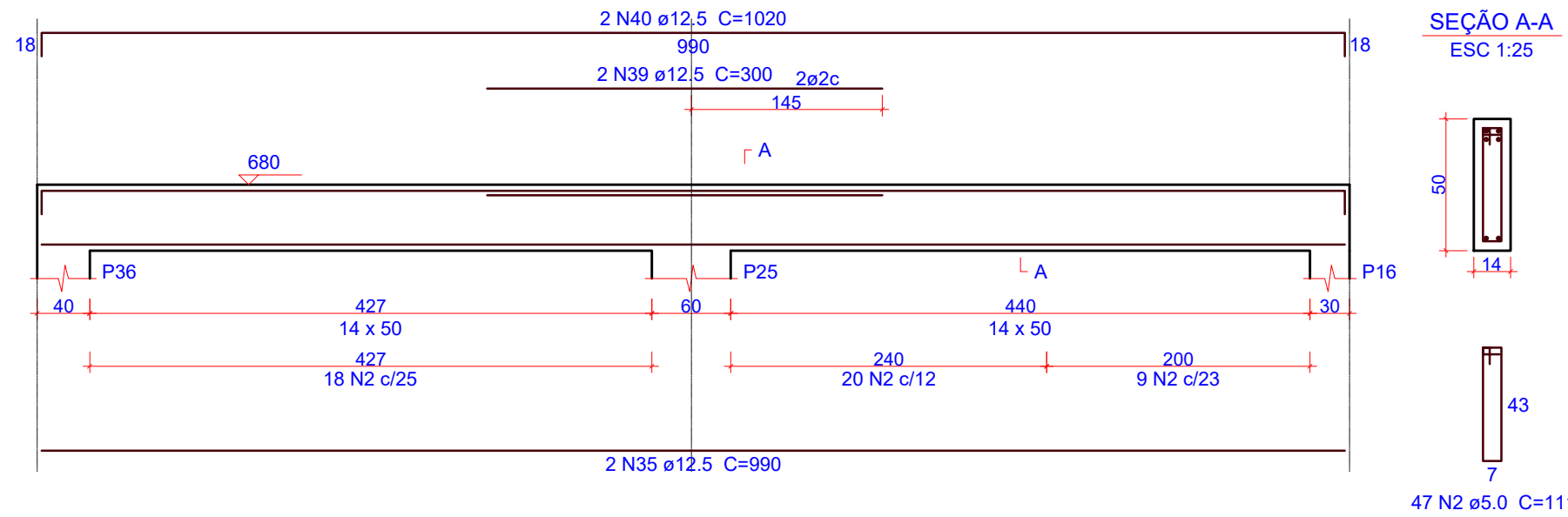
V505

ESC 1:50



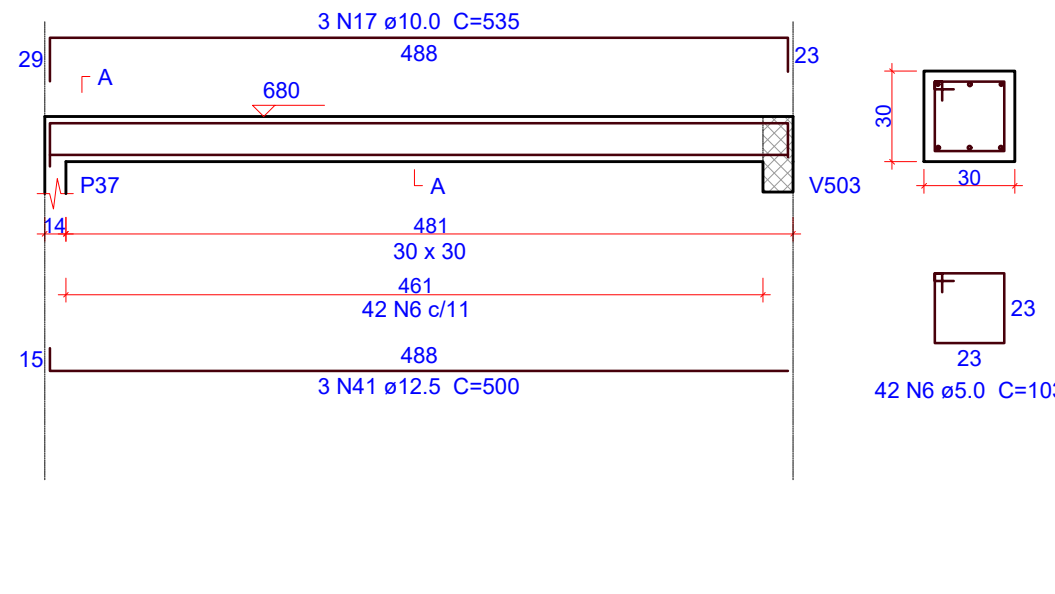
V506

ESC 1:50



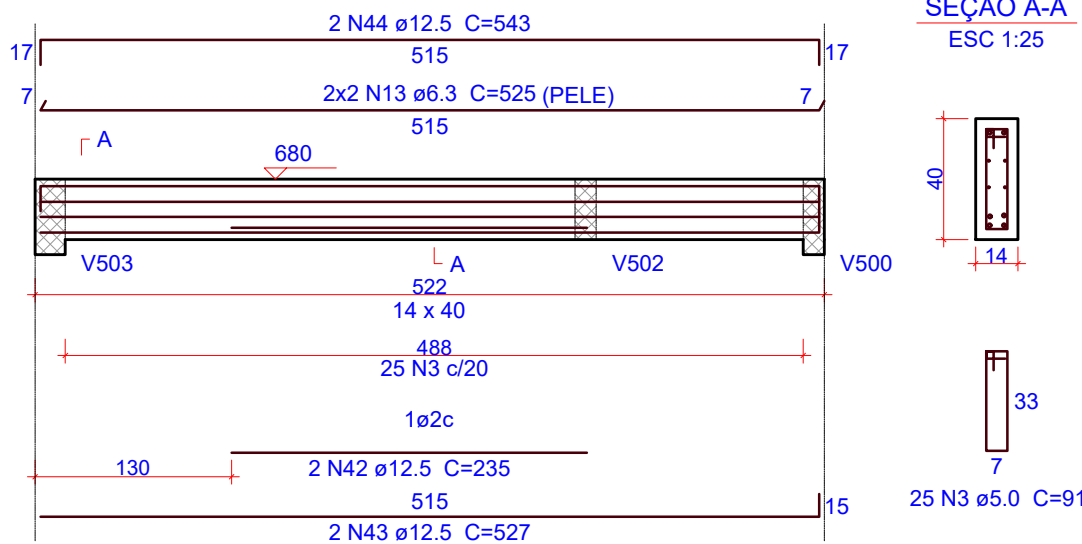
V507

ESC 1:50



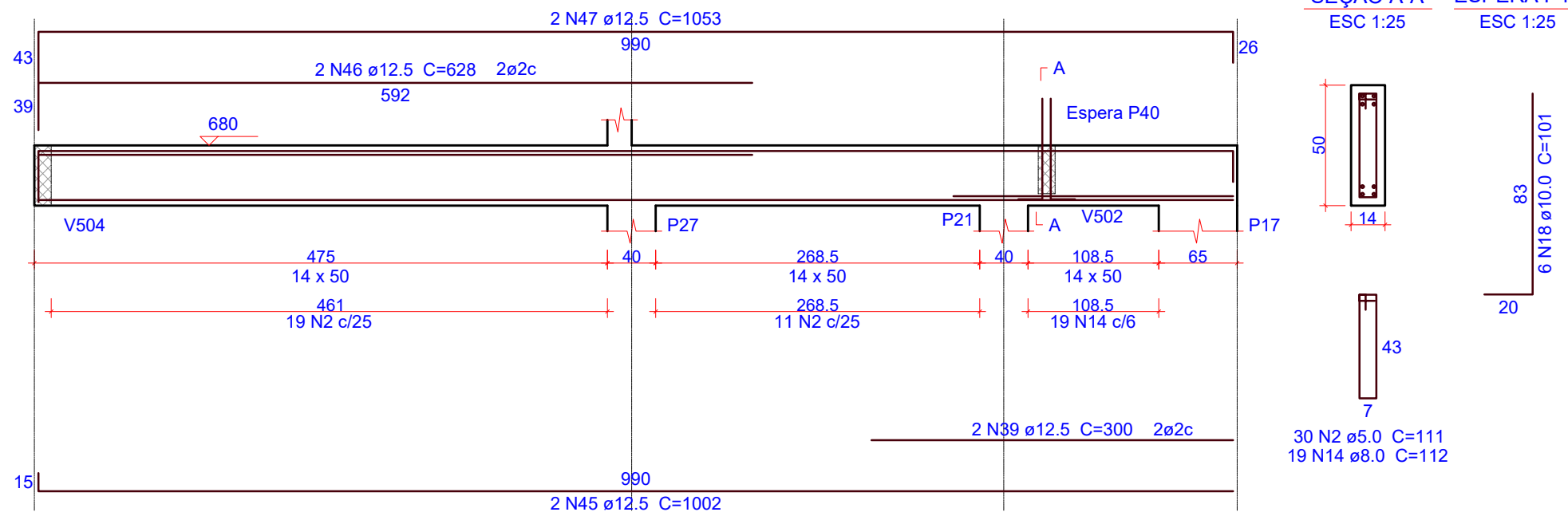
V508

ESC 1:50



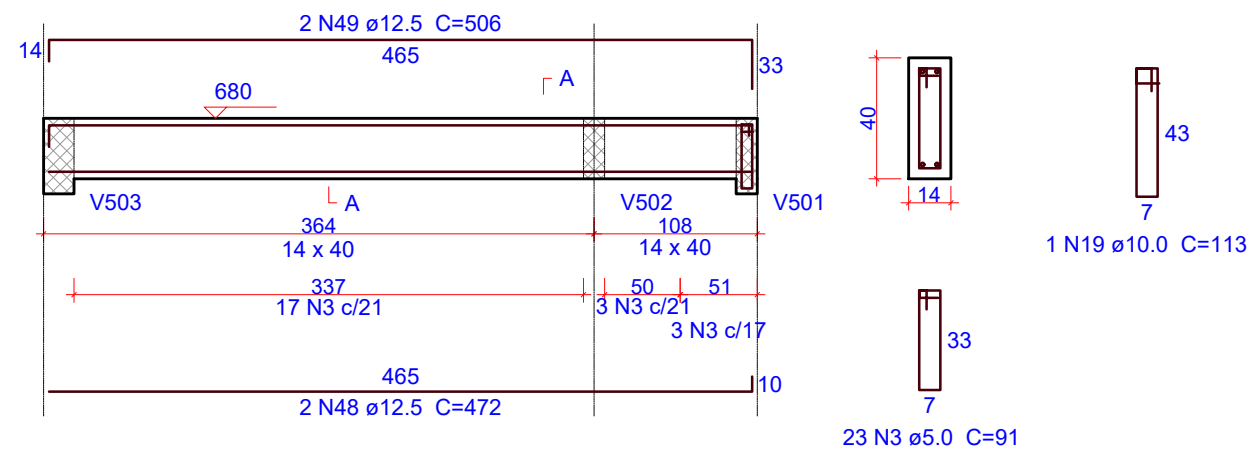
V509

ESC 1:50



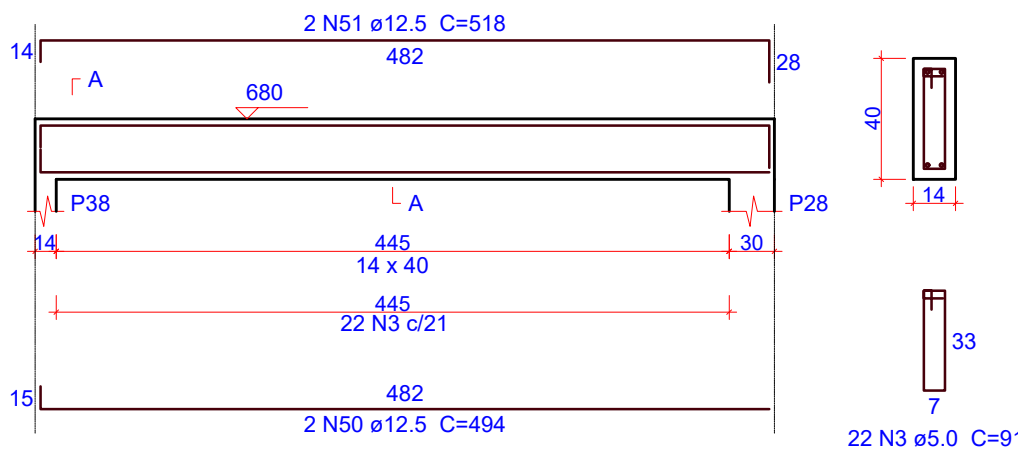
V510

ESC 1:50



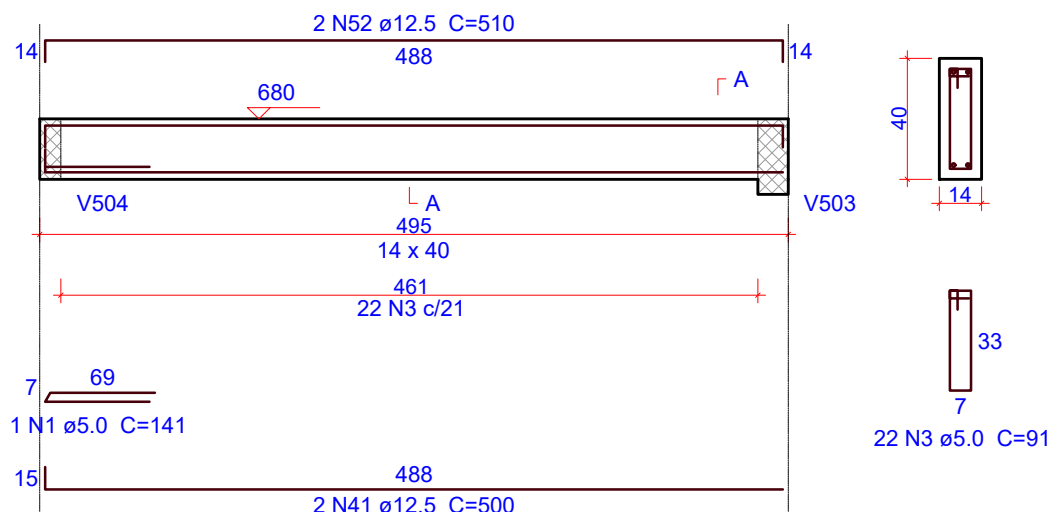
V511

ESC 1:50



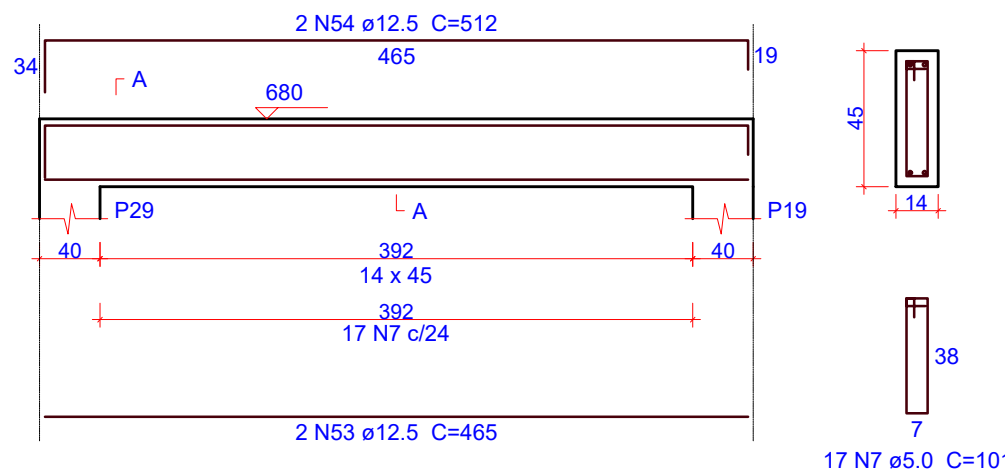
V512

ESC 1:50



V513

ESC 1:50



Relação do aço

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
V500	1	5.0	10	141	1410
V503	2	5.0	218	111	24198
V506	3	5.0	121	91	11011
V509	4	5.0	3	149	447
V512	5	5.0	67	123	8241
	6	5.0	42	103	4326
	7	5.0	17	101	1717
	8	6.3	6	545	3270
	9	6.3	4	165	660
	10	6.3	3	92	276
	11	6.3	3	105	315
	12	6.3	30	124	3720
	13	6.3	4	525	2100
	14	8.0	19	112	2128
	15	10.0	6	1161	6966
	16	10.0	2	434	868
	17	10.0	3	535	1605
	18	10.0	6	101	606
	19	10.0	1	113	113
	20	12.5	2	636	1272
	21	12.5	2	894	1788
	22	12.5	2	938	1876
	23	12.5	2	559	1118
	24	12.5	2	588	1176
	25	12.5	2	274	548
	26	12.5	2	306	612
	27	12.5	2	312	624
	28	12.5	2	320	640
	29	12.5	2	1072	2144
	30	12.5	2	205	410
	31	12.5	2	245	490
	32	12.5	2	310	620
	33	12.5	2	1132	2264
	34	12.5	2	231	462
	35	12.5	4	990	3960
	36	12.5	2	1048	2096
	37	12.5	2	1055	2110
	38	12.5	2	1062	2124
	39	12.5	4	300	1200
	40	12.5	2	1020	2040
	41	12.5	5	500	2500
	42	12.5	2	235	470
	43	12.5	2	527	1054
	44	12.5	2	543	1086
	45	12.5	2	1002	2004
	46	12.5	2	628	1256
	47	12.5	2	1053	2106
	48	12.5	2	472	944
	49	12.5	2	506	1012
	50	12.5	2	494	988
	51	12.5	2	518	1036
	52	12.5	2	510	1020
	53	12.5	2	465	930
	54	12.5	2	512	1024
	55	16.0	2	408	816
	56	16.0	1	493	493
	57	16.0	2	1181	2362
	58	16.0	2	345	690
	59	16.0	2	1189	2378
	60	16.0	2	400	800
	61	16.0	1	600	600

Resumo do aço

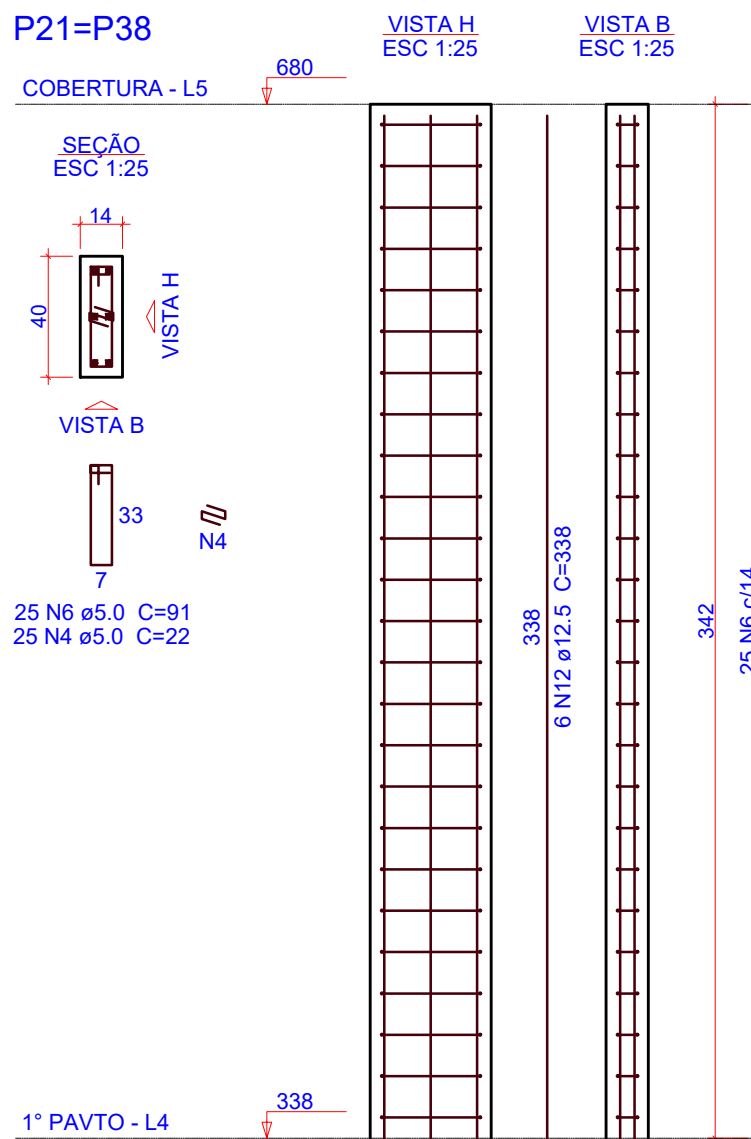
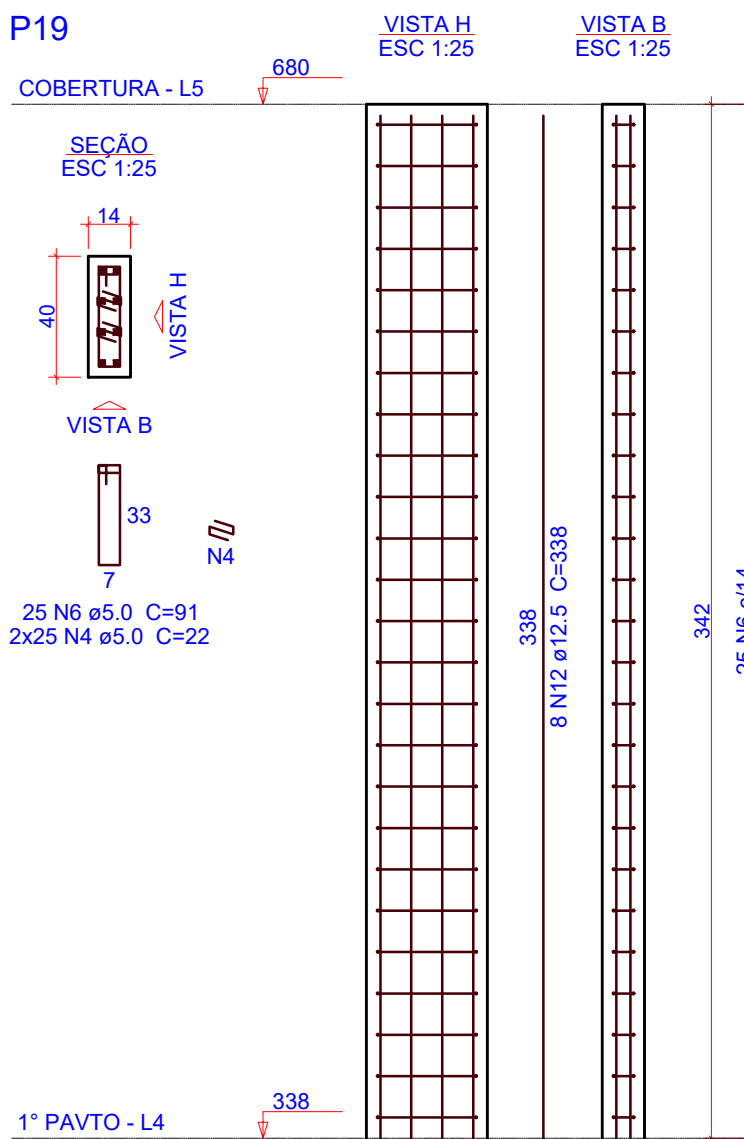
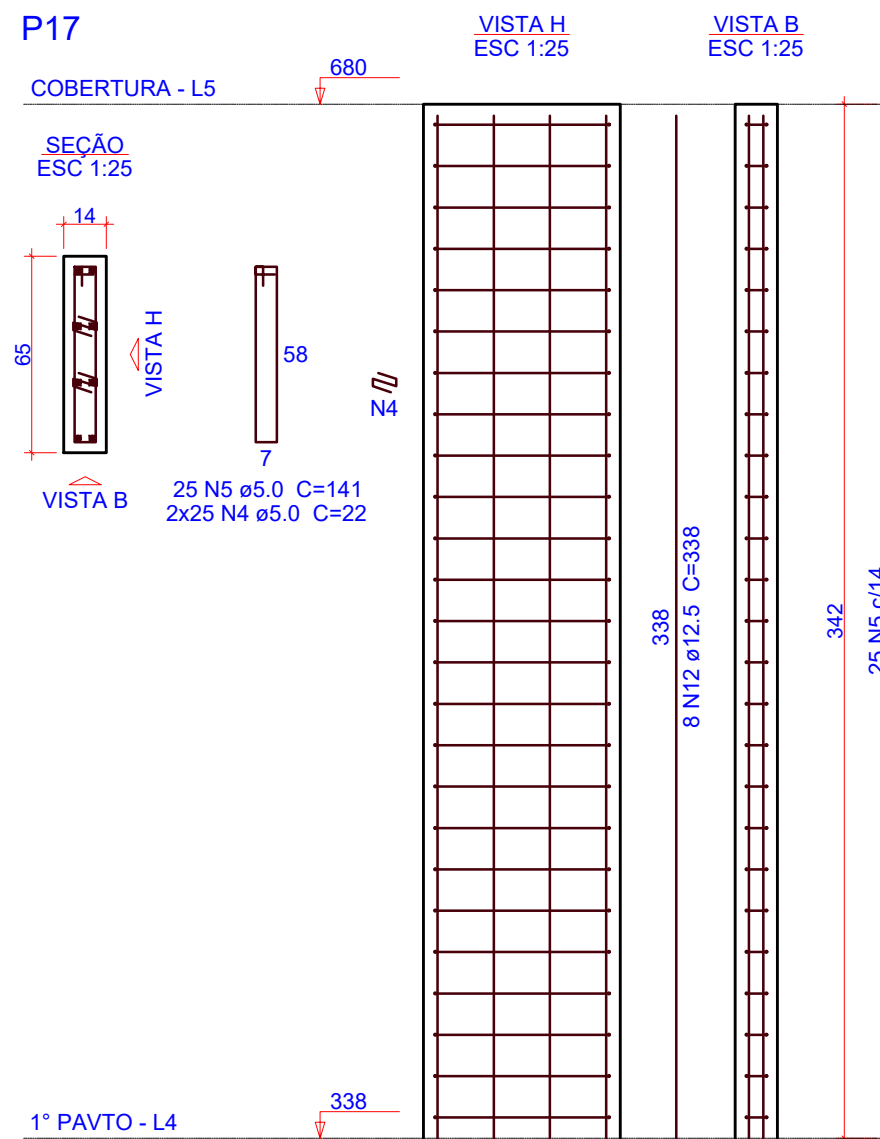
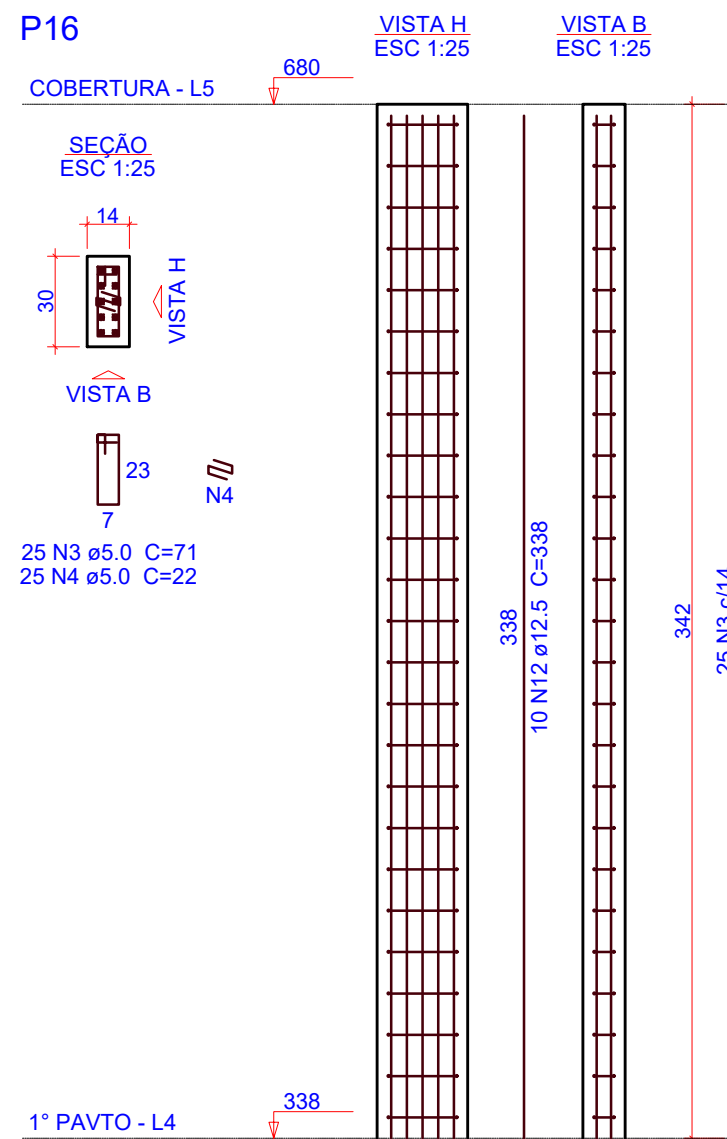
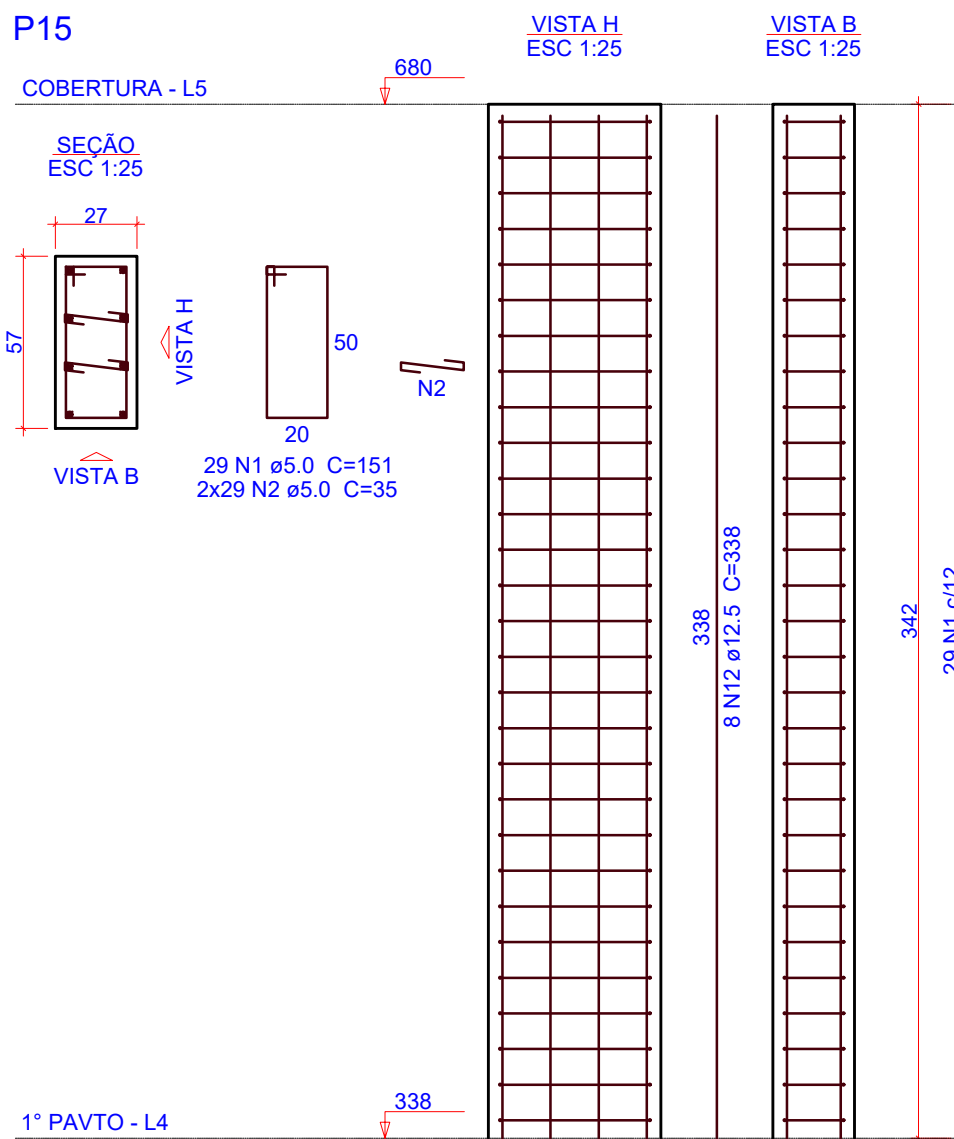
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	103.5	10	27.8
	8.0	21.3	2	9.2
	10.0	101.6	10	68.9
	12.5	470.1	44	498.1
	16.0	81.4	8	141.3
CA60	5.0	513.5	48	87.1
PESO TOTAL (kg)				
CA50	745.4			
CA60				

Volume de concreto (C-30) = 6.5 m³
Área de forma = 75.91 m²

1 DETALHAMENTO VIGAS - COBERTURA

ESCALA INDICADA

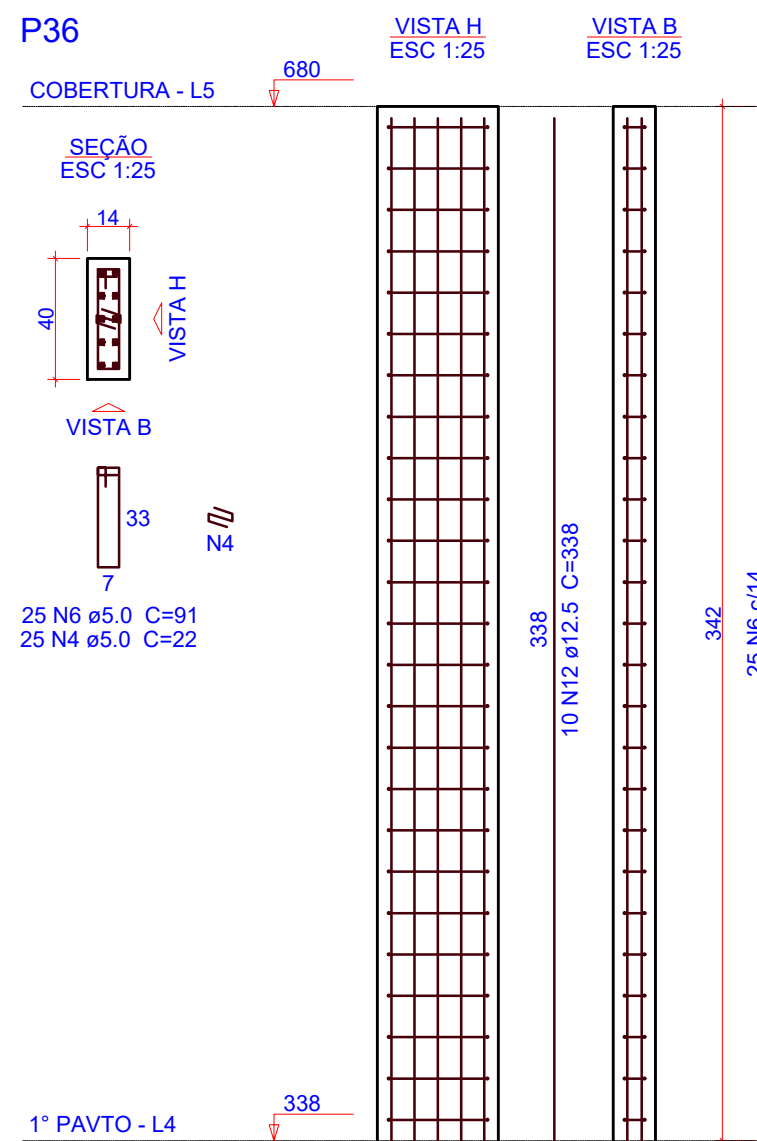
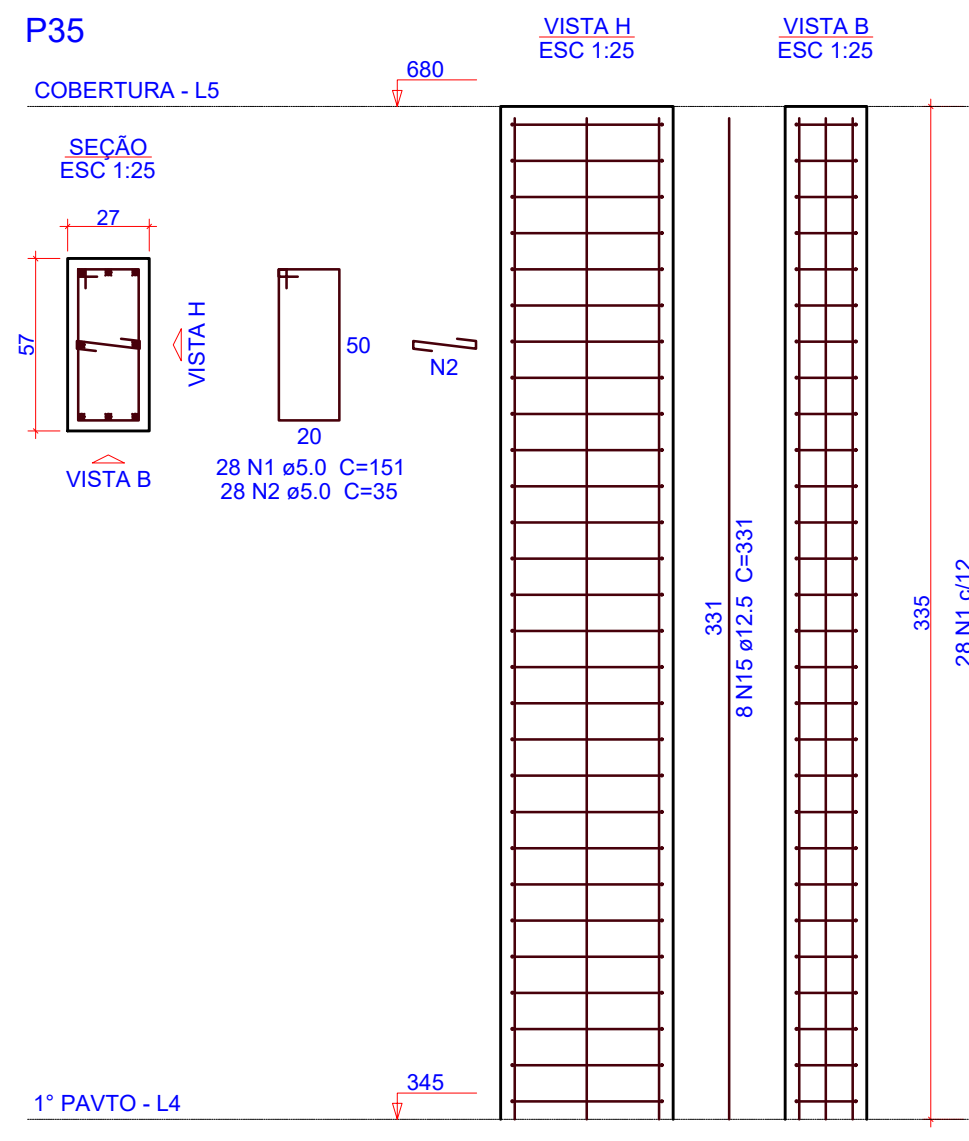
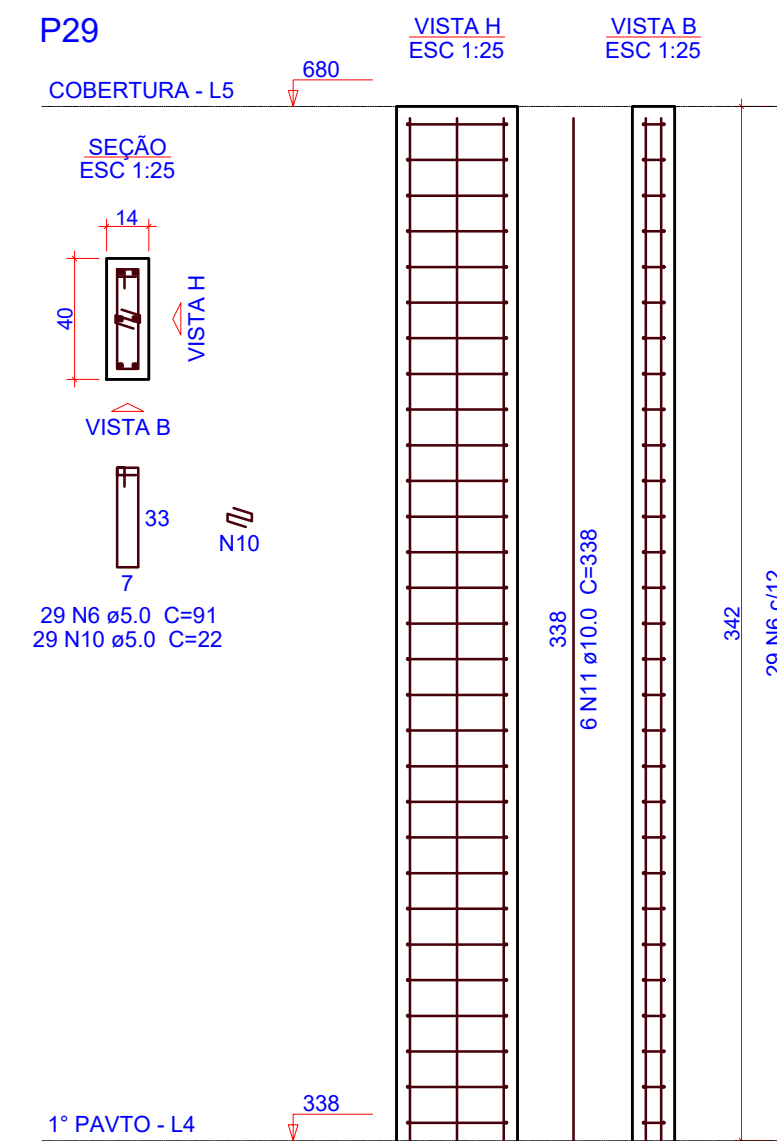
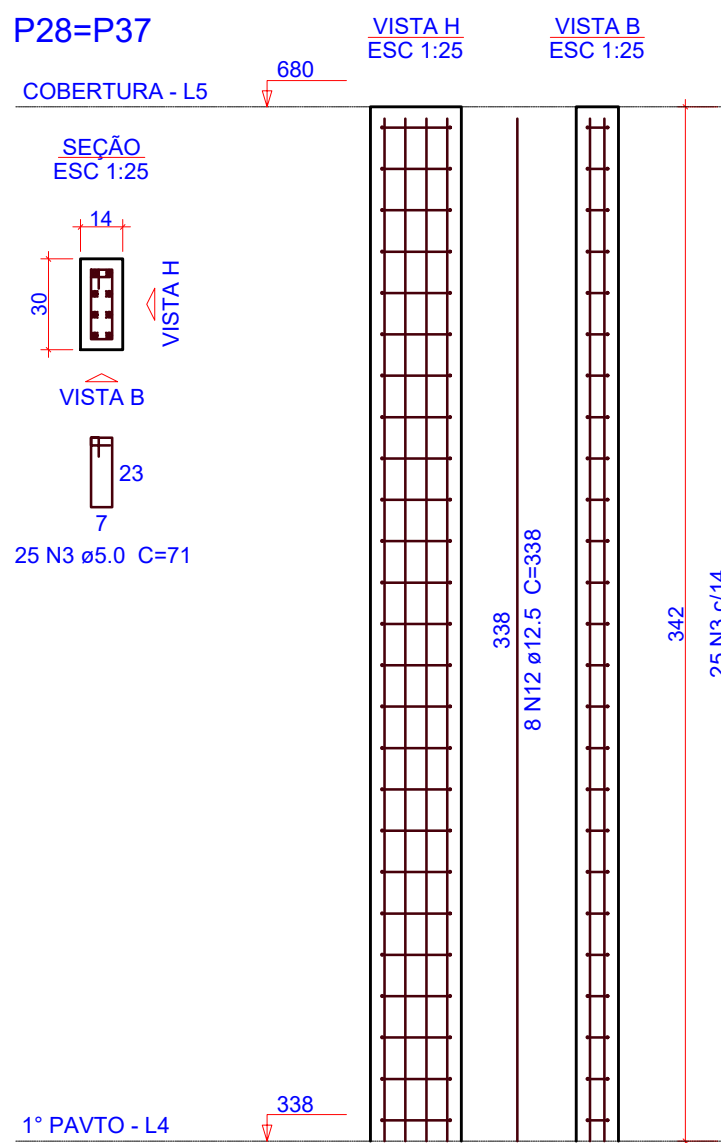
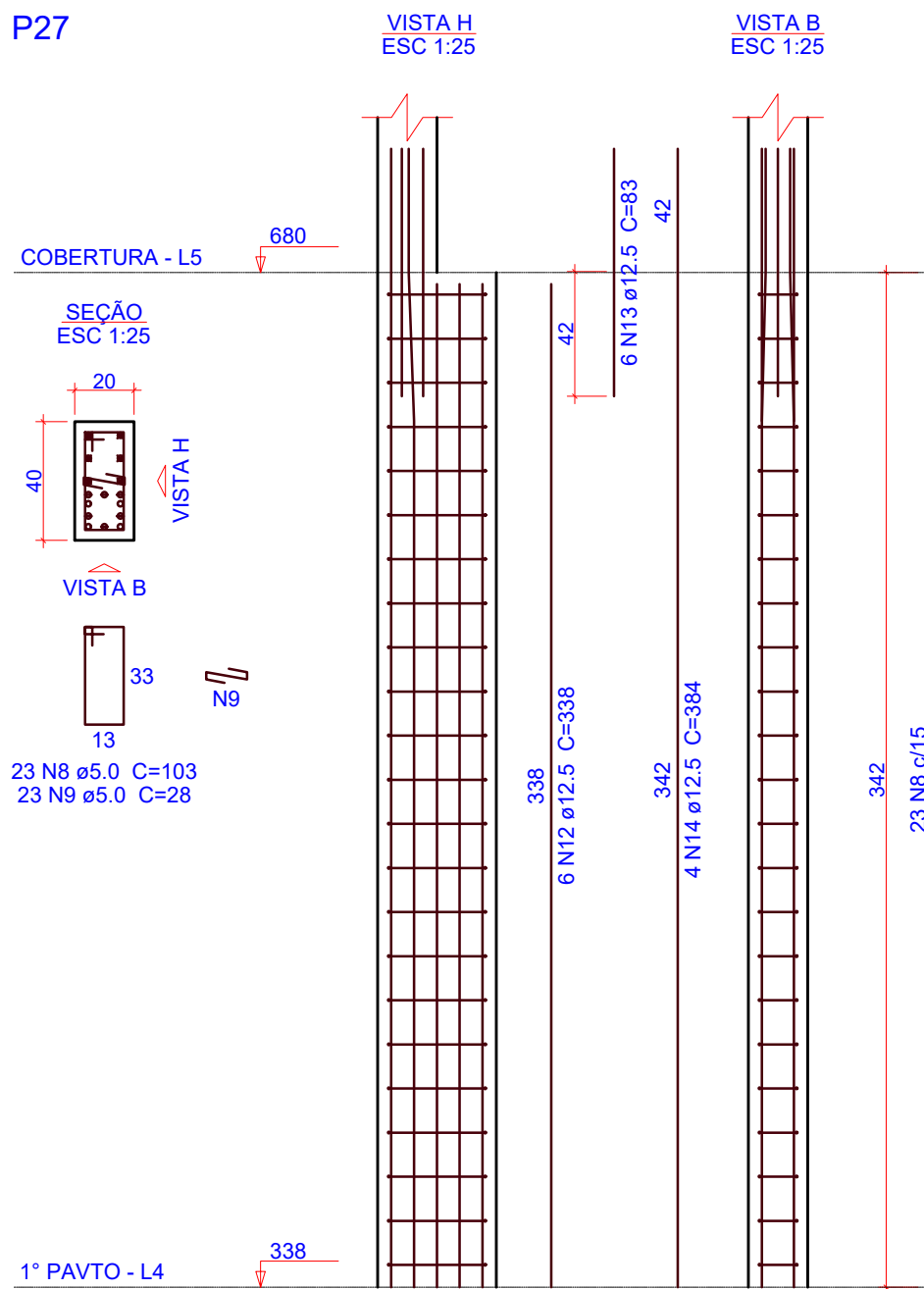
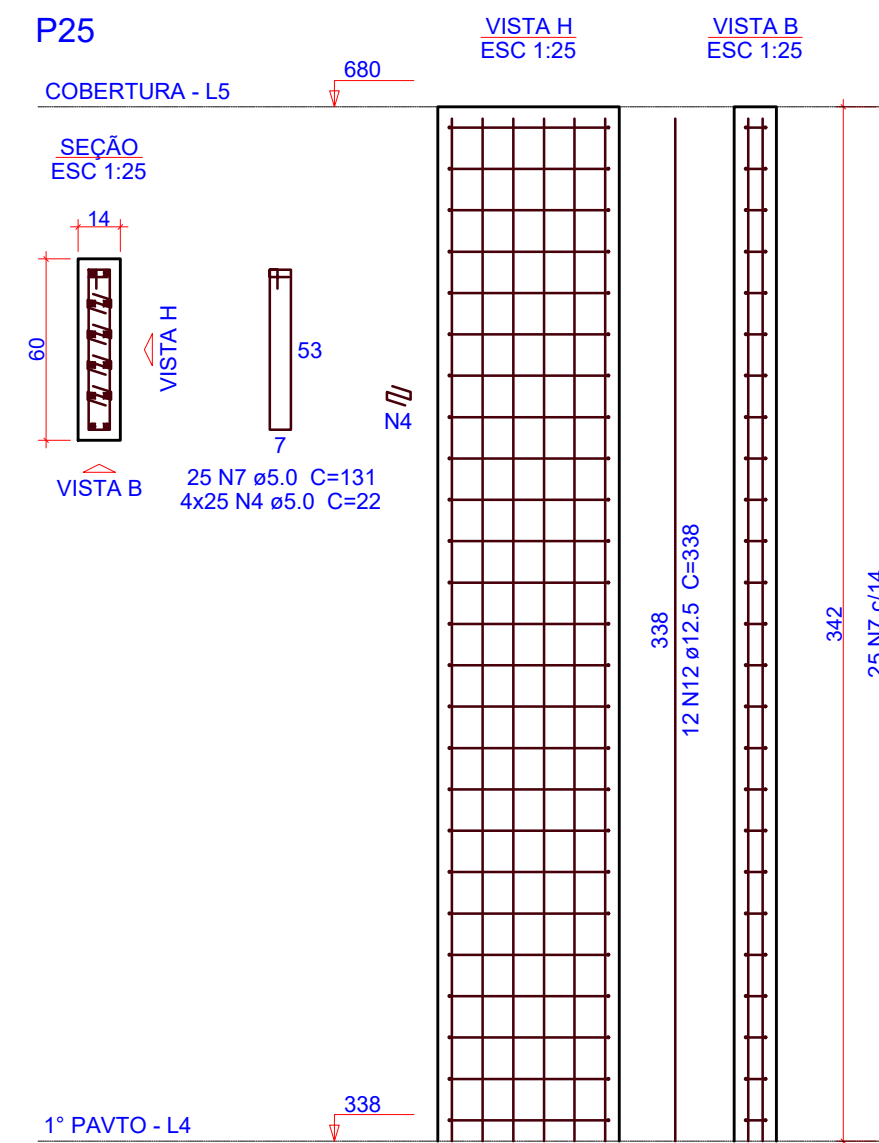
Nº	REVISÃO	DATA	CLIENTE
A	EMISSÃO INICIAL	15/12/2025	JULIANO CORBELLINI
			PORTO ALEGRE - RS
			RUA ALCEBIADES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			DETALHAMENTO VIGAS - COBERTURA
			PROJETO
			Engº Rafael Buss
			CREA/SC 133.073-9
			DESENHO
			MDO
			ESCALA INDICADA
			DATA
			TOP.
			FOLHA Nº
			EST-0250-A



Relação do aço					
AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	57	151	8607
	2	5.0	86	35	3010
	3	5.0	75	71	5325
	4	5.0	300	22	6600
	5	5.0	25	141	3525
	6	5.0	129	91	11739
	7	5.0	25	131	3275
	8	5.0	23	103	2369
	9	5.0	23	28	644
	10	5.0	29	22	638
CA50	11	10.0	6	338	2028
	12	12.5	90	338	30420
	13	12.5	6	83	498
	14	12.5	4	384	1536
	15	12.5	8	331	2648

Resumo do aço				
AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	10.0	20.3	2	13.8
	12.5	351.1	33	372
CA60	5.0	457.4	42	77.5
PESO TOTAL (kg)				
CA50	385.7			
CA60	77.5			

Volume de concreto (C-30) = 3.3 m³
Área de forma = 53.44 m²



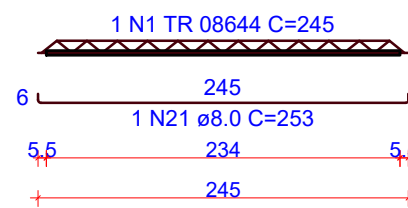
N°	REVISÃO	DATA	CLIENTE
A	EMISSÃO INICIAL	15/12/2025	JULIANO CORBELLINI
			PORTO ALEGRE - RS
			RUA ALCEBIÁDES ANTÔNIO DOS SANTOS- 356 - LOTE 207
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			DETALHAMENTO PILARES - COBERTURA
			PROJETO
			Engº Rafael Buss
			CREA/SC 133.073-9
			DESENHO
			MDO
			ESCALA INDICADA
			DATA 15/12/2025
			DATA TOP.
			FOLHA N°
			EST-0260-A



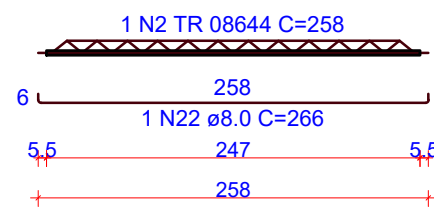
1 PLANTA DE VIGOTAS PRÉ-MOLDADAS - COBERTURA

ESCALA 1:50

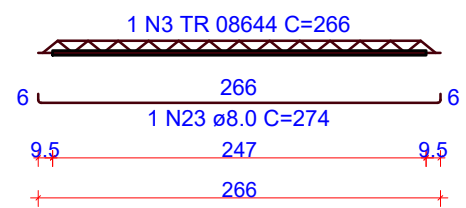
VT500a (1 unidades)
(L500)
ESC 1:50



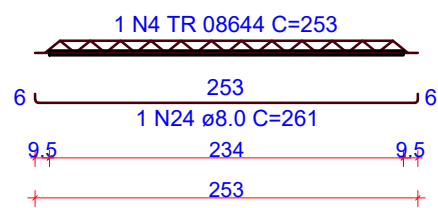
VT500b (11 unidades)
(L500)
ESC 1:50



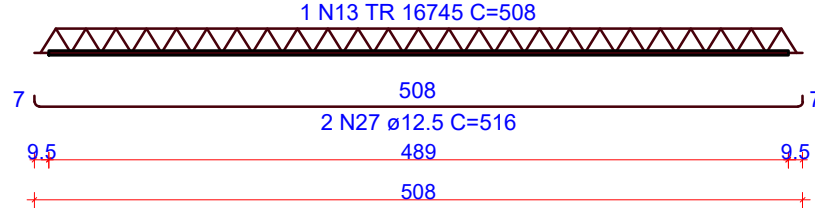
VT500c (10 unidades)
(L500)
ESC 1:50



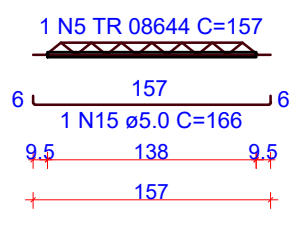
VT500d (1 unidades)
(L500)
ESC 1:50



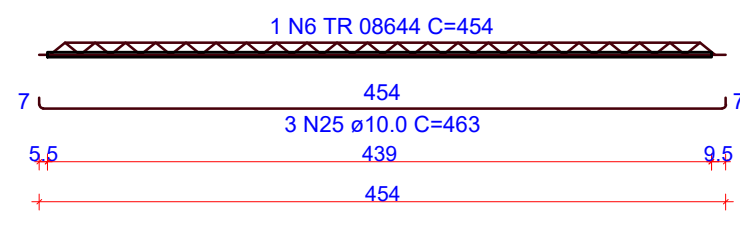
VT501a (10 unidades)
(L501)
ESC 1:50



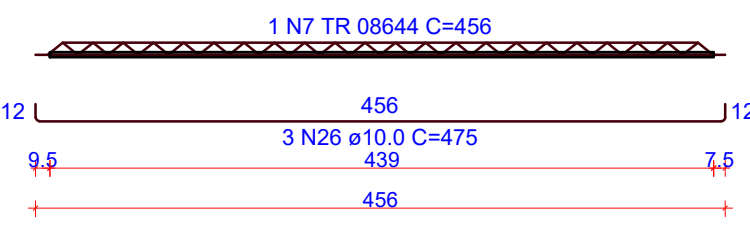
VT502a (3 unidades)
(L502)
ESC 1:50



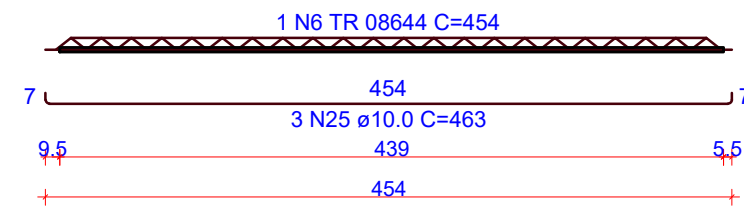
VT504a (5 unidades)
(L504)
ESC 1:50



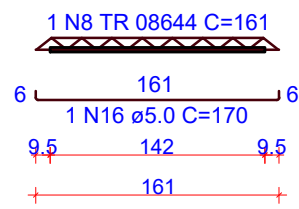
VT504b (3 unidades)
(L504)
ESC 1:50



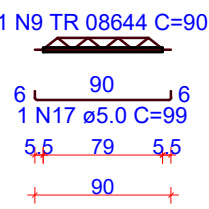
VT504c (2 unidades)
(L504)
ESC 1:50



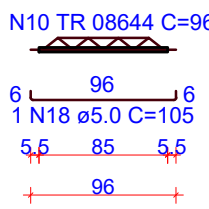
VT505a (8 unidades)
(L505)
ESC 1:50



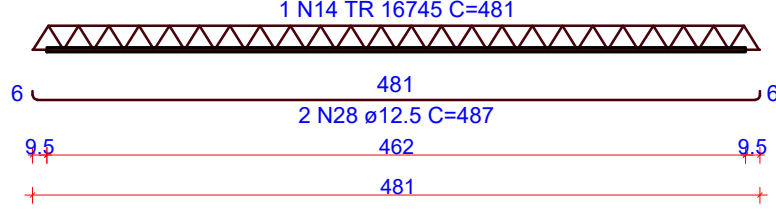
VT506a (1 unidades)
(L506)
ESC 1:50



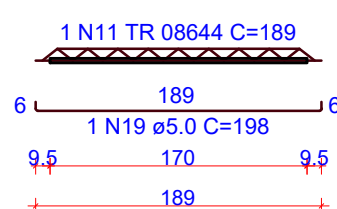
VT506b (7 unidades)
(L506)
ESC 1:50



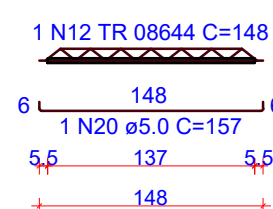
VT507a (14 unidades)
(L507)
ESC 1:50



VT509a (13 unidades)
(L509)
ESC 1:50



VT510a (11 unidades)
(L510)
ESC 1:50



2 DETALHAMENTO DE VIGOTAS PRÉ-MOLDADAS - COBERTURA

ESCALA 1:50

RELAÇÃO DO AÇO

VT500a	11xVT500b	10xVT500c
VT500d	10xVT501a	3xVT502a
5xVT504a	3xVT504b	2xVT504c
8xVT505a	VT505a	7xVT506b
14xVT507a	13xVT509a	11xVT510a

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	TR 08644	1	245	245
	2	TR 08644	11	258	2838
	3	TR 08644	10	266	2660
	4	TR 08644	1	253	253
	5	TR 08644	3	157	471
	6	TR 08644	7	454	3178
	7	TR 08644	3	456	1368
	8	TR 08644	8	161	1288
	9	TR 08644	1	90	90
	10	TR 08644	7	96	672
	11	TR 08644	13	189	2457
	12	TR 08644	11	148	1628
	13	TR 16745	10	508	5080
	14	TR 16745	14	481	6734
	15	5.0	3	166	498
	16	5.0	8	170	1360
	17	5.0	1	99	99
	18	5.0	7	105	735
	19	5.0	13	198	2574
	20	5.0	11	157	1727
CA50	21	8.0	1	253	253
	22	8.0	11	266	2926
	23	8.0	10	274	2740
	24	8.0	1	261	261
	25	10.0	21	463	9723
	26	10.0	9	475	4275
	27	12.5	20	516	10320
	28	12.5	28	487	13636

RESUMO DO AÇO

AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10% (Barras)	PESO + 10% (kg)
CA50	8.0	61.8	6	26.8
	10.0	140	13	94.9
	12.5	239.6	22	253.9
CA60	5.0	69.9	-	11.9
	TR 08644	171.5	-	138.6
	TR 16745	118.1	-	134.1

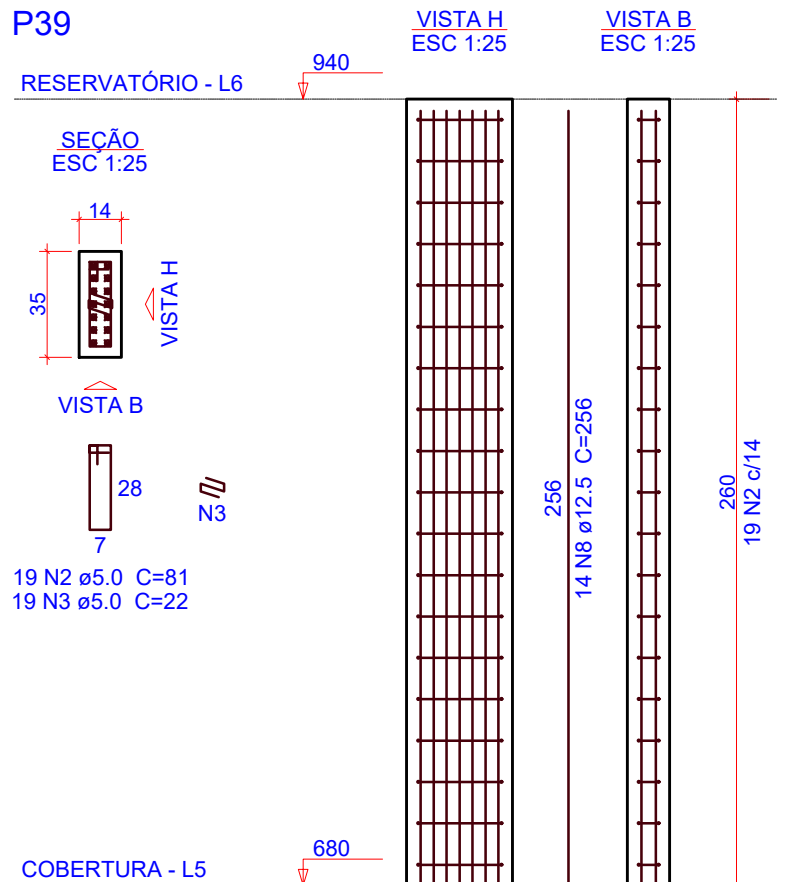
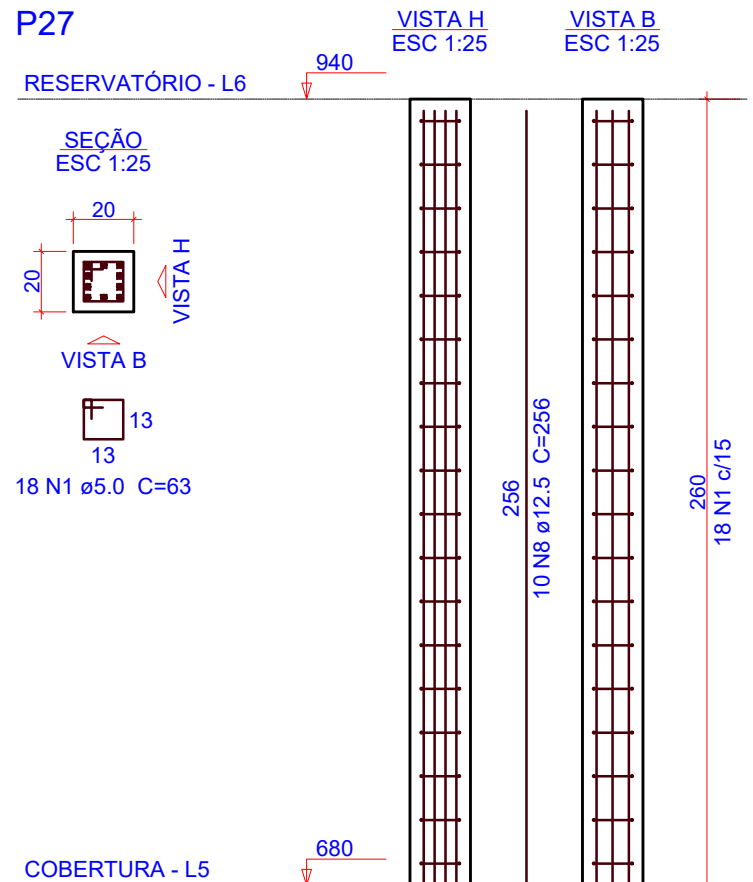
PESO TOTAL (kg)

CA50	375.6
CA60	284.6

Volume de concreto (C-30) = 0.00 m³

Área de forma = 0.00 m²

Nº	REVISÃO	DATA	CLIENTE
A	EMISSÃO INICIAL	15/12/2025	JULIANO CORBELLINI
			LOCALIDADE - MUNICÍPIO
			ENDEREÇO
			PROJETO ESTRUTURAL
			PROJETO ESTRUTURAL
			PLANTA DE ARMADURAS
			PLANTA E DETALHAMENTO VIGOTAS PRÉ-MOLDADAS - COBERTURA
			PROJETO
			Engº Rafael Buss
			CREA/SC
			133.073-9
			DESENHO
			MDO
			ESCALA
			INDICADA
			DATA
			15/12/2025
			DATA TOP.
			FOLHA Nº
			EST-0270-A



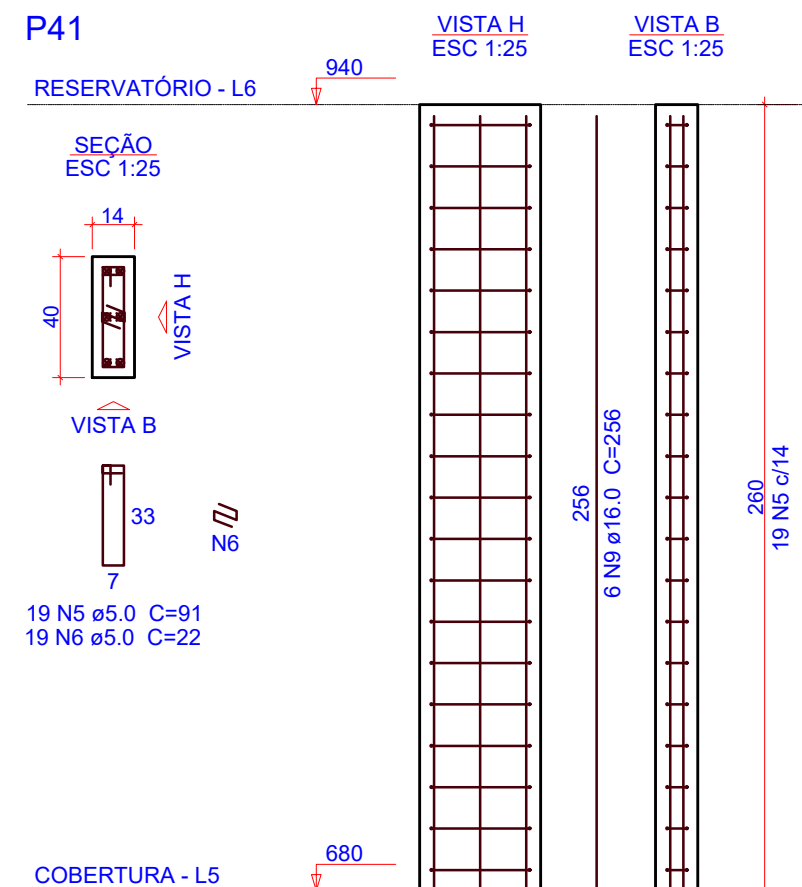
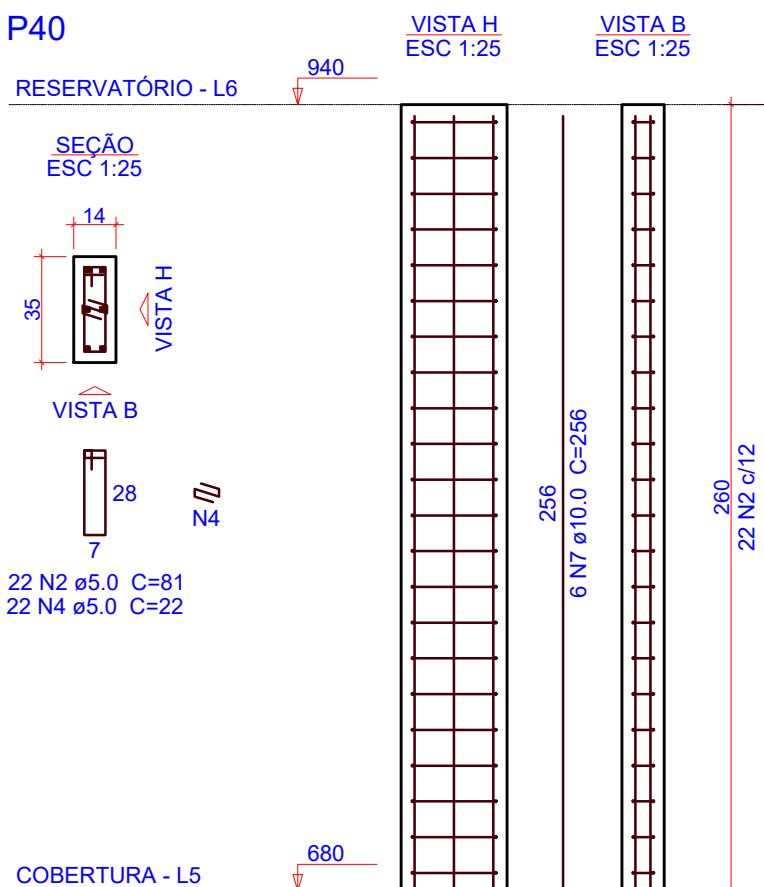
Relação do aço

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	18	63	1134
	2	5.0	41	81	3321
	3	5.0	19	22	418
	4	5.0	22	22	484
	5	5.0	19	91	1729
	6	5.0	19	22	418
CA50	7	10.0	6	256	1536
	8	12.5	24	256	6144
	9	16.0	6	256	1536

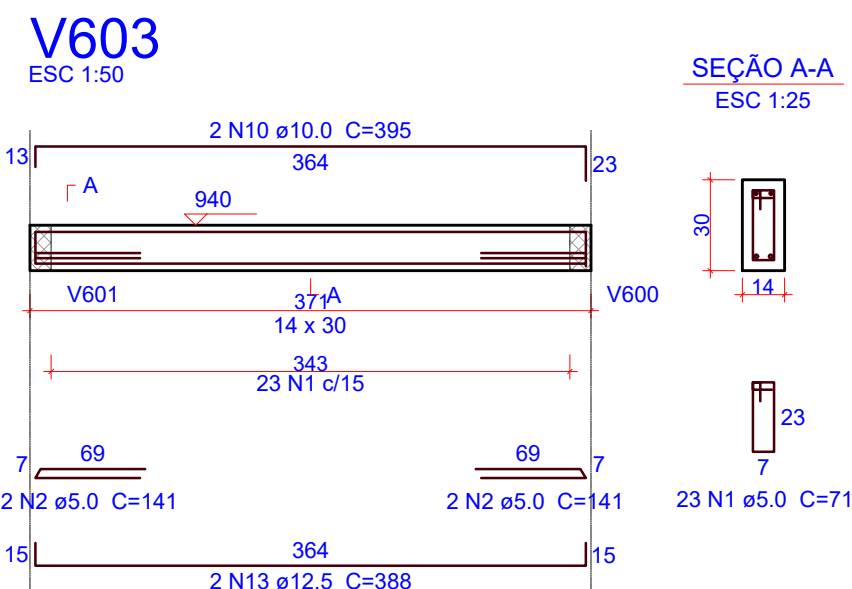
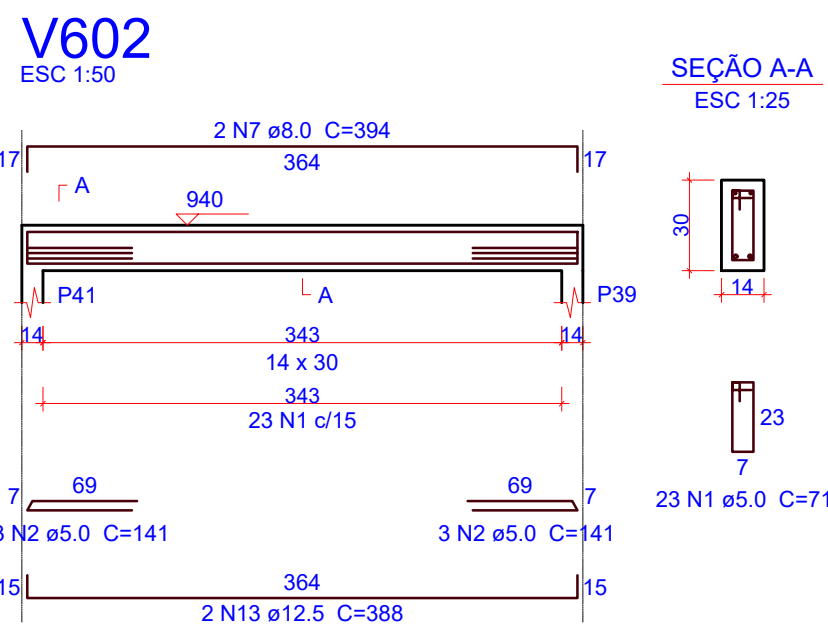
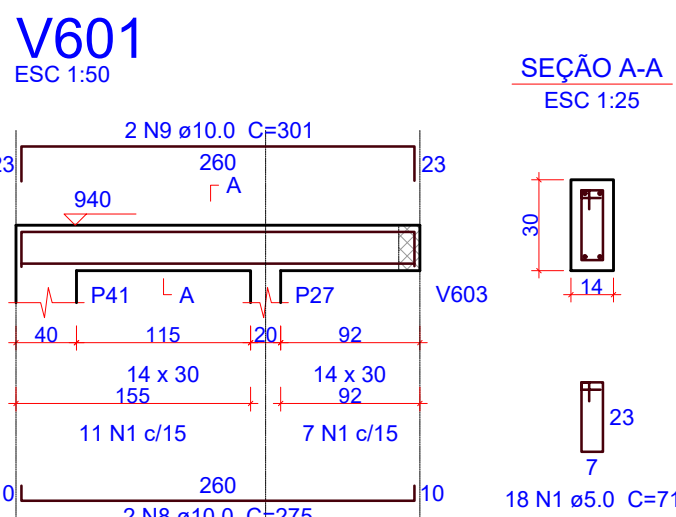
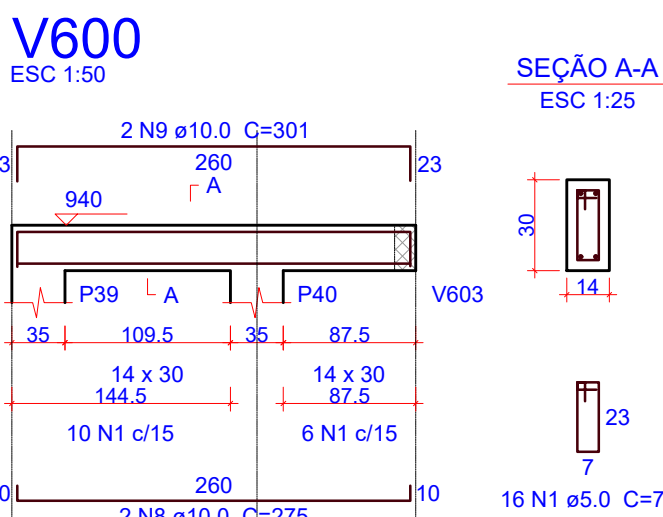
Resumo do aço

AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	10.0	15.4	2	10.4
	12.5	61.5	6	65.1
	16.0	15.4	2	26.7
CA60	5.0	75.1	7	12.7
PESO TOTAL (kg)				
CA50		102.2		
CA60		12.7		

Volume de concreto (C-30) = 0.5 m³
Área de forma = 9.98 m²

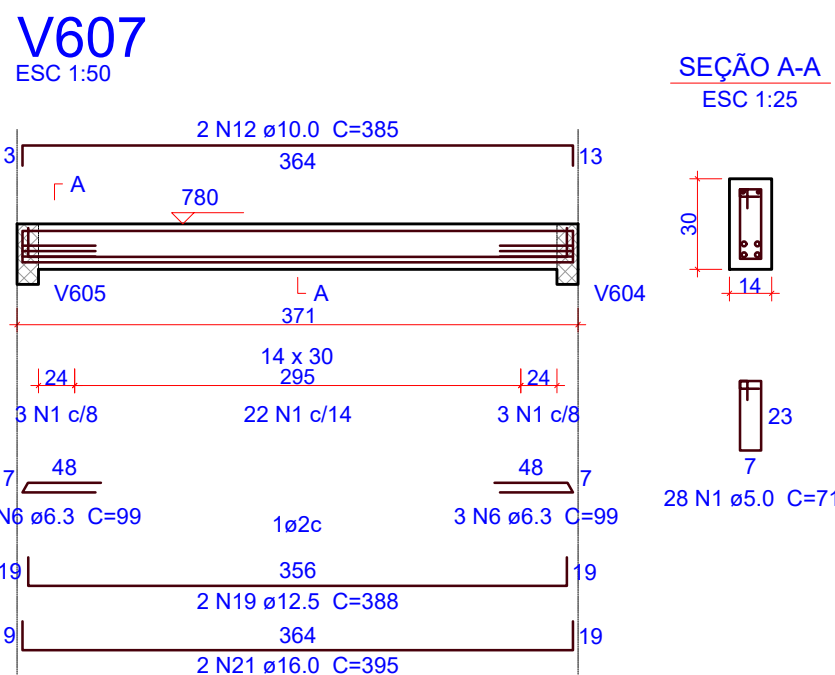
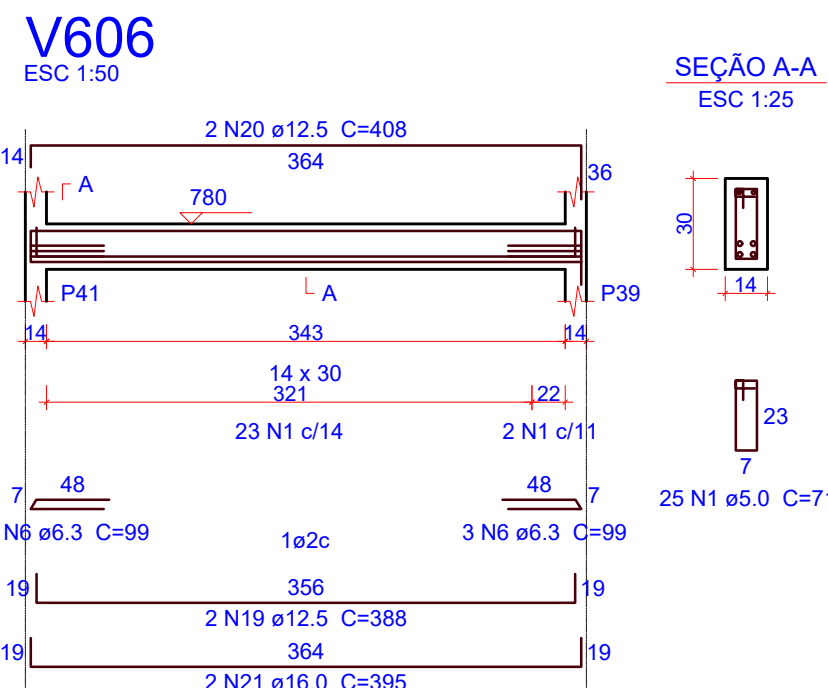
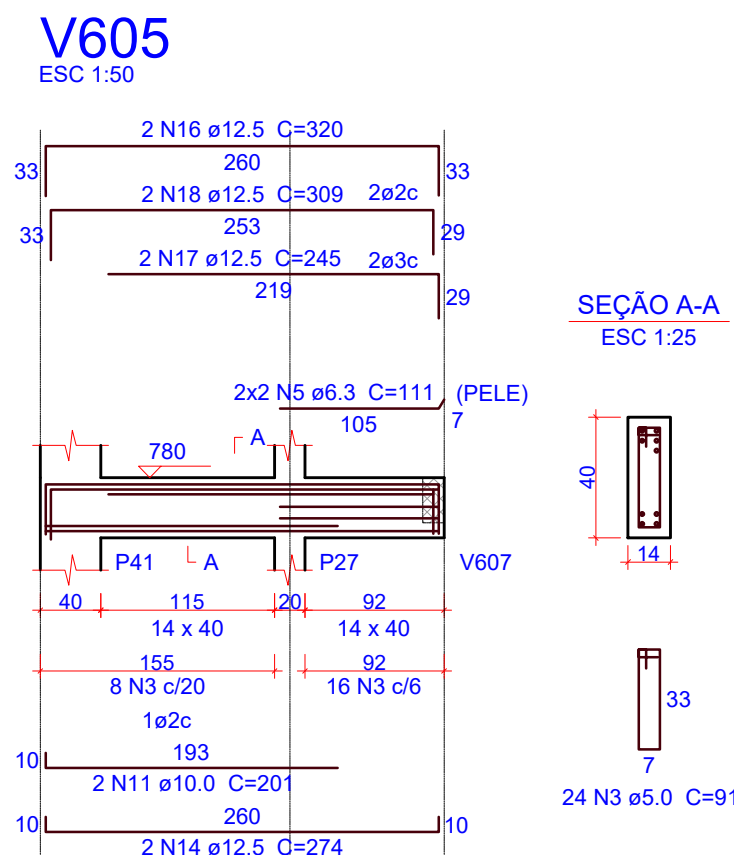
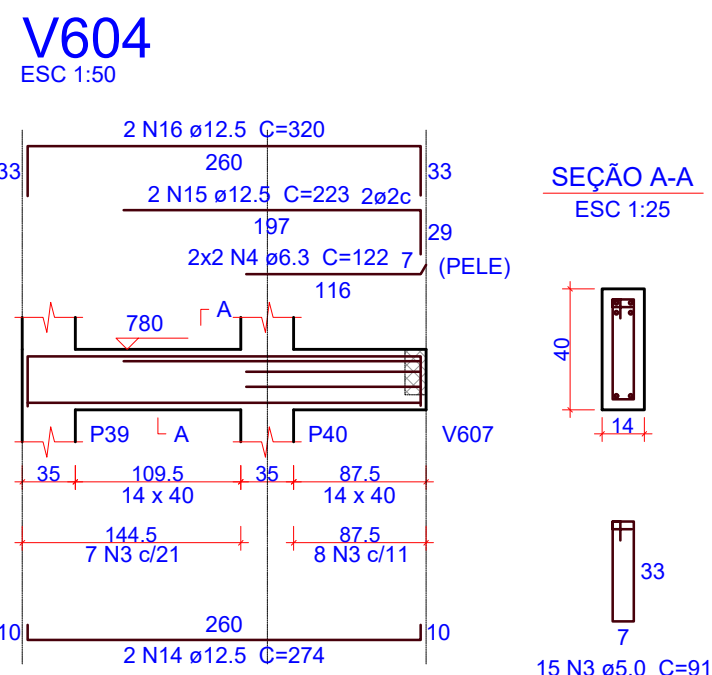


1 DETALHAMENTO PILARES - RESERVATÓRIO



Relação do aço

AÇO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	133	71	9443
	2	5.0	10	141	1410
	3	5.0	39	91	3549
	4	6.3	4	122	488
	5	6.3	4	111	444
	6	6.3	12	99	1188
CA50	7	8.0	2	394	788
	8	10.0	4	275	1100
	9	10.0	4	301	1204
	10	10.0	2	395	790
	11	10.0	2	201	402
	12	10.0	2	385	770
	13	12.5	4	388	1552
	14	12.5	4	274	1096
	15	12.5	2	223	446
	16	12.5	4	320	1280
	17	12.5	2	245	490
	18	12.5	2	309	618
	19	12.5	4	388	1552
	20	12.5	2	408	816
	21	16.0	4	395	1580



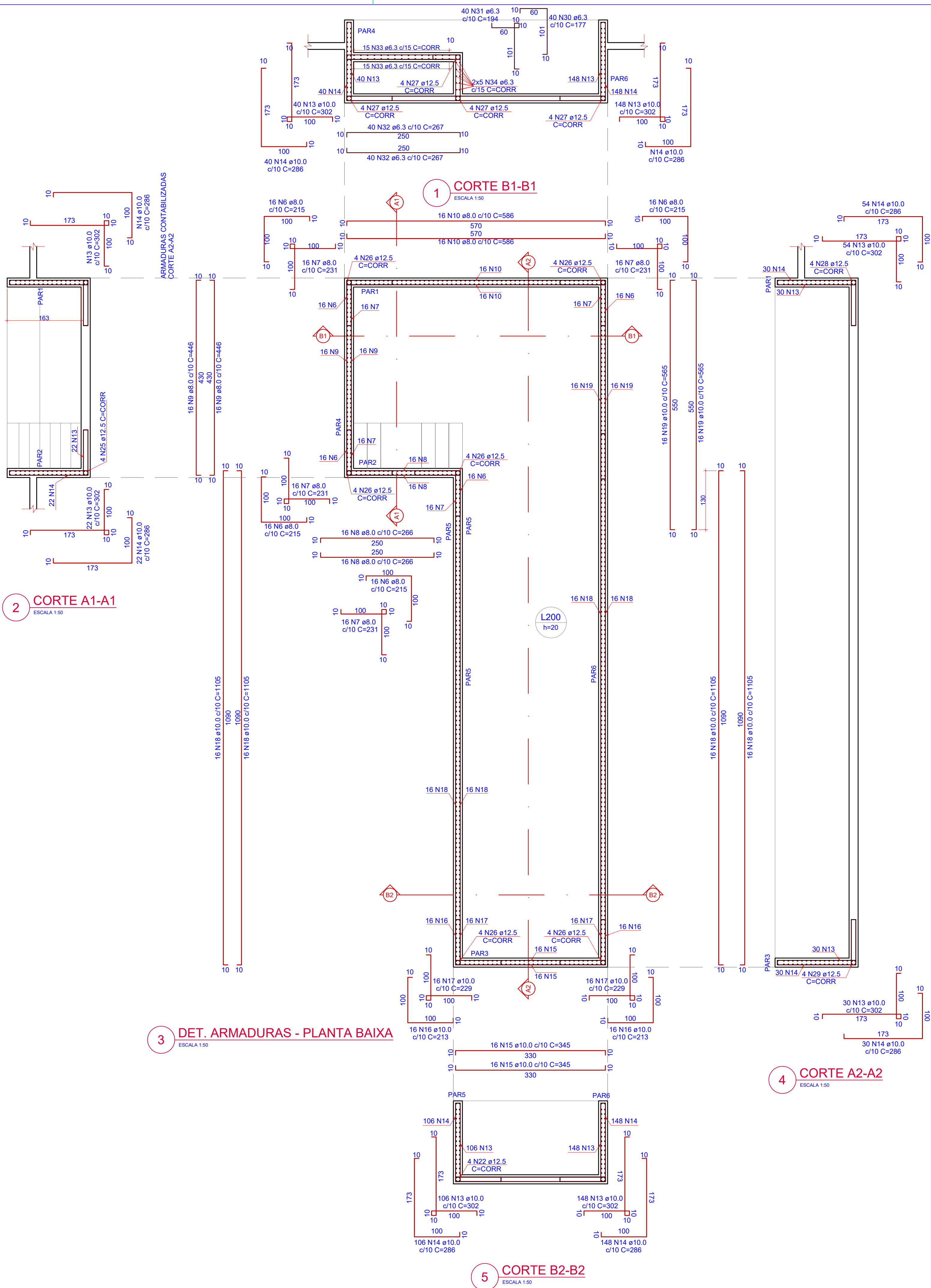
Resumo do aço

AÇO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	21.2	2	5.7
	8.0	7.9	1	3.4
	10.0	42.7	4	28.9
	12.5	78.5	8	83.2
	16.0	15.8	2	27.4
CA60	5.0	144.1	14	24.4
PESO TOTAL (kg)				
CA50		148.7		
CA60		24.4		

Volume de concreto (C-30) = 0.97 m³
Área de forma = 15.52 m²

2 DETALHAMENTO VIGAS - RESERVATÓRIO

Nº	REVISÃO	DATA	CLIENTE
A	EMISSÃO INICIAL	15/12/2025	JULIANO CORBELLINI
			LOCALIDADE - MUNICÍPIO
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			PLANTA DE ARMADURAS
			DETALHAMENTO PILARES E VIGAS - RESERVATÓRIO
			PROJETO
			Engº Rafael Buss
			CREA/SC
			133.073-9
			DESIGNO
			MUD.
			ESCALA INDICADA
			DATA
			15/12/2025
			FOLHA Nº
			EST-0280-A



Relação do aço

AO	N	DIAM (mm)	QUANT	C.UNIT (cm)	C.TOTAL (cm)
CA60	1	5.0	106	347	36782
	2	5.0	40	587	23480
	3	5.0	30	1199	35970
	4	5.0	30	416	12480
CA50	5	5.0	22	447	9834
	6	8.0	64	215	13760
	7	8.0	64	231	14784
	8	8.0	32	266	8512
	9	8.0	32	446	14272
	10	8.0	32	586	18752
	11	8.0	30	1198	35940
	12	8.0	30	415	12450
	13	10.0	400	302	120800
	14	10.0	400	286	114400
	15	10.0	138	345	47610
	16	10.0	32	213	6816
	17	10.0	32	229	7328
	18	10.0	64	1105	70720
	19	10.0	32	565	18080
	20	10.0	40	585	23400
	21	10.0	22	445	9790
	22	12.5	4	1090	4360
	23	12.5	4	1200	4800
	24	12.5	4	445	1780
	25	12.5	4	250	1000
	26	12.5	26	173	4498
	27	12.5	12	430	5160
	28	12.5	4	570	2280
	29	12.5	4	330	1320
	30	6.3	40	177	7080
	31	6.3	40	194	7760
	32	6.3	80	267	21360
	33	6.3	30	447	13410
	34	6.3	10	447	4470

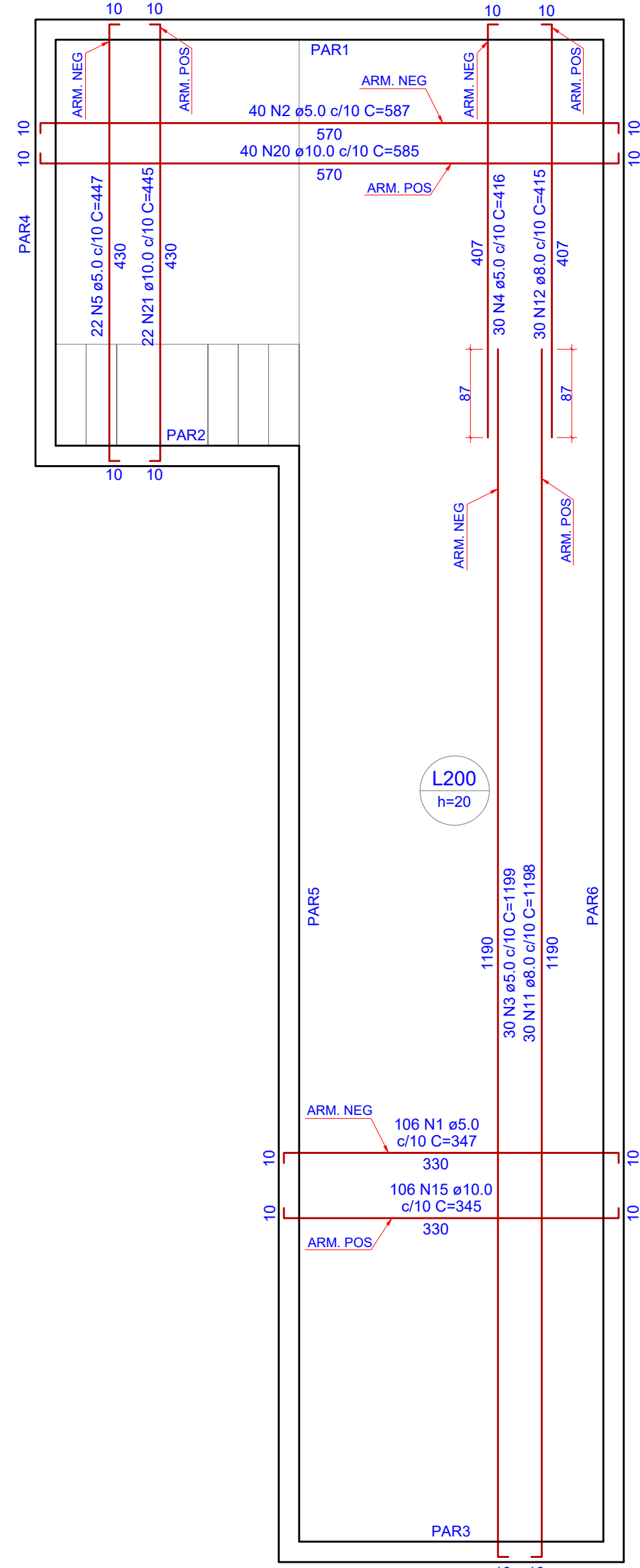
Resumo do aço

AO	DIAM (mm)	C.TOTAL (m)	QUANT + 10 % (Barras)	PESO + 10 % (kg)
CA50	6.3	540.8	50	145.7
	8.0	1184.7	109	514.2
	10.0	4189.5	385	2841.2
	12.5	251.9	22	266.9
CA60	5.0	1185.5	109	201

PESO TOTAL (kg)

CA50 3586
CA60 201

Volume de concreto (C-30) = 27.5 m³
Área de forma = 220 m²



ARMADURA POSITIVA E NEGATIVA DA LAJE - FUNDO PISCINA

Nº	REVISÃO	DATA	CLIENTE
A	EMIÇÃO INICIAL	15/12/2025	JULIANO CORBELLINI
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			PLANTA DE ARMADURAS
			DETALHAMENTO PISCINA
			PROJETO
			DESIGNO
			DATA
			FOLHA Nº
			EST-0290-A